Vector Bundles and K-Theory

— *EYNTKA* Series —

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Contents

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

K-theory begins with a deceptively simple idea: rather than study a space X directly, study the vector bundles that can be laid over it. A vector bundle is a continuously varying family of vector spaces parametrized by X, and the collection of all of them, taken up to isomorphism, already remembers a great deal about the shape of X. Under direct sum these bundles form a commutative monoid; completing that monoid to a group, exactly as the integers are built from the natural numbers, produces the Grothendieck group $K^0(X)$. This single construction turns the geometry of bundles into an algebraic invariant that is computable, functorial, and sensitive to phenomena that ordinary cohomology cannot detect.

What makes topological K-theory more than a convenient bookkeeping device is *Bott periodicity*: the groups $K^n(X)$ repeat with period two, and this periodicity promotes K^0 and K^1 into a full generalized cohomology theory, one satisfying every Eilenberg-Steenrod axiom except the dimension axiom ([14]). A reader who knows ordinary cohomology will find in K-theory a close structural parallel, with its own long exact sequences, products, and computational machinery, but with answers ordinary cohomology cannot see. The same circle of ideas produces the *characteristic classes* (Stiefel-Whitney, Chern, Euler, Pontryagin), cohomology classes attached to a bundle that obstruct sections, detect orientability, and measure twisting, turning qualitative geometry into numbers.

This book develops the topological theory from vector bundles through Bott periodicity to characteristic classes. It is the topological member of a family. Its algebraic counterpart, where rings replace spaces and projective modules replace bundles, is [3]; its operator-algebraic counterpart, where the Serre-Swan theorem reincarnates bundles as projections over $C(X)$, is [4]. The non-periodicity of the algebraic theory and the 2-periodicity shared by the topological and operator theories are two sides of one story, and the reader is encouraged to read across the three.

This book assumes the cohomology, classifying spaces, and generalized-cohomology framework of [14], and uses the connections and curvature of [16], where characteristic classes are computed by the Chern-Weil theory of the Riemannian-geometry volume [7]. Beyond these, the prerequisites are point-set topology and linear algebra.

1

Vector Bundles

1.0.1 Convention: Paracompact Includes Hausdorff Every base space of a bundle in this book is assumed *paracompact*, and throughout the word means *Hausdorff*, together with the condition that every open cover admits a locally finite open refinement. Building the separation axiom into the word is the convention of both standard references for this material, [13, §5.8] and the definition following Proposition 1.2 of [10], and it is not a formality. The constructions this book runs on paracompact bases are partitions of unity: [14, Inner Products on a Vector Bundle] builds an inner product with one and [14, Classification of Vector Bundles over a Paracompact Base] builds a classifying map with one. Partitions of unity subordinate to an arbitrary open cover exist on a paracompact space because it is normal, and normality is what fails without the separation axiom ([8, Paracompact Hausdorff Spaces Are Normal]).

Topology [8] defines paracompactness by the covering condition alone, so a result cited from there is applied here with the Hausdorff hypothesis given separately at the point of use. Manifolds are Hausdorff under the same convention. Schemes are not, and never appear here as bundle bases[a]

[a]See [2] for why only trivial schemes are Hausdorff; the scheme-theoretic side of K-theory is [3].

1.1 Fiber Bundles, Principal Bundles, Vector Bundles

Definition 1.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \to B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F, or in particular
$$\pi^{-1}(b) \cong b \times F$$
2. (local trivialization)For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes
$$\begin{tikzcd} \pi^{-1}(U) \arrow[r, "\sim"] \arrow[d, "\pi"'] & U \times F \arrow[dl, "\pi_1"] \\ U & \end{tikzcd}$$

When clear from context, the fiber bundle is usually denoted as E.

Definition 1.1.2: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle complex vector bundle.*

One clause of definition 1.1.1 is relaxed for vector bundles: no single model fiber is required over all of B, only over each trivializing open set, so the *rank* $b \mapsto \dim \pi^{-1}(b)$ is locally constant and may differ over different connected components of B. Over a connected base the two readings agree.

Definition 1.1.3: Principal Bundle

If $F = G$ is a topological group and $P \to X$ is a fiber bundle over X where there is a continuous right group action $P \times G \to P$ that preserves the fibers of P and is free and transitive on each fiber, then it is called a *principal G-bundle* (or *principal bundle* if G is clear from context).

Each fiber is called a *G-torsor* (which is a space where the action $G \curvearrowright F_x$ is transitive with trivial stabilizers, hence "is" the group G but does not intrinsically have a notion of identity element)

Note that these conditions make each fiber homeomorphic to G. As a consequence the Orbit space P/G is homeomorphic to X.

Example 1.1.4: Fiber Bundle

1. Given a topological B and fiber F, then $E = B \times F$ is the *trivial F-fiber bundle* or simply the *trivial bundle*. Such a bundle has a *global trivialization*, namely by letting $U = B$:

$$\begin{tikzcd} B \times F \arrow[r, "\mathrm{id}"] \arrow[d, "\pi"'] & B \times F \arrow[dl, "\pi_1"] \\ B & \end{tikzcd}$$

2. If M is a n-manifold, then there are various interesting vector bundles defined on it:
   (a) Tangent bundle TM
   (b) cotangent bundle T^*M
   (c) differential forms $\Omega^k(M)$ for $k \leq n$

   If M carries more structure, for instance a Riemannian metric, a Kähler structure, or a complex structure, then further vector bundles record that structure: normal bundles, holomorphic bundles, spin bundles, flat bundles.

3. Let $p : E \to B$ be a fiber bundles and $A \subseteq B$. Then $p : p^{-1}(A) \to A$ is certainly a fiber bundle. This is called the *restriction of E over A*. The same goes through for vector-bundles.

4. Subbundles have complements: if $E \to X$ is a bundle (on a paracompact topological space) and F is a subbundle, then there exists a bundle $F^\perp$ such that $E = F \oplus F^\perp$.

5. The tensor product is well-defined: given vector bundles $E \to X$ and $F \to X$, there exists a vector bundle $E \otimes F \to X$ whose fibers over x are $E_x \otimes F_x$. The bundles built from it follow, such as the *determinant bundle* $\det(E) = \Lambda^n E$ for E of rank n.

6. Let $E = I \times \mathbb{R}/\sim$ where $(0,t) \sim (1,-t)$. Then the projection map $I \times \mathbb{R} \to I$ produces a quotient $p : E \to S^1$ known as the Möbius bundle

7. Let TS^n be the tangent bundle of S^n. Then notice that this can be taken as a sub-bundle of $T\mathbb{R}^{n+1}$.

8. Another important bundle on S^n is the normal bundle, the line bundle whose fiber at $p \in S^n$ is the one-dimensional space spanned by the line perpendicular to S^n at p. Writing NS^n for it, example 1.1.7 identifies NS^n with the trivial line bundle, and $TS^n \oplus NS^n$ is then trivial.

9. Let $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$ be real and complex projective space respectively. Then there is a *canonical line bundle* $p : E \to \mathbb{R}\mathrm{P}^n$ and $p : E \to \mathbb{C}\mathrm{P}^n$ where E is the subspace of $\mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1}$ consisting of pairs (ℓ, v) with $v \in \ell$, and $p(\ell, v) = \ell$. Note that the canonical line bundle over $\mathbb{R}\mathrm{P}^n$ has an orthogonal complement

$$E^\perp = \left\{ (\ell, v) \in \mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1} \;\middle|\; v \perp \ell \right\}$$

10. An important bundle for the future is the infinite real/complex projective space $\mathbb{R}\mathrm{P}^\infty$ and $\mathbb{C}\mathrm{P}^\infty$ taking by $\varinjlim_n \mathbb{R}\mathrm{P}^n$ and $\varinjlim_n \mathbb{C}\mathrm{P}^n$.

11. The Grassmann manifold $G_k(\mathbb{R}^n)$ was shown in [16] to be the space of all k-dimensional planes through the origin of $\mathbb{R}^n$.

Definition 1.1.5: Morphism of Fiber Bundles

If E_1, E_2 are two vector bundles, $\varphi : E_1 \to E_2$ is a vector-bundle morphism is a continuous map where this diagram commutes:

$$\begin{tikzcd} E_1 \arrow[rr, "\varphi"] \arrow[dr] & & E_2 \arrow[dl] \\ & B & \end{tikzcd}$$

and it respects local trivialization; this includes a linear homomorphism on each fiber. An isomorphism would have φ be a homeomorphism and is a linear isomorphism on each fiber.

For each dimension n, let $\mathrm{Vect}^n_k(B)$ be the isomorphism classes of vector bundles over B (with field k).

For a topological group G, let $b_G(B)$ be the isomorphism classes of principal G-bundles over B.

When $k = \mathbb{R}$ the subscript is often dropped, writing $\mathrm{Vect}^n(B)$. Note that $\mathrm{Vect}^n_k(B)$ and $b_G(B)$ are contravariant functors by applying the pullback map, for example:

$$f : X \to Y \qquad \Rightarrow \qquad f^* : b_G(Y) \to b_G(X)$$

Lemma 1.1.6: Morphism is Fiber-local

Let $p_1 : E_1 \to B, p_2 : E_2 \to B$ be two vector bundles over B and $\varphi : E_1 \to E_2$ a continuous map. Then φ is an isomorphism if and only if it takes each fiber $p_1^{-1}(b)$ to the corresponding fiber $p_2^{-1}(b)$ via a linear isomorphism.

***Proof*:**
If φ is an isomorphism of vector bundles then it commutes with the projections and is linear on each fiber, and the same holds of its inverse, so φ carries $p_1^{-1}(b)$ onto $p_2^{-1}(b)$ by a linear isomorphism.

Conversely, suppose φ carries each fiber onto the corresponding fiber by a linear isomorphism. Then φ is a bijection, and what remains is continuity of φ^{-1}, which may be checked locally on the base. Fix $b \in B$ and choose a neighborhood U over which both bundles are trivial, with trivializations $h_i : p_i^{-1}(U) \to U \times k^n$. In these coordinates $h_2 \varphi h_1^{-1}$ has the form $(u, v) \mapsto (u, A(u)v)$, where the j-th column of $A(u)$ is the image of the j-th standard basis vector and therefore depends continuously on u. By hypothesis $A(u)$ is invertible for every u, so A takes values in $\mathrm{GL}_n(k)$. Cramer's rule writes each entry of $A(u)^{-1}$ as a polynomial in the entries of $A(u)$ divided by $\det A(u)$, and $\det A$ is continuous and nowhere zero on U, so $u \mapsto A(u)^{-1}$ is continuous. Hence $(u, w) \mapsto (u, A(u)^{-1}w)$ is continuous, and it is $h_1 \varphi^{-1} h_2^{-1}$, so φ^{-1} is continuous over U. Any two of these local inverses agree on the overlap, both being the inverse of the same bijection, so φ^{-1} is continuous on all of E_2, as we sought to show.

Example 1.1.7: Morphism of Bundles

The normal bundle NS^n, as a subbundle of the trivial bundle $S^n \times \mathbb{R}^{n+1}$, is isomorphic to $S^n \times \mathbb{R}$ via the map:

$$(x, tx) \to (x, t)$$

Only S^0, S^1, S^3, S^7 have trivial tangent bundles, which corresponds to real division algebras existing only in dimensions $1, 2, 4, 8$ (chapter 4). For S^1 the trivialization is written down directly:

$$\varphi : TS^1 \to S^1 \times \mathbb{R} \qquad (e^{i\theta}, ite^{i\theta}) \mapsto (e^{i\theta}, t)$$

Another important example is that the Möbius bundle is isomorphic to the canonical line bundle over $\mathbb{RP}^1$ (note that $\mathbb{RP}^1 \cong S^1$).

Definition 1.1.8: Section

Let $p : E \to B$ be a fiber bundle. Then a *section* is a continuous map $s : B \to E$ such that $p \circ s = \mathrm{id}$.

Sections of a vector bundle carry more structure than sections of a general fiber bundle, because every fiber has a distinguished point.

Definition 1.1.9: Zero Section

Let $p : E \to B$ be a vector bundle. Then the *zero section* is the section $z : B \to E$ given pointwise by $z(p) = 0 \in p^{-1}(b)$.

As each vector-bundle morphism on fibers is linear, the zero section is taken to the zero section. This can be taken advantage of: for example the Möbius bundle is not isomorphic to $S^1 \times \mathbb{R}$ since the complement of the zero section is connected in the Möbius bundle but not for the product bundle. The other type of section is a *non-vanishing section* where no elements of the section are zero. Famously, by the Hairy-ball theorem (or Hedgehog theorem), there does not exist a section of TS^2 (i.e a vector-field) that is everywhere nonzero (see [14, Hairy ball theorem]). Example 3.2.4 shows that S^n carries a nonvanishing vector field if and only if n is odd, so the tangent bundle of S^n cannot be trivial for $n > 0$ even. Forced zeros of a section are what the Euler class measures, and that is the subject of section 3.2.

Lemma 1.1.10: Characterizing Trivial Bundles using Sections

Let $p : E \to B$ be an n-dimensional vector bundle. Then p is trivial if and only if it has n sections $s_1, \ldots, s_n$ such that $s_1(b), \ldots, s_n(b)$ are linearly independent on each fiber $p^{-1}(b)$.

***Proof*:**

A trivial n-dimensional vector bundle certainly respects these properties. Conversely, given such sections $s_1, \ldots, s_n$, define:

$$h : B \times \mathbb{R}^n \to E \qquad h(b, t_1, \ldots, t_n) = \sum_i t_i s_i(b)$$

This is a linear isomorphism on each fiber, hence by lemma 1.1.6 it is an isomorphism, completing the proof.

Note that another way of characterizing a trivial real bundle is to say there is a projection map $E \to \mathbb{R}^n$ that is a linear isomorphism on each fiber.

The lemma shows that S^3 has a trivial tangent bundle, by way of the quaternions. Every quaternion $z \in \mathbb{H}$ has the form:

$$z = x_1 + ix_2 + jx_3 + kx_4$$
$$i^2 = j^2 = k^2 = -1,\ ij = k,\ jk = i, ki = j, ji = -k, kj = -i, ik = -j$$

By identifying $\mathbb{H}$ with $\mathbb{R}^4$ via the coordinates (x_1, x_2, x_3, x_4), then the unit sphere S^3 can be given by the three sections of the tangent bundle:

$$\begin{aligned}
z &\mapsto iz \quad &\text{or} \quad &(x_1, x_2, x_3, x_4) \mapsto (-x_2, x_1, -x_4, x_3)\\
z &\mapsto jz \quad &\text{or} \quad &(x_1, x_2, x_3, x_4) \mapsto (-x_3, x_4, x_1, -x_2)\\
z &\mapsto kz \quad &\text{or} \quad &(x_1, x_2, x_3, x_4) \mapsto (-x_4, -x_3, x_2, x_1)
\end{aligned}$$

The key idea is that $1, i, j, k$ form the standard orthonormal basis of $\mathbb{H} \cong \mathbb{R}^4$. Multiplying by an arbitrary unit quaternion $z \in S^3$ produces another orthonormal basis z, iz, jz, kz, and since $|zw| = |z| \cdot |w|$ (with $| - |$ the Euclidean norm on $\mathbb{R}^4$), multiplication by a quaternion in the unit sphere S^3 is an isometry of $\mathbb{H}$. A similar process can be done for the Cayley Octonions by thinking of $\mathbb{R}^8$ as $\mathbb{H} \times \mathbb{H}$ and multiplying the Octonions by:

$$(z_1, z_2)(w_1, w_2) = (z_1 w_1 - \overline{w_2} z_2, z_2 \overline{w_1} + w_2 z_1)$$

This also defines a division algebra, and also satisfies $|zw| = |z||w|$, giving 7 orthogonal tangent vectors fields on S^7 that are nowhere vanishing, and hence TS^7 is also trivial. Chapter 4 shows that these are the only ones.

Definition 1.1.11: Direct Sum of Vector Bundles

Let $p_1 : E_1 \to B$, $p_2 : E_2 \to B$ be vector bundles over B. Then the *direct sum* or *coproduct* (historically called *Whitney sum*) of E_1, E_2 is:

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

The fibers of the direct sum are the direct sums of the fibers. One way to picture the direct sum is to take the product of the two vector bundles (including the base space)

$$E_1 \times E_2 \to B \times B$$

Then the restriction of that product to the diagonal $\{(b, b) \mid b \in B\}$ is $E_1 \oplus E_2$. As for vector spaces, the categorical product and coproduct agree, though the notation $E_1 \times E_2$ is sometimes used in the literature for the bundle over $B \times B$ as well. That convention is avoided here.

Definition 1.1.12: Stably Trivial

Let E be a vector bundle. Then E is *stably trivial* if there exists a trivial vector bundle T such that $E \oplus T$ is the trivial bundle.

The tangent bundle TS^n is stably trivial, with NS^n as the complementary trivial summand. Similarly, $E \oplus E^{\perp}$ where E is the canonical line bundle with its orthogonal complement is isomorphic to $\mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1}$ via

$$(\ell, v, w) \mapsto (\ell, v + w)$$

In the $n = 1$ case, $E^{\perp}$ is isomorphic to E by rotating each vector in the plane by $\pi/2$ degrees. As mention earlier, the canonical line bundle E in the $n = 1$ case is isomorphic to the Möbius bundle. Hence, the direct sum of the Möbius bundle with itself is trivial. The isomorphism is the one built above, $(\ell, v, w) \mapsto (\ell, v + Rw)$ with R the fixed rotation of $\mathbb{R}^2$ by $\pi/2$, and it is a fiberwise isometry: v and Rw are orthogonal, so $|v + Rw|^2 = |v|^2 + |w|^2$.

Example 1.1.13: The Tangent Bundle of $\mathbb{R}\mathrm{P}^n$ and the Canonical Line Bundle

Start from $S^n \times \mathbb{R}^{n+1} \cong TS^n \oplus NS^n$ and impose the equivalence

$$(x, v) \sim (-x, -v)$$

on both sides. Applied to TS^n the quotient is $T\mathbb{R}\mathrm{P}^n$ (see [16], or even [2]). The isomorphism itself does not descend, but each side does, and comparing the two quotients gives the direct sum of $T\mathbb{R}\mathrm{P}^n$ with a trivial line bundle isomorphic to $n + 1$ copies of the canonical line bundle over $\mathbb{R}\mathrm{P}^n$. The computation is direct on each side.

On NS^n the identification $(x, v) \sim (-x, -v)$ reads $(x, tx) \sim (-x, t(-x))$, and the section $x \mapsto (x, x)$ descends, so this quotient is the product bundle $\mathbb{R}\mathrm{P}^n \times \mathbb{R}$. The identification on $S^n \times \mathbb{R}^{n+1}$ respects the coordinate factors of $\mathbb{R}^{n+1}$, so its quotient is a direct sum of $n + 1$ copies of the line bundle E over $\mathbb{R}\mathrm{P}^n$ obtained from $(x, t) \sim (-x, -t)$ on $S^n \times \mathbb{R}$. That line bundle is the canonical one: transporting along $(x, t) \mapsto (x, tx)$ sends $(-x, -t)$ to $(-x, (-t)(-x)) = (-x, tx)$, which is exactly the gluing defining the canonical line bundle over $\mathbb{R}\mathrm{P}^n$.

Hence the direct sum of $T\mathbb{R}\mathrm{P}^n$ with a trivial line bundle is isomorphic to $n + 1$ copies of the canonical bundle over $\mathbb{R}\mathrm{P}^n$. For $n = 3$ the quaternionic sections $z \mapsto iz, jz, kz$ satisfy $(-z) \mapsto -(iz)$ and so descend to the quotient, making $T\mathbb{R}\mathrm{P}^3$ trivial. Four copies of the canonical line bundle over $\mathbb{R}\mathrm{P}^3$ therefore form a trivial bundle, and the octonionic sections give the same conclusion for eight copies over $\mathbb{R}\mathrm{P}^7$. **??** takes up these questions.

Note what is *not* claimed. Stable triviality in the sense of definition 1.1.12 would require a *trivial* bundle as the complementary summand, and here the complement is $n + 1$ copies of a nontrivial line bundle. For $n = 2$ the conclusion fails outright: the total Stiefel-Whitney class is a stable invariant and $w(T\mathbb{R}\mathrm{P}^2) = (1 + a)^3 = 1 + a + a^2 \neq 1$ in $H^*(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2\mathbb{Z})$, so $T\mathbb{R}\mathrm{P}^2$ is not stably trivial.

A vector bundle carries an inner product, a map

$$\langle -, - \rangle \colon E \oplus E \to \mathbb{R}$$

restricting to a positive-definite symmetric form on each fiber of a real bundle $E \to B$. For a complex bundle the fiberwise forms are positive-definite Hermitian, $\langle -, - \rangle \colon E \oplus E \to \mathbb{C}$. Over a paracompact base such an inner product exists, Euclidean or Hermitian respectively ([14, Inner Products on a Vector Bundle]).

An inner product is what produces orthogonal complements:

Proposition 1.1.14: Sub-bundles Have Orthogonal Complements

Let $E \to B$ be a vector bundle over a paracompact base B, and let $E_0 \subseteq E$ be a vector subbundle. Then there is a vector subbundle $E_0^\perp \subseteq E$ such that $E_0 \oplus E_0^\perp \cong E$.

***Proof*:**
Choose an inner product on E, Euclidean in the real case and Hermitian in the complex case, and let $E_0^\perp$ be the subspace of E whose fiber over each $b \in B$ is the orthogonal complement of the fiber of E_0. If the natural projection $E_0^\perp \to B$ is a vector bundle, then $(v, w) \mapsto v + w$ defines a map $E_0 \oplus E_0^\perp \to E$ that is a linear isomorphism on each fiber, hence an isomorphism by lemma 1.1.6.

For the local triviality of $E_0^\perp$, the question is local in B, so we may assume $E = B \times \mathbb{R}^n$ (in the complex case $B \times \mathbb{C}^n$). Let E_0 have dimension m and fix $b_0 \in B$. Near b_0, choose m independent local sections $s_1, ..., s_m$ of E_0, and extend to n independent local sections $s_1, ..., s_n$ of E by taking $s_{m+1}, ..., s_n$ to be constant sections whose values at b_0 complete a basis of the fiber; independence at b_0 persists for nearby b by continuity of the determinant. Applying the Gram-Schmidt process to $s_1, ..., s_n$ in each fiber yields orthonormal sections $s_1', ..., s_n'$, the first m of which span E_0 in each fiber. The last $n - m$ then span $E_0^\perp$ in each fiber and give a local trivialization of $E_0^\perp$, completing the proof.

Taking $E_0 = E$, the Gram-Schmidt step shows in addition that the local trivializations may be chosen fiberwise isometric. Over a compact base a stronger statement holds: every bundle is a direct summand of a trivial one, which is the topological counterpart of a module being projective.

Proposition 1.1.15: Vector Bundle Over Compact Are Projective

Let $E \to B$ be a vector bundle over a compact Hausdorff space B. Then there exists a vector bundle $E' \to B$ such that $E \oplus E'$ is the trivial bundle

Note this fails for non-compact spaces, as example 3.2.1(3) shows for $\mathbb{R}\mathrm{P}^\infty$.

***Proof*:**
Step 1: what has to be produced. Were E already a subbundle of a trivial bundle $B \times \mathbb{R}^N$, composing the inclusion with the projection onto $\mathbb{R}^N$ would give a map $E \to \mathbb{R}^N$ that is a linear injection on each fiber. The proof runs this backwards: construct such a map first, then show that it embeds E in $B \times \mathbb{R}^N$ as a direct summand.

Step 2: constructing the map. Each point $x \in B$ has an open neighborhood U_x over which E is trivial. By Urysohn's Lemma there is a map $\varphi_x : B \to [0, 1]$ that is nonzero at x and has support contained in U_x. Letting x vary, the sets $\varphi_x^{-1}(0, 1]$ form an open cover of B. By compactness there is a finite subcover. Relabel the corresponding U_x's and φ_x's as U_i and φ_i. Define $g_i : E \to \mathbb{R}^n$ by $g_i(v) = \varphi_i(p(v))(\pi_i h_i(v))$ where p is the projection $E \to B$ and $\pi_i h_i$ is the composition of a local trivialization $h_i : p^{-1}(U_i) \to U_i \times \mathbb{R}^n$ with the projection π_i to $\mathbb{R}^n$. Then g_i is a linear injection on each fiber over $\varphi_i^{-1}(0, 1]$, so making the various g_i's the coordinates of a map $g : E \to \mathbb{R}^N$ with $\mathbb{R}^N$ a product of copies of $\mathbb{R}^n$, then g is a linear injection on each fiber.

Step 3: the image is a direct summand. The map g is the second coordinate of a map $f : E \to$

$B \times \mathbb{R}^N$ with first coordinate p. The image of f is a subbundle of the product $B \times \mathbb{R}^N$ since projection of $\mathbb{R}^N$ onto the ith $\mathbb{R}^n$ factor gives the second coordinate of a local trivialization over $\varphi_i^{-1}(0,1]$. Thus, E isomorphic to a subbundle of $B \times \mathbb{R}^N$ so by proposition 1.1.14 there is a complementary subbundle E' with $E \oplus E'$ isomorphic to $B \times \mathbb{R}^N$. For a complex bundle the same argument runs with $\mathbb{C}$ in place of $\mathbb{R}$ throughout, as we sought to show.

The other operation the rest of the book needs is the tensor product, and it is what will make $K^0(X)$ a ring rather than only a group.

Definition 1.1.16: Tensor Product of Vector Bundles

Let $p_1 : E_1 \to B$, $p_2 : E_2 \to B$ be vector bundles over B. Then *tensor product* is a vector bundle where as the set

$$E_1 \otimes E_2 = \coprod_{x \in B} p_1^{-1}(x) \otimes p_2^{-1}(x)$$

and for the topology, choose an open cover $\{U_\alpha\}$ trivializing both bundles, with local trivializations $h_i : p_i^{-1}(U) \to U \times \mathbb{R}^{n_i}$. The topology on $p_1^{-1}(U) \otimes p_2^{-1}(U)$ is the one making the fiberwise tensor product map:

$$h_1 \otimes h_2 : p_1^{-1}(U) \otimes p_2^{-1}(U) \to U \times (\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2})$$

a homeomorphism.

This topology does not depend on the trivializing cover, since any other choice is obtained by composing with an automorphism of $U \times \mathbb{R}^{n_i}$ of the form:

$$(x, v) \mapsto (x, g_i(x)(v))$$

for a continuous $g_i : U \to \mathrm{GL}_{n_i}(\mathbb{R})$. Then $h_1 \otimes h_2$ changes by composition with the corresponding automorphism of $U \times (\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2})$, whose second coordinate $g_1 \otimes g_2$ is a continuous map

$$U \to \mathrm{GL}_{n_1 n_2}(\mathbb{R})$$

and a homeomorphism composed with a homeomorphism is a homeomorphism.

A reader who knows the fibered product of schemes, or who prefers the cocycle description, may instead build the bundle as the gluing of $\coprod_\alpha (U_\alpha \times \mathbb{R}^n)$ subject on intersections to the gluing condition:

$$g_{\beta\alpha} : U_\alpha \cap U_\beta \to \mathrm{GL}_n(\mathbb{R})$$

subject to the cocycle condition on $U_\alpha \cap U_\beta \cap U_\gamma$

$$g_{\gamma\beta} g_{\beta\alpha} = g_{\gamma\alpha}$$

Then the tensor product between $E_1 \to B$ and $E_2 \to B$ is given by choosing an open cover $\{U_\alpha\}$ that is locally trivial for both E_1, E_2 and getting gluing functions $g^i_{\beta\alpha} : U_\alpha \cap U_\beta \to \mathrm{GL}_{n_i}(\mathbb{R})$ for each E_i. Then the tensor bundle $E_1 \otimes E_2$ is the tensor product of the functions $g^1_{\beta\alpha} \otimes g^2_{\beta\alpha}$ assigning to each$x \in U_\alpha \cap U_\beta$ the tensor product of the matrices.

Because the operation is fiberwise, it inherits the algebra of the tensor product of vector spaces: over a fixed base it is commutative, associative, has the trivial line bundle as identity, and distributes over direct sum. The tensor product of two line bundles is again a line bundle, so $\mathrm{Vect}^1(B)$, the set of isomorphism classes of line bundles, is an abelian group under $\otimes$. The inverse is given by the pointwise inverse matrices $g_{\beta\alpha}(x) \in \mathrm{GL}_1(\mathbb{R})$, and the cocycle condition is preserved because 1×1 matrices commute.

In the real case this group is 2-torsion. Give E an inner product and rescale each trivialization h_α to be a fiberwise isometry, so that vectors in the fibers of E go to vectors of the same length in $\mathbb{R}^1$. Every value of every gluing function $g_{\beta\alpha}$ is then ± 1. The gluing functions of $E \otimes E$ are the squares of these, hence identically 1, so every element of $\mathrm{Vect}^1_{\mathbb{R}}(B)$ is its own inverse. The group is in fact $H^1(B; \mathbb{Z}/2\mathbb{Z})$ whenever B is homotopy equivalent to a CW complex, by a chain of three imports rather than by the definition of w_1: classification identifies $\mathrm{Vect}^1_{\mathbb{R}}(B)$ with $[B, \mathrm{Grass}_1(\mathbb{R}^\infty)]$ ([14, Classification of Vector Bundles over a Paracompact Base]), the base of that Grassmannian is $\mathbb{RP}^\infty$, which is a $K(\mathbb{Z}/2\mathbb{Z}, 1)$ ([14, $\mathbb{RP}^\infty$ as $K(\mathbb{Z}/2, 1)$ and the Classifying Space BG]), and maps into an Eilenberg-MacLane space compute cohomology ([14, Cohomology is Represented by $K(G, n)$], whose CW-homotopy-type hypothesis is the one assumed here). The resulting bijection is the first Stiefel-Whitney class.

The complex case differs at exactly this point. Rescaling makes the gluing functions take values of norm 1 rather than in $\{\pm 1\}$, so their squares need not be 1 and $E \otimes E$ need not be trivial. The group $\mathrm{Vect}^1_{\mathbb{C}}(B)$ is $H^2(B; \mathbb{Z})$ by the same three-step chain through $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ ([14, $BU(n)$ as a Grassmannian and Line Bundles]), again for B homotopy equivalent to a CW complex, with the first Chern class as the bijection. What replaces squaring is conjugation: for the *conjugate bundle* $\overline{E} \to B$, whose gluing functions are the complex conjugates $\overline{g_{\beta\alpha}}$, the identity $z\overline{z} = 1$ for $|z| = 1$ makes $E \otimes \overline{E}$ trivial, so $\overline{E}$ is the inverse of E.

The same fiberwise recipe produces the exterior and symmetric powers.

Definition 1.1.17: Exterior and Symmetric Powers of a Vector Bundle

Let $p\colon E \to B$ be a vector bundle over $k = \mathbb{R}$ or $\mathbb{C}$, given by a trivializing cover $\{U_\alpha\}$ with gluing functions $g_{\beta\alpha}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_n(k)$, where n is the rank of E on the trivializing set in question. For $j \geq 0$ the *j-th exterior power* $\Lambda^j E \to B$ is the vector bundle whose fiber over x is $\Lambda^j\big(p^{-1}(x)\big)$, glued by the functions $\Lambda^j g_{\beta\alpha}$, which are continuous because the entries of $\Lambda^j A$ are the $j \times j$ minors of A, which are polynomials in the entries of A. The construction is local, so it does not need the rank of E to be constant on B. Where the rank is n, the rank of $\Lambda^j E$ is $\binom{n}{j}$, which is again locally constant, as definition 1.1.2 allows. The *j-th symmetric power* $\mathrm{Sym}^j E$ is defined the same way with Sym^j in place of Λ^j. If E has constant rank n, the top exterior power $\det(E) = \Lambda^n E$ is the *determinant bundle*, a line bundle with gluing functions $\det g_{\beta\alpha}$.

The identities of multilinear algebra hold on each fiber, being statements about vector spaces ([5, Exterior Powers of a Direct Sum]). Passing from there to an isomorphism of *bundles* is a further step, since it asks the fiberwise maps to be continuous. Two such passages are made now, because both are needed the moment exterior powers are used for anything: that Λ^k does not distinguish isomorphic bundles, and the Whitney formula for exterior powers.

The first is what makes Λ^k an operation on isomorphism classes at all, and it is not automatic from definition 1.1.17, which builds $\Lambda^k E$ out of a chosen trivializing cover.

Lemma 1.1.18: Exterior Powers Respect Isomorphism

Let $E, F \to B$ be vector bundles over $k = \mathbb{R}$ or $\mathbb{C}$. If $E \cong F$, then $\Lambda^j E \cong \Lambda^j F$ for every $j \geq 0$, and likewise $\mathrm{Sym}^j E \cong \mathrm{Sym}^j F$.

***Proof*:**
Let $u \colon E \to F$ be an isomorphism. Over each $b \in B$ the linear map $\Lambda^j(u_b) \colon \Lambda^j(E_b) \to \Lambda^j(F_b)$ is an isomorphism, with inverse $\Lambda^j(u_b^{-1})$, by functoriality of Λ^j on vector spaces. These assemble into a map $\Lambda^j E \to \Lambda^j F$ of total spaces, and it is continuous: over a set trivializing both bundles u is given by a continuous map $a \colon U \to \mathrm{GL}_n(k)$, and the induced map by $\Lambda^j \circ a$, whose entries are the $j \times j$ minors of a and so are polynomials in the entries of a. A continuous map that is a linear isomorphism on each fiber is an isomorphism of bundles (lemma 1.1.6). The symmetric case is identical, with $\mathrm{Sym}^j a$ in place of $\Lambda^j a$.

The second is the Whitney formula. Its fiberwise content is upstream. What is proved here is that the fiberwise isomorphism is continuous, which the algebra does not give.

Proposition 1.1.19: The Whitney Formula for Exterior Powers

Let $E, F \to B$ be vector bundles over $k = \mathbb{R}$ or $\mathbb{C}$. For each $j \geq 0$ there is an isomorphism of vector bundles

$$\Lambda^j(E \oplus F) \;\cong\; \bigoplus_{a+b=j} \Lambda^a E \otimes \Lambda^b F$$

natural in E and F.

***Proof*:**
Over $x \in B$ define

$$\Phi_x \colon \bigoplus_{a+b=j} \Lambda^a(E_x) \otimes \Lambda^b(F_x) \longrightarrow \Lambda^j(E_x \oplus F_x)$$

on decomposables by $(e_1 \wedge \cdots \wedge e_a) \otimes (f_1 \wedge \cdots \wedge f_b) \mapsto e_1 \wedge \cdots \wedge e_a \wedge f_1 \wedge \cdots \wedge f_b$, reading E_x and F_x as subspaces of $E_x \oplus F_x$. The right-hand side is alternating and multilinear in the e's and in the f's separately, so Φ_x is well defined, and it is a linear isomorphism by [5, Exterior Powers of a Direct Sum], which is stated for exactly this map and asserts naturality in both arguments.

Taking the Φ_x together already defines a map Φ of total spaces, fiberwise a linear isomorphism, so all that is left is continuity. Nothing has to be glued: the fiberwise formula is given globally to begin with.

Continuity is local, so fix U trivializing both bundles, with trivializations $h_E \colon E|_U \to U \times k^m$ and $h_F \colon F|_U \to U \times k^n$, and give the exterior powers the trivializations these induce, which are the ones definition 1.1.17 constructs. In those coordinates $\Phi|_U$ is, at the point x, the composite

$$\Lambda^j\big(h_{E,x} \oplus h_{F,x}\big) \circ \Phi_x \circ \Big(\bigoplus_{a+b=j} \Lambda^a h_{E,x}^{-1} \otimes \Lambda^b h_{F,x}^{-1}\Big)$$

and naturality of Φ in its two arguments says precisely that this composite equals the single map $\Phi_0 \colon \bigoplus_{a+b=j} \Lambda^a(k^m) \otimes \Lambda^b(k^n) \to \Lambda^j(k^m \oplus k^n)$ of fixed vector spaces, the *same* map for every $x \in U$. So $\Phi|_U$ is the constant map $\mathrm{id}_U \times \Phi_0$, which is continuous. Since such U cover

B, the map Φ is continuous, and being a fiberwise linear isomorphism it is an isomorphism of bundles (lemma 1.1.6).

1.2 Classifying Vector Bundles

Proposition 1.2.1: Pullback of Vector bundle

Let $f : A \to B$ be a continuous map and $p : E \to B$ a vector bundle. Then there exists a vector bundle $p' : E' \to A$ with a map $f' : E' \to E$ taking the fibers of E' over each $a \in A$ isomorphically onto the fibers of E over $f(a)$. The vector bundle E' is unique up to isomorphism.

The proposition gives the pullback map $f^* : \mathrm{Vect}(B) \to \mathrm{Vect}(A)$, sending the isomorphism class of E to the isomorphism class of $f^*(E) = E'$.

***Proof*:**
Let

$$E' = \{(a, v) \in A \times E \big| f(a) = p(v)\}$$

This subspace of $A \times E$ fits into the following commutative diagram

$$\begin{tikzcd} E' \arrow[r, "f'"] \arrow[d, "p'"'] & E \arrow[d, "p"] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

where $p'(a, v) = a$ and $f'(a, v) = v$. Let Γ_f denote the graph of f in $A \times B$. Then p' factors as the composition $E' \to \Gamma_f \to A$, $(a, v) \mapsto (a, p(v)) = (a, f(a)) \mapsto a$. The first of these two maps is the restriction of the vector bundle $\nVdash \times p : A \times E \to A \times B$ over the graph Γ_f, so it is a vector bundle, and the second map is a homeomorphism, so their composition $p' : E' \to A$ is a vector bundle. The map f' takes the fiber E' over a isomorphically onto the fiber of E over $f(a)$, showing existence.

For uniqueness, construct an isomorphism from an arbitrary E' satisfying the conditions in the proposition to the particular one just constructed by sending $v' \in E'$ to the pair $(p'(v'), f'(v'))$. This map takes each fiber of E' to the corresponding fiber of $f^*(E)$ by a vector space isomorphism, so by lemma 1.1.6 it is an isomorphism of vector bundles, completing the proof.

When restricting to Vect^1, notice the trivial bundle pulls back to the trivial bundle and the tensor product is well-behaved; hence this pullback becomes a group homomorphism.

Example 1.2.2: Vector Bundle Pullbacks

1. Let $A \subseteq B$ so that $\iota : A \hookrightarrow B$. Then the restriction of E over B to A is the pullback $\iota^*(E)$.

2. Let $f : A \to B$ be a constant map: $f(A) = b$. Then

$$f^*(E) = A \times p^{-1}(b)$$

3. The tangent bundle TS^n is the pullback of $T\mathbb{RP}^n$ via he map $S^n \to \mathbb{RP}^n$.

4. Consider the möbius bundle $E \to S^1$, and take $f : S^1 \to S^1$ given by $z \mapsto z^2$. By thinking of the Möbius bundle as $S^1 \times \mathbb{R}$ under $(z,t) \sim (-z,-t)$, then $f^*(E)$ is the trivial bundle (think about this for awhile!). If $f(z) = z^n$, then E_n is the trivial bundle if n is even, and the Möbius bundle is n is odd.

Proposition 1.2.3: Properties of Vector-bundle Pullback

Let E be a vector bundle and f, g continuous maps, everything over the appropriate domains. Then:

1. $(fg)^*(E) \cong g^*(f^*(E))$
2. $\mathrm{id}^*(E) \cong E$
3. $f^*(E_1 \oplus E_2) \cong f^*(E_1) \oplus f^*(E_2)$
4. $f^*(E_1 \otimes E_2) \cong f^*(E_1) \otimes f^*(E_2)$

***Proof*:**
Throughout, write $f^*(E) = \{(a,e) \in A \times E \mid f(a) = p(e)\}$ with projection to the first factor, so that the fiber of $f^*(E)$ over a is carried isomorphically onto $E_{f(a)}$ by the second projection. Each map below is a continuous fiberwise linear bijection with continuous fiberwise inverse, so lemma 1.1.6 promotes it to a bundle isomorphism, and only the map itself is named.

1. For $g\colon A \to B$ and $f\colon B \to C$ and $p\colon E \to C$, send $(a,e) \in (fg)^*(E)$ to $\big(a, (g(a), e)\big)$. This lands in $g^*(f^*(E))$ because $f(g(a)) = p(e)$, it is continuous with continuous inverse $(a,(b,e)) \mapsto (a,e)$, and on the fiber over a both sides are the identity of $E_{f(g(a))}$.

2. Send $(b,e) \in \mathrm{id}^*(E)$ to e. This is a homeomorphism onto E with inverse $e \mapsto (p(e), e)$, and it is the identity on each fiber.

3. The fiber of $f^*(E_1 \oplus E_2)$ over a is $(E_1)_{f(a)} \oplus (E_2)_{f(a)}$, which is the fiber of $f^*(E_1) \oplus f^*(E_2)$ over a. Send $\big(a, (e_1, e_2)\big)$ to $\big((a,e_1), (a,e_2)\big)$. This is continuous, with the evident continuous inverse, and is the identity on fibers.

4. Identical to (3) with $\otimes$ in place of $\oplus$: the fiber of $f^*(E_1 \otimes E_2)$ over a is $(E_1)_{f(a)} \otimes (E_2)_{f(a)}$, and the map $\big(a, e_1 \otimes e_2\big) \mapsto (a,e_1) \otimes (a,e_2)$ is a continuous fiberwise isomorphism.

This establishes all four, completing the proof.

The next goal is homotopy invariance of the isomorphism class of a vector bundle. It needs a technical lemma first:

Lemma 1.2.4: Homotopy Invariant Vector Bundle

Let $E \to X \times I$ be a vector bundle where X is paracompact. Then the restriction over $X \times \{0\}$ and $X \times \{1\}$ is isomorphic.

***Proof*:**
First two smaller proofs:

1. A vector bundle $p : E \to X \times [a, b]$ is trivial if its restrictions over $X \times [a, c]$ and $X \times [c, b]$ are both trivial for some $c \in (a, b)$. Indeed, let these restrictions be $E_1 = p^{-1}(X \times [a, c])$ and $E_2 = p^{-1}(X \times [c, b])$, and let $h_1 : E_1 \to X \times [a, c] \times \mathbb{R}^n$ and $h_2 : E_2 \to X \times [c, b] \times \mathbb{R}^n$ be isomorphisms. These isomorphisms may not agree on $p^{-1}(X \times \{c\})$, but they can be made to agree by replacing h_2 by its composition with the isomorphism $X \times [c, b] \times \mathbb{R}^n \to X \times [c, b] \times \mathbb{R}^n$ which on each slice $X \times \{x\} \times \mathbb{R}^n$ is given by $h_1 h_2^{-1} : X \times \{c\} \times \mathbb{R}^n \to X \times \{c\} \times \mathbb{R}^n$. Once h_1 and h_2 agree on $E_1 \cap E_2$, they define a trivialization of E.

2. For a vector bundle $p : E \to X \times I$, there exists an open cover $\{U_\alpha\}$ of X where each restriction $p^{-1}(U_\alpha \times I) \to U_\alpha \times I$ is trivial. Indeed, for each $x \in X$ find open neighborhoods $U_{x,1}, \dots, U_{x,k}$ in X and a partition $0 = t_0 < t_1 < \cdots < t_k = 1$ of $[0, 1]$ such that the bundle is trivial over $U_{x,i} \times [t_{i-1}, t_i]$, using compactness of $[0, 1]$. Then by (1) the bundle is trivial over $U_\alpha \times I$ where $U_\alpha = U_{x,1} \cap \cdots \cap U_{x,k}$.

Now for the proposition, by (2), choose an open cover $\{U_\alpha\}$ of X so that E is trivial over each $U_\alpha \times I$. Let's assume for now the simpler case that X is compact Hausdorff. Then a finite number of U_α's cover X. Relabel these as U_i for $i = 1, 2, \dots, m$. As shown in [8, Existence of Subordinate Partitions of Unity], there is a corresponding partition of unity by functions φ_i with the support of φ_i contained in U_i. For $i \geq 0$, let $\psi_i = \varphi_1 + \cdots + \varphi_i$, so in particular $\psi_0 = 0$ and $\psi_m = 1$. Let X_i be the graph of ψ_i, the subspace of $X \times I$ consisting of points of the form $(x, \psi_i(x))$, and let $p_i : E_i \to X_i$ be the restriction of the bundle E over X_i. Since E is trivial over $U_i \times I$, the natural projection homeomorphism $X_i \to X_{i-1}$ lifts to a homeomorphism $h_i : E_i \to E_{i-1}$ which is the identity outside $p^{-1}(U_i \times I)$ and which takes each fiber of E_i isomorphically onto the corresponding fiber of E_{i-1}. Explicitly, on points in $p^{-1}(U_i \times I) = U_i \times I \times \mathbb{R}^n$ let $h_i(x, \psi_i(x), v) = (x, \psi_{i-1}(x), v)$. The composition $h = h_1 h_2 \cdots h_m$ is then an isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$.

In the general case that X is only paracompact, there is a countable cover $\{V_i\}_{i \geq 1}$ of X and a partition of unity $\{\varphi_i\}$ with φ_i supported in V_i, such that each V_i is a disjoint union of open sets each contained in some U_α; this is the countable-cover argument of [10, Lemma 1.21], and runs as follows. Repeating one member of the cover countably often, we may assume the index set is infinite; let $\{\varphi_\alpha\}$ be a partition of unity subordinate to $\{U_\alpha\}$ ([8, Existence of Subordinate Partitions of Unity]). For each finite set S of indices let V_S be the set of points x with $\varphi_\alpha(x) > \varphi_\beta(x)$ for all $\alpha \in S$ and $\beta \notin S$. Near any point all but finitely many φ_β vanish identically, so V_S is locally cut out by finitely many strict inequalities between continuous functions, hence open. For $\alpha \in S$ we have $\varphi_\alpha > \varphi_\beta \geq 0$ on V_S for any $\beta \notin S$, so $V_S \subseteq \mathrm{supp}\varphi_\alpha \subseteq U_\alpha$. If $S \neq S'$ with $|S| = |S'|$, then choosing $\alpha \in S \setminus S'$ and $\beta \in S' \setminus S$, on $V_S \cap V_{S'}$ both $\varphi_\alpha > \varphi_\beta$ and $\varphi_\beta > \varphi_\alpha$ would hold, so $V_S \cap V_{S'} = \emptyset$. Thus V_i, the union of the V_S with $|S| = i$, is a disjoint union of open sets each contained in some U_α, and the V_i cover X since each x lies in $V_{S(x)}$ for the finite set $S(x) = \{\alpha \mid \varphi_\alpha(x) > 0\}$. A second application of the same theorem, now to the cover $\{V_i\}$, gives the partition of unity $\{\varphi_i\}$. Then E is trivial over each $V_i \times I$. As before we let $\psi_i = \varphi_1 + \cdots + \varphi_i$ and let $p_i : E_i \to X_i$ be the restriction of E over the graph of ψ_i. Also as before we construct $h_i : E_i \to E_{i-1}$ using the fact that E is trivial over $V_i \times I$. The infinite composition $h = h_1 h_2 \cdots$ is then a well-defined isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$ since near each point $x \in X$ only finitely many φ_i's are nonzero, so there is a neighborhood of x in which all but finitely many h_i's are the identity.

Theorem 1.2.5: Homotopy Invariance Vector Bundles

Let $p : E \to B$ a vector bundle and $f_0, f_1 : A \to B$ be homotopic maps. Then if A is paracompact,
$$f_0^*(E) \cong f_1^*(E)$$

Note this result holds for fiber bundles more generally.

***Proof*:**
Let $F : A \times I \to B$ be a homotopy from f_0 to f_1. Then the restriction $f^*(E)$ over $A \times \{0\}$ and $A \times \{1\}$ are $f_0^*(E)$ and $f_1^*(E)$, and we get the result by the above lemma.

Corollary 1.2.6: Homotopy For N-vector Bundles

Let $f : A \to B$ be a homotopy equivalence of paracompact spaces. Then
$$f^* : \mathrm{Vect}^n(B) \to \mathrm{Vect}^n(A)$$
is a bijection. In particular, vector bundles over a contractible paracompact base is trivial.

***Proof*:**
Let g be the homotopy inverse of f. Then
$$f^*g^* = \mathrm{id}^* = \mathrm{id} \qquad g^*f^* = \mathrm{id}^* = \mathrm{id}$$
completing the proof.

1.2.1 Classifying Vector Bundles on Spheres

Using the tensor product, we can construct some non-trivial bundles on the sphere S^k. The clutching function is a way of creating a bundle on the sphere S^k by gluing a bundle on the upper and lower hemisphere with a compatibility condition on the “equator subspace” (this is the “clutch”). In particular, let D^k+, D^k_- be the upper and lower hemisphere so that $D^k_+ \cap D^k_- \cong S^{k-1}$. Given a map $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$, we can define:
$$E_f = \left((D^k_+ \times \mathbb{R}^n) \sqcup (D^k_- \times \mathbb{R}^n)\right) / \sim$$
$$\partial D^k_- \times \mathbb{R}^n \ni (x, v) \sim (x, f(x)(v)) \in \partial D^k_+ \times \mathbb{R}^n$$

Then we get a natural n-vector bundle map $f : E_f \to S^k$. Naturally, replacing $\mathbb{R}$ with $\mathbb{C}$ would require $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$.

Example 1.2.7: Clutching Function

1. Let us create TS^2 using a clutching function. Let v_+ be a vector field on D^2_+ given by choosing a tangent vector at the north pole and parallel-transporting it along each meridian circle (see [16] for parallel-transports). Here is a visual on only a quarter of the upper hemisphere:

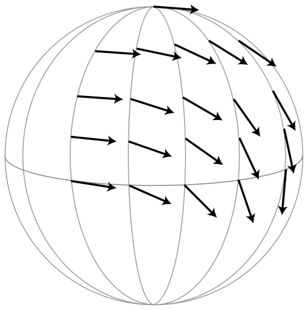

Figure 1.1: v_+ on D^2_+ (from Hatcher [10])

Reflect across the equation to create the vector field v_- on D^2_-. Then take w_+ and w_- to be v_+ and v_- respectively each rotate by $\pi/2$ degree when viewed from outside the sphere. Then these for vector fields give a trivialization of TS^2 over $D^2_\pm$: the clutching function f read off the coordinates of v_-, w_- in the v_+, w_+ coordinate system, in particular f rotates (v_+, w_+) to (v_-, w_-). Going around the equator S^1 counter-clockwise, the angle runs from 0 to 4π.

2. The same construction for $\mathbb{C}\mathrm{P}^1 = S^2$ produces the canonical complex line bundle, whose clutching function is $f\colon S^1 \to \mathrm{GL}_1(\mathbb{C})$, $f(z) = z^{-1}$.

The advantage of clutching functions is that homotopic clutching functions give isomorphic bundles, and conversely: $E_f \cong E_g$ if and only if f and g are homotopic. Over $\mathbb{C}$ this is a bijection. The real case needs one modification, because $\mathrm{GL}_n(\mathbb{R})$ is disconnected, and it is treated after the complex statement.

Proposition 1.2.8: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \to \mathrm{Vect}^n_{\mathbb{C}}(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ to complex vector bundles E_f is a bijection

***Proof*:**
Let $[X, Y]$ denote the set of homotopy classes of continuous maps $X \to Y$. If $f \simeq g$, then there is a homotopy $F : S^{k-1} \times I \to \mathrm{GL}_n(\mathbb{C})$. Then the same sort of clutching construction can be used to produce a vector bundle $E_F \to S^k \times I$ that restricts to E_f over $S^k \times \{0\}$ and E_g over $S^k \times \{1\}$, hence $E_f \cong E_g$ by theorem 1.2.5. We thus have a map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \to \mathrm{Vect}^n_{\mathbb{C}}(S^k)$$

Let us find an inverse Ψ. Let $p : E \to S^k$ be an n-vector bundle, and let E_+, E_- be its restriction to D^k_+ and D^k_-. $D^k_\pm$ is contractible, by theorem 1.2.5 these bundles are trivial. Choose trivializations $h_\pm : E_\pm \to D^k_\pm \times \mathbb{C}^n$. Then we get that $h_+ h_-^{-1}$ defines a map $S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ whose homotopy class is by construction

$$\Psi(E) \in [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

It remains to check that $\Psi(E)$ is well-defined, that is that the trivializations are unique up to homotopy. Any two choices of $h_\pm$ differ by a map $D^k_\pm \to \mathrm{GL}_n(\mathbb{C})$, and $D^k_\pm$ is contractible, so that map is homotopic to a constant map $c_\pm$. The group $\mathrm{GL}_n(\mathbb{C})$ is path-connected, being the complement in $\mathbb{C}^{n^2}$ of the zero locus of the polynomial det, and the complement of a proper algebraic subset of $\mathbb{C}^N$ is connected. (Diagonalizability is neither available nor needed here: a general $M \in \mathrm{GL}_n(\mathbb{C})$ has Jordan blocks.) A path from $c_\pm$ to the identity therefore exists, and composing the two homotopies shows $h_\pm$ is unique up to homotopy. Thus $h_+h_-^{-1} : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ is unique up to homotopy and $\Psi : \mathrm{Vect}^n_{\mathbb{C}}(S^k) \to [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$ is well-defined. Finally $\Psi\Phi = \mathrm{id}$ because the clutching construction produces a bundle whose two evident trivializations have transition f, and $\Phi\Psi = \mathrm{id}$ because reassembling E_+ and E_- along $h_+h_-^{-1}$ returns E, completing the proof.

The correspondence has immediate consequences. Every complex vector bundle over S^1 is trivial, since $\mathrm{Vect}^n_{\mathbb{C}}(S^1)$ corresponds to $[S^0, \mathrm{GL}_n(\mathbb{C})]$, which has a single element because $\mathrm{GL}_n(\mathbb{C})$ is path-connected. A second consequence is that the canonical line bundle $H \to \mathbb{C}\mathrm{P}^1$ satisfies the relation:

$$(H \otimes H) \oplus 1 \cong H \oplus H$$

where 1 is the trivial one-dimensional bundle (see [10, p.24]).

In the real case $\mathrm{GL}_n(\mathbb{R})$ has two path-connected components, the preimages of the two signs of det ([18, Connected Components of the General Linear Group]). The argument nonetheless goes through for *oriented real vector bundles*, meaning those whose transition maps all have positive determinant, because the proof used only path-connectedness of the relevant component:

Proposition 1.2.9: Clutching Function Correspondence for Oriented Bundles

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}^+_n(\mathbb{R})] \to \mathrm{Vect}^n_+(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \to \mathrm{GL}^+_n(\mathbb{R})$ to oriented real vector bundles E_f is a bijection.

***Proof*:**

The argument for proposition 1.2.8 used exactly two facts about $\mathrm{GL}_n(\mathbb{C})$: that $D^k_\pm$ is contractible, so the restrictions $E_\pm$ are trivial and any two trivializations differ by a null-homotopic map into the group, and that the group is path-connected, so that constant map is homotopic to the identity. Both survive verbatim with $\mathrm{GL}^+_n(\mathbb{R})$ in place of $\mathrm{GL}_n(\mathbb{C})$: the first mentions the group only as a target, and the second is [18, Connected Components of the General Linear Group], which states that $\mathrm{GL}^+_n(\mathbb{R})$ is path-connected.

Only one point is particular to the oriented setting. A trivialization of an oriented bundle over $D^k_\pm$ is required to be orientation-preserving, so the transition $h_+h_-^{-1}$ takes values in $\mathrm{GL}^+_n(\mathbb{R})$ rather than in $\mathrm{GL}_n(\mathbb{R})$, and conversely a clutching function valued in $\mathrm{GL}^+_n(\mathbb{R})$ glues the two orientations of $D^k_\pm \times \mathbb{R}^n$ into an orientation of E_f. The two constructions are inverse to each other for the same reason as in the complex case, completing the proof.

This is enough to determine $\mathrm{Vect}^n_{\mathbb{R}}(S^k)$ for small n and k. The comparison that organizes the computation is a map $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k) \to \mathrm{Vect}^n_{\mathbb{R}}(S^k)$, where $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k)$ denotes isomorphism classes carrying an orientation specified at the point $x_0 \in S^{k-1}$. By the correspondence above, $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k)$ is in bijection with homotopy classes of maps $S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$ sending x_0 into $\mathrm{GL}^+_n(\mathbb{R})$, and forgetting the orientation gives the map

$$\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k) \to \mathrm{Vect}^n_{\mathbb{R}}(S^k)$$

that forgets the orientation over x_0. It is surjective, and 2-to-1 at all points except where a bundle has an automorphism that reverses the orientation of the fiber over x_0; on such points it is 1-to-1.

When $k = 1$ and $n \in \mathbb{N}_{>0}$ there are two homotopy classes of maps $S^0 \to \mathrm{GL}_n(\mathbb{R})$: the point x_0 goes to $\mathrm{id} \in \mathrm{GL}^+_n(\mathbb{R})$, and the other point goes to either the identity or a reflection. For $n = 1$ these give the trivial bundle and the Möbius bundle. For $n > 1$ both bundles admit automorphisms reversing the orientations of the fibers. Hence $\mathrm{Vect}^n_{\mathbb{R}}(S^1)$ has two elements for every $n \geq 1$, the nontrivial one being nonorientable.

When $k > 1$, then S^{k-1} is path-connected, and hence $S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}^+_n(\mathbb{R})$ must take all of S^{k-1} into $\mathrm{GL}^+_n(\mathbb{R})$, which gives a bijection

$$\mathrm{Vect}^n_+(S^k) \xrightarrow{\sim} \mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k)$$

and so every vector bundle over S^k is orientable and has (exactly) two different orientations. Thus, the map:

$$\mathrm{Vect}^n_+(S^k) \to \mathrm{Vect}^n(S^k)$$

is surjective, being 2-to-1 on bundles not having an orientation-reversing automorphism, and is injective otherwise.

Over a paracompact base an inner product is available, and it deformation-retracts $\mathrm{GL}_n(\mathbb{R})$ onto the orthogonal subgroup $O(n) \leqslant \mathrm{GL}_n(\mathbb{R})$, where:

$$O(n) = \left\{ A \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; AA^T = I \right\}$$

The retraction runs the Gram-Schmidt process and interpolates linearly at each stage, exactly as in the proof of proposition 1.1.14. Restricting it to the positive-determinant component deformation-retracts $\mathrm{GL}^+_n(\mathbb{R})$ onto the special orthogonal group $SO(n)$, where

$$SO(n) = \{A \in O(n) \mid \det(A) = 1\}$$

In the complex case the same process retracts $\mathrm{GL}_n(\mathbb{C})$ onto the unitary group $U(n)$, where

$$U(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid AA^* = I\}$$

and can be characterized by having column forming an orthonormal basis.

Since the inclusions $SO(n) \hookrightarrow \mathrm{GL}^+_n(\mathbb{R})$ and $U(n) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$ are homotopy equivalences, they induce natural bijections:

$$[S^{k-1}, SO(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}^+_n(\mathbb{R})] \qquad \text{and} \qquad [S^{k-1}, U(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

This simplifies the classification, $SO(n)$ being a much smaller group. For instance $SO(2)$ is a circle, the orientation-preserving isometries of $\mathbb{R}^2$ being the rotations. Taking $k = 2$ and $n = 2$, each integer $m \in \mathbb{Z}$ gives a map $f_m : S^1 \to S^1$, $z \mapsto z^m$, which is a clutching function for an oriented 2-dimensional vector bundle $E_m \to S^2$. Note that E_2 is TS^2 and E_{-1} is the real vector bundle underlying the canonical complex line bundle over $\mathbb{CP}^1 = S^2$. Since every continuous map $f : S^1 \to S^1$ is homotopic to $z \mapsto z^m$ for a unique $m \in \mathbb{Z}$ (see [14]), this gives:

$$\mathrm{Vect}^2_+(S^2) \cong \mathbb{Z}$$

Reversing the orientation of E_m gives E_{-m}, and hence:

$$\mathrm{Vect}^2_{\mathbb{R}}(S^2) \cong \mathbb{N}$$

Since $U(1)$ is homeomorphic to S^1, the same count gives:

$$\mathrm{Vect}^1_{\mathbb{C}}(S^2) \cong \mathbb{Z}$$

For $k > 2$, recall that any two maps $S^{k-1} \to S^1$ are homotopic if $k > 2$ (such a map lifts to a covering space $\mathbb{R}$ of S^1 and $\mathbb{R}$ is contractible, see [14, Conditions for Maps being Lifted, Fundamental Group of n-sphere]). Hence, every 2-dimensional vector bundle over S^k for $k \geq 3$ is trivial:

$$\mathrm{Vect}^2_{\mathbb{R}}(S^k) \cong \{0\} \qquad k \geq 3$$

The same argument applies to complex *line* bundles, $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ being homotopy equivalent to S^1:

$$\mathrm{Vect}^1_{\mathbb{C}}(S^k) \cong \{0\} \qquad k \geq 3$$

It does *not* extend to higher rank: $\mathrm{Vect}^2_{\mathbb{C}}(S^4) \leftrightarrow \pi_3(U(2)) = \pi_3(SU(2)) = \pi_3(S^3) = \mathbb{Z}$, so there are infinitely many rank-two complex bundles on S^4. Two further facts sharpen the picture before the examples resume. First, for any space X, the sets $[X, SO(n)]$ and $[X, U(n)]$ carry a natural group structure given by pointwise multiplication, and therefore:

$$[S^i, SO(n)] \cong_{\mathrm{Grp}} \pi_i(SO(n)) \qquad \text{and} \qquad [S^i, U(n)] \cong_{\mathrm{Grp}} \pi_i(U(n))$$

where homotopies preserve base points.

Proposition 1.2.10: Vector Bundles and Suspension

Let SX denote the suspension of a compact Hausdorff space X. Then given a choice of base-point $x \in X$, there is a 1-1 correspondence between the set $\mathrm{Vect}^n(SX)$ of isomorphism classes of n-dimensional (resp. real, complex, or quaternion) vector bundles over SX and the respective set

$$[X, O_n]_* / \pi_0(O_n), \quad [X, U_n]_*, \quad \text{or} \quad [X, Sp_n]_*$$

of based homotopy classes of maps from X to the orthogonal group O_n, unitary group U_n, or symplectic group Sp_n, the real case taken modulo the conjugation action of $\pi_0(O_n)$.

***Proof*:**
Write $SX = (X \times [0,1])/(X \times \{0\},\, X \times \{1\})$, each factor collapsed to a point, and let C_1 and C_2 be the images of $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$. Each is a cone on X, hence contractible, and $C_1 \cap C_2 = X \times \{\frac{1}{2}\} \cong X$. This is a direct consequence of the definition and needs no hypothesis on X beyond the compact Hausdorff standing assumption. Each C_i is a quotient of the compact Hausdorff $X \times [0, \frac{1}{2}]$ by a closed equivalence relation, hence itself compact Hausdorff and therefore paracompact, which is the hypothesis theorem 1.2.5 carries, so a vector bundle on a contractible paracompact base is trivial and every vector bundle on SX is assembled from an isomorphism of trivial bundles over the intersection X. This is the clutching construction run on the cover $\{C_1, C_2\}$ rather than on the two hemispheres, and it is carried out here rather than quoted: proposition 1.2.8 is stated for complex bundles over a sphere and proposition 1.2.9 for oriented real ones, so neither covers the unoriented real and quaternionic cases claimed here. What

the construction needs is only that C_1, C_2 and $C_1 \cap C_2$ be closed with $SX = C_1 \cup C_2$, which holds by construction. Such an isomorphism is given by a homotopy class of maps from X to GL_n, or equivalently, to the appropriate deformation retract (O_n, U_n, or Sp_n) of GL_n. Then the indeterminacy in the resulting map from $[X, GL_n]$ to classes of vector bundles is reduced by insisting that the base point of X map to $1 \in GL_n$; what survives is that changing the trivializations over the two cones by a constant automorphism conjugates the clutching map. This conjugation acts trivially on classes for the connected groups U_n and Sp_n, and gives the stated $\pi_0(O_n)$-quotient in the real case.

What the correspondence needs is therefore the low-degree homotopy of the classical groups, and those groups are close enough to spheres that the sphere fibrations compute them.

Proposition 1.2.11: Low-Degree Homotopy of the Compact Classical Groups

For every $n \geq 1$:

1. $\pi_0(O(n)) \cong \mathbb{Z}/2\mathbb{Z}$, while $U(n)$ and $\mathrm{Sp}(n)$ are path-connected.
2. $\pi_1(O(1)) = 0$, $\pi_1(O(2)) \cong \mathbb{Z}$, and $\pi_1(O(n)) \cong \mathbb{Z}/2\mathbb{Z}$ for $n \geq 3$. Also $\pi_1(U(n)) \cong \mathbb{Z}$ for all $n \geq 1$, and $\pi_1(\mathrm{Sp}(n)) = 0$.
3. $\pi_2(O(n)) = \pi_2(U(n)) = \pi_2(\mathrm{Sp}(n)) = 0$.

***Proof*:**
Throughout, each quotient below is a fiber bundle and hence a Serre fibration ([14, Fiber Bundles are Serre Fibrations]). Wherever its base is path-connected, [14, Long Exact Sequence of a Fibration] then applies, that being the hypothesis the cited theorem carries. Exactly one base used below fails it, the $\{\pm 1\}$ of Step 1, and that step is argued without the sequence. The others are spheres, the circle, and $\mathrm{SO}(3)$, each path-connected. Throughout as well, $\pi_k(S^m) = 0$ for $k < m$ by [14, Connectivity and Bottom Homotopy of Spheres].

Step 1: reduction to the identity component and to SU. The group $O(n)$ has $\mathrm{SO}(n)$ as its identity component, with det surjecting onto the discrete $\{\pm 1\}$ ([18, Connectivity of the Compact Classical Groups]), which gives (1) for $O(n)$. No exact sequence is used here, and none may be: the base $\{\pm 1\}$ is not path-connected. Instead, $\mathrm{SO}(n)$ is the path-component of the identity in $O(n)$, and a based homotopy class of maps from a path-connected S^i lands in a single path-component, so the inclusion induces $\pi_i(\mathrm{SO}(n)) \cong \pi_i(O(n))$ for every $i \geq 1$ directly. The determinant $U(n) \to S^1$ is a fiber bundle with fiber $\mathrm{SU}(n)$, and $\mathrm{SU}(n)$ is path-connected (same citation), so from the long exact sequence

$$\pi_i(\mathrm{SU}(n)) \to \pi_i(U(n)) \to \pi_i(S^1)$$

together with $\pi_i(S^1) = 0$ for $i \geq 2$ one gets $\pi_i(U(n)) \cong \pi_i(\mathrm{SU}(n))$ for $i \geq 2$. At $i = 1$ the sequence reads $0 = \pi_1(\mathrm{SU}(n)) \to \pi_1(U(n)) \to \pi_1(S^1) \to \pi_0(\mathrm{SU}(n)) = 0$, using $\pi_1(\mathrm{SU}(n)) = 0$ from [18, The Fundamental Groups of SU(2), SO(3), and SU(n)], so $\pi_1(U(n)) \cong \pi_1(S^1) \cong \mathbb{Z}$. That is the U clause of (2).

Step 2: the three sphere fibrations. The unit spheres carry transitive actions with the smaller group as stabilizer, so by [18, Smooth Orbit-Stabilizer]

$$S^{n-1} \cong \mathrm{SO}(n)/\mathrm{SO}(n-1), \qquad S^{2n-1} \cong \mathrm{SU}(n)/\mathrm{SU}(n-1), \qquad S^{4n-1} \cong \mathrm{Sp}(n)/\mathrm{Sp}(n-1)$$

the first two being [18, Spheres as Quotients of Orthogonal and Unitary Groups] and the third the same argument for the action of $\mathrm{Sp}(n)$ on the unit sphere of $\mathbb{H}^n \cong \mathbb{R}^{4n}$, whose stabilizer of a unit vector is the copy of $\mathrm{Sp}(n-1)$ acting on the orthogonal $\mathbb{H}^{n-1}$.

Step 3: π_1. $O(1) = \{\pm 1\}$ is discrete, so $\pi_1(O(1)) = 0$, and $\mathrm{SO}(2) \cong S^1$ gives $\pi_1(O(2)) \cong \mathbb{Z}$. For $n \geq 4$ the fibration of Step 2 gives exactness of

$$\pi_2(S^{n-1}) \to \pi_1(\mathrm{SO}(n-1)) \to \pi_1(\mathrm{SO}(n)) \to \pi_1(S^{n-1})$$

in which both outer terms vanish because $n-1 \geq 3$, so $\pi_1(\mathrm{SO}(n)) \cong \pi_1(\mathrm{SO}(n-1))$. Descending to the base case, $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2\mathbb{Z}$ by [18, The Fundamental Group of SO(3)], which gives $\pi_1(O(n)) \cong \mathbb{Z}/2\mathbb{Z}$ for every $n \geq 3$. For Sp, the fibration gives $\pi_1(\mathrm{Sp}(n-1)) \to \pi_1(\mathrm{Sp}(n)) \to \pi_1(S^{4n-1}) = 0$, so each $\pi_1(\mathrm{Sp}(n))$ is a quotient of its predecessor. The base $\mathrm{Sp}(1) \cong S^3$ has $\pi_1 = 0$, so $\pi_1(\mathrm{Sp}(n)) = 0$ for all $n \geq 1$, and the same sequence one step lower shows each $\mathrm{Sp}(n)$ is path-connected, completing (1).

Step 4: π_2. In each of the three families the fibration gives

$$\pi_2(G_{n-1}) \to \pi_2(G_n) \to \pi_2(S^d)$$

with $d = n-1,\ 2n-1,\ 4n-1$ respectively. Each of those three exponents is at least 3 in the range being treated, since $2n-1 \geq 3$ for $n \geq 2$, $4n-1 \geq 3$ for $n \geq 1$, and $n-1 \geq 3$ for $n \geq 4$, so the right-hand term vanishes and $\pi_2(G_n)$ is a quotient of $\pi_2(G_{n-1})$. It therefore suffices to kill the base cases. $\mathrm{SU}(1)$ and $\mathrm{Sp}(0)$ are trivial. For the orthogonal family, $\mathrm{SO}(1)$ is trivial and $\mathrm{SO}(2) \cong S^1$ has $\pi_2 = 0$. For $\mathrm{SO}(3)$, the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of [18, The Fundamental Group of SO(3)] is a fiber bundle with discrete fiber, so its long exact sequence reads $0 \to \pi_2(S^3) \to \pi_2(\mathrm{SO}(3)) \to 0$ and $\pi_2(\mathrm{SO}(3)) \cong \pi_2(S^3) = 0$. Induction upward now gives $\pi_2 = 0$ in all three families, and Step 1 transports the conclusion from $\mathrm{SO}(n)$ to $O(n)$ and from $\mathrm{SU}(n)$ to $U(n)$, completing the proof.

The vanishing in (3) is not special to the classical groups: $\pi_2(G) = 0$ for every compact Lie group G, a theorem whose general proof runs through the cell structure of the flag manifold G/T rather than through sphere fibrations. Only the classical case is needed here, and only the classical case is proved above.

Example 1.2.12: Vector Bundles On Spheres

Each item reads a clause of proposition 1.2.11 through proposition 1.2.10 with $X = S^{m-1}$, so that $SX = S^m$ and

$$\mathrm{Vect}^n_{\mathbb{C}}(S^m) \leftrightarrow \pi_{m-1}(U_n), \qquad \mathrm{Vect}^n_{\mathbb{H}}(S^m) \leftrightarrow \pi_{m-1}(Sp_n), \qquad \mathrm{Vect}^n_{\mathbb{R}}(S^m) \leftrightarrow \pi_{m-1}(O_n)/\pi_0(O_n)$$

the real case taken modulo the conjugation action.

1. $X = S^0$. By proposition 1.2.11(1), $|\pi_0(O_n)| = 2$ for every $n \geq 1$, so there are exactly two real bundles of each rank on S^1: the trivial bundle and, for rank one, the Möbius bundle. Direct sums of copies of these two give every real vector bundle on S^1. Since $U(n)$ is path-connected, $|\pi_0(U_n)| = 1$ and every complex vector bundle on S^1 is trivial.

2. $X = S^1$. By proposition 1.2.11(2), $\pi_1(O_1) = 0$, so S^2 carries just the trivial real line bundle. Next, $\pi_1(O_2) \cong \mathbb{Z}$, so there are infinitely many rank-two real bundles on S^2, indexed by the absolute value of the degree of the clutching map as in proposition 1.2.9. Finally

$\pi_1(O_n) \cong \mathbb{Z}/2\mathbb{Z}$ for $n \geq 3$, so there are exactly two real bundles on S^2 of each rank $n \geq 3$. Complex line bundles on S^2 show the same phenomenon in the other direction, $\pi_1(U_1) \cong \mathbb{Z}$ giving the infinite family computed earlier in this section.

3. $X = S^2$. By proposition 1.2.11(3), π_2 vanishes for all three families, so every real, complex and quaternionic vector bundle on S^3 is trivial.

4. $X = S^3$. Both line-bundle cases on S^4 are trivial, since $\pi_3(O_1) = \pi_3(U_1) = 0$ ($O(1)$ is discrete and $U(1) \cong S^1$ has vanishing higher homotopy). In higher rank the answer is $\mathbb{Z}$ in each family,

$$\pi_3(O_n) \cong \mathbb{Z} \quad (n \geq 5), \qquad \pi_3(U_n) \cong \mathbb{Z} \quad (n \geq 2), \qquad \pi_3(Sp_n) \cong \mathbb{Z} \quad (n \geq 1)$$

so each gives an infinite family of rank-n bundles on S^4. These three are quoted, not proved here: the sphere-fibration induction that settled π_1 and π_2 stalls at π_3, where the fiber sphere's own π_3 no longer vanishes, and the standard arguments need either the explicit quaternionic maps $S^3 \to \mathrm{SO}(4)$ or the periodicity theorem of section 2.2. See Husemoller, *Fibre Bundles*, §8.12.6 and §8.12.11 for the three computations and §8.4 for the stability that fixes the ranges [11]. The stable values agree with theorem 2.2.16: $\pi_3(U) \cong \mathbb{Z}$ since 3 is odd, and $\pi_3(O) \cong \mathbb{Z}$ from the eight-fold table.

1.2.2 Universal Bundles

In this section, we shall construct the canonical vector bundle on the Grassmannian G_n over $\mathbb{R}, \mathbb{C}$, or $\mathbb{H}$; the theorem that every real or complex vector bundle over a paracompact base is a pullback of it is [14, Classification of Vector Bundles over a Paracompact Base]. As a quick reminder of the Grassmannian, let V be a finite dimensional vector space and let $\mathrm{Grass}_k(V)$ be all k-dimensional subspaces of V; i.e. the *Grassmannian manifold* of all sub-spaces of V . Let's define a topology on $\mathrm{Grass}_n(\mathbb{R}^k)$ by first defining the *Stiefel Manifold* $V_n(\mathbb{R}^k)$ which is the space of orthonormal n-frames in $\mathbb{R}^k$. This is a subspace of the product of n copies of the unit sphere S^{k-1}, namely the subspace of orthonormal n-tuples. It is a closed subspace, orthonormality being given by the equations $\langle v_i, v_j \rangle = \delta_{ij}$. Hence $V_n(\mathbb{R}^k)$ is compact, being a closed subspace of a compact product, not that product itself. Then there is a natural surjection $V_n(\mathbb{R}^k) \to \mathrm{Grass}_n(\mathbb{R}^k)$ sending n-frames to the subspace it spans. This gives the quotient topology on $G_n(\mathbb{R}^k)$, and it is easy to verify it is also compact. To show its a manifold, see see [16]. The *infinite Grassmannian* Grass_n is the union all $\mathrm{Grass}_n(V)$ as V ranges over all finite dimensional subspace of a fixed infinite dimensional vector space (usually $\mathbb{R}^\infty$, $\mathbb{C}^\infty$, or $\mathbb{H}^\infty$). With a bit of work, we see that it is an infinite-dimensional CW complex (it is the limit of CW complexes respecting the appropriate directed mappings). These are rather natural, for example if taking $n = 1$, we get that Grass_1 is $\mathbb{RP}^\infty$, $\mathbb{CP}^\infty$, or $\mathbb{HP}^\infty$ respectively.

Just as for $\mathbb{RP}^n$, $\mathbb{CP}^n$ and $\mathbb{HP}^n$, the Grassmannian carries a tautological bundle, of rank n rather than one: let

$$E_n(\mathbb{R}^k) = \left\{ (\ell, v) \in \mathrm{Grass}_n(\mathbb{R}^k) \times \mathbb{R}^k \;\middle|\; v \in \ell \right\}$$

and similarly for $\mathbb{C}^k, \mathbb{H}^k$. Then the inclusion $\mathbb{R}^k \subseteq \mathbb{R}^{k+1} \subseteq \cdots$ induces inclusions $E_n(\mathbb{R}^k) \subseteq E_n(\mathbb{R}^{k+1}) \subseteq \cdots$, and hence

$$E_n(\mathbb{R}^\infty) = \cup_k E_n(\mathbb{R}^k)$$

where it is given the colimit topology, and similarly for the complex and quaternion case. Then:

Proposition 1.2.13: Grassmannian Bundle is Vector Bundle

The projection $p : E_n(\mathbb{R}^k) \to \mathrm{Grass}_n(\mathbb{R}^k)$ where $p(\ell, v) = \ell$ is a vector bundle for finite and infinite k.

***Proof*:**
By proving the finite case, the infinite case follows by local trivializations working over every $k \in \mathbb{N}$, hence we prove the finite case. For $\ell \in G_n(\mathbb{R}^k)$, let $\pi_\ell : \mathbb{R}^k \to \ell$ be orthogonal projection and let

$$U_\ell = \left\{\ell' \in G_n(\mathbb{R}^k) \;\middle|\; \pi_\ell(\ell') \text{ has dimension } n\right\}$$

so that $\ell \in U_\ell$. We will show that U_ℓ is open in $G_n(\mathbb{R}^k)$ and that the map $h : p^{-1}(U_\ell) \to U_\ell \times \ell \cong U_\ell \times \mathbb{R}^n$ defined by $h(\ell', v) = (\ell', \pi_\ell(v))$ is a local trivialization of $E_n(\mathbb{R}^k)$.

For U_ℓ to be open is equivalent to its preimage in $V_n(\mathbb{R}^k)$ being open. This preimage consists of orthonormal frames v_i, $1 \leq i \leq n$, such that $\pi_\ell(v_1), \dots, \pi_\ell(v_n)$ are independent. Let A be the matrix of π_ℓ with respect to the standard basis in the domain $\mathbb{R}^k$ and any fixed basis in the range ℓ. The condition on v_i is then that the $n \times n$ matrix with columns Av_i has nonzero determinant. Since the value of this determinant is a continuous function of v_i, it follows that the frames v_i yielding a nonzero determinant form an open set in $V_n(\mathbb{R}^k)$.

Then it's clear that h is a bijection which is a linear isomorphism on each fiber. Furthermore h is continuous since its two coordinate functions are continuous. To see that h^{-1} is continuous, for $\ell' \in U_\ell$ there is a unique invertible linear map $L_{\ell'} : \mathbb{R}^k \to \mathbb{R}^k$ restricting to π_ℓ on ℓ' and the identity on $\ell^\perp = \mathrm{Ker}\ \pi_\ell$. Then $L_{\ell'}$, regarded as a $k \times k$ matrix, depends continuously on ℓ'. Indeed, we can write L_ℓ as a product AB^{-1} where:

- B sends the standard basis to $v_1, \dots, v_n, v_{n+1}, \dots, v_k$ with $v_1, \dots, v_n$ an orthonormal basis for ℓ' and $v_{n+1}, \dots, v_k$ a fixed basis for $\ell^\perp$.
- A sends the standard basis to $\pi_\ell(v_1), \dots, \pi_\ell(v_n), v_{n+1}, \dots, v_k$.

Both A and B depend continuously on $v_1, \dots, v_n$. Since matrix multiplication and matrix inversion are continuous operations, it follows that the product $L_{\ell'} = AB^{-1}$ depends continuously on $v_1, \dots, v_n$. But since $L_{\ell'}$ depends only on ℓ', not on the basis $v_1, \dots, v_n$ for ℓ', it follows that $L_{\ell'}$ depends continuously on ℓ' since $G_n(\mathbb{R}^k)$ has the quotient topology from $V_n(\mathbb{R}^k)$. We have $h^{-1}(\ell', w) = (\ell', L_{\ell'}^{-1}(w))$ with $L_{\ell'}^{-1}$ depending continuously on ℓ' since matrix inversion is continuous, so h^{-1} is continuous, and so we have a local trivialization, as we sought to show.

Definition 1.2.14: Universal Bundle

The tautological bundles over the Grassmannians Grass_n are called the *universal bundles*. The Grassmannians themselves serve as the classifying spaces, and for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO_n, BU_n, and BSp_n respectively.

Every vector bundle carries a natural structure group, namely O_n, U_n or Sp_n according as the bundle is real, complex or quaternionic. The infinite Grassmannian is for that reason called the classifying space of O_n, U_n or Sp_n, and is denoted BO_n, BU_n or BSp_n respectively.

1.3 The Stabilization Theorem

The classification of [14, Classification of Vector Bundles over a Paracompact Base] reduces rank-n real or complex bundles over X to homotopy classes of maps into the Grassmannian Grass_n, one classification for each rank separately. That is not yet a single invariant of the space: the sets $[X, \mathrm{Grass}_n]$ are indexed by n, and bundles of different rank have no common home. The remedy is to let the rank grow. Write $\mathbf{VB}_n(X)$ for the set $\mathrm{Vect}_k^n(X)$ of isomorphism classes of rank-n bundles over X (definition 1.1.5), the field k being whichever of $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ is in play, and $\mathbf{VB}(X)$ for the union over all n. Adding a trivial line maps $\mathbf{VB}_n(X)$ to $\mathbf{VB}_{n+1}(X)$, and the colimit along these stabilization maps discards the rank while keeping everything about a bundle that survives stabilization. The theorem below identifies that colimit, for complex bundles, with the reduced K-theory of chapter 2.

Write $\widetilde{K}(X)$ for that reduced group (the kernel of the rank map); it is captured by maps into the stable classifying space.

Theorem 1.3.1: Stabilization Theorem of Classifying Spaces

Let X be a connected compact Hausdorff space, and take the field to be $\mathbb{C}$. Then

$$\widetilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X) \cong \varinjlim_n [X, \mathrm{Grass}_n] = \varinjlim_n [X, BU_n]$$

Over $\mathbb{R}$ the same holds, with the real reduced group $\widetilde{KO}(X)$ in place of $\widetilde{K}(X)$ and BO_n in place of BU_n.

***Proof*:**
Over a compact Hausdorff base the trivial bundles are cofinal in $\mathbf{VB}(X)$: every bundle E is a summand of a trivial bundle (proposition 1.1.15), so $E \oplus E' \cong T^N$ for some complement E' and some N. Cofinality lets every reduced class be written in the normal form $[E] - n$, and by theorem 2.1.8 $[E] - n = [F] - n$ holds in $\widetilde{K}(X)$ exactly when $E \oplus T^k \cong F \oplus T^k$ for some k, that is, when E and F are stably isomorphic. Hence $\widetilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X)$, the colimit taken along the stabilization $E \mapsto E \oplus T^1$. The classification of rank-n bundles by homotopy classes into the Grassmannian ([14, Classification of Vector Bundles over a Paracompact Base]) identifies $\mathbf{VB}_n(X) \cong [X, \mathrm{Grass}_n]$ compatibly with stabilization, giving $\varinjlim_n [X, \mathrm{Grass}_n]$; as the field is $\mathbb{C}$, $\mathrm{Grass}_n = BU_n$. Over $\mathbb{R}$ every input covers real bundles as well (proposition 1.1.15, the real clause of theorem 2.1.8, and the classification), so the same argument gives the real clause with $\mathrm{Grass}_n = BO_n$, completing the proof.

Nothing in the argument is particular to $\mathbb{C}$ except its inputs. Over $\mathbb{R}$ both inputs hold, [14, Classification of Vector Bundles over a Paracompact Base] being stated over $\mathbb{R}$ as well and theorem 2.1.8 carrying a real clause, which is why the theorem carries one too. Over $\mathbb{H}$ the corresponding inputs are not developed here, so the quaternionic analogue $\varinjlim_n [X, BSp_n]$ is only recorded. The stabilization theorem is what makes K-theory representable: the colimit over n replaces the unstable Grassmannians by the single space BU, and chapter 2 restores the rank component to obtain $K(X) \cong [X, \mathbb{Z} \times BU]$.

2

K-Theory for Vector Bundles

The previous chapter solved the classification problem for vector bundles over a paracompact base: by [14, Classification of Vector Bundles over a Paracompact Base], the isomorphism classes of n-dimensional bundles over X are the homotopy classes $[X, \mathrm{Grass}_n]$, and by theorem 1.2.5 this set is a homotopy invariant of X. Yet the collection of all vector bundles over X, taken together under the direct sum of definition 1.1.11, carries more structure than a bare set of homotopy classes: it is a commutative monoid. The single defect of this monoid is that it has no inverses, since a nonzero bundle can never sum to the zero bundle. K-theory is what results from repairing exactly this defect. One formally adjoins inverses to the monoid of vector bundles, in the same way that the integers are built from the natural numbers, and the resulting group $K^0(X)$ becomes a powerful and computable invariant of the space.

This chapter develops topological K-theory from that single idea. The group $K^0(X)$ is introduced here as the group completion of the monoid of bundles, with its reduced companion $\widetilde{K}^0(X)$ recording bundles up to the addition of trivial summands. The stabilization theorem of the previous chapter (theorem 1.3.1) already hints that this group is representable: stable equivalence classes of bundles are homotopy classes of maps into the infinite Grassmannian. Making this precise identifies $K^0(X)$ with the homotopy functor $[X, \mathbb{Z} \times BU]$ in the complex case, and the tensor product of definition 1.1.16 upgrades the group to a ring. The chapter then turns these structural results into a genuine cohomology theory: Bott periodicity (theorem 2.2.16) gives the periodicity isomorphism that lets one define negative-degree groups, and the resulting theory satisfies the generalized cohomology axioms. We close with explicit computations, including the K-theory of spheres and projective spaces.

2.1 The Group $K^0(X)$

Throughout this section X denotes a compact Hausdorff space, and $\mathrm{Vect}(X)$ denotes the set of isomorphism classes of (finite-dimensional, complex) vector bundles over X. The direct sum of definition 1.1.11 descends to isomorphism classes: if $E \cong E'$ and $F \cong F'$, then $E \oplus F \cong E' \oplus F'$, so $\oplus$ is a well-defined binary operation on $\mathrm{Vect}(X)$. It is commutative and associative up to isomorphism, and the zero bundle (the rank-0 bundle, whose total space is X itself) is a two-sided identity. Thus $(\mathrm{Vect}(X), \oplus)$ is a commutative monoid. The obstruction to it being a group is total: the only invertible element is 0, because the rank of a direct sum is the sum of the ranks, and ranks are non-negative integers. The construction that follows forces inverses to exist by a purely formal device, exactly the device that produces $\mathbb{Z}$ from $\mathbb{N}$.

Definition 2.1.1: The Grothendieck Group $K^0(X)$

Let X be a compact Hausdorff space and let $(\mathrm{Vect}(X), \oplus)$ be the commutative monoid of isomorphism classes of complex vector bundles over X. The *Grothendieck group* (or *topological K-theory group*) $K^0(X)$ is the abelian group with one generator $[E]$ for each isomorphism class of bundle E and one relation

$$[E \oplus F] = [E] + [F]$$

for each pair of bundles E, F. Equivalently, $K^0(X)$ is the quotient of the free abelian group on isomorphism classes of bundles by the subgroup generated by all elements $[E \oplus F] - [E] - [F]$.

The elements of $K^0(X)$ are *formal differences* $[E] - [F]$ of bundles, called *virtual bundles*, and two such differences $[E] - [F]$ and $[E'] - [F']$ are equal precisely when $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some bundle G. The auxiliary bundle G is unavoidable: it encodes the equivalence relation built into the Grothendieck construction, where one identifies pairs that become equal only after adding a common summand. The class $[E]$ of a bundle is the image of the monoid element E under the canonical map $\mathrm{Vect}(X) \to K^0(X)$, and every element of $K^0(X)$ is a difference of two such classes. The universal property that pins down $K^0(X)$ is that any monoid homomorphism from $\mathrm{Vect}(X)$ into an abelian group factors uniquely through $K^0(X)$; this is what makes K-theory the receptacle for every additive invariant of vector bundles.

Example 2.1.2: K^0 of a Point

Let $X = \{*\}$ be the one-point space. A vector bundle over a point is the same data as its single fiber, a finite-dimensional complex vector space, so isomorphism classes of bundles are classified by dimension. Hence $\mathrm{Vect}(\{*\}) \cong \mathbb{N}$ as monoids, with $\oplus$ corresponding to addition of dimensions and $[\mathbb{C}^n]$ corresponding to n. The Grothendieck group of $(\mathbb{N}, +)$ is $\mathbb{Z}$: the formal differences $m - n$ of natural numbers are exactly the integers, with $m - n = m' - n'$ in $\mathbb{Z}$ iff $m + n' = m' + n$, which is the Grothendieck relation with trivial auxiliary term. Therefore

$$K^0(\{*\}) \cong \mathbb{Z}$$

with the class $[\mathbb{C}^n] - [\mathbb{C}^m]$ corresponding to the integer $n - m$. The isomorphism sends a virtual bundle to its rank, the difference of the dimensions of its positive and negative parts.

The rank computation in this example is the universal one. Over a connected base, every bundle has a well-defined rank (a single non-negative integer), and rank is additive under direct sum, so it defines a monoid homomorphism $\mathrm{Vect}(X) \to \mathbb{N} \hookrightarrow \mathbb{Z}$. By the universal property this extends to a group homomorphism, which is the central tool for splitting off the trivial part of $K^0(X)$.

Proposition 2.1.3: The Rank Homomorphism

Let X be a compact Hausdorff space with connected components $X_1, \dots, X_r$. Assigning to a bundle its rank on each component defines a surjective group homomorphism

$$\mathrm{rk}\colon K^0(X) \longrightarrow \mathbb{Z}^r, \qquad [E] \longmapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$$

which is split by the homomorphism $\mathbb{Z}^r \to K^0(X)$ sending the i-th standard generator to the class of the line bundle that is trivial of rank 1 on X_i and rank 0 elsewhere. In particular, when X is connected, there is a split surjection $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}$.

***Proof*:**
On each component X_i the rank of a bundle is a single non-negative integer, because the rank function $X_i \to \mathbb{N}$ is locally constant by the local triviality of definition 1.1.2 and X_i is connected. Hence $E \mapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$ is a well-defined map $\mathrm{Vect}(X) \to \mathbb{N}^r$, and it is a monoid homomorphism since rank is additive under $\oplus$. By the universal property of definition 2.1.1 it extends uniquely to a group homomorphism $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}^r$.

For the splitting, let T_i denote the bundle that restricts to the trivial line bundle on X_i and to the rank-0 bundle on the remaining components; this is a genuine bundle over X since the components are open and closed. Then $\mathrm{rk}[T_i]$ is the i-th standard generator $e_i \in \mathbb{Z}^r$, so the homomorphism $s\colon \mathbb{Z}^r \to K^0(X)$ defined on generators by $s(e_i) = [T_i]$ satisfies $\mathrm{rk} \circ s = \mathrm{id}$. This exhibits rk as a split surjection, and surjectivity is immediate from the existence of the splitting, completing the proof.

The kernel of the rank homomorphism is the part of $K^0(X)$ that sees genuine twisting rather than the dimension count. Isolating it gives the reduced theory. For the rest of the section it is convenient to assume X is connected so that rank is a single integer; the general statement over r components differs only by replacing $\mathbb{Z}$ with $\mathbb{Z}^r$.

Definition 2.1.4: Reduced K-Theory $\widetilde{K}^0(X)$

Let (X, x_0) be a pointed compact Hausdorff space. The *reduced K-theory* of X is the kernel of restriction to the basepoint,

$$\widetilde{K}^0(X) = \ker\bigl(x_0^*\colon K^0(X) \to K^0(\{x_0\}) = \mathbb{Z}\bigr)$$

Restriction to x_0 records the rank of a virtual bundle at the basepoint, so when X is *connected* this is the kernel of the rank homomorphism $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}$ and $\widetilde{K}^0(X)$ is the subgroup of virtual bundles of rank zero. Over a disconnected base the two differ, and it is the basepoint version that the rest of the book uses: for $X = S^0$ the rank-at-basepoint kernel is $\widetilde{K}^0(S^0) \cong \mathbb{Z}$, generated by the class $[\mathbb{C}_-]$ of the line bundle supported at the non-basepoint, whereas the total-rank kernel is 0.

Reduced K-theory discards the information that any space, however trivial, already carries: the integer recording the dimension of a bundle. What remains in $\widetilde{K}^0(X)$ is the obstruction to a virtual bundle being a difference of trivial bundles. The splitting of proposition 2.1.3 shows that this discarding loses nothing beyond the dimension count, since the full group decomposes as the reduced group together with a copy of the integers.

Corollary 2.1.5: The Splitting $K^0 = \widetilde{K}^0 \oplus \mathbb{Z}$

For X a connected compact Hausdorff space there is a natural isomorphism

$$K^0(X) \cong \widetilde{K}^0(X) \oplus \mathbb{Z}$$

where the $\mathbb{Z}$ summand is generated by the class $[\underline{\mathbb{C}}]$ of the trivial line bundle and the projection to $\mathbb{Z}$ is the rank.

***Proof*:**
The rank homomorphism $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}$ is split surjective by proposition 2.1.3, with splitting $s\colon \mathbb{Z} \to K^0(X)$, $s(1) = [\underline{\mathbb{C}}]$. For a split short exact sequence of abelian groups

$$0 \to \widetilde{K}^0(X) \to K^0(X) \xrightarrow{\mathrm{rk}} \mathbb{Z} \to 0$$

the splitting lemma gives $K^0(X) \cong \ker(\mathrm{rk}) \oplus \mathbb{Z} = \widetilde{K}^0(X) \oplus \mathbb{Z}$, the isomorphism being $\xi \mapsto (\xi - s(\mathrm{rk}\,\xi), \mathrm{rk}\,\xi)$ with inverse $(\eta, n) \mapsto \eta + n[\underline{\mathbb{C}}]$. Naturality in X holds because rank, the trivial bundle, and the splitting are all preserved by pullback along continuous maps, completing the proof.

To make $\widetilde{K}^0(X)$ usable one needs a concrete description of its elements, and this is where the compactness of X enters decisively. The Grothendieck construction in general produces formal differences $[E] - [F]$ with two genuine bundles in play, and the equivalence relation involves an unknown auxiliary summand G. Over a compact base both of these complications collapse. The mechanism is that trivial bundles are *cofinal* in $\mathrm{Vect}(X)$: every bundle embeds in a trivial one as a direct summand, by proposition 1.1.15. The next two results extract from cofinality first a normal form for elements of $K^0(X)$ and then a clean criterion for two bundles to be equal in the reduced group.

Lemma 2.1.6: Cofinality and the Normal Form $[E] - n$

Let X be a compact Hausdorff space and write $\underline{\mathbb{C}}^n$ for the trivial bundle of rank n.

1. For every bundle F over X there is a bundle F' and an integer n with $F \oplus F' \cong \underline{\mathbb{C}}^n$.
2. Every element of $K^0(X)$ can be written in the form $[E] - n := [E] - [\underline{\mathbb{C}}^n]$ for some bundle E and some $n \geq 0$.

***Proof*:**
Part (1): cofinality. By proposition 1.1.15, for the bundle F over the compact Hausdorff space X there is a complementary bundle F' with $F \oplus F'$ isomorphic to a trivial bundle, say $F \oplus F' \cong \underline{\mathbb{C}}^n$ where $n = \mathrm{rk}F + \mathrm{rk}F'$. This is the cofinality statement: every bundle is a summand of a trivial bundle.

Part (2): normal form. A general element of $K^0(X)$ is a formal difference $[E_0] - [F]$ of two bundle classes, since by definition 2.1.1 the group is generated by the classes $[E]$ under addition and subtraction, and any such combination collapses to a single difference using $[E] + [E'] = [E \oplus E']$.

Apply part (1) to F: choose F' with $F \oplus F' \cong \underline{\mathbb{C}}^n$. Then in $K^0(X)$,

$$[E_0] - [F] = [E_0] - [F] + \big([F'] - [F']\big) = [E_0] + [F'] - \big([F] + [F']\big) = [E_0 \oplus F'] - [\underline{\mathbb{C}}^n]$$

using $[F]+[F'] = [F \oplus F'] = [\underline{\mathbb{C}}^n]$. Setting $E = E_0 \oplus F'$ gives the normal form $[E_0] - [F] = [E] - n$, completing the proof.

The normal form $[E] - n$ replaces a formal difference of two unknown bundles by a single bundle minus a trivial summand, which is a substantial simplification: it says every virtual bundle is, up to a shift by trivial bundles, an actual bundle. The same cofinality argument now eliminates the auxiliary summand G from the equivalence relation, pinning down exactly when two normal forms agree. This is the telescoping argument.

Definition 2.1.7: Stable Isomorphism and Stable Equivalence

Two vector bundles E and F over X are *stably isomorphic*, written $E \cong_s F$, if there is an integer $k \geq 0$ with

$$E \oplus \underline{\mathbb{C}}^k \cong F \oplus \underline{\mathbb{C}}^k$$

(the *same* trivial summand on each side). They are *stably equivalent*, written $E \approx_s F$, if there exist integers $m, n \geq 0$ with

$$E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$$

(possibly *different* trivial summands). A bundle is *stably trivial* (in the sense of definition 1.1.12) precisely when it is stably equivalent to a trivial bundle.

The two notions differ by exactly the rank: stable isomorphism preserves rank ($\mathrm{rk}E = \mathrm{rk}F$, since adding $\underline{\mathbb{C}}^k$ to both sides forces equal ranks), whereas stable equivalence allows the ranks to differ. Stable isomorphism is the relation that equality in $K^0(X)$ detects, and stable equivalence is the coarser relation that equality in $\widetilde{K}^0(X)$ detects. The next theorem proves both statements, with the cofinality lemma giving the telescoping that turns the abstract Grothendieck congruence into these concrete bundle isomorphisms.

Theorem 2.1.8: Reduced K-Theory Classifies Stable Equivalence

Let X be a connected compact Hausdorff space, and let E, F be vector bundles over X.

1. In $K^0(X)$ one has $[E] = [F]$ if and only if E and F are stably isomorphic, $E \cong_s F$.
2. Sending a bundle E to the class $[E] - \mathrm{rk}E \in \widetilde{K}^0(X)$ induces a bijection between stable equivalence classes of bundles and elements of $\widetilde{K}^0(X)$. Under this bijection $\widetilde{K}^0(X)$ is the set of stable equivalence classes, made into a group by direct sum.

The real case holds verbatim: running the constructions of this section on real bundles, with $\underline{\mathbb{R}}$ for $\underline{\mathbb{C}}$, produces the real groups $KO^0(X)$ and $\widetilde{KO}^0(X)$, and both parts hold for them.

***Proof*:**
Part (1). Suppose first that $E \cong_s F$, so $E \oplus \underline{\mathbb{C}}^k \cong F \oplus \underline{\mathbb{C}}^k$ for some k. Passing to classes in

$K^0(X)$ and using definition 2.1.1,

$$[E] + k[\underline{\mathbb{C}}] = [E \oplus \underline{\mathbb{C}}^k] = [F \oplus \underline{\mathbb{C}}^k] = [F] + k[\underline{\mathbb{C}}]$$

Cancelling $k[\underline{\mathbb{C}}]$ in the abelian group $K^0(X)$ gives $[E] = [F]$.

For the converse, suppose $[E] = [F]$ in $K^0(X)$. By construction of the Grothendieck group, the relation $[E] = [F]$ holds exactly when E and F become isomorphic after adding a common bundle G, that is $E \oplus G \cong F \oplus G$ for some bundle G. This is the defining congruence of definition 2.1.1: equality of $[E]$ and $[F]$ in the group completion of the monoid $\mathrm{Vect}(X)$ means precisely that there exists G with $E \oplus G \cong F \oplus G$. Now apply cofinality (lemma 2.1.6(1)) to G: choose G' with $G \oplus G' \cong \underline{\mathbb{C}}^k$. Adding G' to both sides of $E \oplus G \cong F \oplus G$,

$$E \oplus G \oplus G' \cong F \oplus G \oplus G' \quad \Longrightarrow \quad E \oplus \underline{\mathbb{C}}^k \cong F \oplus \underline{\mathbb{C}}^k$$

This is exactly stable isomorphism, $E \cong_s F$. The telescoping step (replacing the unknown summand G by a trivial summand $\underline{\mathbb{C}}^k$) is what makes the equivalence concrete.

Part (2). Define Φ on bundles by $\Phi(E) = [E] - \mathrm{rk}E \cdot [\underline{\mathbb{C}}] \in K^0(X)$. Since $\mathrm{rk}\big([E] - \mathrm{rk}E\,[\underline{\mathbb{C}}]\big) = \mathrm{rk}E - \mathrm{rk}E = 0$, the image lies in $\widetilde{K}^0(X)$. First, $\Phi(E) = \Phi(F)$ if and only if $E \approx_s F$. Indeed,

$$\Phi(E) = \Phi(F) \iff [E] + \mathrm{rk}F\,[\underline{\mathbb{C}}] = [F] + \mathrm{rk}E\,[\underline{\mathbb{C}}] \iff [E \oplus \underline{\mathbb{C}}^{\mathrm{rk}F}] = [F \oplus \underline{\mathbb{C}}^{\mathrm{rk}E}]$$

and by part (1) the last equality means $E \oplus \underline{\mathbb{C}}^{\mathrm{rk}F} \cong_s F \oplus \underline{\mathbb{C}}^{\mathrm{rk}E}$, that is $E \oplus \underline{\mathbb{C}}^{\mathrm{rk}F+k} \cong F \oplus \underline{\mathbb{C}}^{\mathrm{rk}E+k}$ for some k; setting $m = \mathrm{rk}F + k$ and $n = \mathrm{rk}E + k$, this is precisely $E \approx_s F$. Conversely any stable equivalence $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ runs the chain backwards, since it gives $[E] + m[\underline{\mathbb{C}}] = [F] + n[\underline{\mathbb{C}}]$ and hence $\Phi(E) = [E] - \mathrm{rk}E[\underline{\mathbb{C}}] = [F] - \mathrm{rk}F[\underline{\mathbb{C}}] = \Phi(F)$ after using $\mathrm{rk}E + m = \mathrm{rk}F + n$. Thus Φ is injective on stable equivalence classes.

For surjectivity, let $\eta \in \widetilde{K}^0(X)$. By lemma 2.1.6(2) write $\eta = [E] - n$ for a bundle E; applying rank gives $0 = \mathrm{rk}\eta = \mathrm{rk}E - n$, so $n = \mathrm{rk}E$ and $\eta = [E] - \mathrm{rk}E\,[\underline{\mathbb{C}}] = \Phi(E)$. Hence Φ is onto. The group operation $\Phi(E) + \Phi(F) = \Phi(E \oplus F)$ holds by additivity of rank and of the bracket, so the bijection transports $\oplus$ to the group law of $\widetilde{K}^0(X)$. Nothing in either part is particular to $\mathbb{C}$: with $\underline{\mathbb{R}}$ for $\underline{\mathbb{C}}$ the same proof runs on real bundles, cofinality over $\mathbb{R}$ given by proposition 1.1.15, giving the real clause and completing the proof.

The content of theorem 2.1.8 is that the reduced group $\widetilde{K}^0(X)$ is not an abstract quotient but a concrete object: its elements are the stable equivalence classes of bundles, two bundles representing the same element exactly when they agree after stabilizing by trivial summands. This is the form in which K-theory becomes computable, because stable equivalence of bundles over a compact space is governed by maps into the infinite Grassmannian. Indeed the stabilization theorem of the previous chapter (theorem 1.3.1) identifies $\widetilde{K}^0(X)$ with $\varinjlim_n [X, \mathrm{Grass}_n] = [X, BU]$, and assembling this with the rank splitting gives a representing space for the full group.

Theorem 2.1.9: Representability of K^0

Let X be a connected compact Hausdorff space. There are natural isomorphisms

$$\widetilde{K}^0(X) \cong [X, BU], \qquad K^0(X) \cong [X, \mathbb{Z} \times BU]$$

where $BU = \varinjlim_n BU_n$ is the infinite complex Grassmannian and $[X, \mathbb{Z} \times BU]$ denotes homotopy classes of maps into $\mathbb{Z} \times BU$. In the real case the same holds with BU replaced by BO, giving $\widetilde{KO}^0(X) \cong [X, BO]$ and $KO^0(X) \cong [X, \mathbb{Z} \times BO]$.

***Proof*:**
Step 1: the reduced group as maps into BU. By theorem 2.1.8, $\widetilde{K}^0(X)$ is the set of stable equivalence classes of bundles over X. By the stabilization theorem (theorem 1.3.1), for X compact the natural maps assemble into an isomorphism

$$\widetilde{K}^0(X) \cong \varinjlim_n \mathrm{Vect}^n(X) \cong \varinjlim_n [X, BU_n]$$

where the colimit is taken along the stabilization maps $\mathrm{Vect}^n(X) \to \mathrm{Vect}^{n+1}(X)$, $E \mapsto E \oplus \underline{\mathbb{C}}$, which on classifying maps correspond to the inclusions $BU_n \hookrightarrow BU_{n+1}$ given by $V \mapsto V \oplus \mathbb{C}$. Each identification $\mathrm{Vect}^n(X) \cong [X, BU_n]$ is [14, Classification of Vector Bundles over a Paracompact Base] (in its complex form, with $\mathrm{Grass}_n = BU_n$ the complex Grassmannian of definition 1.2.14). Since X is compact, every map $X \to BU = \varinjlim_n BU_n$ has image in some BU_n (a compact subset of the colimit $\varinjlim_n BU_n$ of the closed inclusions $BU_n \hookrightarrow BU_{n+1}$ meets only finitely many of the cells adjoined at each stage, hence lies in a single finite stage BU_n), and likewise the compact image of any homotopy $X \times [0,1] \to BU$ between two such maps lands in some BU_N. Therefore

$$\varinjlim_n [X, BU_n] \cong [X, \varinjlim_n BU_n] = [X, BU]$$

Combining the displays gives $\widetilde{K}^0(X) \cong [X, BU]$. This isomorphism is natural in X: a map $f\colon X' \to X$ induces $f^*\colon \mathrm{Vect}^n(X) \to \mathrm{Vect}^n(X')$ by pullback (proposition 1.2.1), which corresponds to precomposition with f on classifying maps, hence to $f^*\colon [X, BU] \to [X', BU]$.

Step 2: the full group as maps into $\mathbb{Z} \times BU$. Since X is connected, $[X, \mathbb{Z}] \cong \mathbb{Z}$, the homotopy classes of maps into the discrete space $\mathbb{Z}$ being the constant maps. The splitting of corollary 2.1.5 reads $K^0(X) \cong \widetilde{K}^0(X) \oplus \mathbb{Z}$. Combining with Step 1,

$$K^0(X) \cong [X, BU] \oplus \mathbb{Z} \cong [X, BU] \times [X, \mathbb{Z}] \cong [X, BU \times \mathbb{Z}] = [X, \mathbb{Z} \times BU]$$

where the last identification uses that homotopy classes of maps into a product are pairs of homotopy classes, $[X, Y \times Z] \cong [X, Y] \times [X, Z]$, and that for connected X the factor $[X, \mathbb{Z}]$ recovers the rank. Naturality is inherited from Step 1 together with naturality of the rank splitting. The real case is identical, replacing BU_n by $BO_n = \mathrm{Grass}_n(\mathbb{R}^\infty)$ throughout and invoking the real form of [14, Classification of Vector Bundles over a Paracompact Base], completing the proof.

The representability theorem reframes K-theory entirely in homotopy-theoretic terms: computing $K^0(X)$ is computing homotopy classes of maps from X into the fixed space $\mathbb{Z} \times BU$. This is the first sign that K-theory will be a generalized cohomology theory, since cohomology theories are exactly the functors represented by spaces (or spectra). The space $\mathbb{Z} \times BU$ is the degree-zero representing space; Bott periodicity, taken up in the next section, will show that the degree-one space is U and that these two spaces suffice to represent the whole theory. For now the value of the theorem is computational, as the closing examples illustrate.

So far $K^0(X)$ has only an additive structure. The tensor product of definition 1.1.16 gives a multiplication, turning $K^0(X)$ into a commutative ring. The verification that the tensor product descends to the Grothendieck group is a matter of checking that the bilinearity and distributivity of $\otimes$ over $\oplus$ at the level of bundles pass to formal differences.

Proposition 2.1.10: The Ring Structure on $K^0(X)$

Let X be a compact Hausdorff space, connected where the rank clause below is used. The tensor product of vector bundles (definition 1.1.16) induces on $K^0(X)$ a well-defined product

$$\big([E_1] - [F_1]\big) \cdot \big([E_2] - [F_2]\big) = [E_1 \otimes E_2] + [F_1 \otimes F_2] - [E_1 \otimes F_2] - [F_1 \otimes E_2]$$

making $K^0(X)$ a commutative ring with unit $[\underline{\mathbb{C}}]$, the class of the trivial line bundle, for every compact Hausdorff X. For *connected* X the reduced group $\widetilde{K}^0(X)$ is the augmentation ideal of this ring, the kernel of the ring homomorphism $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}$. Connectedness is what makes the rank a single integer rather than one per component, and corollary 2.1.5 is the disconnected statement. Pullback $f^*\colon K^0(X) \to K^0(X')$ along a continuous map is a ring homomorphism.

***Proof*:**
Step 1: the product is well-defined. The assignment $(E_1, E_2) \mapsto [E_1 \otimes E_2]$ is a map $\mathrm{Vect}(X) \times \mathrm{Vect}(X) \to K^0(X)$. It is additive in each variable: by the distributivity of tensor over direct sum at the bundle level, $(E_1 \oplus E_1') \otimes E_2 \cong (E_1 \otimes E_2) \oplus (E_1' \otimes E_2)$ and symmetrically in the second variable (these isomorphisms are recorded in the discussion following definition 1.1.16, where the tensor product is shown to be commutative, associative, unital, and distributive over $\oplus$ fiberwise). Passing to classes,

$$[(E_1 \oplus E_1') \otimes E_2] = [E_1 \otimes E_2] + [E_1' \otimes E_2]$$

so the map is biadditive. A biadditive map from $\mathrm{Vect}(X) \times \mathrm{Vect}(X)$ into an abelian group factors through the tensor product of the Grothendieck groups, equivalently extends uniquely to a $\mathbb{Z}$-bilinear map $K^0(X) \times K^0(X) \to K^0(X)$. Concretely, this forces the formula in the statement: expanding $([E_1] - [F_1])([E_2] - [F_2])$ by bilinearity gives $[E_1][E_2] - [E_1][F_2] - [F_1][E_2] + [F_1][F_2]$, which is the displayed expression. Independence of the representatives of each virtual bundle is exactly the assertion that the formula respects the relation $[E \oplus F] = [E] + [F]$, which biadditivity guarantees.

Step 2: ring axioms. Commutativity, associativity, and the unit law for the product on $K^0(X)$ follow from the corresponding properties of $\otimes$ on bundles, since the product is determined by its values $[E_1] \cdot [E_2] = [E_1 \otimes E_2]$ on the generating classes and these properties are stated for the bundle-level tensor product. Explicitly, $E_1 \otimes E_2 \cong E_2 \otimes E_1$ gives $[E_1][E_2] = [E_2][E_1]$; $(E_1 \otimes E_2) \otimes E_3 \cong E_1 \otimes (E_2 \otimes E_3)$ gives associativity; and $\underline{\mathbb{C}} \otimes E \cong E$ gives $[\underline{\mathbb{C}}][E] = [E]$, so $[\underline{\mathbb{C}}]$ is the unit. Distributivity of the product over addition in $K^0(X)$ is the bilinearity established in Step 1. Hence $(K^0(X), +, \cdot)$ is a commutative ring with unit $[\underline{\mathbb{C}}]$.

Step 3: rank is a ring map and $\widetilde{K}^0$ is its kernel. For bundles over a connected base, $\mathrm{rk}(E_1 \otimes E_2) = \mathrm{rk}E_1 \cdot \mathrm{rk}E_2$ since the fiber of $E_1 \otimes E_2$ over a point is the tensor product of the fibers, of dimension the product of the dimensions. Thus rk is multiplicative as well as additive, and $\mathrm{rk}[\underline{\mathbb{C}}] = 1$, so $\mathrm{rk}\colon K^0(X) \to \mathbb{Z}$ is a unital ring homomorphism. Its kernel is $\widetilde{K}^0(X)$ by definition 2.1.4, which is therefore an ideal, the augmentation ideal of the ring. (Over a base with r components, rk lands in the product ring $\mathbb{Z}^r$.)

Step 4: naturality. For a continuous $f\colon X' \to X$, the pullback satisfies $f^*(E_1 \otimes E_2) \cong f^*E_1 \otimes f^*E_2$ and $f^*(E_1 \oplus E_2) \cong f^*E_1 \oplus f^*E_2$ by proposition 1.2.3, and $f^*\underline{\mathbb{C}} \cong \underline{\mathbb{C}}$. Hence the induced map $f^*\colon K^0(X) \to K^0(X')$ preserves products, sums, and the unit, so it is a ring homomorphism, completing the proof.

With the ring structure in hand, K^0 is a contravariant functor from compact Hausdorff spaces to commutative rings, refining the homotopy-classification of bundles into an algebraic invariant that multiplies. The reduced group sits inside as the augmentation ideal, and one checks readily that the product of two reduced classes is again reduced (the product of two elements of an ideal lies in the ideal), so $\widetilde{K}^0(X)$ is a non-unital ring in its own right. The two closing computations below show the smallest nontrivial values of the theory and confirm that the formal machinery agrees with direct calculation.

Example 2.1.11: K^0 of a Contractible Space

Let X be a contractible compact Hausdorff space, for instance a disk D^n or the interval $[0, 1]$. By corollary 1.2.6, every vector bundle over a contractible paracompact base is trivial, so $\mathrm{Vect}^n(X)$ has a single element for each n and $\mathrm{Vect}(X) \cong \mathbb{N}$ as monoids, with $E \mapsto \mathrm{rk}E$. Exactly as in example 2.1.2, the Grothendieck group is

$$K^0(X) \cong \mathbb{Z}, \qquad \widetilde{K}^0(X) = 0$$

with the rank map an isomorphism onto $\mathbb{Z}$ and the reduced group trivial. This also follows from theorem 2.1.9: a contractible X admits only nullhomotopic maps into BU, so $\widetilde{K}^0(X) = [X, BU] = 0$, and $K^0(X) = [X, \mathbb{Z} \times BU] \cong [X, \mathbb{Z}] = \mathbb{Z}$. The ring structure is the unique unital ring structure on $\mathbb{Z}$. Thus K-theory cannot distinguish a contractible space from a point, as a homotopy invariant must.

The contractible case shows that all the content of K-theory is in the reduced group, which vanishes whenever X has the homotopy type of a point. The smallest space with nontrivial reduced K-theory is the 2-sphere, where the canonical line bundle over $\mathbb{C}\mathrm{P}^1 = S^2$ generates an infinite cyclic group. This computation is the prototype for every later calculation and the first concrete appearance of Bott periodicity.

Example 2.1.12: Reduced K-Theory of S^2

Let $X = S^2$, identified with the complex projective line $\mathbb{C}\mathrm{P}^1$. By the clutching correspondence (proposition 1.2.8) and the stabilization theorem (theorem 1.3.1),

$$\widetilde{K}^0(S^2) \cong \varinjlim_n \pi_1(U(n)) = \pi_1(U)$$

where $\pi_1(U) \cong \mathbb{Z}$ is computed with no appeal to periodicity: the determinant $\det\colon U(n) \to U(1)$ is a surjective homomorphism with kernel $\mathrm{SU}(n)$, which is simply connected for every $n \geq 2$ ([18, The Fundamental Groups of SU(2), SO(3), and SU(n)]), so the long exact sequence of the fibration $\mathrm{SU}(n) \to U(n) \to U(1)$ gives $\pi_1(U(n)) \cong \pi_1(U(1)) = \pi_1(S^1) \cong \mathbb{Z}$. These isomorphisms are compatible with the inclusions $U(n) \hookrightarrow U(n+1)$, since det is, so $\pi_1(U) \cong \mathbb{Z}$ in the colimit. Therefore

$$\widetilde{K}^0(S^2) \cong \mathbb{Z}$$

An explicit generator is the reduced class of the canonical (tautological) complex line bundle $H \to \mathbb{C}\mathrm{P}^1$ introduced in section 1.2: the element $[H] - [\underline{\mathbb{C}}] \in \widetilde{K}^0(S^2)$ generates. To see that it is a generator and not a proper multiple of one, compute its clutching function. Cover $\mathbb{C}\mathrm{P}^1$ by $U_0 = \{z_0 \neq 0\}$ and $U_1 = \{z_1 \neq 0\}$ with coordinates $w = z_1/z_0$ and $v = z_0/z_1$, and trivialize H over them by the nowhere-zero sections $s_0[1 : w] = (1, w)$ and $s_1[v : 1] = (v, 1)$. On the overlap $v = w^{-1}$, so $s_1 = (w^{-1}, 1) = w^{-1}s_0$, and the transition function from the U_1-trivialization to the U_0-trivialization is $w \mapsto w^{-1}$. With $D_+ = \{|w| \leq 1\}$ and $D_- = \{|w| \geq 1\} \cup \{\infty\}$, the convention fixed in section 2.2, this says $H = [\underline{\mathbb{C}}, w^{-1}]$.

Now $\mathrm{Vect}^1_{\mathbb{C}}(S^2) \cong [S^1, \mathrm{GL}_1(\mathbb{C})]$ by proposition 1.2.8, and since $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ is path-connected with abelian fundamental group, free homotopy classes agree with $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$, detected by winding number. The class of H is the loop $w \mapsto w^{-1}$, of winding number -1, hence a generator. Passing between $\mathrm{GL}_n(\mathbb{C})$ and $U(n)$ is harmless: polar decomposition writes every invertible complex matrix uniquely as $g = up$ with u unitary and p positive-definite Hermitian ([18, The Classical Polar Decomposition]), and scaling p to the identity deformation retracts $\mathrm{GL}_n(\mathbb{C})$ onto $U(n)$, so the inclusion induces an isomorphism on π_1. Stabilizing then preserves the generator: writing $\iota\colon U(1) \to U(n)$ for $u \mapsto \mathrm{diag}(u, 1, \ldots, 1)$, one has $\det \circ \iota = \mathrm{id}_{U(1)}$, so $\det_* \iota_* = \mathrm{id}$ on π_1, and since $\det_*$ is an isomorphism so is ι_*. Therefore the class of H generates $\pi_1(U(n))$ for every n, and by theorem 1.3.1 it generates the colimit $\widetilde{K}^0(S^2)$. So $[H] - [\underline{\mathbb{C}}]$ generates. By corollary 2.1.5 the full group is

$$K^0(S^2) \cong \widetilde{K}^0(S^2) \oplus \mathbb{Z} \cong \mathbb{Z} \oplus \mathbb{Z}$$

with the rank accounting for the second summand. As a ring, $K^0(S^2) \cong \mathbb{Z}[H]/(H-1)^2$, where H is shorthand for the class $[H]$ and the relation $(H-1)^2 = 0$ records that the reduced generator $h = [H] - 1$ squares to zero. This relation comes from an isomorphism of bundles: the canonical line bundle over $\mathbb{C}\mathrm{P}^1$ satisfies $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$, recorded in section 1.2. Passing to classes in the ring $K^0(S^2)$ and using proposition 2.1.10, this reads $[H]^2 + 1 = 2[H]$, that is $[H]^2 - 2[H] + 1 = ([H]-1)^2 = 0$, so $h^2 = 0$. Since $\{1, h\}$ generates $K^0(S^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ additively and $h^2 = 0$, the ring is exactly $\mathbb{Z}[h]/(h^2)$.

The relation $([H]-1)^2 = 0$ in the last example is worth isolating, because it is the K-theoretic form of the geometric identity $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$ over $\mathbb{C}\mathrm{P}^1$. In ordinary cohomology the analogous generator squares to zero for dimensional reasons; here it squares to zero because of an isomorphism of vector bundles. This pattern, where a relation among bundles produces a relation in the K-theory ring, recurs in the computation of $K^0(\mathbb{C}\mathrm{P}^n)$ in section 2.4, and the squaring-to-zero phenomenon over S^2 is the seed of the Bott periodicity isomorphism studied next.

Exercise 2.1.1

1. Let X be a compact Hausdorff space with r connected components, and write $\mathrm{rk}_{\mathrm{tot}}\colon K^0(X) \to \mathbb{Z}^r$ for the total rank homomorphism, recording the rank on each component. Show that $K^0(X) \cong \ker(\mathrm{rk}_{\mathrm{tot}}) \oplus \mathbb{Z}^r$. Then take $X = S^0 = \{p, q\}$ pointed at p and compute both $\ker(\mathrm{rk}_{\mathrm{tot}})$ and the reduced group $\widetilde{K}^0(S^0)$ of definition 2.1.4, and say why they differ. (Only the second is what the rest of the book means by $\widetilde{K}^0$.)

2. Let A be a commutative monoid. Its Grothendieck group $G(A)$ is the universal abelian group receiving a monoid homomorphism from A. Prove that the canonical map $A \to G(A)$ is injective if and only if A is cancellative (that is, $a+c = b+c$ implies $a = b$). Using lemma 2.1.6 and theorem 2.1.8, explain why the monoid $\mathrm{Vect}(X)$ need not be cancellative, and identify exactly when two bundle classes $[E], [F]$ become equal in $K^0(X)$.
3. Show that the tangent bundle TS^2 represents the trivial element of the reduced real K-theory $\widetilde{KO}^0(S^2)$, even though TS^2 is not the trivial bundle. (Use that $TS^2 \oplus NS^2 \cong \underline{\mathbb{R}}^3$ for the normal bundle NS^2, which is trivial, and pass to the appropriate reduced group.)
4. Let $f\colon X \to Y$ be a homotopy equivalence of compact Hausdorff spaces. Prove directly from theorem 2.1.9 that $f^*\colon K^0(Y) \to K^0(X)$ is a ring isomorphism. Conclude that K^0 is a homotopy-invariant functor.
5. Verify the ring isomorphism $K^0(S^2) \cong \mathbb{Z}[h]/(h^2)$, where $h = [H] - 1$ is the reduced class of the canonical line bundle. Specifically, show that $\{1, h\}$ is a $\mathbb{Z}$-basis of $K^0(S^2)$ as an abelian group, that $h^2 = 0$, and that the rank homomorphism is the quotient $\mathbb{Z}[h]/(h^2) \to \mathbb{Z}$ sending $h \mapsto 0$.
6. Let $X = \{*\}$ and recompute $K^0(\{*\})$ as a ring using proposition 2.1.10. Then explain why, for any nonempty compact Hausdorff X, the unique map $X \to \{*\}$ and any choice of point $* \to X$ exhibit $\mathbb{Z} = K^0(\{*\})$ as a direct summand (indeed a unital subring) of $K^0(X)$, recovering the rank splitting of corollary 2.1.5 functorially.

2.2 Bott Periodicity

The group $\widetilde{K}^0(X)$ of section 2.1 is one contravariant homotopy functor. What turns it into a cohomology theory is a periodicity: smashing a space with S^2 leaves its reduced K-theory unchanged. This section proves that statement and derives from it the periodicity of the stable homotopy groups of the unitary group.

Two formulations are in circulation. Bott's original theorem is a statement about homotopy groups, $\pi_k(U) \cong \pi_{k+2}(U)$, proved by Morse theory for the energy functional on the loop space of a symmetric space. The form K-theory uses is multiplicative: external multiplication by a single class β on S^2 is an isomorphism $\widetilde{K}^0(X) \to \widetilde{K}^0(\Sigma^2 X)$ for every compact X. The argument below establishes the multiplicative form directly, running on the clutching-function description of bundles over a sphere from section 1.2, and obtains the homotopy-group form from it as a corollary.

This section works throughout in complex K-theory, writing $K = KU$ and $\widetilde{K} = \widetilde{KU}$ unless stated otherwise, and restricts to compact Hausdorff spaces, where every bundle has a complement (proposition 1.1.15) and the representability theorem (theorem 2.1.9) applies. Throughout, X_+ denotes X with a disjoint basepoint adjoined, $\Sigma X = S^1 \wedge X$ is the reduced suspension of a pointed space, and X/A is the quotient of X by a closed subspace A, basepointed at the image of A. The smash product $X \wedge Y = (X \times Y)/(X \vee Y)$ is the reduced product. For a pointed space the reduced group $\widetilde{K}^0(X)$ is the kernel of restriction to the basepoint (definition 2.1.4).

2.2.1 Convention: Where a CW Hypothesis Is Needed Two different hypotheses are in play from here on, and it is worth separating them once.
Everything proved by the clutching construction or by the bundle argument of lemma 2.2.3 needs only that the spaces are *compact Hausdorff* and the subspaces closed. This covers the whole of the product theorem theorem 2.2.15 in its unreduced ring form, and Axioms 1 to 3 of theorem 2.3.6.
Extending a cofiber sequence to the left is different: it needs the Puppe sequence, which [14, The Puppe Sequence and its Coexactness] establishes for *pointed CW complexes*. Everything resting on that extension is stated for finite CW complexes and finite CW pairs, whose basepoints are taken to be 0-cells. The complete list is: the reduced half of definition 2.2.6, hence definition 2.2.7; proposition 2.2.5; the reduced Bott-map clause of theorem 2.2.15; box 2.3.5 and with it the whole $\mathbb{Z}/2$-graded family; Axiom 4 of theorem 2.3.6; and theorems 2.3.7 and 2.4.8 and corollaries 2.3.8 and 2.3.9; and within theorem 2.3.6, the periodicity clause of Axiom 5 and the multiplicativity claim, both of which use periodicity. Axioms 1 to 3, and Axiom 5 in degrees ≤ 0, need only compact Hausdorff.
Spheres, projective spaces and every space in section 2.4 are finite CW complexes, with one exception: box 2.4.6 concerns $\mathbb{C}\mathrm{P}^\infty$, which is neither compact nor finite, so this book's K^0 is not defined on it. That remark records what the theory gives once it is extended to infinite complexes by representability, which is [14, The K-theory Spectrum KU], together with [14, The Milnor Exact Sequence]; the computation there is a report on that extension rather than a consequence of anything proved in this book.
The restriction is to the hypothesis the cited theorem carries. A weaker condition would do, since what the Puppe argument uses is the homotopy extension property ([14, Homotopy Extension Property]), which a CW pair has ([14, Homotopy Extension For CW Pair]); the CW form is taken because it is the hypothesis under which the Puppe sequence is stated there.

There is a second route to the leftward extension, and this book now takes it as well. Section 2.5 rebuilds the exact sequence for an arbitrary *compact* pair without the Puppe sequence at all: lemma 2.5.7 constructs it from lemmas 2.2.3 and 2.2.4 alone, and theorem 2.5.11 and corollary 2.5.10 carry the long exact sequence and the periodic grading to compact pairs. Everything listed in this box keeps the finite CW hypothesis it has, and those are the versions used outside section 2.5. The compact versions exist because the projective-bundle theorem's induction runs over spaces that carry no CW structure. On a finite CW pair both apply. They are not compared here, and nothing in this book depends on their agreeing (box 2.5.12).

Before the product can be defined, one point about reduced groups has to be settled. The external product of two reduced classes will be a class on $X \times Y$ that vanishes on the wedge $X \vee Y$, and the natural home for such a class is the smash product $X \wedge Y = (X \times Y)/(X \vee Y)$. That the reduced K-theory of the smash is exactly the group of such classes is a statement about the collapse map, and it is proved below. No statement in this group of lemmas uses periodicity, and the general exact sequences of section 2.3 rest on lemma 2.2.3.

Lemma 2.2.2: Reduced K-Theory of a Wedge

Let A and B be compact Hausdorff pointed spaces. Restriction along the two inclusions gives an isomorphism
$$\widetilde{K}^0(A \vee B) \xrightarrow{\cong} \widetilde{K}^0(A) \oplus \widetilde{K}^0(B)$$

***Proof*:**
A vector bundle on $A \vee B$ is a bundle E_A on A, a bundle E_B on B, and an identification of their fibres over the common basepoint, since $A \vee B$ is the pushout of $A \leftarrow \{*\} \rightarrow B$ and a bundle on a pushout of compact Hausdorff spaces along a closed inclusion is exactly this gluing data. Reduced classes are those trivial at the basepoint, for which the gluing datum carries no information, so the assignment $E \mapsto (E|_A, E|_B)$ is a bijection on reduced classes. It is additive, so it induces the displayed isomorphism of groups. The inverse sends (a, b) to the class glued from representatives trivialized at the basepoint, completing the proof.

Lemma 2.2.3: Exactness at the Middle Term

Let X be compact Hausdorff and $A \subseteq X$ a closed nonempty subspace containing the basepoint of X, with $i\colon A \hookrightarrow X$ the inclusion and $q\colon X \rightarrow X/A$ the collapse; X/A carries the image of A as its basepoint. The nonemptiness and the basepoint are what make all three reduced groups defined, $\widetilde{K}^0$ being a functor of pointed spaces (definition 2.1.4). Then

$$\widetilde{K}^0(X/A) \xrightarrow{q^*} \widetilde{K}^0(X) \xrightarrow{i^*} \widetilde{K}^0(A)$$

is exact at $\widetilde{K}^0(X)$.

***Proof*:**
Step 1: $\operatorname{im} q^* \subseteq \ker i^*$. The composite $q \circ i$ sends all of A to the basepoint of X/A, so it factors through a point. Since $\widetilde{K}^0$ of a point vanishes (definition 2.1.4), $i^* q^* = 0$.

Step 2: A class trivial on A is trivial near A. Let $\alpha \in \widetilde{K}^0(X)$ with $i^*\alpha = 0$. Writing $\alpha = [E] - \operatorname{rk} E$ and adding a trivial bundle, one may assume $E|_A$ is trivial. Fix a trivialization, equivalently sections $s_1, \ldots, s_n$ of $E|_A$ forming a basis of each fibre over A. Cover A by finitely many open sets $U_1, \ldots, U_r$ of X over which E is trivial. Shrink the cover first: $\{U_1, \ldots, U_r, X \setminus A\}$ is an open cover of the compact Hausdorff, hence normal, space X, so it admits a shrinking ([8, Shrinking Lemma]); let $V_1, \ldots, V_r$ be the shrunken members with $\overline{V_j} \subseteq U_j$. Since the last member is unchanged on A, the V_j still cover A. In a trivialization over U_j, the restriction $s_i|_{A \cap \overline{V_j}}$ is a continuous $\mathbb{C}^n$-valued function on a subspace closed in X, so the Tietze extension theorem ([8, Tietze Extension Theorem]), whose hypothesis of normality is met by X itself, extends it coordinatewise in real and imaginary parts to a continuous $\mathbb{C}^n$-valued function on X; multiplying by a continuous bump function that is 1 on $\overline{V_j}$ and vanishing outside U_j ([8, Urysohn]) and reading the result back through the trivialization gives a section s_{ij} of E over U_j agreeing with s_i on $A \cap \overline{V_j}$. Let $\{\varphi_1, \ldots, \varphi_r, \varphi\}$ be a partition of unity subordinate to the cover $\{V_1, \ldots, V_r, X \setminus A\}$ of X, which exists because a compact Hausdorff space is paracompact ([8, Compact Spaces, Discrete Spaces, and Euclidean Space](1)) and a paracompact Hausdorff space admits subordinate partitions of unity ([8, Existence of Subordinate Partitions of Unity]), and set $\tilde{s}_i = \sum_j \varphi_j s_{ij}$, a global section of E restricting to s_i on A, because φ vanishes on A, the φ_j sum to 1 there, and each s_{ij} agrees with s_i on the support of φ_j.

Over each U_j the sections $\tilde{s}_1, \ldots, \tilde{s}_n$ form a matrix-valued function whose determinant is nonzero at every point of $A \cap U_j$, hence nonzero on an open neighbourhood of $A \cap U_j$. The union of these neighbourhoods is an open $U \supseteq A$ over which $\tilde{s}_1, \ldots, \tilde{s}_n$ trivialize E.

Step 3: Descending to the quotient. Let h denote the trivialization of $E|_A$ and let E/h be the quotient of E by the identifications $h^{-1}(x, v) \sim h^{-1}(y, v)$ for $x, y \in A$ and $v \in \mathbb{C}^n$. The projection $E \to X$ descends to $E/h \to X/A$. Away from the basepoint this is E itself, and over the image U/A of the neighbourhood U from Step 2 the trivialization of E over U descends to a trivialization of E/h, because it agrees with h along A. Hence E/h is a vector bundle on X/A. The quotient map $E \to E/h$ is an isomorphism on each fibre and covers q, so $q^*(E/h) \cong E$, giving $\alpha \in \operatorname{im} q^*$ and completing the proof.

Lemma 2.2.4: Collapsing a Contractible Subspace

Let X be compact Hausdorff and $A \subseteq X$ a closed contractible subspace containing the basepoint of X, so that A is in particular nonempty and all the reduced groups below are defined. Then $q^* \colon \widetilde{K}^0(X/A) \to \widetilde{K}^0(X)$ is an isomorphism.

***Proof*:**
A bundle E on X is trivial over A, since A is contractible and homotopy invariance (theorem 1.2.5) makes $E|_A$ isomorphic to the pullback of a fibre. Choosing a trivialization h, Step 3 of lemma 2.2.3 produces a bundle E/h on X/A with $q^*(E/h) \cong E$, so q^* is surjective on isomorphism classes and hence on $\widetilde{K}^0$. For injectivity, two trivializations of $E|_A$ differ by a map $A \to \mathrm{GL}_n(\mathbb{C})$, which is null-homotopic because A is contractible, and a null-homotopy gives an isomorphism between the two descended bundles by theorem 1.2.5 applied over $X/A \times [0, 1]$. So the descended bundle is well defined up to isomorphism, and $E \mapsto E/h$ is a two-sided inverse to q^*, completing the proof.

Exactness extends one step to the left by the standard cone construction, and that one step is what makes the smash identification an isomorphism rather than merely a surjection.

Proposition 2.2.5: The Smash Product Carries the Reduced Product

Let X and Y be finite pointed CW complexes (box 2.2.1), let $r \colon X \vee Y \hookrightarrow X \times Y$ be the inclusion and $q \colon X \times Y \to X \wedge Y$ the collapse. Then

$$q^* \colon \widetilde{K}^0(X \wedge Y) \longrightarrow \widetilde{K}^0(X \times Y)$$

is injective, with image exactly $\ker r^*$.

***Proof*:**
The image is $\ker r^*$ by lemma 2.2.3 applied to the pair $(X \times Y, X \vee Y)$, so only injectivity is at issue.

Step 1: Extending the sequence. Let $A = X \vee Y$ and $Z = X \times Y$, and let $C_i = Z \cup CA$ be the mapping cone of $i \colon A \hookrightarrow Z$, formed with the *reduced* cone $CA = (A \times [0, 1]) \big/ \big(A \times \{1\} \cup \{a_0\} \times [0, 1]\big)$

glued along $A \times \{0\} = A$, where a_0 is the basepoint. The reduced cone is used rather than the unreduced one because CA/A is then the reduced suspension $\Sigma A = S^1 \wedge A$ on the nose, which is what Σ denotes throughout; the unreduced cone would give the unreduced suspension, for which the identification $\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$ used in Step 2 fails. The reduced cone is contractible by the straight-line contraction to the cone point. Every space occurring below is built from finite CW complexes by attaching cones and collapsing subcomplexes, so all are again finite CW complexes, in particular compact Hausdorff, and lemmas 2.2.3 and 2.2.4 apply throughout.

Collapsing CA gives $C_i/CA = Z/A = X \wedge Y$, so lemma 2.2.4 makes $\widetilde{K}^0(X \wedge Y) \to \widetilde{K}^0(C_i)$ an isomorphism. Collapsing Z gives $C_i/Z = CA/A = \Sigma A$. Applying lemma 2.2.3 to the pair (C_i, Z) and transporting along the first isomorphism yields exactness at $\widetilde{K}^0(X \wedge Y)$ in

$$\widetilde{K}^0(\Sigma A) \xrightarrow{p^*} \widetilde{K}^0(X \wedge Y) \xrightarrow{q^*} \widetilde{K}^0(Z)$$

where $p \colon C_i \to C_i/Z = \Sigma A$ is the collapse. So $\ker q^* = \mathrm{im} p^*$, and it suffices to show $p^* = 0$.

Set $V = C_i \cup CZ$, the mapping cone of $Z \hookrightarrow C_i$. Then $V/C_i = CZ/Z = \Sigma Z$ and $V/CZ = C_i/Z = \Sigma A$, so lemma 2.2.3 applied to (V, C_i) and lemma 2.2.4 applied to (V, CZ) give exactness at $\widetilde{K}^0(\Sigma A)$ in

$$\widetilde{K}^0(\Sigma Z) \xrightarrow{\alpha} \widetilde{K}^0(\Sigma A) \xrightarrow{p^*} \widetilde{K}^0(C_i)$$

the outgoing map being p^* because the composite $C_i \hookrightarrow V \to V/CZ$ is p. It follows that $p^* = 0$, and hence that q^* is injective, as soon as α is surjective.

The map α is the transported collapse $V \to V/C_i$, and identifying it is the one point that does not follow from the two lemmas above. It is the content of the Puppe sequence ([14, The Puppe Sequence and its Coexactness]): every three consecutive terms of $A \to Z \to C_i \xrightarrow{p} \Sigma A \xrightarrow{\Sigma i} \Sigma Z \to \cdots$ form, up to based homotopy equivalence, the cofiber sequence of the first map, so the triple $(C_i, \Sigma A, \Sigma Z)$ identifies α with $(\Sigma i)^*$ up to the self-equivalence of ΣA reversing the suspension coordinate. That reversal is an isomorphism, so it does not affect surjectivity, though it does move the image. The cited proposition applies on its own hypothesis here: $A = X \vee Y$ and $Z = X \times Y$ are finite CW complexes, as are all the cones and quotients formed from them. Therefore q^* is injective as soon as $(\Sigma i)^*$ is surjective.

Step 2: The restriction is surjective. Suspension carries the wedge to a wedge, $\Sigma A = \Sigma X \vee \Sigma Y$, so by lemma 2.2.2 an element of $\widetilde{K}^0(\Sigma A)$ is a pair (a, b) with $a \in \widetilde{K}^0(\Sigma X)$ and $b \in \widetilde{K}^0(\Sigma Y)$. The projections $p_X : Z \to X$ and $p_Y : Z \to Y$ suspend to maps $\Sigma Z \to \Sigma X$ and $\Sigma Z \to \Sigma Y$, and the class $(\Sigma p_X)^* a + (\Sigma p_Y)^* b \in \widetilde{K}^0(\Sigma Z)$ restricts to (a, b): on the ΣX summand, p_X composed with the inclusion $X \hookrightarrow Z$ is the identity, while p_Y composed with it is the constant map to the basepoint, whose suspension is constant and so pulls every reduced class back to zero. The same computation holds on the ΣY summand. Hence the restriction is surjective and q^* is injective, completing the proof.

K-theory carries a multiplication finer than the internal ring structure of proposition 2.1.10: an external product that combines classes on two different spaces into a class on their product. This is what upgrades K-theory from a sequence of abelian groups to a multiplicative cohomology theory, and it defines the Bott map.

Definition 2.2.6: External Product

Let X and Y be compact Hausdorff spaces. The *external product* is the homomorphism

$$\mu\colon K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y)$$

defined on generators by $[E]\otimes[F] \mapsto [p_X^* E \otimes p_Y^* F]$, where $p_X : X\times Y \to X$ and $p_Y : X\times Y \to Y$ are the projections and $\otimes$ on the right is the tensor product of bundles (definition 1.1.16). On reduced groups, the external product descends to a map

$$\widetilde{\mu}\colon \widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \longrightarrow \widetilde{K}^0(X \wedge Y)$$

when X and Y are finite pointed CW complexes: the product of two reduced classes restricts to zero on $X \vee Y$, so it is q^* of a class on the smash by lemma 2.2.3, and that class is unique by the injectivity in proposition 2.2.5, which is what needs the CW hypothesis (box 2.2.1). Then $\widetilde{\mu}$ is defined to be that class.

The external product is well-defined because pullback commutes with tensor product (proposition 1.2.3) and because both operations respect direct sums, so the assignment is additive in each variable and passes to the Grothendieck group of definition 2.1.1. Its reduction to the smash product is the algebraic form of the geometric fact that a class trivial on either factor restricts trivially to the wedge $X \vee Y \subseteq X \times Y$, hence factors through the quotient $X \wedge Y = (X \times Y)/(X \vee Y)$. The next definition extracts from this product the construction periodicity is stated in terms of.

Definition 2.2.7: The Bott Generator and Bott Map

Let $H \to \mathbb{C}\mathrm{P}^1 = S^2$ be the canonical complex line bundle (section 1.2). The *Bott generator* is the reduced class

$$\beta = [H] - 1 \in \widetilde{K}^0(S^2)$$

The *Bott map* for a finite pointed CW complex X is the homomorphism

$$\beta_X : \widetilde{K}^0(X) \longrightarrow \widetilde{K}^0(S^2 \wedge X) = \widetilde{K}^0(\Sigma^2 X), \qquad \alpha \longmapsto \widetilde{\mu}(\beta \otimes \alpha)$$

given by external multiplication with β, using the homeomorphism $S^2 \wedge X \cong \Sigma^2 X$.

The Bott map is the comparison morphism whose invertibility is periodicity. It takes a virtual bundle on X and tensors it, externally, with the Bott class on S^2. The resulting class lives on the double suspension $\Sigma^2 X$ because β is a reduced class on S^2 and the external product of reduced classes lands in the smash. That β_X is an isomorphism is the periodicity theorem, and the rest of this section proves it.

The argument computes $K^0(X \times S^2)$ outright, by presenting every bundle over $X \times S^2$ through clutching data over X and then normalizing that data until only two kinds remain. Section 1.2 carried out the unparametrized case, where the base was a sphere and a clutching function was a map $S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$. What is needed now is the version with a parameter space X carried along, where the constant group $\mathrm{GL}_n(\mathbb{C})$ is replaced by the automorphisms of a bundle over X.

Write $S^2 = D_+ \cup D_-$ for the decomposition into closed upper and lower hemispheres, meeting in the equator $S^1 = D_+ \cap D_-$. Write $\pi_\pm\colon X \times D_\pm \to X$ and $\pi\colon X \times S^1 \to X$ for the projections to X from a hemisphere and from the equator, and $\pi_X : X \times S^2 \to X$, $\pi_{S^2} : X \times S^2 \to S^2$ for the two projections off the product.

Definition 2.2.8: Clutching Data over $X \times S^2$

Let $E \to X$ be a complex vector bundle over a compact Hausdorff space. A *clutching function for E* is a continuous section f of the bundle $\mathrm{Aut}(\pi^*E) \to X \times S^1$, so that $f(x,z)$ is a linear automorphism of the fibre E_x for every $x \in X$ and $z \in S^1$. The *clutched bundle* is

$$[E,f] = \big(\pi_+^*E \ \sqcup \ \pi_-^*E\big)/\sim, \qquad (x,z,v)_- \sim (x,z,f(x,z)v)_+ \quad (z \in S^1,\ v \in E_x)$$

a complex vector bundle over $X \times S^2$ of the same rank as E.

Two clutching functions for the same E are *homotopic* when they are connected by a path of clutching functions, that is by a section of $\mathrm{Aut}(\pi^*E)$ over $X \times S^1 \times [0,1]$. The next proposition collects everything about this construction that the proof uses.

Proposition 2.2.9: Properties of Parametrized Clutching

Let X be compact Hausdorff and let $E, E' \to X$ be complex vector bundles.

1. If $f \simeq g$ as clutching functions for E, then $[E,f] \cong [E,g]$.
2. $[E,f] \oplus [E',f'] \cong [E \oplus E',\ f \oplus f']$ and $[E,f] \otimes [E',f'] \cong [E \otimes E',\ f \otimes f']$.
3. $[E,\mathrm{id}] \cong \pi_X^*E$, so its class in $K^0(X \times S^2)$ lies in the image of π_X^*.
4. Every complex vector bundle over $X \times S^2$ is isomorphic to $[E,f]$ for some bundle $E \to X$ and some clutching function f for E.
5. If α is a section of $\mathrm{Aut}(\pi_+^*E)$ over $X \times D_+$ and γ a section of $\mathrm{Aut}(\pi_-^*E)$ over $X \times D_-$, then $[E, \alpha|_{X\times S^1}\, f\, \gamma|_{X \times S^1}] \cong [E,f]$.
6. The presentation in (4) may be taken *normalized*, meaning $f(x,1) = \mathrm{id}_{E_x}$ for every x, where $1 \in S^1$ is the basepoint. If f_0 and f_1 are normalized clutching functions for the same E with $[E,f_0] \cong [E,f_1]$, then f_1 is homotopic through normalized clutching functions to $\psi f_0 \psi^{-1}$ for some section ψ of $\mathrm{Aut}(E)$ over X.

***Proof*:**

1. A homotopy F of clutching functions over $X \times S^1 \times [0,1]$ is itself a clutching function for the bundle $E \times [0,1]$ over $X \times [0,1]$, so it clutches to a bundle $[E \times [0,1], F]$ over $X \times [0,1] \times S^2$ restricting to $[E,f]$ at $t=0$ and to $[E,g]$ at $t=1$. Homotopy invariance (theorem 1.2.5) applied to the two inclusions $X \times S^2 \rightrightarrows X \times [0,1] \times S^2$, which are homotopic, gives $[E,f] \cong [E,g]$.
2. Both follow from the construction: the gluing relation for a direct sum is the direct sum of the gluing relations, and likewise for tensor products, because $\pi_\pm^*$ commutes with $\oplus$ and $\otimes$ (proposition 1.2.3).
3. With $f = \mathrm{id}$ the relation identifies $(x,z,v)_-$ with $(x,z,v)_+$, so the quotient is the pullback of E along the projection, which is π_X^*E.

4. Let $F \to X \times S^2$ be a bundle. Each hemisphere $D_\pm$ is contractible, so the inclusion $X \times \{n_\pm\} \hookrightarrow X \times D_\pm$ at the pole $n_\pm$ is a deformation retract, and homotopy invariance (theorem 1.2.5) gives isomorphisms $h_\pm\colon F|_{X\times D_\pm} \xrightarrow{\cong} \pi_\pm^* E_\pm$ where $E_\pm = F|_{X\times\{n_\pm\}}$. The poles lie in the path-connected space S^2, so $E_+ \cong E_-$ by homotopy invariance again. Fix such an isomorphism and call the common bundle E. Over the equator the two trivializations differ by $f = h_+h_-^{-1}$, a section of $\mathrm{Aut}(\pi^*E)$ over $X \times S^1$, and by construction $F \cong [E, f]$.

5. Over $X \times D_+$ the automorphism α changes the chosen isomorphism h_+ to αh_+, and over $X \times D_-$ the automorphism γ^{-1} changes h_- to $\gamma^{-1}h_-$. The clutched bundle is unchanged because it is the quotient by the relation these isomorphisms transport, and the new comparison over the equator is $(\alpha h_+)(\gamma^{-1}h_-)^{-1} = \alpha f \gamma$.

6. Given any $h_\pm$ as in the proof of (4), replace $h_\pm$ by $\left(h_\pm|_{X\times\{1\}}\right)^{-1} \circ h_\pm$, which is again an isomorphism onto $\pi_\pm^*E$ and restricts to the identity over $X \times \{1\}$. The resulting $f = h_+h_-^{-1}$ then satisfies $f(x, 1) = \mathrm{id}$, so a normalized presentation exists.

   For the uniqueness statement, let $f_0 = h_+^0(h_-^0)^{-1}$ and $f_1 = h_+^1(h_-^1)^{-1}$ be normalized clutching functions for E and let $\Theta\colon [E, f_0] \to [E, f_1]$ be an isomorphism. Both clutching functions are the identity over $X \times \{1\}$, so the fibre of either clutched bundle over $(x, 1)$ is canonically E_x and $\psi := \Theta|_{X\times\{1\}}$ is a section of $\mathrm{Aut}(E)$ over X. Set $h_\pm = (\pi_\pm^*\psi)^{-1}\, h_\pm^1\, \Theta$, an isomorphism $[E, f_0]|_{X\times D_\pm} \to \pi_\pm^*E$ restricting to id over $X \times \{1\}$ and with $h_+h_-^{-1} = \psi^{-1}f_1\psi$ over the equator. So it suffices to compare two normalized choices of $h_\pm$ for the single bundle $[E, f_0]$. Two such choices differ by $g_\pm = h_\pm(h_\pm^0)^{-1}$, a section of $\mathrm{Aut}(\pi_\pm^*E)$ over $X \times D_\pm$ with $g_\pm|_{X\times\{1\}} = \mathrm{id}$. Since $D_\pm$ is a disk and 1 lies on its boundary circle, there is a deformation retraction $r_s\colon D_\pm \to D_\pm$ of $D_\pm$ onto $\{1\}$ with $r_0 = \mathrm{id}$ and $r_1 \equiv 1$. Then $s \mapsto g_\pm \circ (\mathrm{id}_X \times r_s)$ is a homotopy from $g_\pm$ to the constant id, through sections of $\mathrm{Aut}(\pi_\pm^*E)$ that restrict to id over $X \times \{1\}$ at every stage. Restricting these homotopies to the equator and composing as in (5) joins f_0 to $\psi^{-1}f_1\psi$ through clutching functions, each normalized because each $g_\pm$ stage is the identity at $z = 1$. Conjugating that homotopy by the z-independent ψ joins $\psi f_0\psi^{-1}$ to f_1, again through normalized clutching functions.

establishing each of the six claims in turn and completing the proof.

Part (5) is what every simplification below uses: a clutching function may be modified on the left by a section of $\mathrm{Aut}(\pi_+^*E)$ over $X \times D_+$ and on the right by a section of $\mathrm{Aut}(\pi_-^*E)$ over $X \times D_-$ without changing the clutched bundle. The two sides are not interchangeable, and what is required of the factor is continuity over the relevant hemisphere, not holomorphy. A factor independent of z extends over both hemispheres and so may be absorbed on either side.

Fix from now on the identification $S^2 = \mathbb{C} \cup \{\infty\}$ with $D_+ = \{|z| \le 1\}$ and $D_- = \{|z| \ge 1\} \cup \{\infty\}$, so that a factor polynomial in z extends over D_+ and a factor polynomial in z^{-1} extends over D_-. With this convention the canonical line bundle $H \to \mathbb{C}\mathrm{P}^1 = S^2$ has clutching function $z \mapsto z^{-1}$ (example 1.2.7), so $H = [\underline{\mathbb{C}}, z^{-1}]$ and its dual is $H^\vee = [\underline{\mathbb{C}}, z]$. The dual is written $H^\vee$, not H^*, throughout: in this book H^* always means cohomology. By part (2), multiplying a clutching function by z^m tensors the clutched bundle with $(H^\vee)^{\otimes m}$:

$$[E,\, z^m f] \cong [E, f] \otimes \pi_{S^2}^*\big((H^\vee)^{\otimes m}\big) \tag{2.1}$$

In $K^0(S^2)$ the class $[H] = 1 + \beta$ is a unit, with inverse $[H^\vee] = 1 - \beta$ because $\beta^2 = 0$ (example 2.1.12), so twisting by a power of H is an invertible operation on $K^0(X \times S^2)$ and nothing is lost by multiplying a clutching function by a power of z.

The proof now proceeds in three normalizations. An arbitrary clutching function is first replaced by one whose entries are Laurent polynomials in z; the Laurent polynomial is then cleared to a polynomial and the polynomial reduced to degree one by enlarging E with summands that clutch trivially; and finally a degree-one clutching function is shown to split the bundle into a part on which it is homotopic to z and a part on which it is homotopic to the identity. Only the first step is analytic.

Lemma 2.2.10: Laurent Approximation of Clutching Functions

Let $E \to X$ be a complex vector bundle over a compact Hausdorff space and let f be a clutching function for E. Then f is homotopic, through clutching functions for E, to one of the form

$$\ell(x, z) = \sum_{k=-N}^{N} a_k(x)\, z^k$$

for some $N \geq 0$, where each a_k is a continuous section of $\mathrm{End}(E)$ over X. Moreover, if two Laurent polynomial clutching functions for E are homotopic through clutching functions, then they are homotopic through Laurent polynomial clutching functions.

***Proof*:**
Step 0: A norm. Choose a Hermitian metric on E ([14, Inner Products on a Vector Bundle]) and let $\|\cdot\|_x$ be the associated operator norm on $\mathrm{End}(E_x)$. Fix a finite cover of X by open sets over which E is trivial, together with compact sets $X_1, \ldots, X_r$ covering X with X_i inside the i-th open set. This is possible because X is compact Hausdorff, hence normal. Over each X_i a trivialization identifies $\mathrm{End}(E)$ with $X_i \times M_n(\mathbb{C})$, and on the compact X_i the transported norm and any fixed norm on $M_n(\mathbb{C})$ are comparable with constants independent of the point. Statements below of the form "uniformly on $X \times S^1$" are meant with respect to $\|\cdot\|$, and are checked on each $X_i \times S^1$ in a trivialization, where ordinary uniform continuity of a matrix-valued function on a compact metric space applies.

Step 1: Fourier coefficients. For $k \in \mathbb{Z}$ define a section a_k of $\mathrm{End}(E)$ over X by the fibrewise integral

$$a_k(x) = \frac{1}{2\pi} \int_0^{2\pi} f\big(x, e^{i\theta}\big)\, e^{-ik\theta}\, d\theta$$

For fixed x the integrand takes values in the single finite-dimensional vector space $\mathrm{End}(E_x)$, so the integral is the coordinatewise Riemann integral in any basis. This is independent of the choice of trivialization for the following reason, which is the whole content of the step: π^*E has transition functions pulled back from X, so they do not depend on z, and a change of trivialization conjugates the integrand by a matrix $g(x)$ constant in θ. Conjugation is linear, so it passes through the integral:

$$\int_0^{2\pi} g(x)\, f(x, e^{i\theta})\, g(x)^{-1} e^{-ik\theta} d\theta = g(x) \Big(\int_0^{2\pi} f(x, e^{i\theta}) e^{-ik\theta} d\theta \Big) g(x)^{-1}$$

Continuity of a_k in x is checked on each X_i, where f is a matrix-valued function, uniformly continuous on the compact $X_i \times S^1$, so the integrals converge uniformly in x.

Step 2: Cesàro means converge uniformly. Let

$$\sigma_N f(x,z) = \sum_{|k|\le N} \Big(1 - \frac{|k|}{N+1}\Big) a_k(x) z^k$$

be the N-th Cesàro mean of the Fourier series of f. Substituting the definition of a_k and summing the resulting geometric sum gives $\sigma_N f(x,e^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} f(x, e^{i(\theta-\varphi)}) K_N(\varphi)\,d\varphi$, where K_N is the Fejér kernel. Fejér's theorem ([17, Fejér's Theorem]) states $K_N \ge 0$ and that K_N has total mass 1, and its closed form

$$K_N(\varphi) = \frac{1}{N+1}\left(\frac{\sin\big((N+1)\varphi/2\big)}{\sin(\varphi/2)}\right)^2$$

gives the third property the estimate needs: for $\delta \le |\varphi| \le \pi$ the denominator is bounded below by $\sin(\delta/2) > 0$ and the numerator by 1, so the mass of K_N outside $|\varphi| < \delta$ is at most $\big((N+1)\sin^2(\delta/2)\big)^{-1}$, which tends to 0. (That source normalizes on $\mathbb{R}/\mathbb{Z}$ and defines σ_N as the average of the partial sums, and the displayed weights $1 - |k|/(N+1)$ are what that average produces, and the change of variable to $[0, 2\pi]$ rescales both the kernel and the measure, leaving the total mass at 1.)

Now for fixed x and θ, both $\sigma_N f(x,e^{i\theta})$ and $f(x,e^{i\theta})$ lie in the same space $\mathrm{End}(E_x)$, so their difference is

$$\sigma_N f(x,e^{i\theta}) - f(x,e^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} \Big(f(x,e^{i(\theta-\varphi)}) - f(x,e^{i\theta})\Big) K_N(\varphi)\,d\varphi$$

using that K_N has total mass 1. Taking norms and using $\|\int g\| \le \int \|g\|$, valid for a continuous integrand in a finite-dimensional normed space because it holds for each Riemann sum and passes to the limit, together with $K_N \ge 0$, bounds the left side by $\frac{1}{2\pi}\int_0^{2\pi} \|f(x,e^{i(\theta-\varphi)}) - f(x,e^{i\theta})\| K_N(\varphi) d\varphi$. Given $\varepsilon > 0$, Step 0 gives a δ with $\|f(x,e^{i(\theta-\varphi)}) - f(x,e^{i\theta})\| < \varepsilon/2$ whenever $|\varphi| < \delta$, uniformly in x and θ; splitting the integral at $|\varphi| = \delta$ and using the mass bound above on the outer part gives $\sigma_N f \to f$ uniformly on $X \times S^1$.

Step 3: Nearby sections are automorphisms. The set of automorphisms is open in End: for $g \in \mathrm{Aut}(E_x)$ and $u \in \mathrm{End}(E_x)$ with $\|u\| < \|g^{-1}\|^{-1}$, the endomorphism $g + u = g(\mathrm{id} + g^{-1}u)$ is invertible because $\|g^{-1}u\| < 1$ and the Neumann series converges. The function $(x,z) \mapsto \|f(x,z)^{-1}\|^{-1}$ is continuous and positive on the compact $X \times S^1$, so it has a positive minimum ε. Choose N with $\|\sigma_N f - f\| < \varepsilon$ uniformly. Then $\ell = \sigma_N f$ is a clutching function for E, and the straight-line homotopy $f_t = (1-t)f + t\ell$ satisfies $\|f_t - f\| \le \|\ell - f\| < \varepsilon$ for every t, so each f_t is an automorphism. Thus $f \simeq \ell$ through clutching functions, and ℓ has the stated Laurent form.

Step 4: Homotopies may be taken Laurent. Let ℓ_0, ℓ_1 be Laurent polynomial clutching functions for E and let F be a homotopy of clutching functions between them, that is a section of $\mathrm{Aut}(\pi^* E)$ over $X \times S^1 \times [0,1]$. Regard F as a clutching function for the bundle $E \times [0,1]$ over the compact Hausdorff space $X \times [0,1]$. Steps 0 to 3 apply verbatim with X replaced by $X \times [0,1]$, producing a Laurent polynomial clutching function $L(x,t,z) = \sum_{|k|\le M} c_k(x,t) z^k$ for $E \times [0,1]$, with $\|L - F\| < \varepsilon$ uniformly, where ε is the minimum of $\|F^{-1}\|^{-1}$ over the compact $X \times S^1 \times [0,1]$.

Then $t \mapsto L(\cdot, t, \cdot)$ is a homotopy through Laurent polynomial clutching functions for E, from L_0 to L_1. Its endpoints satisfy $\|L_0 - \ell_0\| < \varepsilon$ and $\|L_1 - \ell_1\| < \varepsilon$, since $F_0 = \ell_0$ and $F_1 = \ell_1$, so the straight-line homotopies $(1-s)\ell_0 + sL_0$ and $(1-s)L_1 + s\ell_1$ consist of automorphisms by the estimate of Step 3. Each stage of a straight-line homotopy between Laurent polynomials of degree at most $\max(M, \deg \ell_0, \deg \ell_1)$ is again such a Laurent polynomial. Concatenating $\ell_0 \to L_0$, $L_0 \to L_1$, $L_1 \to \ell_1$ gives a homotopy from ℓ_0 to ℓ_1 through Laurent polynomial clutching functions, completing the proof.

Multiplying by z^N clears the negative powers, which by (2.1) only twists the clutched bundle by a power of $H^\vee$. So it is enough to normalize *polynomial* clutching functions, and the next step removes all but the linear term.

Lemma 2.2.11: Linearization of Polynomial Clutching Functions

Let $p(x,z) = \sum_{k=0}^n a_k(x) z^k$ be a clutching function for E whose coefficients are sections of $\operatorname{End}(E)$. Then there is a clutching function Lp for $E^{\oplus(n+1)}$, of the form $Lp(x,z) = B(x)\,z + A(x)$ with A, B sections of $\operatorname{End}(E^{\oplus(n+1)})$, such that

$$\left[E^{\oplus(n+1)},\, Lp\right] \cong [E, p] \;\oplus\; \pi_X^*\left(E^{\oplus n}\right)$$

***Proof*:**
Define, in block form with respect to $E^{\oplus(n+1)} = E \oplus \cdots \oplus E$,

$$Lp(x,z) = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_n \\ -z & 1 & 0 & \cdots & 0 \\ 0 & -z & 1 & & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -z & 1 \end{pmatrix}$$

which is of the form $Bz + A$ with A the matrix of a_k's and 1's and B the matrix with -1 in the subdiagonal blocks.

Step 1: Lp is invertible on the equator. Perform the column operations $C_1 \mapsto C_1 + zC_2$, then $C_1 \mapsto C_1 + z^2 C_3$, and so on up to $C_1 \mapsto C_1 + z^n C_{n+1}$. The first replaces the column-1 entries $(a_0, -z, 0, \dots)^T$ by $(a_0 + za_1,\, 0,\, -z^2,\, 0, \dots)^T$; the second replaces these by $(a_0 + za_1 + z^2 a_2,\, 0,\, 0,\, -z^3, \dots)^T$; after n such operations column 1 is $(p(x,z), 0, \dots, 0)^T$. Next clear the first row from the bottom up: for $j = n+1, n, \dots, 2$ in that order, the row operation $R_1 \mapsto R_1 - (\text{entry in position } (1,j)) \cdot R_j$ removes the $(1,j)$ entry, and since R_j has zeros in columns 1 and $j+1, \dots, n+1$ it disturbs only the $(1, j-1)$ entry, which is cleared at the next step. The result is

$$p(x,z) \;\oplus\; T(z), \qquad T(z) = \begin{pmatrix} 1 & & & \\ -z & 1 & & \\ & \ddots & \ddots & \\ & & -z & 1 \end{pmatrix}$$

an $n \times n$ lower triangular block matrix with 1 on the diagonal and $-z$ on the subdiagonal. The row and column operations clear the first row and column only. The subdiagonal entries of T are never touched, and clearing them by further row operations would reintroduce a $-z^2$ one column to the left at each stage.

Every operation used is an elementary block operation whose multiplier is a polynomial in z with coefficients in $\mathrm{End}(E)$, so each is realized by a block-elementary matrix, which is invertible with inverse of the same shape. Since $T(z)$ is lower triangular with 1 on the diagonal it is invertible for every z. Hence Lp is invertible at (x, z) precisely when $p(x, z)$ is, and p is a clutching function, so Lp is invertible on all of $X \times S^1$ and is a clutching function for $E^{\oplus(n+1)}$.

Step 2: The elementary operations are homotopies. A block-elementary matrix $I + \lambda e_{ij}$ with $i \neq j$ is connected to the identity through invertible matrices by $t \mapsto I + t\lambda e_{ij}$, $t \in [0, 1]$, each of which is invertible since $e_{ij}^2 = 0$ for $i \neq j$. When λ depends on z this is a homotopy through clutching functions, because invertibility holds at every (x, z, t). Applying proposition 2.2.9(1) to the composite of these homotopies,

$$\left[E^{\oplus(n+1)}, Lp\right] \cong \left[E^{\oplus(n+1)}, p \oplus T\right]$$

Step 3: Contracting the triangular block. For $s \in [0, 1]$ let $T_s(z)$ be $T(z)$ with the subdiagonal entries $-z$ replaced by $-sz$. Each $T_s(z)$ is lower triangular with 1 on the diagonal, hence invertible for every s, x and z, so $s \mapsto p \oplus T_s$ is a homotopy of clutching functions from $p \oplus T$ to $p \oplus T_0 = \mathrm{diag}(p, 1, \ldots, 1)$. By proposition 2.2.9(1) again,

$$\left[E^{\oplus(n+1)}, Lp\right] \cong \left[E^{\oplus(n+1)}, \mathrm{diag}(p, 1, \ldots, 1)\right]$$

Step 4: Splitting the block-diagonal form. By proposition 2.2.9(2) the right-hand side is $[E, p] \oplus [E^{\oplus n}, \mathrm{id}]$, and by (3) the second summand is $\pi_X^*(E^{\oplus n})$, which is the claim, completing the proof.

Two recursions relate the linearizations at consecutive degrees. They are what makes the invariant constructed at the end of this section independent of the degree chosen to linearize at, and they are proved by the same column operations as lemma 2.2.11.

Lemma 2.2.12: Recursions for the Linearization

Let p be a polynomial clutching function for E of degree at most n, regarded also as one of degree at most $n + 1$ by setting $a_{n+1} = 0$. Then

1. $\left[E^{\oplus(n+2)},\, L^{n+1}p\right] \cong \left[E^{\oplus(n+1)},\, L^n p\right] \oplus [E, \mathrm{id}]$;
2. $\left[E^{\oplus(n+2)},\, L^{n+1}(zp)\right] \cong \left[E^{\oplus(n+1)},\, L^n p\right] \oplus [E, z]$.

***Proof*:**
Throughout, an elementary block operation with multiplier $\lambda(z)$ is realized by the homotopy $t \mapsto I + t\lambda(z)e_{ij}$, $i \neq j$, which is invertible at every (x, z, t) because $e_{ij}^2 = 0$, and proposition 2.2.9(1) then gives an isomorphism of clutched bundles. No appeal to part (5) is needed, so no hemisphere condition on λ arises.

Each homotopy below stays inside the *linear* clutching functions, which is what proposition 2.2.14(1) requires in order to compare the two splittings. Writing $L^{n+1}p = A + zB$ as in lemma 2.2.11, the two products used are

$$(A+zB)\big(I+tz\, e_{n+2,n+1}\big) = A+z\big(B+tA\, e_{n+2,n+1}\big), \qquad (A+zB)\big(I+tz^{-1}e_{12}\big) = \big(A-t\, e_{22}\big)+zB$$

The first holds because $Be_{n+2,n+1} = 0$, the subdiagonal of B having no entry in the last column; the second because $Ae_{12} = 0$, the first column of A vanishing when the constant coefficient b_0 of

zp is zero. In particular the z^{-1} multiplier produces no z^{-1} term, and both products are again linear in z at every t.

(1). With $a_{n+1} = 0$ the last column of $L^{n+1}p$ is $(0, \dots, 0, 1)^T$ and the last row is $(0, \dots, 0, -z, 1)$. Apply $C_{n+1} \mapsto C_{n+1} + z\,C_{n+2}$. Column $n+1$ has $-z$ in row $n+2$, and adding z times the last column replaces that entry by $-z + z = 0$, leaving every other entry unchanged. The last row and column now meet the rest of the matrix nowhere, so the matrix is $L^n p \oplus 1$, and proposition 2.2.9(2),(3) give the claim. For $n = 0$ this reads $\left(\begin{smallmatrix} a_0 & 0 \\ -z & 1 \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} a_0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and for $n = 1$ it reads $\left(\begin{smallmatrix} a_0 & a_1 & 0 \\ -z & 1 & 0 \\ 0 & -z & 1 \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} a_0 & a_1 & 0 \\ -z & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$.

(2). The coefficients of zp are $b_0 = 0$ and $b_k = a_{k-1}$ for $1 \le k \le n+1$, so the first row of $L^{n+1}(zp)$ is $(0, a_0, a_1, \dots, a_n)$ and the first column is $(0, -z, 0, \dots, 0)^T$. Apply $C_2 \mapsto C_2 + z^{-1}C_1$, legitimate on $X \times S^1$ where z^{-1} is defined. Column 1 scaled by z^{-1} is $(0, -1, 0, \dots, 0)^T$, and column 2 is $(a_0, 1, -z, 0, \dots)^T$, so the entry in row 2 becomes $1 - 1 = 0$. Row 2 is now $(-z, 0, \dots, 0)$ and column 1 is $(0, -z, 0, \dots, 0)^T$: they meet the rest of the matrix nowhere and isolate the single entry $-z$ at position $(2, 1)$. Deleting that row and column leaves, on rows $1, 3, 4, \dots, n+2$ and columns $2, \dots, n+2$, the matrix with first row $(a_0, \dots, a_n)$ and $-z$, 1 on the two diagonals below it, which is $L^n p$.

Bringing the isolated entry onto the diagonal is a transposition of two indices, realized by a permutation matrix P. Since $\mathrm{GL}_{n+2}(\mathbb{C})$ is path-connected, P is joined to the identity by a path of constant invertible matrices, and $t \mapsto P_t\, L^{n+1}(zp)$ is therefore a homotopy through *linear* clutching functions; each P_t is also constant in z, so it extends over both hemispheres and proposition 2.2.9(5) applies as well. Therefore $\left[E^{\oplus(n+2)}, L^{n+1}(zp)\right] \cong \left[E^{\oplus(n+1)}, L^n p\right] \oplus [E, -z]$. Finally $[E, -z] \cong [E, z]$, since $-\mathrm{id}$ is a z-independent automorphism of E and part (5) absorbs it. For $n = 0$ the computation reads $\left(\begin{smallmatrix} 0 & a_0 \\ -z & 1 \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} 0 & a_0 \\ -z & 0 \end{smallmatrix}\right)$, which is $\mathrm{diag}(a_0, -z)$ after the transposition, completing the proof.

What remains is the case of a clutching function linear in z. Such a function $az + b$ is invertible on the unit circle and degenerates at the roots of $\det(a(x)z + b(x))$, a polynomial in z of degree at most the rank of E, so for each x there are finitely many bad points and they may lie anywhere off the circle. The treatment has two stages: first normalize so that the coefficient of z is the identity, which is a homotopy argument; then split the remaining endomorphism according to whether its eigenvalues lie inside or outside the unit circle, which is the spectral projection of a contour integral. The projection is taken of the *normalized* endomorphism and not of the pencil $az + b$ itself, which in general commutes with neither a nor b.

Lemma 2.2.13: Eigenvalue Splitting of an Endomorphism

Let $E \to X$ be a complex vector bundle over a compact Hausdorff space and let b be a section of $\mathrm{End}(E)$ such that $b(x)$ has no eigenvalue on the unit circle, for every $x \in X$. Then there are subbundles $E_+, E_- \subseteq E$ with

1. $E = E_+ \oplus E_-$;
2. $b(E_\pm) \subseteq E_\pm$;
3. every eigenvalue of $b|_{E_+}$ lies outside S^1, and every eigenvalue of $b|_{E_-}$ lies inside S^1.

The splitting is unique, and it respects direct sums: applied to $b_1 \oplus b_2$ on $E_1 \oplus E_2$ it gives $(E_1 \oplus E_2)_\pm = (E_1)_\pm \oplus (E_2)_\pm$.

***Proof*:**
Step 1: The projection. For $x \in X$ and z on the unit circle, $z - b(x)$ is invertible, since otherwise z would be an eigenvalue of $b(x)$ on S^1. Define a section Π of $\mathrm{End}(E)$ by the contour integral

$$\Pi(x) = \frac{1}{2\pi i} \oint_{|z|=1} \big(z - b(x)\big)^{-1}\, dz$$

taken fibrewise in the finite-dimensional space $\mathrm{End}(E_x)$. The resolvent $(z - b(x))^{-1}$ is jointly continuous in (x, z) on the compact set $X \times S^1$, being the inverse of a continuous family of invertible endomorphisms, so Π is a continuous section of $\mathrm{End}(E)$.

Step 2: Π is idempotent. Write $R(z) = (z - b(x))^{-1}$ for fixed x. Since R is holomorphic on the annulus where $z - b(x)$ stays invertible, which by compactness contains $1 - \varepsilon < |z| < 1 + \varepsilon$ for some $\varepsilon > 0$ uniformly in x, Cauchy's theorem ([15, Cauchy's (Integral) Theorem]) lets the contour be taken on any circle in that annulus without changing Π. Choose radii $r_1 < r_2$ in the annulus, and use the resolvent identity

$$R(z)R(w) = \frac{R(w) - R(z)}{z - w} \qquad (z \neq w)$$

which follows from $R(z)R(w)\big[(w - b) - (z - b)\big] = R(z) - R(w)$. Then

$$\Pi^2 = \frac{1}{(2\pi i)^2} \oint_{|z|=r_1} \oint_{|w|=r_2} \frac{R(w) - R(z)}{z - w}\, dw\, dz$$

In the term containing $R(w)$, integrating first in z over the smaller circle gives $\oint_{|z|=r_1} (z-w)^{-1} dz = 0$, because w lies outside that circle. In the term containing $R(z)$, integrating first in w over the larger circle gives $-\oint_{|w|=r_2} (z - w)^{-1} dw = \oint_{|w|=r_2} (w - z)^{-1} dw = 2\pi i$, because z lies inside it. Both evaluations are Cauchy's integral formula applied to the constant function 1. What survives is $\Pi^2 = \frac{1}{2\pi i} \oint_{|z|=r_1} R(z)\, dz = \Pi$.

Step 3: The subbundles. Set $E_- = \operatorname{im} \Pi$ and $E_+ = \ker \Pi$. A continuous idempotent has locally constant rank, since $\operatorname{rank} \Pi(x) = \operatorname{tr} \Pi(x)$ is both continuous and integer-valued, so E_- and E_+ are subbundles and $E = E_+ \oplus E_-$, which is (1). Each is b-invariant because b commutes with $(z - b)^{-1}$, hence with Π, which is (2).

Step 4: Locating the eigenvalues. Fix x and decompose E_x into generalized eigenspaces of $b(x)$. On the generalized eigenspace for an eigenvalue λ, the resolvent $(z-b)^{-1}$ has a pole at $z=\lambda$ and no other singularity, so by the residue theorem ([15, Residue Theorem]) the integral Π restricts to the identity when $|\lambda|<1$ and to zero when $|\lambda|>1$. Hence E_- is the sum of the generalized eigenspaces for eigenvalues inside S^1 and E_+ the sum of those outside, which is (3), and this description also gives uniqueness: any splitting satisfying (1)-(3) must assign each generalized eigenspace to the side its eigenvalue dictates. Compatibility with direct sums is immediate from the same description, since the generalized eigenspaces of $b_1 \oplus b_2$ are the sums of those of b_1 and b_2, which completes the proof.

Proposition 2.2.14: Linear Clutching Functions Split

Let $\ell(x,z)=a(x)\,z+b(x)$ be a clutching function for E, with a,b sections of $\mathrm{End}(E)$. Then there is a splitting $E \cong E_+ \oplus E_-$ into subbundles with

$$[E,\ell] \cong [E_+,\mathrm{id}] \ \oplus\ [E_-,z]$$

and the splitting respects direct sums of linear clutching functions. Moreover:

1. the isomorphism classes of E_+ and E_- are unchanged along a homotopy through *linear* clutching functions, and in particular do not depend on the choice of t_0 made in the proof;
2. for $\ell=\mathrm{id}$ one has $E_-=0$ and $E_+=E$; for $\ell = z\cdot\mathrm{id}$ one has $E_+=0$ and $E_-=E$.

***Proof*:**
Step 1: Normalizing the coefficient of z. For $t\in[0,1)$ consider the Möbius substitution $z\mapsto (z+t)/(1+tz)$ with t real, which carries S^1 to itself: if $|z|=1$ then $|1+tz|=|z|\,|\overline{z}+t| = |\overline{z}+t| = |\overline{z+t}| = |z+t|$, using $|z|=1$, $z\overline{z}=1$ and $\bar{t}=t$, so the quotient has modulus 1. Multiplying by the scalar $1+tz$, which is nonzero for $|z|=1$ and $t<1$, gives

$$(1+tz)\left[a(x)\,\frac{z+t}{1+tz}+b(x)\right] = \big(a(x)+t\,b(x)\big)z + \big(t\,a(x)+b(x)\big) \tag{2.2}$$

At $t=0$ the right-hand side is $a(x)z+b(x)$. For each $t\in[0,1)$ the left-hand side is invertible on $X\times S^1$, being the composite of the substitution, which preserves S^1, with ℓ and a nonzero scalar. So (2.2) is a homotopy of clutching functions as t runs from 0 to any $t_0<1$.

At $t=1$ the coefficient $a(x)+b(x)$ equals $\ell(x,1)$, which is invertible. The function $t\mapsto \inf_{x\in X}|\det(a(x)+t\,b(x))|$ is continuous, and X compact makes the infimum attained, so it is nonzero at $t=1$ and hence nonzero for $t=t_0$ close enough to 1. Fix such a t_0. Then $a+t_0 b$ is a section of $\mathrm{Aut}(E)$, independent of z, so it extends over both hemispheres and proposition 2.2.9(5) permits multiplying the clutching function by its inverse on the right. Doing so replaces (2.2) at $t=t_0$ by

$$z+c(x), \qquad c=\big(t_0 a+b\big)\big(a+t_0 b\big)^{-1}$$

Hence $[E,\ell]\cong[E,z+c]$ with c a section of $\mathrm{End}(E)$.

Step 2: Splitting. Since $z+c(x)$ is invertible for $|z|=1$, the endomorphism $-c(x)$, and therefore $c(x)$, has no eigenvalue on the unit circle. Apply lemma 2.2.13 to c to get $E=E_+\oplus E_-$ preserved by c, with the eigenvalues of $c_+ = c|_{E_+}$ outside S^1 and those of $c_- = c|_{E_-}$ inside. Because the splitting is c-invariant, the clutching function decomposes, and proposition 2.2.9(2) gives

$$[E, z+c] \cong [E_+,\, z+c_+] \,\oplus\, [E_-,\, z+c_-]$$

Step 3: Contracting each summand. On E_+ every eigenvalue of c_+ lies outside S^1, so for $s \in [0,1]$ and $|z| = 1$ the endomorphism $sz + c_+$ is invertible: it fails to be so only if $-sz$ is an eigenvalue of c_+, and $|-sz| = s \leq 1$ while every eigenvalue has modulus greater than 1. Thus $s \mapsto sz + c_+$ is a homotopy of clutching functions from $z + c_+$ to c_+, and c_+ is a z-independent automorphism, so a further application of proposition 2.2.9(5) and (1) gives $[E_+, z + c_+] \cong [E_+, \mathrm{id}]$.

On E_- every eigenvalue of c_- lies inside S^1, so for $s \in [0,1]$ and $|z| = 1$ the endomorphism $z + s\,c_-$ is invertible: it fails only if $-z/s$ is an eigenvalue of c_-, and $|-z/s| = 1/s \geq 1$ while every eigenvalue has modulus less than 1 (for $s = 0$ the endomorphism is $z \cdot \mathrm{id}$, invertible). Thus $s \mapsto z + s\,c_-$ is a homotopy of clutching functions from $z + c_-$ to $z \cdot \mathrm{id}$, giving $[E_-, z + c_-] \cong [E_-, z]$.

Combining the three steps yields $[E, \ell] \cong [E_+, \mathrm{id}] \oplus [E_-, z]$. Compatibility with direct sums holds because Step 1 is applied coordinatewise to a direct sum of linear clutching functions and the splitting of lemma 2.2.13 respects direct sums.

Step 4: Homotopy invariance and the two special cases. Let $\ell_s = a_s z + b_s$, $s \in [0,1]$, be a homotopy through linear clutching functions for E. Running Step 1 with the parameter space $X \times [0,1]$ in place of X produces a single endomorphism c of $E \times [0,1]$ over $X \times [0,1]$, and lemma 2.2.13 applied there gives subbundles $(E \times [0,1])_\pm$ restricting to the splittings at each s. A subbundle of $E \times [0,1]$ restricts to isomorphic bundles at $s = 0$ and $s = 1$ by theorem 1.2.5, so the isomorphism classes of $E_\pm$ are constant along the homotopy. The choice of t_0 is covered by this. Step 1 produces an interval $(\tau, 1)$ on which $\inf_x |\det(a + tb)|$ stays positive, and any two values in that interval are joined inside it, along which (2.2) varies continuously through linear clutching functions.

For $\ell = \mathrm{id}$, that is $a = 0$ and $b = \mathrm{id}$, Step 1 gives $c = (t_0 \cdot 0 + \mathrm{id})(0 + t_0\mathrm{id})^{-1} = t_0^{-1}\mathrm{id}$, whose only eigenvalue t_0^{-1} exceeds 1; hence $E_+ = E$ and $E_- = 0$. For $\ell = z \cdot \mathrm{id}$, that is $a = \mathrm{id}$ and $b = 0$, Step 1 gives $c = (t_0\mathrm{id})(\mathrm{id})^{-1} = t_0\mathrm{id}$, whose only eigenvalue $t_0 < 1$ lies inside the circle; hence $E_- = E$ and $E_+ = 0$, completing the proof.

Theorem 2.2.15: Bott Periodicity Isomorphism

Let X be a compact Hausdorff space. The external product

$$\mu\colon K^0(X) \otimes_{\mathbb{Z}} K^0(S^2) \longrightarrow K^0(X \times S^2)$$

is an isomorphism of rings. Equivalently, $K^0(X \times S^2)$ is a free $K^0(X)$-module of rank 2 with basis $\{1, [H]\}$, and as a $K^0(X)$-algebra it is $K^0(X)[H]/\big(([H]-1)^2\big)$.
Consequently, for every finite pointed CW complex X (the hypothesis of proposition 2.2.5, which the next clause uses through the injectivity of q^*) the Bott map

$$\beta_X : \widetilde{K}^0(X) \longrightarrow \widetilde{K}^0(\Sigma^2 X)$$

of definition 2.2.7 is an isomorphism of abelian groups, natural in X.

***Proof*:**
Write $\beta = [H] - 1 \in \widetilde{K}^0(S^2)$, so that $K^0(S^2) = \mathbb{Z}\{1, \beta\}$ with $\beta^2 = 0$ and $[H] = 1 + \beta$ is a unit with inverse $[H^\vee] = 1 - \beta$ (example 2.1.12). For a bundle $F \to X$ and $k \in \mathbb{Z}$ abbreviate $F * [H]^k = \mu\big([F] \otimes [H]^k\big) = \pi_X^*[F] \cdot \pi_{S^2}^*[H]^k$.

Step 1: Surjectivity. Let F be a bundle over $X \times S^2$. By proposition 2.2.9(4) write $F \cong [E, f]$, and by lemma 2.2.10 and proposition 2.2.9(1) replace f by a Laurent polynomial clutching function. Multiplying by a power of z, which by (2.1) twists the clutched bundle by a power of H, write that function as $z^{-m}q$ with q polynomial of degree at most n, so that $[E, z^{-m}q] \cong [E, q] \otimes \pi_{S^2}^* H^{\otimes m}$. By lemma 2.2.11, in $K^0(X \times S^2)$,

$$[E, q] = \big[E^{\oplus(n+1)}, L^n q\big] - \pi_X^*\big[E^{\oplus n}\big]$$

and by proposition 2.2.14 the first term splits as $\big[G_+, \mathrm{id}\big] + \big[G_-, z\big]$, where $G = E^{\oplus(n+1)}$ and $G_\pm$ are the two summands produced there. Finally $[G_+, \mathrm{id}] = \pi_X^*[G_+]$ by proposition 2.2.9(3), and $[G_-, z] = \pi_X^*[G_-] \cdot \pi_{S^2}^*[H^\vee]$ by (2.1). Assembling,

$$[F] = \Big(G_+ * 1 \;+\; G_- * [H^\vee] \;-\; E^{\oplus n} * 1\Big) \cdot [H]^m \tag{2.3}$$

every term of which lies in the image of μ. Since $K^0(X \times S^2)$ is generated by classes of bundles, μ is surjective.

Step 2: An inverse on the level of clutching data. Define, for a bundle presented as $[E, z^{-m}q]$ with $\deg q \leq n$,

$$\nu\big([E, z^{-m}q]\big) = -\big[G_-\big] \otimes \beta \;+\; [E] \otimes [H]^m \;\in\; K^0(X) \otimes K^0(S^2)$$

where $G_- = \big(E^{\oplus(n+1)}\big)_-$ is the minus-summand of the splitting of $L^n q$. Five checks make this well defined, the last because the formula names E and not only n, m, t_0 and the clutching function.

Independence of n. Regarding q as having degree at most $n + 1$, lemma 2.2.12(1) gives $\big[E^{\oplus(n+2)}, L^{n+1}q\big] \cong \big[E^{\oplus(n+1)}, L^n q\big] \oplus [E, \mathrm{id}]$, an isomorphism realized by a homotopy of linear clutching functions. By proposition 2.2.14 the splitting respects direct sums and is homotopy invariant, and the minus-summand of $[E, \mathrm{id}]$ is 0 by clause (2) there. Hence $\big(E^{\oplus(n+2)}\big)_- \cong \big(E^{\oplus(n+1)}\big)_-$, so G_- does not depend on n.

Independence of m. Replacing $z^{-m}q$ by $z^{-m-1}(zq)$ presents the same bundle. By lemma 2.2.12(2), $\big[E^{\oplus(n+2)}, L^{n+1}(zq)\big] \cong \big[E^{\oplus(n+1)}, L^n q\big] \oplus [E, z]$, and the minus-summand of $[E, z]$ is E by proposition 2.2.14(2). So the minus-summand for zq is $G_- \oplus E$, and

$$\nu\big([E, z^{-m-1}(zq)]\big) = -\big([G_-] + [E]\big) \otimes \beta + [E] \otimes [H]^{m+1}$$

Since $[H]^k = 1 + k\beta$ for every k, one has $[H]^{m+1} - \beta = 1 + (m+1)\beta - \beta = [H]^m$, so the last two terms combine to $[E] \otimes [H]^m$ and the whole expression equals $\nu\big([E, z^{-m}q]\big)$.

Independence of the presenting bundle E. The term $[E] \otimes [H]^m$ refers to E itself, so ν would depend on the presentation if two non-isomorphic bundles over X could clutch to the same bundle over $X \times S^2$. They cannot: restricting $[E, f]$ to $X \times \{s\}$ for s the center of either hemisphere returns E, so an isomorphism $[E, f] \cong [E', f']$ over $X \times S^2$ restricts to an isomorphism $E \cong E'$ over X, and $[E] = [E']$ in $K^0(X)$.

Independence of the normalization constant. The constant t_0 in the proof of proposition 2.2.14 may be any value sufficiently close to 1, and proposition 2.2.14(1) records that the splitting is unchanged along a homotopy through linear clutching functions, so G_- does not depend on it.

Independence of the presenting clutching function. By proposition 2.2.9(6) the presentation may be taken normalized, and two normalized clutching functions f_0, f_1 for isomorphic bundles satisfy $f_1 \simeq \psi f_0 \psi^{-1}$ through clutching functions, for some section ψ of $\mathrm{Aut}(E)$ over X. Conjugation by ψ leaves G_- unchanged: if $z^{-m}q$ is a Laurent polynomial clutching function then so is $\psi(z^{-m}q)\psi^{-1} = z^{-m}(\psi q \psi^{-1})$, and writing $\Psi = \psi^{\oplus(n+1)}$ one has $L^n(\psi q \psi^{-1}) = \Psi\,(L^n q)\,\Psi^{-1}$, since ψ commutes with the entries $-z$ and 1 off the first row. Step 1 of proposition 2.2.14 then replaces c by $\Psi c \Psi^{-1}$, whose generalized eigenspaces are the images under Ψ of those of c, so lemma 2.2.13 returns $\Psi(G_\pm) \cong G_\pm$. It therefore suffices to treat two clutching functions homotopic through clutching functions. By lemma 2.2.10 their Laurent approximations are then homotopic through Laurent polynomial clutching functions. Running lemma 2.2.11 and proposition 2.2.14 over $X \times [0,1]$ with that homotopy as clutching function produces a bundle over $X \times [0,1]$ whose minus-summand restricts to the two candidate G_- at the ends, so they are isomorphic by theorem 1.2.5.

Finally ν is additive: $L^n(q_1 \oplus q_2) = L^n q_1 \oplus L^n q_2$ directly from the matrix, and the splitting respects direct sums, so ν carries a direct sum of clutching data to the sum of its values. Since every bundle over $X \times S^2$ has such a presentation, ν extends to a group homomorphism $K^0(X \times S^2) \to K^0(X) \otimes K^0(S^2)$.

Step 3: $\nu\mu = \mathrm{id}$. The group $K^0(S^2) = \mathbb{Z}\{1, \beta\}$ is generated by $[H]^0 = 1$ and $[H]^1 = 1+\beta$, so it is generated by the classes $[H]^m$ with $m \geq 0$, and it suffices to evaluate $\nu\mu$ on elements $[E] \otimes [H]^m$. Now $\mu\big([E] \otimes [H]^m\big) = \big[E, z^{-m}\big]$, whose polynomial part is $q = \mathrm{id}$ of degree 0. Taking $n = 0$, the linearization $L^0\mathrm{id}$ is id on E, and by proposition 2.2.14(2) its minus-summand is 0. Hence

$$\nu\mu\big([E] \otimes [H]^m\big) = -\,0 \otimes \beta + [E] \otimes [H]^m = [E] \otimes [H]^m$$

So $\nu\mu = \mathrm{id}$, μ is injective, and with Step 1 it is an isomorphism. It is a ring map because the external product is multiplicative (definition 2.2.6), and the module description follows since $K^0(S^2) = \mathbb{Z}\{1, [H]\}$ with $([H]-1)^2 = \beta^2 = 0$.

Step 4: The reduced form. Let X be pointed. Decompose $K^0(X) = \widetilde{K}^0(X) \oplus \mathbb{Z}$ and $K^0(S^2) = \widetilde{K}^0(S^2) \oplus \mathbb{Z}$. This uses only the basepoint, not connectedness: for the basepoint inclusion $i\colon \{*\} \to X$ and the collapse $c\colon X \to \{*\}$ one has $c \circ i = \mathrm{id}_{\{*\}}$, so $i^*c^* = \mathrm{id}$ on $K^0(\{*\}) = \mathbb{Z}$. Hence restriction to the basepoint is a split surjection, its kernel is $\widetilde{K}^0(X)$ by definition, and c^* splits it. Under μ these give

$$K^0(X \times S^2) \cong \big(\widetilde{K}^0(X) \otimes \widetilde{K}^0(S^2)\big) \oplus \widetilde{K}^0(X) \oplus \widetilde{K}^0(S^2) \oplus \mathbb{Z}$$

Restriction to $X \vee S^2$ kills exactly the first summand: a class $\pi_X^* a \cdot \pi_{S^2}^* b$ with a, b both reduced restricts on $X \times \{s_0\}$ to $\pi_X^* a$ times the pullback of $b|_{s_0} = 0$, and symmetrically on $\{x_0\} \times S^2$, while on the other three summands restriction is injective, being the identity on each factor. Hence

$$\ker\Big(\widetilde{K}^0(X \times S^2) \to \widetilde{K}^0(X \vee S^2)\Big) = \widetilde{K}^0(X) \cdot \beta$$

and μ restricts to an isomorphism $\widetilde{K}^0(X) \otimes \widetilde{K}^0(S^2) \to$ that kernel. Since $\widetilde{K}^0(S^2) = \mathbb{Z}\beta$, the map $a \mapsto \mu(a \otimes \beta)$ is an isomorphism of $\widetilde{K}^0(X)$ onto the kernel. By proposition 2.2.5 the kernel is q^* of $\widetilde{K}^0\big((X \times S^2)/(X \vee S^2)\big) = \widetilde{K}^0(X \wedge S^2) = \widetilde{K}^0(\Sigma^2 X)$, using the homeomorphism

$X \wedge S^2 \cong S^2 \wedge X$ that interchanges the smash factors, and q^* is injective, so composing with $(q^*)^{-1}$ gives the Bott map β_X of definition 2.2.7. Therefore β_X is an isomorphism, natural in X because every construction used is, completing the proof.

The homotopy-theoretic form of periodicity is a consequence of the theorem rather than an input to it, obtained by evaluating the theorem on spheres and reading the answer through the clutching description of bundles over a sphere.

Theorem 2.2.16: Bott Periodicity

The stable homotopy groups of the unitary group are 2-periodic: $\pi_k(U) \cong \pi_{k+2}(U)$ for all $k \geq 0$, and

$$\pi_k(U) = \begin{cases} 0 & k \text{ even} \\ \mathbb{Z} & k \text{ odd} \end{cases}$$

The orthogonal group is 8-periodic, $\pi_k(O) \cong \pi_{k+8}(O)$, with

$$\big(\pi_k(O)\big)_{k \bmod 8} = \big(\mathbb{Z}/2,\ \mathbb{Z}/2,\ 0,\ \mathbb{Z},\ 0,\ 0,\ 0,\ \mathbb{Z}\big)$$

The complex statement is proved here. The real statement is *not* proved in this section; it is recorded for comparison, with the argument in Atiyah's *K-Theory* [1], and the period 8 is derived from the mod-8 periodicity of the real Clifford algebras when real K-theory is developed.

***Proof*:**
Only the complex statement is proved here. Take $X = S^k$ in theorem 2.2.15, in its reduced form: $\widetilde{K}^0(S^k) \cong \widetilde{K}^0(\Sigma^2 S^k) = \widetilde{K}^0(S^{k+2})$, since $\Sigma S^k = S^{k+1}$. So $\widetilde{K}^0(S^k)$ depends only on the parity of k. Its two values are computed directly: $\widetilde{K}^0(S^1) = 0$, because every complex bundle over S^1 is trivial by the clutching correspondence (proposition 1.2.8) and the path-connectedness of $\mathrm{GL}_n(\mathbb{C})$, and $\widetilde{K}^0(S^2) \cong \mathbb{Z}$ generated by β (example 2.1.12). Hence $\widetilde{K}^0(S^k)$ is $\mathbb{Z}$ for k even and 0 for k odd.

The clutching correspondence (proposition 1.2.8) presents $\mathrm{Vect}^n_{\mathbb{C}}(S^k)$ as the *free* homotopy classes $[S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$, and two identifications turn these into homotopy groups. First, $\mathrm{GL}_n(\mathbb{C})$ is a path-connected topological group and so a simple space, so its π_1 acts trivially on every π_m and the forgetful map $\pi_{k-1}(\mathrm{GL}_n(\mathbb{C})) \to [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$ is a bijection ([14, Free and Based Homotopy Classes of Maps from a Sphere]). This is the general form of the winding-number count used for $n = 1$ in example 2.1.12. Second, the polar-decomposition retraction recorded there deforms $\mathrm{GL}_n(\mathbb{C})$ onto $U(n)$, so $\pi_{k-1}(\mathrm{GL}_n(\mathbb{C})) \cong \pi_{k-1}(U(n))$. With the stabilization theorem (theorem 1.3.1), applicable since S^k is connected compact Hausdorff for $k \geq 1$,

$$\pi_{k-1}(U) = \varinjlim_n \pi_{k-1}(U(n)) \cong \widetilde{K}^0(S^k)$$

which is $\mathbb{Z}$ for k even and 0 for k odd, that is $\pi_j(U)$ is $\mathbb{Z}$ for j odd and 0 for j even. Periodicity $\pi_j(U) \cong \pi_{j+2}(U)$ follows, completing the proof of the complex statement.

2.2.17 Note: The Loop-Space Form Periodicity is often stated as a homotopy equivalence $\mathbb{Z} \times BU \simeq \Omega^2(\mathbb{Z} \times BU)$, and in the real case $\mathbb{Z} \times BO \simeq \Omega^8(\mathbb{Z} \times BO)$. That form says more than the table above: it asserts an equivalence of spaces, not merely an isomorphism of homotopy groups, and producing it requires representing the Bott map by an actual map of spaces and then invoking Whitehead's theorem. The representability theorem available here (theorem 2.1.9) is stated on compact spaces, which are not the objects at which such an argument must be evaluated, so the equivalence is recorded rather than derived; see Atiyah's *K-Theory* [1]. Nothing later in this book uses the loop-space form, the table and the isomorphism β_X being what the computations use.

2.2.18 Historical Note: Bott's Original Route Bott's proof was not this one. He studied the loop space of a compact symmetric space by Morse theory for the energy functional, whose critical points are the geodesics joining two fixed points. An index computation identifies the space of minimal geodesics well enough to give $\pi_k(M^\nu) \cong \pi_{k+1}(M)$, and iterating along the chain U, O, U/O, Sp/U, $Sp, \ldots$ produces both periodicities at once. That argument is in Milnor's *Morse Theory* [12]. It needs infinite-dimensional Morse theory and the Morse index theorem, neither of which this book develops, and it proves the homotopy-group statement first, so the multiplicative form used throughout K-theory has to be recovered from it afterwards. The route taken above reverses that order and stays inside the material of chapter 1.

Periodicity collapses the doubly-infinite sequence of groups $\widetilde{K}^{-n}$ to two, $\widetilde{K}^0$ and $\widetilde{K}^{-1}$, the latter conventionally written $\widetilde{K}^1$. This is what is meant by saying complex K-theory is $\mathbb{Z}/2$-graded, and the next section makes the whole family into a cohomology theory.

Exercise 2.2.1

1. Let f be a clutching function for E that extends to a section α of $\mathrm{Aut}(\pi_+^* E)$ over $X \times D_+$, meaning $f = \alpha|_{X \times S^1}$. Show that $[E, f] \cong \pi_X^* E$. Deduce that a clutching function whose entries are polynomials in z with invertible value at every point of D_+ presents a pullback.

2. Using proposition 2.2.9(2), show that $[\underline{\mathbb{C}}, z] \otimes [\underline{\mathbb{C}}, z^{-1}]$ is the trivial line bundle over $X \times S^2$, and conclude that $[H]$ is a unit in $K^0(S^2)$ with $[H]^{-1} = [H^\vee]$.

3. Show that $[\underline{\mathbb{C}}, z^k] \cong (H^\vee)^{\otimes k}$ for every $k \in \mathbb{Z}$, and deduce that the class of $[\underline{\mathbb{C}}, z^k]$ in $K^0(S^2)$ is $1 - k\beta$. Which values of k give a bundle with vanishing reduced class?

4. Let $S_n f = \sum_{|k| \le n} a_k z^k$ be the partial sums of the Fourier series of a clutching function. Verify that the Cesàro mean $\frac{1}{N+1} \sum_{n=0}^{N} S_n f$ equals $\sum_{|k| \le N} \left(1 - \frac{|k|}{N+1}\right) a_k z^k$, the expression used in the proof of lemma 2.2.10.

5. Write out the matrix Lp of lemma 2.2.11 for a polynomial clutching function $p = a_0 + a_1 z + a_2 z^2$ of degree 2, and carry out the column and row operations of Step 1 explicitly. Exhibit the intermediate matrix $p \oplus T$ and check that T is not the identity, so that the homotopy of Step 3 is needed.

6. Let $E = X \times \mathbb{C}^2$ be trivial and let b be the constant endomorphism $\mathrm{diag}(\lambda, \mu)$ with $|\lambda| < 1 < |\mu|$. Compute the projection Π of lemma 2.2.13 directly from the contour integral and identify E_+ and E_-.

7. Take $a = \left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$ and $b = \left(\begin{smallmatrix}0&1\\1&2\end{smallmatrix}\right)$, so that $\ell(z) = az + b$ is invertible on S^1. Compute $P = \frac{1}{2\pi i}\oint_{|z|=1} \ell(z)^{-1} a\, dz$ and verify that $P^2 = P$ but $Pa \neq aP$. Explain why this does not contradict proposition 2.2.14, and identify which step of that proof the normalization to $z + c$ is doing.

8. Use theorem 2.2.15 with $X = S^2$ to compute $K^0(S^2 \times S^2)$ as a ring, and give its rank as an abelian group. Compare with $H^{\text{even}}(S^2 \times S^2; \mathbb{Z})$.

9. Path-connectedness of $\mathrm{GL}_n(\mathbb{C})$ is used twice in this section: once in the proof of lemma 2.2.12, and once in the proof of theorem 2.2.16. Identify both, and identify where the argument uses that every endomorphism of a complex vector space has an eigenvalue. Then explain why the section does not establish a 2-periodicity for real vector bundles, and state the period that replaces it.

2.3 K-Theory as a Generalized Cohomology Theory

A cohomology theory is more than a single functor: it is a graded family $\{h^n\}_{n\in\mathbb{Z}}$ tied together by suspension isomorphisms and by the long exact sequence of a pair. This section assembles topological K-theory into such a theory. The negative groups $K^{-n}(X)$ are defined by iterated reduced suspension, the relative groups $K^0(X, A)$ by collapsing the subspace A, and periodicity (theorem 2.2.15) is what collapses the doubly-infinite family down to two groups.

The passage from a single functor to a graded family rests on one construction: suspension. For ordinary cohomology the suspension isomorphism $\widetilde{H}^n(X) \cong \widetilde{H}^{n+1}(\Sigma X)$ shifts degree by smashing with a circle. K-theory has no a priori notion of degree, so the strategy is reversed: one *defines* the negative groups by suspending the space. The reduced suspension is forced, rather than the unreduced cone-based one, because $\widetilde{K}^0$ is a functor on pointed spaces.

Definition 2.3.1: Negative K-Groups

Let X be a compact Hausdorff space and let $n \geq 0$ be an integer. The *n-th negative reduced K-group* of a pointed space X is

$$\widetilde{K}^{-n}(X) = \widetilde{K}^0(\Sigma^n X)$$

where $\Sigma^n X$ is the n-fold reduced suspension and $\Sigma^0 X = X$. For an unpointed compact space X, the *n-th negative (unreduced) K-group* is

$$K^{-n}(X) = \widetilde{K}^0(\Sigma^n(X_+)) = \widetilde{K}^{-n}(X_+)$$

With $n = 0$ this recovers $K^0(X) = \widetilde{K}^0(X_+)$ and $\widetilde{K}^0(X)$ for the unreduced and reduced groups respectively.

The two are reconciled by the splitting $X_+ = X \sqcup \{*\}$. Adding a disjoint basepoint converts an unpointed space into a pointed one without losing information, and the reduced theory of X_+ is the unreduced theory of X. Concretely, the inclusion of the added basepoint and the collapse $X_+ \to S^0$ give a natural splitting $K^{-n}(X) \cong \widetilde{K}^{-n}(X) \oplus \widetilde{K}^{-n}(S^0)$ when X itself is pointed; since $\widetilde{K}^{-n}(S^0) = \widetilde{K}^0(\Sigma^n S^0) = \widetilde{K}^0(S^n)$, the discrepancy between reduced and unreduced is the K-theory of a sphere. This is the K-theoretic analogue of the $\widetilde{H}^*$ versus H^* bookkeeping in ordinary cohomology, and it is the reason every formula below comes in a reduced and an unreduced version.

Example 2.3.2: Negative Groups of a Point

Take $X = \{*\}$ the one-point space, regarded as pointed. Then $\Sigma^n\{*\} = \{*\}$ for every n, since suspending a point gives a point: $\Sigma\{*\} = S^1 \wedge \{*\} = \{*\}$. Hence

$$\widetilde{K}^{-n}(\{*\}) = \widetilde{K}^0(\{*\}) = 0$$

for all $n \geq 0$, the reduced group of a point being trivial by definition 2.1.4. For the unreduced groups, $\{*\}_+ = S^0$, so

$$K^{-n}(\{*\}) = \widetilde{K}^0(\Sigma^n S^0) = \widetilde{K}^0(S^n)$$

The right-hand side is $\mathbb{Z}$ when n is even and 0 when n is odd. This is example 2.4.1, and the division of labour is worth stating, because the clutching correspondence alone does not give it. That correspondence (proposition 1.2.8) identifies $\widetilde{K}^0(S^m)$ with a quotient of $\pi_{m-1}(U)$, which settles S^1 outright, $\pi_0(U)$ being trivial. For S^{2k+1} with $k \geq 1$ it reduces the claim to $\pi_{2k}(U) = 0$, and that is not a separate fact: it is the homotopy-group half of theorem 2.2.16. Quoting the correspondence for those spheres would therefore be circular, and the vanishing is quoted from the periodicity theorem instead. In the even case $\widetilde{K}^0(S^2) \cong \mathbb{Z}$ is generated by the reduced class of the canonical line bundle. Thus the coefficient groups $K^{-n}(\{*\})$ exhibit the 2-periodic pattern $\mathbb{Z}, 0, \mathbb{Z}, 0, \ldots$ that periodicity imposes on every space.

Relative K-theory measures the K-theory of X *rel* a subspace A. The model is the quotient X/A, which collapses A to a single basepoint; classes in the relative group are classes on X that are trivialized over A. This is the geometric content of the cofiber: X/A is the cofiber of the inclusion $A \hookrightarrow X$, and reduced K-theory of the cofiber is the receptacle for relative classes.

Definition 2.3.3: Relative K-Theory

Let (X, A) be a compact pair, meaning X is compact Hausdorff and $A \subseteq X$ is a closed (hence compact) subspace. For $n \geq 0$ the *relative K-groups* are

$$K^{-n}(X, A) = \widetilde{K}^{-n}(X/A) = \widetilde{K}^0(\Sigma^n(X/A))$$

where X/A is basepointed at the collapsed image of A. When $A = \emptyset$ one collapses the empty set, which by convention adjoins a disjoint basepoint, so $X/\emptyset = X_+$ and $K^{-n}(X, \emptyset) = K^{-n}(X)$ recovers the absolute groups.

The convention $X/\emptyset = X_+$ is what makes the relative theory a generalization of the absolute one. It encodes the principle that "no subspace collapsed" should leave the absolute group unchanged, and adding a disjoint basepoint is the operation that turns the absolute unreduced group into a reduced group on which the machinery operates. With this convention the long exact sequence of the pair $(X, \emptyset)$ degenerates correctly and excision reduces to a tautology.

Example 2.3.4: Relative K-Theory of a Disk Rel its Boundary

Let $X = D^2$ be the closed 2-disk and $A = \partial D^2 = S^1$ its boundary circle. The quotient collapses the boundary to a point, and

$$D^2/S^1 \cong S^2$$

since a disk with its boundary identified to a single point is a 2-sphere. Therefore

$$K^0(D^2, S^1) = \widetilde{K}^0(D^2/S^1) = \widetilde{K}^0(S^2) \cong \mathbb{Z}$$

the generator being the reduced class $[H] - 1$ of the canonical line bundle $H \to \mathbb{C}\mathrm{P}^1 = S^2$. This is the relative group housing the Bott generator of definition 2.2.7. More generally $D^{2k}/\partial D^{2k} \cong S^{2k}$, so $K^0(D^{2k}, \partial D^{2k}) \cong \widetilde{K}^0(S^{2k}) \cong \mathbb{Z}$, while collapsing the boundary of an odd-dimensional disk gives an odd sphere and the relative group vanishes.

2.3.5 Convention: Periodic Grading Let X be a finite pointed CW complex (box 2.2.1), and, by corollary 2.5.10, more generally every compact Hausdorff pointed space and every compact pair. By theorem 2.2.15 the groups $\widetilde{K}^{-n}(X)$ depend only on the parity of n. One therefore extends the indexing to all $n \in \mathbb{Z}$ by declaring $\widetilde{K}^n(X) = \widetilde{K}^{n-2}(X) = \widetilde{K}^{n+2}(X)$, so that $\widetilde{K}^n = \widetilde{K}^0$ for n even and $\widetilde{K}^n = \widetilde{K}^1$ for n odd, where $\widetilde{K}^1(X) = \widetilde{K}^0(\Sigma X) = \widetilde{K}^{-1}(X)$. The same convention applies to the unreduced and relative groups. Henceforth $\widetilde{K}^*$ denotes this $\mathbb{Z}/2$-graded family.

A generalized cohomology theory on compact pairs is, by the axioms developed in the generalized-cohomology chapter of [14, Eilenberg-Steenrod Cohomology Theory, Generalized Cohomology Theory], a sequence of contravariant functors h^n together with natural coboundary maps $\delta \colon h^n(A) \to h^{n+1}(X, A)$ satisfying homotopy invariance, exactness for pairs, excision, and additivity. The K-groups satisfy all of these, with the suspension isomorphism given by periodicity. The next theorem records this; its proof verifies each axiom, drawing the exactness from the cofiber sequence and excision from the homeomorphism type of the relevant quotients.

Theorem 2.3.6: K-Theory is a Generalized Cohomology Theory

The family $\{K^{-n}, \widetilde{K}^{-n}\}_{n\geq 0}$ of definition 2.3.1, together with the relative groups of definition 2.3.3, satisfies Axioms 1 to 3 and Axiom 5 below on compact Hausdorff pairs. On finite CW pairs, periodicity folds that family into the $\mathbb{Z}/2$-graded family $\{K^n, \widetilde{K}^n\}_{n\in\mathbb{Z}}$ of box 2.3.5, Axiom 4 holds, and together with the long exact sequence of theorem 2.3.7, whose proof uses only Axiom 2, the result satisfies the axioms of [14, Generalized Cohomology Theory], multiplicatively, *restricted to compact Hausdorff pairs and finite CW pairs.* That restriction narrows the claim: the cited definition asks for functors on all CW pairs with additivity over arbitrary disjoint unions, and Axiom 5 below is proved only for finite unions. The unrestricted theory is the spectrum-level one, [14, The K-theory Spectrum KU]. The division of hypotheses is box 2.2.1. Explicitly:

1. *(Homotopy invariance.)* If $f_0 \simeq f_1 \colon (X, A) \to (Y, B)$ are homotopic maps of pairs, then $f_0^* = f_1^* \colon K^n(Y, B) \to K^n(X, A)$.
2. *(Exactness.)* For a compact pair (X, A) with inclusions $i\colon A \hookrightarrow X$ and $j\colon (X, \emptyset) \hookrightarrow (X, A)$, the sequence
$$K^n(X, A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A)$$
is exact at $K^n(X)$.
3. *(Excision.)* If (X, A) is a compact pair and $U \subseteq A$ is open with $\overline{U} \subseteq \mathrm{int}A$, then the inclusion $(X \setminus U, A \setminus U) \hookrightarrow (X, A)$ induces an isomorphism $K^n(X, A) \cong K^n(X \setminus U, A \setminus U)$.
4. *(Suspension.)* There are natural isomorphisms $\widetilde{K}^n(X) \cong \widetilde{K}^{n+1}(\Sigma X)$, and the periodicity $\widetilde{K}^n \cong \widetilde{K}^{n+2}$ holds.
5. *(Additivity.)* For a finite disjoint union, restriction to the pieces is an isomorphism $K^n\big(\coprod_{j=1}^m X_j\big) \cong \bigoplus_{j=1}^m K^n(X_j)$.

Moreover the external product of definition 2.2.6 makes K^* a multiplicative theory.

***Proof*:**
Axiom 1: Homotopy invariance. For absolute groups this is theorem 1.2.5: homotopic maps induce the same map on isomorphism classes of bundles, hence the same map on Grothendieck groups K^0 (definition 2.1.1), and the same on $\widetilde{K}^0$. For negative groups, a homotopy f_t induces a homotopy $\Sigma^n f_t$ of the suspended maps, so $f_0^* = f_1^*$ on $\widetilde{K}^{-n}(Y) = \widetilde{K}^0(\Sigma^n Y)$. For relative groups, a homotopy of pairs $(X, A) \to (Y, B)$ induces a homotopy of the quotient maps $X/A \to Y/B$, since the homotopy carries A into B at every time and hence descends to the quotients; homotopy invariance of $\widetilde{K}^0$ on the quotients gives the claim.

Axiom 2: Exactness. The inclusion $A \overset{i}{\hookrightarrow} X$ has cofiber X/A, and there is a cofiber sequence
$$A \xrightarrow{i} X \xrightarrow{q} X/A$$
where q is the collapse. The content is that the composite $A \xrightarrow{i} X \xrightarrow{q} X/A$ is the constant map to the basepoint, so $i^*q^* = 0$, and conversely any class on X restricting to zero on A extends over the quotient. Concretely, work with the reduced groups via $K^n(X) = \widetilde{K}^n(X_+)$ and

$K^n(X,A) = \widetilde{K}^n(X/A)$. The collapse $q\colon X_+ \to X/A$ (sending A and the added basepoint to the basepoint) and the inclusion $i\colon A_+ \to X_+$ give

$$\widetilde{K}^n(X/A) \xrightarrow{q^*=j^*} \widetilde{K}^n(X_+) \xrightarrow{i^*} \widetilde{K}^n(A_+)$$

Exactness at the middle term is lemma 2.2.3, proved in section 2.2 because the smash-product identification there needed it. The same argument applied to Σ^n yields exactness in every degree.

Axiom 3: Excision. For a compact pair the excision isomorphism is a tautology in this model, because the relative group is defined through the quotient and the quotient is unchanged by excising an open set buried inside A. If $U \subseteq A$ is open with $\overline{U} \subseteq \mathrm{int}A$, then collapsing A in X and collapsing $A \setminus U$ in $X \setminus U$ produce homeomorphic quotients:

$$(X \setminus U)/(A \setminus U) \cong X/A$$

The map is induced by the inclusion $X \setminus U \hookrightarrow X$; it is a continuous bijection of compact Hausdorff spaces (every point of A maps to the single basepoint on both sides, and points outside A are untouched), hence a homeomorphism. Applying $\widetilde{K}^n$ to this homeomorphism gives the excision isomorphism $K^n(X,A) \cong K^n(X \setminus U, A \setminus U)$.

Axiom 5: Additivity. A vector bundle on a disjoint union is exactly a tuple of vector bundles on the pieces, since the pieces are open and closed, and the same holds for isomorphisms; so $\mathrm{Vect}(\coprod_j X_j) \cong \prod_j \mathrm{Vect}(X_j)$ as monoids, and applying the Grothendieck completion of definition 2.1.1, which preserves finite products, gives the claim in degree 0. For the negative degrees the substitution $X_j \mapsto \Sigma^n((X_j)_+)$ is not available, because $(\coprod_j X_j)_+$ is the wedge $\bigvee_j (X_j)_+$ rather than a disjoint union. Instead

$$K^{-n}\Big(\coprod_j X_j\Big) = \widetilde{K}^0\Big(\Sigma^n \bigvee_j (X_j)_+\Big) = \widetilde{K}^0\Big(\bigvee_j \Sigma^n((X_j)_+)\Big) \cong \bigoplus_j \widetilde{K}^0(\Sigma^n((X_j)_+)) = \bigoplus_j K^{-n}(X_j)$$

using that suspension carries a wedge to a wedge and, for the middle isomorphism, lemma 2.2.2 applied inductively to the finitely many summands. On finite CW pairs periodicity carries the statement to the $\mathbb{Z}/2$-graded family. Only finite unions occur in this book, all spaces being compact.

Axiom 4: Suspension and periodicity. For a finite pointed CW complex X and any integer $m \geq 0$ the definition of the negative groups gives, by the very meaning of suspension,

$$\widetilde{K}^{-(m+1)}(X) = \widetilde{K}^0(\Sigma^{m+1}X) = \widetilde{K}^0(\Sigma^m(\Sigma X)) = \widetilde{K}^{-m}(\Sigma X)$$

which is the suspension isomorphism $\widetilde{K}^n(X) \cong \widetilde{K}^{n+1}(\Sigma X)$ at $n = -(m+1)$, holding tautologically for all negative degrees. Periodicity (theorem 2.2.15) extends this identity to every integer n by shifting degrees in steps of 2 until they become negative, so the suspension isomorphism holds in the $\mathbb{Z}/2$-graded family of box 2.3.5. The substantive content beyond the tautology is the periodicity $\widetilde{K}^n \cong \widetilde{K}^{n+2}$ itself, which is exactly theorem 2.2.15. Together these identify the abstract suspension isomorphism required by the generalized-cohomology axioms of [14, Reduced Cohomology Theory] with the maps in hand.

Multiplicativity. The external product μ of definition 2.2.6 is associative and commutative up to natural isomorphism, since the bundle tensor product is, and the diagonal $X \to X \times X$ pulls μ back to the internal ring product of proposition 2.1.10. The Bott class β is the unit for the periodicity action, so the graded group $K^*(X)$ is a module over the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$, a Laurent polynomial ring on the invertible degree -2 generator β. This is the multiplicative structure, completing the verification of all axioms.

The theorem places K-theory on the same axiomatic footing as ordinary cohomology, but with one decisive difference: the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$ is not concentrated in degree 0. The dimension axiom of Eilenberg and Steenrod, which would force $h^n(\{*\}) = 0$ for $n \neq 0$, fails, and it is this failure that makes K-theory a *generalized* cohomology theory rather than ordinary cohomology in disguise. Every consequence of the axioms valid for an arbitrary generalized theory, the long exact sequence of a pair, Mayer-Vietoris, and the Atiyah-Hirzebruch spectral sequence of [14, The Atiyah-Hirzebruch Spectral Sequence] computing K^* from ordinary cohomology H^*, now applies to K-theory.

The exactness axiom is local in the sense that it relates three terms; iterating it along the suspensions of a pair produces a long exact sequence running infinitely in both directions. Because of periodicity this long sequence wraps around into a six-term cyclic sequence, the topological precursor of the six-term sequence for C^*-algebras developed in [4]. The next theorem assembles it.

Theorem 2.3.7: The Long Exact Sequence of a Pair

Let (X, A) be a finite CW pair (box 2.2.1) with inclusions $i\colon A \hookrightarrow X$ and $j\colon (X, \emptyset) \hookrightarrow (X, A)$. There is a long exact sequence

$$\cdots \to K^n(X,A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A) \xrightarrow{\delta} K^{n+1}(X,A) \xrightarrow{j^*} K^{n+1}(X) \to \cdots$$

natural in the pair, where δ is the coboundary map. By Bott periodicity (theorem 2.2.15) this sequence is 2-periodic and collapses to the six-term exact sequence

$$\begin{tikzcd}
K^0(X,A) \arrow[r, "j^*"] & K^0(X) \arrow[r, "i^*"] & K^0(A) \arrow[d, "\delta"] \\
K^1(A) \arrow[u, "\delta"] & K^1(X) \arrow[l, "i^*"] & K^1(X,A) \arrow[l, "j^*"]
\end{tikzcd}$$

***Proof*:**
Step 1: The Puppe sequence of the inclusion. The inclusion $i\colon A \hookrightarrow X$ of a subcomplex extends to its Puppe (cofiber) sequence of pointed spaces ([14, The Puppe Sequence and its Coexactness], applicable because (X, A) is a CW pair)

$$A_+ \xrightarrow{i} X_+ \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A_+ \xrightarrow{\Sigma i} \Sigma X_+ \xrightarrow{\Sigma q} \Sigma(X/A) \to \cdots$$

where each space is the cofiber of the preceding map and $\partial\colon X/A \to \Sigma A_+$ is the connecting map, identifying the cofiber of q with the suspension ΣA_+, as constructed in [14, The Puppe Sequence and its Coexactness]. The construction in [14, The Long Exact Sequence of a Cofiber Sequence] establishes that applying any reduced cohomology functor to a cofiber sequence produces an exact sequence: for the reduced functor $\widetilde{K}^0$ each consecutive pair of maps in the Puppe sequence gives an exact triple, the exactness being the case $n = 0$ of Axiom 2 in theorem 2.3.6.

Step 2: Applying $\widetilde{K}^0$. Apply $\widetilde{K}^0$ to the Puppe sequence. Each three-term stretch $W' \to W \to W''$ of cofiber and cofiber-of-cofiber yields exactness of $\widetilde{K}^0(W'') \to \widetilde{K}^0(W) \to \widetilde{K}^0(W')$, and assembling all of them gives the exact sequence

$$\cdots \to \widetilde{K}^0(\Sigma(X/A)) \to \widetilde{K}^0(\Sigma X_+) \to \widetilde{K}^0(\Sigma A_+) \xrightarrow{\partial^*} \widetilde{K}^0(X/A) \to \widetilde{K}^0(X_+) \to \widetilde{K}^0(A_+)$$

Translating with the definitions $\widetilde{K}^0(X_+) = K^0(X)$, $\widetilde{K}^0(X/A) = K^0(X,A)$, and $\widetilde{K}^0(\Sigma X_+) = \widetilde{K}^{-1}(X_+) = K^{-1}(X)$, this reads

$$\cdots \to K^{-1}(X,A) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(A) \xrightarrow{\delta} K^0(X,A) \xrightarrow{j^*} K^0(X) \xrightarrow{i^*} K^0(A)$$

where the coboundary δ is induced by the Puppe connecting map ∂. Iterating the Puppe sequence (each further suspension adds three more terms) extends this to a sequence infinite in both directions, exactly the displayed long exact sequence with K^n written for $\widetilde{K}^0 \circ \Sigma^{-n}$ in the periodic sense.

Step 3: Naturality. A map of pairs $(X,A) \to (Y,B)$ induces a map of Puppe sequences $A \to X \to X/A$ to $B \to Y \to Y/B$, hence a map of the associated Puppe sequences, hence a commuting ladder after applying $\widetilde{K}^0$. Thus the long exact sequence is natural in the pair, and in particular δ is a natural transformation.

Step 4: Collapsing by periodicity. By theorem 2.2.15 the groups satisfy $K^n \cong K^{n-2}$, so the doubly-infinite sequence has only two distinct rows of groups, those in even and odd degree. Identifying $K^{-1} = K^1$ via periodicity folds the long sequence into the cyclic six-term sequence

$$K^0(X,A) \to K^0(X) \to K^0(A) \xrightarrow{\delta} K^1(X,A) \to K^1(X) \to K^1(A) \xrightarrow{\delta} K^0(X,A)$$

which is the hexagon displayed in the statement. Exactness at each of the six vertices is inherited from the long exact sequence, completing the proof.

The six-term sequence is the computational engine of K-theory. Where ordinary cohomology gives a long exact sequence with infinitely many distinct groups, periodicity compresses the K-theory sequence into six terms arranged in a hexagon, and a single coboundary δ in each parity carries all the connecting information. Two standard specializations deserve to be recorded. The first relates the reduced theory of X, A, and X/A; the second is the Mayer-Vietoris sequence for a cover.

Corollary 2.3.8: Reduced Cofiber Sequence

For a finite CW pair (X,A) with A nonempty there is an exact sequence of reduced groups

$$\widetilde{K}^0(X/A) \xrightarrow{q^*} \widetilde{K}^0(X) \xrightarrow{i^*} \widetilde{K}^0(A) \xrightarrow{\delta} \widetilde{K}^1(X/A) \xrightarrow{q^*} \widetilde{K}^1(X) \xrightarrow{i^*} \widetilde{K}^1(A) \xrightarrow{\delta} \widetilde{K}^0(X/A)$$

***Proof*:**

Choose the basepoint of X in A. Then $\widetilde{K}^n(X) = K^n(X, \mathrm{pt})$ and the splitting $K^n(X) = \widetilde{K}^n(X) \oplus K^n(\mathrm{pt})$ is natural in the based maps $A \to X \to X/A$, all of which preserve the basepoint. Subtracting the constant-point summand from the six-term sequence of theorem 2.3.7 for (X,A) leaves exactly the displayed sequence of reduced groups, since $K^n(X,A) = \widetilde{K}^n(X/A)$ by definition 2.3.3 and the relative term is already reduced, completing the proof.

Corollary 2.3.9: Mayer-Vietoris in K-Theory

Let $X = X_1 \cup X_2$ be a finite CW complex written as the union of two subcomplexes, and set $X_0 = X_1 \cap X_2$, assumed nonempty. There is a six-term exact Mayer-Vietoris sequence

$$\begin{tikzcd} K^0(X) \arrow[r] & K^0(X_1) \oplus K^0(X_2) \arrow[r] & K^0(X_0) \arrow[d, "\delta"] \\ K^1(X_0) \arrow[u, "\delta"] & K^1(X_1) \oplus K^1(X_2) \arrow[l] & K^1(X) \arrow[l] \end{tikzcd}$$

with the horizontal maps the differences of restrictions.

***Proof*:**
Apply the long exact sequence of theorem 2.3.7 to the pairs (X, X_2) and (X_1, X_0). The two are compared by a homeomorphism $X_1/X_0 \cong X/X_2$, proved directly rather than by excision: the inclusion $X_1 \hookrightarrow X$ carries $X_0 = X_1 \cap X_2$ into X_2, so it descends to a continuous map $X_1/X_0 \to X/X_2$. This map is a bijection, because $X = X_1 \cup X_2$ makes every point of $X \setminus X_2$ lie in $X_1 \setminus X_0$ and both quotients collapse their second subspace to a single point; and it is a continuous bijection of compact Hausdorff spaces, hence a homeomorphism. Note that Axiom 3 of theorem 2.3.6 does *not* apply here: it requires an open U with $\overline{U} \subseteq \mathrm{int}A$, and for $U = X \setminus X_1$, $A = X_2$ this fails already for $X = S^1$ covered by two closed arcs. Applying $\widetilde{K}^n$ to the homeomorphism gives an isomorphism of relative groups

$$K^n(X, X_2) \cong K^n(X_1, X_0)$$

The Mayer-Vietoris sequence is the standard consequence of this excision isomorphism together with the two long exact sequences of pairs: splicing the pair sequences for (X, X_2) and (X_1, X_0) along the common relative term, and using periodicity to fold the result into six terms, produces the displayed hexagon, completing the proof.

These two corollaries are the tools used in section 2.4 to compute the K-theory of spheres, projective spaces, and other spaces built by gluing. The reduced cofiber sequence handles spaces presented as a quotient, and Mayer-Vietoris handles spaces presented as a union; periodicity guarantees that in either case only two groups, $\widetilde{K}^0$ and $\widetilde{K}^1$, ever appear.

Exercise 2.3.1

1. Using definition 2.3.1 and the computation of $\widetilde{K}^0(S^n)$ in example 2.3.2, determine $\widetilde{K}^{-n}(S^1)$ for all $n \geq 0$. Then identify $K^1(\{*\})$ and explain why it vanishes.

2. Let $\beta \in \widetilde{K}^0(S^2)$ be the Bott generator and let β_X be the Bott map of definition 2.2.7. Show directly from the external product that $\beta_{S^0} : \widetilde{K}^0(S^0) \to \widetilde{K}^0(S^2)$ carries the generator $\gamma = [\mathbb{C}_-]$ of $\widetilde{K}^0(S^0)$ to β, and deduce that β_{S^0} is an isomorphism without invoking theorem 2.2.15. (Note that neither the unit $1 = [\underline{\mathbb{C}}]$ nor $[\mathbb{C}_-] - [\mathbb{C}_+]$ lies in the reduced group: both have nonzero rank at the basepoint, and $\widetilde{K}^0(S^0)$ is the kernel of that rank.)

3. Use the reduced cofiber sequence (corollary 2.3.8) for the pair (D^2, S^1) to recompute $\widetilde{K}^0(S^2)$ and $\widetilde{K}^1(S^2)$, given that $\widetilde{K}^*(D^2) = 0$ (contractibility) and $\widetilde{K}^0(S^1) = 0$, $\widetilde{K}^1(S^1) \cong \mathbb{Z}$.

4. Verify the excision homeomorphism $(X \setminus U)/(A \setminus U) \cong X/A$ used in the proof of theorem 2.3.6 in the explicit case $X = [0, 2]$, $A = [0, 1]$, $U = [0, \frac{1}{2})$, by exhibiting the homeomorphism of quotients.
5. Let $X = S^2 \vee S^2$. Using Mayer-Vietoris (corollary 2.3.9) or the reduced cofiber sequence, compute $\widetilde{K}^0(S^2 \vee S^2)$ and $\widetilde{K}^1(S^2 \vee S^2)$.
6. Show that the coefficient ring $K^*(\{*\})$ of complex K-theory is the Laurent polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$ with β in degree -2, and contrast this with the coefficient ring of ordinary cohomology. Explain in one sentence why this difference is exactly the failure of the dimension axiom.

2.4 Computations in K-Theory

The machinery assembled in the previous three sections (the ring $K^0(X)$ of section 2.1, the Bott periodicity isomorphism, and the long exact sequence that turns K^* into a cohomology theory in section 2.3) is worth very little until it produces numbers. This section computes K^* for the spaces a topologist meets first: spheres, complex projective spaces, and products. Every input is exhibited explicitly. The reduced groups $\widetilde{K}^0(S^n)$ fall out of the suspension and clutching descriptions together with 2-periodicity; the ring $K^0(\mathbb{C}\mathrm{P}^n)$ is computed by induction over the cell structure, with its two multiplicative facts isolated as lemmas beforehand; and the external product yields a Künneth statement for $K^*(X \times Y)$. The recurring theme is that K-theory, unlike ordinary cohomology, sees the multiplicative geometry of line bundles directly: the ring generator of $K^0(\mathbb{C}\mathrm{P}^n)$ is literally $[H] - 1$ for the tautological line bundle H, and the single relation $([H] - 1)^{n+1} = 0$ encodes the whole ring.

Throughout, $K^0(X) = KU(X)$ denotes complex topological K-theory unless stated otherwise, and $\widetilde{K}^0(X)$ is the reduced group, the kernel of the rank map $K^0(X) \to \mathbb{Z}$ at a chosen basepoint (definition 2.1.4). For a finite CW pair the negative groups $K^{-n}(X)$ and the relative groups $K^0(X, A)$ are those of definitions 2.3.1 and 2.3.3, and Bott periodicity gives the isomorphism $\widetilde{K}^0(X) \cong \widetilde{K}^0(\Sigma^2 X)$ used at every turn (theorem 2.2.15).

The spheres are the natural starting point, since their bundle theory is already in hand. The clutching description of section 1.2 already computed the unreduced bundle sets $\mathrm{Vect}^n_{\mathbb{C}}(S^k)$; what remains is to package them into the K-groups and to read off the ring structure. The cleanest route is through reduced K-theory and the identity $\widetilde{K}^0(S^k) \cong \widetilde{K}^0(\Sigma^2 S^k) = \widetilde{K}^0(S^{k+2})$ given by Bott periodicity, which reduces every sphere to one of two base cases. The base cases are S^0 and S^1, both of which are immediate from the bundle classifications already in hand.

Example 2.4.1: The K-Theory of Spheres

For complex K-theory the reduced groups of the spheres are

$$\widetilde{K}^0(S^n) \cong \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases} \qquad \widetilde{K}^{-1}(S^n) \cong \begin{cases} 0 & n \text{ even} \\ \mathbb{Z} & n \text{ odd} \end{cases}$$

and for $k \geq 1$ the full unreduced ring of the even sphere is $K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$, where $\eta = \beta^{\times k} \in \widetilde{K}^0(S^{2k})$ is the k-fold *external* power of the Bott class, a generator of the reduced group, and the generator squares to zero.

Step 1: The two base cases. The space S^1 is the suspension ΣS^0, and every complex vector bundle over it is trivial: by proposition 1.2.8 the isomorphism classes of n-dimensional complex bundles over S^1 are in bijection with $[S^0, \mathrm{GL}_n(\mathbb{C})] = \pi_0(\mathrm{GL}_n(\mathbb{C}))$, which is a single point because $\mathrm{GL}_n(\mathbb{C})$ is path-connected (this is the content of example 1.2.12, item (1), on the complex side). Hence $\mathrm{Vect}^n_{\mathbb{C}}(S^1)$ is a single class for each n, every element of $K^0(S^1)$ is a difference of trivial bundles, and $\widetilde{K}^0(S^1) = 0$. For $S^0 = \{+, -\}$ a two-point space, a complex bundle is just a pair of vector spaces, so $K^0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}$ by rank at each point, and the reduced group (kernel of rank at $+$) is $\widetilde{K}^0(S^0) \cong \mathbb{Z}$, generated by the class $[\mathbb{C}_-]$ of the line bundle supported at the non-basepoint, which has rank 0 at the basepoint and so lies in the kernel of restriction there.

Step 2: Suspension and clutching for S^2. For $k \geq 2$ the correspondence proposition 1.2.8 identifies $\mathrm{Vect}^n_{\mathbb{C}}(S^k)$ with $[S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \cong \pi_{k-1}(\mathrm{GL}_n(\mathbb{C}))$ (free classes agree with based ones because $\mathrm{GL}_n(\mathbb{C})$, being a topological group, is a simple space, so π_1 acts trivially on π_{k-1}). In the stable range, and after deformation-retracting $\mathrm{GL}_n(\mathbb{C})$ onto $U(n)$, the reduced group $\widetilde{K}^0(S^k)$ is the colimit $\varinjlim_n \pi_{k-1}(U(n)) = \pi_{k-1}(U)$, exactly the stabilization theorem 1.3.1. For $k = 2$ this is $\pi_1(U) \cong \mathbb{Z}$, computed by winding number in example 2.1.12 with no appeal to periodicity (theorem 2.2.16), so

$$\widetilde{K}^0(S^2) \cong \pi_1(U) \cong \mathbb{Z}$$

A generator is the class $\beta = [H] - 1$, where $H \to \mathbb{C}\mathrm{P}^1 = S^2$ is the tautological line bundle and 1 is the trivial line bundle; this β is the *Bott class*, the same element implementing the periodicity isomorphism theorem 2.2.15.

Step 3: Periodicity propagates the answer. Each sphere is a double suspension of one two lower, $S^n = \Sigma^2 S^{n-2}$, so Bott periodicity theorem 2.2.15 gives

$$\widetilde{K}^0(S^n) \cong \widetilde{K}^0(S^{n-2})$$

Starting from $\widetilde{K}^0(S^0) \cong \mathbb{Z}$ and $\widetilde{K}^0(S^1) = 0$, induction yields $\widetilde{K}^0(S^{2k}) \cong \mathbb{Z}$ and $\widetilde{K}^0(S^{2k+1}) = 0$. The statement for $\widetilde{K}^{-1}$ follows by the same periodicity from the suspension identity $\widetilde{K}^{-1}(S^n) = \widetilde{K}^0(\Sigma S^n) = \widetilde{K}^0(S^{n+1})$.

Step 4: The generator is the iterated Bott class. Write $\beta^{\times k} \in \widetilde{K}^0(S^{2k})$ for the k-fold reduced external product of definition 2.2.6, under the identification $S^{2k} \cong S^2 \wedge \cdots \wedge S^2$. The cross keeps this apart from the internal power: $\beta^2 = 0$ in $\widetilde{K}^0(S^2)$, while $\beta^{\times 2}$ generates $\widetilde{K}^0(S^4)$. (In the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$ of box 2.3.5 the powers of β are the graded ones, which correspond to the external powers under $K^{-2k}(\{*\}) \cong \widetilde{K}^0(S^{2k})$.) It is a generator: $\beta^{\times 1} = \beta$ generates $\widetilde{K}^0(S^2)$ by example 2.1.12, and $\beta^{\times k} = \beta_{S^{2k-2}}\big(\beta^{\times(k-1)}\big)$ is the image of $\beta^{\times(k-1)}$ under the Bott map definition 2.2.7, an isomorphism by theorem 2.2.15, so induction on k carries the generator along.

Step 5: The ring structure. It remains to see that the generator of $\widetilde{K}^0(S^{2k})$ squares to zero. Write $\eta = \beta^{\times k}$. The product η^2 lies in $\widetilde{K}^0(S^{2k}) \cdot \widetilde{K}^0(S^{2k})$, and the cup product on a reduced group factors through the smash product: multiplication $\widetilde{K}^0(S^{2k}) \otimes \widetilde{K}^0(S^{2k}) \to \widetilde{K}^0(S^{2k})$ is induced by the diagonal $\Delta : S^{2k} \to S^{2k} \times S^{2k}$ followed by the projection to the smash $S^{2k} \wedge S^{2k} = S^{4k}$. The composite $S^{2k} \xrightarrow{\Delta} S^{2k} \times S^{2k} \to S^{2k} \wedge S^{2k}$ is null-homotopic, because the diagonal of a sphere lands in the wedge $S^{2k} \vee S^{2k}$ up to homotopy and the wedge maps to a point in the smash. Hence $\eta^2 = 0$, giving

$$K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$$

with $\eta = \beta^{\times k}$ the generator of the reduced group, completing the computation.

The reduction to S^0 and S^1 is the whole story: K-theory of spheres is $\mathbb{Z}$ in even degrees and 0 in odd degrees, with all higher structure governed by the single Bott class. The vanishing $\eta^2 = 0$ is forced by dimension once one knows that products of reduced classes live on smash products: there is simply no room for $\beta^{\times 2k}$, which lives on S^{4k}, inside $\widetilde{K}^0(S^{2k})$. This is the K-theoretic form of the fact that $H^*(S^n)$ has trivial cup products in positive degrees, and it is the reason the even spheres are "square-zero" extensions of $\mathbb{Z}$. The contrast with $\mathbb{C}\mathrm{P}^n$ below is instructive: there the generator does *not* square to zero until the $(n+1)$-st power, and the difference is exactly that $\mathbb{C}\mathrm{P}^n$ has cohomology in many even degrees while S^{2k} has it in only one.

A remark on the real theory rounds out the picture. The same suspension-and-clutching argument applied to KO uses $\pi_{k-1}(O)$ in place of $\pi_{k-1}(U)$, and the 8-fold periodicity of theorem 2.2.16 produces the well-known pattern $\widetilde{KO}^0(S^n) \cong \mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$ for $n \equiv 0, 1, \ldots, 7 \pmod 8$. The two $\mathbb{Z}/2$'s in dimensions 1 and 2 are precisely the Möbius-type phenomena recorded in example 1.2.12; they have no analogue in the complex theory, where $\pi_{2j}(U) = 0$ kills the odd spheres outright.

2.4.2 Remark: The Index-Theoretic Meaning of β The generator $\beta \in \widetilde{K}^0(S^2)$ is not an arbitrary choice. Under the identification $S^2 = \mathbb{C}\mathrm{P}^1$ it is $[H] - 1$, and under the isomorphism $\widetilde{K}^0(S^2) \cong K^{-2}(\{*\})$ it is the periodicity element, multiplication by which *is* the Bott map of definition 2.2.7. When one later builds the analytic index, which no book in this series yet develops, β becomes the K-theory class of the symbol of the Dolbeault operator on $\mathbb{C}\mathrm{P}^1$, and the equation $\beta^2 = 0$ on the sphere is the statement that this symbol class has no room to interact with itself. The reader should keep the shape $[L] - 1$ in mind: every generator produced in this section is of that form, for a line bundle L.

The spheres gave a ring that is essentially trivial ($\mathbb{Z}$ with a square-zero generator). Complex projective space is the first multiplicative example, and it is the model computation of the subject. The target is the statement that $K^0(\mathbb{C}\mathrm{P}^n)$ is a truncated polynomial ring on one generator, with the truncation degree equal to $n+1$. The additive answer is an induction over the cell structure of $\mathbb{C}\mathrm{P}^n$ using only the long exact sequence of theorem 2.3.7 and the K-theory of spheres. The ring structure needs two inputs that the induction does not produce, and both are isolated first: the relation $t^{n+1} = 0$, which comes from an isomorphism of bundles over $\mathbb{C}\mathrm{P}^n$, and the fact that t^n generates the top summand rather than a proper multiple of it, which comes from comparing the diagonal of $\mathbb{C}\mathrm{P}^n$ with the cup product in ordinary cohomology. The Atiyah-Hirzebruch spectral sequence is recorded afterwards. It explains *why* the additive answer has the same shape as $H^*(\mathbb{C}\mathrm{P}^n)$, and it is not what proves the ring structure here.

The first input generalizes an identity already seen. On $\mathbb{C}\mathrm{P}^1$ the relation $\beta^2 = 0$ was the K-theoretic form of the bundle isomorphism $\underline{\mathbb{C}} \oplus (H \otimes H) \cong H \oplus H$ recorded after example 2.1.12. Over $\mathbb{C}\mathrm{P}^n$ the same phenomenon occurs with $n+1$ summands, and it is visible in the exterior algebra of $H^{\oplus(n+1)}$. Exterior powers of a complex vector bundle are built exactly as the tensor product of definition 1.1.16 was, and are used here for the first time: $\Lambda^k E$ is the bundle with fiber $\Lambda^k(E_x)$ and transition functions $\Lambda^k(g_{\beta\alpha})$, with $\Lambda^0 E = \underline{\mathbb{C}}$, $\Lambda^1 E = E$ and $\Lambda^k E = 0$ for $k > \mathrm{rk} E$. Write

$$\lambda_{-1}(E) = \sum_{k \geq 0} (-1)^k [\Lambda^k E] \in K^0(X)$$

a finite sum. It depends only on the class $[E] \in K^0(X)$, and not on E itself: if $[E] = [F]$ then $E \oplus G \cong F \oplus G$ for some G, so $\Lambda^k(E \oplus G) \cong \Lambda^k(F \oplus G)$ by lemma 1.1.18, and expanding both sides by proposition 1.1.19 and inducting on k gives $[\Lambda^k E] = [\Lambda^k F]$ for every k. The two sums also run over the same range, since $[E]$ determines the rank (proposition 2.1.3). What λ_{-1} does *not* do is extend to virtual classes: an extension would carry sums to products, forcing $\lambda_{-1}(\underline{\mathbb{C}}) = 0$ to be invertible. Chapter 4 makes the family $\{\lambda^k\}$ into the λ-ring structure on K^0. Nothing here needs more than the definition just given.

Lemma 2.4.3: The Tautological Relation on $\mathbb{C}\mathrm{P}^n$

Let $H \to \mathbb{C}\mathrm{P}^n$ be the tautological line bundle and $t = [H] - 1 \in K^0(\mathbb{C}\mathrm{P}^n)$. Then

$$t^{n+1} = 0$$

***Proof*:**
Step 1: λ_{-1} is multiplicative and a trivial summand kills it. For bundles E, F over X the Whitney formula $\Lambda^k(E \oplus F) \cong \bigoplus_{i+j=k} \Lambda^i E \otimes \Lambda^j F$ is an isomorphism of bundles (proposition 1.1.19). Summing over k with signs gives

$$\lambda_{-1}(E \oplus F) = \lambda_{-1}(E) \cdot \lambda_{-1}(F)$$

For the trivial line bundle $\lambda_{-1}(\underline{\mathbb{C}}) = [\Lambda^0 \underline{\mathbb{C}}] - [\Lambda^1 \underline{\mathbb{C}}] = 1 - 1 = 0$. Hence $\lambda_{-1}(E) = 0$ for any E with a trivial line summand.

Step 2: The bundle and its class. Let $V = \underline{\mathbb{C}}^{n+1}$ be the trivial bundle over $\mathbb{C}\mathrm{P}^n$ and let $W = \mathrm{Hom}(V, H)$, whose fiber over a line $\ell \subset \mathbb{C}^{n+1}$ is $\mathrm{Hom}(\mathbb{C}^{n+1}, \ell)$. Then $W \cong V^\vee \otimes H \cong H^{\oplus(n+1)}$. A line bundle has $\Lambda^0 H = \underline{\mathbb{C}}$, $\Lambda^1 H = H$ and $\Lambda^k H = 0$ for $k \geq 2$, so $\lambda_{-1}(H) = 1 - [H]$, and Step 1 applied n times gives

$$\lambda_{-1}(W) = \lambda_{-1}(H)^{n+1} = \big(1 - [H]\big)^{n+1}$$

Step 3: W has a trivial line summand. Give $\mathbb{C}^{n+1}$ its standard Hermitian inner product and let $H^\perp \subset V$ have fiber $\ell^\perp$. Fiberwise linear algebra alone would not make this a bundle, so fix ℓ_0, let v_0 span ℓ_0, and work over the open set U of lines ℓ not orthogonal to ℓ_0, which contains ℓ_0. On a neighbourhood in U choose a continuous nowhere-zero section u of H, which exists because H is a bundle. The orthogonal projection $\pi\colon \mathbb{C}^{n+1} \to \ell_0^\perp$ has kernel ℓ_0, and ℓ_0 meets $\ell^\perp$ only in 0 on U, so π restricts to an injection $\ell^\perp \to \ell_0^\perp$, an isomorphism because both spaces have dimension n. Its inverse is continuous in (ℓ, w), being given by the explicit formula

$$w \longmapsto w - \frac{\langle w, u(\ell)\rangle}{\langle v_0, u(\ell)\rangle} v_0$$

whose denominator is nonzero exactly on U. These inverses are the local trivializations, so $H^\perp$ is a subbundle of V. The fiberwise orthogonal decomposition $\mathbb{C}^{n+1} = \ell \oplus \ell^\perp$ is then a continuous bundle map $H \oplus H^\perp \to V$ that is an isomorphism on each fiber, hence an isomorphism of bundles by lemma 1.1.6. Applying $\mathrm{Hom}(-, H)$, which decomposes over direct sums by the same fiberwise-natural gluing as in Step 1,

$$W = \mathrm{Hom}(V, H) \cong \mathrm{Hom}(H, H) \oplus \mathrm{Hom}(H^\perp, H) \cong \underline{\mathbb{C}} \oplus \mathrm{Hom}(H^\perp, H)$$

the summand $\mathrm{Hom}(H, H)$ being a line bundle with the continuous nowhere-zero section $\ell \mapsto \mathrm{id}_\ell$, hence trivial by lemma 1.1.10.

Step 4: Conclusion. Steps 1 and 3 give $\lambda_{-1}(W) = 0$, and Step 2 evaluates that class as $(1 - [H])^{n+1} = (-1)^{n+1} t^{n+1}$. Hence $t^{n+1} = 0$, as we sought to show.

For $n = 1$ the proof is the identity quoted above: $V \cong H \oplus H^\perp$ has both summands lines and V trivial of rank 2, so $H \otimes H^\perp$ is trivial and $H^\perp \cong H^\vee$. Hence $\mathrm{Hom}(H^\perp, H) \cong H \otimes H$ and $W = H \oplus H$ splits as $\underline{\mathbb{C}} \oplus (H \otimes H)$. The complement $\mathrm{Hom}(H^\perp, H)$ in Step 3 is in general the *holomorphic* cotangent bundle of $\mathbb{C}\mathrm{P}^n$, of complex rank n, so the splitting is the dual of the Euler sequence appearing in example 3.2.7. That identification is not needed here, only the trivial summand is.

The second input is the one the cell induction cannot see. That induction places t^n in the top summand $\widetilde{K}^0(S^{2n}) \cong \mathbb{Z}$ but says nothing about which element of it t^n is, and no argument available *at this point* settles it. The K-theoretic projective-bundle theorem would, and theorem 2.5.15 proves it, but that theorem is not available here: its proof takes the present theorem as an input, through theorem 2.5.14's fiberwise hypothesis, so quoting it would be circular. The answer is instead a statement about how the diagonal of $\mathbb{C}\mathrm{P}^n$ treats the top cell, and it is fixed by the corresponding statement in ordinary cohomology, where x^n is known to generate $H^{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$. The next lemma performs that comparison once and packages it as a statement about spaces, so that the theorem's proof can quote it in K-theory without redoing any cohomology.

Lemma 2.4.4: The Reduced Diagonal on the Top Cell of $\mathbb{C}\mathrm{P}^n$

Let $n \geq 2$ and give $\mathbb{C}\mathrm{P}^n$ its standard cell structure, with $2k$-skeleton $\mathbb{C}\mathrm{P}^k$, based at the 0-cell. Write $Y = \mathbb{C}\mathrm{P}^n \wedge \cdots \wedge \mathbb{C}\mathrm{P}^n$ for the smash of n copies, a finite CW complex, and let

$$\bar{\Delta}\colon \mathbb{C}\mathrm{P}^n \longrightarrow Y$$

be the n-fold reduced diagonal: the diagonal into $(\mathbb{C}\mathrm{P}^n)^{\times n}$ followed by the collapse of the subcomplex of points having some coordinate at the basepoint. Let $c\colon \mathbb{C}\mathrm{P}^n \to \mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1}$ be the collapse and $\iota\colon (\mathbb{C}\mathrm{P}^1)^{\wedge n} \hookrightarrow Y$ the inclusion. Then $(\mathbb{C}\mathrm{P}^1)^{\wedge n}$ is the $2n$-skeleton of Y, both it and $\mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1}$ are homeomorphic to S^{2n}, and there is a homotopy equivalence $h\colon \mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1} \to (\mathbb{C}\mathrm{P}^1)^{\wedge n}$ with

$$\bar{\Delta} \simeq \iota \circ h \circ c$$

as pointed maps.

***Proof*:**
Step 1: The cells of Y. The cells of a product of finite CW complexes are the products of cells, and the cells of a smash are those product cells no factor of which is the basepoint cell, together with the basepoint. Since $\mathbb{C}\mathrm{P}^n$ has one cell in each even dimension $0, 2, \ldots, 2n$, the non-basepoint cells of Y are indexed by tuples $(a_1, \ldots, a_n)$ with $1 \leq a_i \leq n$ and have dimension $2(a_1 + \cdots + a_n)$. The smallest such dimension is $2n$, attained only by $(1, \ldots, 1)$, and the next dimension occurring is $2n + 2$. So Y has a single cell in dimensions $\leq 2n$ besides the basepoint, and its attaching map is constant. That cell is the one indexed by $(1, \ldots, 1)$, the top cell of the subcomplex $(\mathbb{C}\mathrm{P}^1)^{\wedge n}$: the product $(\mathbb{C}\mathrm{P}^1)^{\times n}$ is a subcomplex of $(\mathbb{C}\mathrm{P}^n)^{\times n}$ meeting the collapsed fat wedge in its own fat wedge, so it descends to the subcomplex of Y spanned by the cells with every $a_i = 1$, of which there is exactly one. Hence

$$Y^{(2n)} = Y^{(2n+1)} = (\mathbb{C}\mathrm{P}^1)^{\wedge n} \cong S^{2n}$$

Step 2: Compression onto the bottom cell. By cellular approximation ([9, Theorem 4.8]) the map $\bar{\Delta}$ is homotopic to a cellular map g, and the homotopy may be taken stationary on any subcomplex on which $\bar{\Delta}$ is already cellular, in particular on the basepoint. So g is pointed and pointed-homotopic to $\bar{\Delta}$. Being cellular, g carries $\mathbb{C}\mathrm{P}^n = (\mathbb{C}\mathrm{P}^n)^{(2n)}$ into $Y^{(2n)} = (\mathbb{C}\mathrm{P}^1)^{\wedge n}$ and $\mathbb{C}\mathrm{P}^{n-1} = (\mathbb{C}\mathrm{P}^n)^{(2n-1)}$ into $Y^{(2n-1)} = \{*\}$. A map collapsing $\mathbb{C}\mathrm{P}^{n-1}$ to the basepoint factors uniquely through the quotient, so $g = \iota \circ h \circ c$ for a pointed map $h\colon \mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1} \to (\mathbb{C}\mathrm{P}^1)^{\wedge n}$. Both source and target are S^{2n}: the target by Step 1, the source because $\mathbb{C}\mathrm{P}^n$ is obtained from $\mathbb{C}\mathrm{P}^{n-1}$ by attaching a single $2n$-cell. Each carries a preferred basepoint, the collapsed subcomplex, so the two homeomorphisms may be taken pointed. Fix such a pair once and for all, making h a pointed self-map of S^{2n} with a degree. That degree depends on the choice only up to sign, which is all that is used.

Step 3: A cohomology class detecting the top cell. Write $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$ and $x = c_1(H^\vee)$ the standard positive generator ([14, The First Chern Class of a Line Bundle]). This is the truncation of $H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) = \mathbb{Z}[x]$ ([14, Computing Cohomology Rings by Multiplicativity]) along the inclusion of $\mathbb{C}\mathrm{P}^n$ as the $2n$-skeleton: both complexes have cells only in even dimensions, so their cellular cochain complexes carry zero differentials and agree through dimension $2n$, making restriction an isomorphism on H^k for $k \le 2n$, while $H^k(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = 0$ for $k > 2n$ by dimension. Restriction is a ring map, so x^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ for $k \le n$. We claim there is $\xi \in H^{2n}(Y;\mathbb{Z})$ with $\bar{\Delta}^*\xi = x^n$. Write Y_m for the smash of m copies of $\mathbb{C}\mathrm{P}^n$ and $\bar{\Delta}_m$ for the m-fold reduced diagonal, so $Y_1 = \mathbb{C}\mathrm{P}^n$ and $\bar{\Delta}_1 = \mathrm{id}$, and take $\xi_1 = x$. Given $\xi_{m-1} \in H^{2(m-1)}(Y_{m-1};\mathbb{Z})$ with $\bar{\Delta}^*_{m-1}\xi_{m-1} = x^{m-1}$, consider on $\mathbb{C}\mathrm{P}^n \times Y_{m-1}$ the cross product ([14, Cohomology Cross Product])

$$x \times \xi_{m-1} = p_1^*x \smile p_2^*\xi_{m-1}$$

Its restriction to $\mathbb{C}\mathrm{P}^n \vee Y_{m-1}$ vanishes: on the summand $\mathbb{C}\mathrm{P}^n$ the map p_2 is constant and ξ_{m-1} has positive degree, so $p_2^*\xi_{m-1}$ restricts to 0, and symmetrically on the other summand. A class on a wedge of two based CW complexes restricting to zero on both summands is zero: apply [14, The Long Exact Sequence of a Cofiber Sequence] to the pair $(\mathbb{C}\mathrm{P}^n \vee Y_{m-1},\ \mathbb{C}\mathrm{P}^n)$, whose quotient is Y_{m-1}, so a class killed by restriction to $\mathbb{C}\mathrm{P}^n$ is $q_\vee^*\gamma$ for some γ on Y_{m-1}; the inclusion j of Y_{m-1} satisfies $q_\vee \circ j = \mathrm{id}$, so $\gamma = j^*(q_\vee^*\gamma)$ is the restriction to the other summand, which vanishes as well. By the exact sequence of the based CW pair $(\mathbb{C}\mathrm{P}^n \times Y_{m-1},\ \mathbb{C}\mathrm{P}^n \vee Y_{m-1})$ in reduced ordinary cohomology ([14, The Long Exact Sequence of a Cofiber Sequence]), it is therefore $q^*\xi_m$ for some $\xi_m \in H^{2m}(Y_m;\mathbb{Z})$, where q is the collapse onto $\mathbb{C}\mathrm{P}^n \wedge Y_{m-1}$. Two separate things are used here. First, $Y_m = \mathbb{C}\mathrm{P}^n \wedge Y_{m-1}$: the two quotients of $(\mathbb{C}\mathrm{P}^n)^{\times m}$ collapse exactly the points having some coordinate at the basepoint, so the natural continuous bijection between them is a homeomorphism, the source being compact and the target Hausdorff. Second, under that identification the reduced diagonals satisfy

$$\bar{\Delta}_m = q \circ (\mathrm{id} \times \bar{\Delta}_{m-1}) \circ \Delta$$

with Δ the diagonal of $\mathbb{C}\mathrm{P}^n$, both sides sending z to the class of $(z, \ldots, z)$. Since $p_1 \circ (\mathrm{id} \times \bar{\Delta}_{m-1}) = p_1$ and $p_2 \circ (\mathrm{id} \times \bar{\Delta}_{m-1}) = \bar{\Delta}_{m-1} \circ p_2$, and $p_i \circ \Delta = \mathrm{id}$,

$$\bar{\Delta}^*_m\xi_m = \Delta^*\big(p_1^*x \smile p_2^*\bar{\Delta}^*_{m-1}\xi_{m-1}\big) = x \smile x^{m-1} = x^m$$

Taking $m = n$ gives the claim.

Step 4: h has degree ± 1. The collapse induces an isomorphism $c^*\colon H^{2n}(\mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) \to H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$: in the exact sequence of the pair $(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1})$ ([14, The Long Exact Sequence of a Cofiber Sequence]) the neighbouring groups $H^{2n-1}(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ and $H^{2n}(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ both vanish, the first for parity and the second because $\mathbb{C}\mathrm{P}^{n-1}$ has dimension $2n-2$. Write $u \in H^{2n}\big((\mathbb{C}\mathrm{P}^1)^{\wedge n};\mathbb{Z}\big)$ and $u' \in H^{2n}(\mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ for the generators corresponding to one another under the identifications with S^{2n} fixed in Step 2, so that $h^*u = \deg(h)\, u'$. That identity is where the homological definition of degree ([14, Degree], $f_* = \deg f$ on H_{2n}) passes into cohomology, and it has two inputs. Evaluation against the fundamental class is an isomorphism $H^{2n}(S^{2n};\mathbb{Z}) \to \mathbb{Z}$, because $H_{2n-1}(S^{2n};\mathbb{Z}) = 0$ leaves no Ext term in [14, Universal Coefficient Theorem For Cohomology]; and $\langle f^*\varphi, c\rangle = \langle \varphi, f_*c\rangle$ holds already on cochains, f^* being precomposition with $f_\#$. Write $\iota^*\xi = k\, u$. Then by Step 2 and Step 3,

$$x^n = \bar{\Delta}^*\xi = c^*h^*\iota^*\xi = k\deg(h)\, c^*u'$$

Both x^n and c^*u' generate $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$, so $k\deg(h) = \pm 1$ and in particular $\deg(h) = \pm 1$.

Step 5: Degree ± 1 forces a homotopy equivalence. By [14, Connectivity and Bottom Homotopy of Spheres] the group $\pi_{2n}(S^{2n})$ is infinite cyclic, with the isomorphism to $\mathbb{Z}$ carrying the class of a pointed self-map to its degree. Hence $[h] = \pm[\mathrm{id}]$. A reflection of S^{2n} in a hyperplane through the basepoint is a pointed homeomorphism of degree -1 ([14, Properties Of Degree](5)), so it represents $-[\mathrm{id}]$. In either case h is pointed-homotopic to a homeomorphism, hence a homotopy equivalence, completing the proof.

Nothing in the lemma is special to K-theory: it says that the reduced diagonal of $\mathbb{C}\mathrm{P}^n$ hits the bottom cell of the n-fold smash with degree ± 1, which is the same fact that makes x^n a generator of $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$. That is where the multiplicative input sits. A multiplicative Atiyah-Hirzebruch spectral sequence would package it as a statement about products on the pages, but the package is heavier than the single instance needed, and [14, The Atiyah-Hirzebruch Spectral Sequence] states existence, naturality, the second page and convergence only.

Theorem 2.4.5: K-Theory of Complex Projective Space

Let $H \to \mathbb{C}\mathrm{P}^n$ be the tautological (canonical) complex line bundle and set $t = [H] - 1 \in K^0(\mathbb{C}\mathrm{P}^n)$. Then $t^{n+1} = 0$ and

$$K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

as a ring, a free abelian group of rank $n+1$ on the basis $1, t, t^2, \ldots, t^n$. The odd group vanishes: $K^{-1}(\mathbb{C}\mathrm{P}^n) = 0$. The reduced group $\widetilde{K}^0(\mathbb{C}\mathrm{P}^n)$ is the ideal (t), free of rank n on $t, \ldots, t^n$.

The interpretive content is that the entire ring is generated by the single line bundle H, and that the only relation is the vanishing of $t^{n+1} = ([H]-1)^{n+1}$. Geometrically, $([H]-1)^{n+1} = 0$ is a K-theoretic Cayley-Hamilton identity: the line bundle H satisfies a degree-$(n+1)$ "polynomial", and lemma 2.4.3 exhibits that polynomial as an isomorphism of bundles. The dimension count the relation invites, that $\mathbb{C}\mathrm{P}^n$ is $2n$-dimensional so there is no room for a nonzero class in $\widetilde{K}^0$ beyond the n-th power, is a heuristic rather than an argument; turning it into one is what would need a multiplicative filtration on $\widetilde{K}^0$. The ring is abstractly isomorphic to the cohomology ring $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$, but the isomorphism is not the identity on generators: since $x = c_1(H^\vee)$ and hence $c_1(H) = -x$, the class t corresponds under the Chern character to $e^{-x} - 1 = -x + x^2/2 - \cdots$, not to x itself.

***Proof*:**
The additive structure is an induction over the cell structure of $\mathbb{C}\mathrm{P}^n$. The multiplicative input is the two lemmas above. An alternative account of the additive answer, via the spectral sequence, is given afterward.

Step 1: The cohomological input. Recall that $\mathbb{C}\mathrm{P}^n = P(\mathbb{C}^{n+1})$ is the projectivization of the trivial bundle $\mathbb{C}^{n+1} \to \{*\}$. The tautological line bundle H has fiber over a line $\ell \in \mathbb{C}\mathrm{P}^n$ equal to ℓ itself. Its dual has first Chern class $x = c_1(H^\vee) \in H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$, the standard positive generator, and x generates the cohomology ring,

$$H^*(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$$

a free abelian group with basis $1, x, \ldots, x^n$, concentrated in even degrees. The provenance is the one given in lemma 2.4.4 Step 3: truncate $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[x]$ ([14, Computing Cohomology Rings by Multiplicativity]) along the $2n$-skeleton, with the normalization $x = c_1(H^\vee)$ from [14, The First Chern Class of a Line Bundle]. Not from the projective bundle formula: over a point that formula degenerates, its Leray-Hirsch input being the cohomology of the fiber, which here *is* $\mathbb{C}\mathrm{P}^n$. This ring is used only through lemma 2.4.4.

Step 2: The candidate basis. The class $t = [H] - 1 \in \widetilde{K}^0(\mathbb{C}\mathrm{P}^n)$ is reduced, and the powers $1, t, \ldots, t^n$ are the candidate basis. That they are one is proved by the induction of Step 3 over the cell structure together with Step 4. Neither of the two standard shortcuts is open: [14, The Leray-Hirsch Theorem] is a statement about singular cohomology, and the projective-bundle theorem in K-theory, which would give the result directly, is proved in this book as theorem 2.5.15 but is not available here, its hypothesis (2) in theorem 2.5.14 being given by the present theorem.

Step 3: Induction via the subcomplex $\mathbb{C}\mathrm{P}^{n-1} \hookrightarrow \mathbb{C}\mathrm{P}^n$. The pair $(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1})$ has quotient $\mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1} = S^{2n}$, the top cell. This yields the cofiber sequence and hence the long exact sequence of theorem 2.3.7 (with the relative group definition 2.3.3)

$$\cdots \to \widetilde{K}^{-1}(\mathbb{C}\mathrm{P}^{n-1}) \to \widetilde{K}^0(S^{2n}) \to \widetilde{K}^0(\mathbb{C}\mathrm{P}^n) \to \widetilde{K}^0(\mathbb{C}\mathrm{P}^{n-1}) \to \widetilde{K}^1(S^{2n}) \to \cdots$$

By example 2.4.1, $\widetilde{K}^0(S^{2n}) \cong \mathbb{Z}$ and $\widetilde{K}^{-1}(S^{2n}) = \widetilde{K}^1(S^{2n}) = 0$, and by the inductive hypothesis $\widetilde{K}^{-1}(\mathbb{C}\mathrm{P}^{n-1}) = 0$ and $\widetilde{K}^0(\mathbb{C}\mathrm{P}^{n-1})$ is free of rank $n-1$. The sequence therefore collapses to the short exact sequence

$$0 \to \widetilde{K}^0(S^{2n}) \to \widetilde{K}^0(\mathbb{C}\mathrm{P}^n) \to \widetilde{K}^0(\mathbb{C}\mathrm{P}^{n-1}) \to 0$$

with the two outer groups free of ranks 1 and $n-1$. Since the right-hand group is free, the sequence splits and $\widetilde{K}^0(\mathbb{C}\mathrm{P}^n)$ is free of rank n; together with the rank-1 summand $K^0(\{*\}) = \mathbb{Z}$ this gives $K^0(\mathbb{C}\mathrm{P}^n)$ free of rank $n+1$. The induction uses three clauses at $n-1$, and all three hold at $n = 1$: $\widetilde{K}^0(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}$ generated by $t = \beta$ (example 2.1.12), $\widetilde{K}^{-1}(\mathbb{C}\mathrm{P}^1) = 0$ (example 2.4.1), and $t^2 = 0$ (lemma 2.4.3), so $K^0(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}[t]/(t^2)$. Concretely the base case is $\widetilde{K}^0(\mathbb{C}\mathrm{P}^1) = \widetilde{K}^0(S^2) \cong \mathbb{Z}$ from example 2.4.1. The vanishing of the odd groups propagates the same way, giving $K^{-1}(\mathbb{C}\mathrm{P}^n) = 0$ at every stage.

Step 4: t^n generates the top summand. Write $c\colon \mathbb{C}\mathrm{P}^n \to \mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1} = S^{2n}$ for the collapse, so that Step 3 exhibits $\widetilde{K}^0(S^{2n})$ as the summand $\operatorname{im} c^* = \ker\big(\widetilde{K}^0(\mathbb{C}\mathrm{P}^n) \to \widetilde{K}^0(\mathbb{C}\mathrm{P}^{n-1})\big)$, with c^* injective. For $n = 1$ the claim is example 2.1.12: $t = \beta$ generates $\widetilde{K}^0(\mathbb{C}\mathrm{P}^1)$. Let $n \geq 2$, let $\bar{\Delta}\colon \mathbb{C}\mathrm{P}^n \to Y = (\mathbb{C}\mathrm{P}^n)^{\wedge n}$ be the n-fold reduced diagonal, and write $t^{\times n} \in \widetilde{K}^0(Y)$ for the n-fold reduced external product of definition 2.2.6. Three facts combine.

First, $\bar{\Delta}^*(t^{\times n}) = t^n$. By construction the reduced external product satisfies $q^*\widetilde{\mu}(a \otimes b) = \mu(a \otimes b) = p_X^* a \cdot p_Y^* b$, and pulling that back along the diagonal turns the right-hand side into the internal product ab, since p_X and p_Y both become the identity. Since $\bar{\Delta}$ is the diagonal followed by q, this reads $\bar{\Delta}^*\widetilde{\mu}(a \otimes b) = ab$. Iterating along the recursion $\bar{\Delta}_m = q \circ (\mathrm{id} \times \bar{\Delta}_{m-1}) \circ \Delta$, established in Step 3 of the proof of lemma 2.4.4, gives the n-fold statement. The n-fold product itself is unambiguous because $\widetilde{\mu}$ is associative, which follows from the associativity of μ together with the injectivity of q^* in proposition 2.2.5.

Second, $\iota^*(t^{\times n}) = \beta^{\times n}$, a generator of $\widetilde{K}^0(S^{2n})$, where $\iota\colon (\mathbb{C}\mathrm{P}^1)^{\wedge n} = S^{2n} \hookrightarrow Y$ is the inclusion of the $2n$-skeleton. The reduced external product is natural, $(f \wedge g)^*\widetilde{\mu}(a \otimes b) = \widetilde{\mu}(f^*a \otimes g^*b)$ for pointed f, g: applying the injective q^* of proposition 2.2.5 reduces the identity to the naturality of μ, which is immediate from definition 2.2.6 because pullback commutes with tensor product. Now ι is the n-fold smash of the inclusion $j\colon \mathbb{C}\mathrm{P}^1 \hookrightarrow \mathbb{C}\mathrm{P}^n$, and j^*H is the tautological bundle of $\mathbb{C}\mathrm{P}^1$, so $j^*t = \beta$. That $\beta^{\times n}$ generates $\widetilde{K}^0(S^{2n})$ is example 2.4.1.

Third, $\bar{\Delta} \simeq \iota \circ h \circ c$ with h a homotopy equivalence (lemma 2.4.4). Combining the three,

$$t^n = \bar{\Delta}^*(t^{\times n}) = c^*h^*\iota^*(t^{\times n}) = c^*h^*(\beta^{\times n})$$

and $h^*(\beta^{\times n})$ generates $\widetilde{K}^0(S^{2n})$ because h^* is an isomorphism. So t^n generates $\mathrm{im} c^*$.

Step 5: Assembling the ring. Restriction to $\mathbb{C}\mathrm{P}^{n-1}$ is a ring map sending t to t, so by the inductive hypothesis it carries $t, \dots, t^{n-1}$ to a basis of $\widetilde{K}^0(\mathbb{C}\mathrm{P}^{n-1})$. In the short exact sequence of Step 3, a family mapping onto a basis of the quotient together with a generator of the kernel is a basis, and by Step 4 that generator is t^n. Hence $t, \dots, t^n$ is a $\mathbb{Z}$-basis of $\widetilde{K}^0(\mathbb{C}\mathrm{P}^n)$ and $1, t, \dots, t^n$ one of $K^0(\mathbb{C}\mathrm{P}^n)$. The relation $t^{n+1} = 0$ is lemma 2.4.3, so the ring homomorphism $\mathbb{Z}[t] \to K^0(\mathbb{C}\mathrm{P}^n)$ sending $t \mapsto t$ factors through $\mathbb{Z}[t]/(t^{n+1})$ and carries the basis $1, t, \dots, t^n$ of the source to the basis $1, t, \dots, t^n$ of the target. It is therefore an isomorphism,

$$K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

completing the proof.

***Proof Second proof of the additive statement*:**
The Atiyah-Hirzebruch spectral sequence explains the shape of the groups structurally. It gives the additive answer only, which is all that is claimed here. The ring structure stays with lemma 2.4.3 and lemma 2.4.4.

Let X be a finite CW complex. The construction of [14, The Atiyah-Hirzebruch Spectral Sequence] runs on the skeletal filtration. Its exact-couple and convergence machinery is upstream and stays there. What it asks of the theory K^* is available on a finite complex. The long exact sequence of a skeletal pair is theorem 2.3.7, which holds on finite CW pairs. Homotopy invariance is Axiom 1 of theorem 2.3.6, and the suspension isomorphism, in every degree, is its Axiom 4 read through box 2.3.5. The relative group of a skeletal pair *is* the reduced group of the quotient, which is how definition 2.3.3 defines it. The splitting over the wedge $X^{(p)}/X^{(p-1)} = \bigvee_{J_p} S^p$, finite because X is, is lemma 2.2.2 applied inductively, in whichever degree is wanted. The last input is the identification of the first differential with the cellular coboundary, which needs that a self-map of S^p of degree d act on $\widetilde{K}^*(S^p)$ as multiplication by d; for $p \geq 1$ that follows from [14, Connectivity and Bottom Homotopy of Spheres], which makes $\pi_p(S^p)$ infinite cyclic on the identity with degree as the isomorphism, together with the additivity of $\widetilde{K}^*$ along the pinch map

$S^p \to S^p \vee S^p$, which is lemma 2.2.2. Every complex the sequence is run on below, in the text and in the exercises alike, has cells in even dimensions only, so its 1-skeleton is a single point. The two such complexes are $\mathbb{C}\mathrm{P}^n$ here and $S^2 \times S^4$ in exercise 2.4.1 (3). This is why the spectral sequence is available here even though theorem 2.3.6 stops short of the unrestricted class of [14, Generalized Cohomology Theory]. For that theorem in its stated generality the relevant theory is the spectrum-level one, [14, The K-theory Spectrum KU]. The spectral sequence has

$$E_2^{p,q} = H^p(X; K^q(\{*\})) \Longrightarrow K^{p+q}(X)$$

where $K^q(\{*\}) = \mathbb{Z}$ for q even and 0 for q odd, by Bott periodicity theorem 2.2.15. For $X = \mathbb{C}\mathrm{P}^n$ the cohomology $H^p(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ is $\mathbb{Z}$ for p even with $0 \le p \le 2n$ and 0 otherwise. Thus on the E_2 page every nonzero entry $E_2^{p,q}$ has both p and q even, so $p+q$ is always even. Each differential $d_r \colon E_r^{p,q} \to E_r^{p+r,q-r+1}$ raises the total degree by 1, so it sends an even-total-degree class into an odd-total-degree group, and every odd-total-degree group here vanishes. Therefore every differential is zero and the spectral sequence collapses at E_2:

$$E_\infty^{p,q} = E_2^{p,q} = H^p(\mathbb{C}\mathrm{P}^n; K^q(\{*\}))$$

The associated graded of $K^0(\mathbb{C}\mathrm{P}^n)$ is $\bigoplus_p E_\infty^{p,-p} = \bigoplus_{k=0}^n H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}^{n+1}$, and that of $K^{-1}(\mathbb{C}\mathrm{P}^n)$ is 0 since all odd cohomology vanishes. As each filtration quotient is free, the filtration of $K^0(\mathbb{C}\mathrm{P}^n)$ splits and $K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}^{n+1}$, with $K^{-1}(\mathbb{C}\mathrm{P}^n) = 0$, as we sought to show.

What the spectral sequence does *not* give here is the ring. Reading the products off the filtration would need the abutment filtration to be multiplicative, $F^p \cdot F^q \subseteq F^{p+q}$, with the induced product on E_∞ identified with the cup product on E_2, and an edge identification carrying $[L]-1$ to $c_1(L)$. The cited theorem asserts none of these: it gives existence, naturality, the second page and convergence. Nor is the identification available from the edge: [14, The Edge Homomorphisms of the Atiyah-Hirzebruch Spectral Sequence] states the leading-term edge for a *connective* theory, and K^* is not connective, its coefficient groups populating every even negative degree. The multiplicative input is lemma 2.4.4, proved directly.

The two arguments illuminate different features. The induction over cells is hands-on: it builds the answer one cell at a time and pins the generator t^k to the $2k$-cell. The spectral sequence is conceptual: the collapse is forced purely by parity ($K^*(\{*\})$ is concentrated in even degrees and $\mathbb{C}\mathrm{P}^n$ has cohomology only in even degrees, so no differential can connect two nonzero groups), and it shows that the abstract shape of $K^0(\mathbb{C}\mathrm{P}^n)$ is dictated by $H^*(\mathbb{C}\mathrm{P}^n)$. Neither is the source of the ring, which is the two lemmas. That $t = [H]-1$ rather than some other lift of x is what makes the ring computable, and it is why the multiplicative statement is sharper than "$K^0(\mathbb{C}\mathrm{P}^n)$ is a free group of rank $n+1$". The relation $([H]-1)^{n+1} = 0$ should be compared with the Chern-class relation $c_1(H)^{n+1} = 0$: the latter holds in H^*, the former in K^0, and the Chern character intertwines them. The comparison of the two rings is carried out in example 3.3.9, once the Chern character has been constructed in section 3.3.

Taken along the whole tower, that same relation is what makes $K^0(\mathbb{C}\mathrm{P}^\infty)$ a power series ring, in the spectrum-level theory rather than in this book's K^0, as the next remark is careful to say. This is the generating case behind the splitting principle and the theory of Chern classes.

2.4.6 Remark: $K^0(\mathbb{C}\mathrm{P}^\infty)$ as a Power Series Ring Passing to the colimit $\mathbb{C}\mathrm{P}^\infty = \varinjlim_n \mathbb{C}\mathrm{P}^n$ requires care, because theorem 2.4.5 computes K^0 of the finite stages only. The bridge is the Milnor exact sequence [14, The Milnor Exact Sequence], which for a based CW complex written as an increasing union of subcomplexes fits $\widetilde{h}^n(X)$ into a short exact sequence between $\varprojlim^1_m \widetilde{h}^{n-1}(X_m)$ and $\varprojlim_m \widetilde{h}^n(X_m)$; here $\widetilde{h}^*$ must be a reduced cohomology theory on *all* based CW complexes, which this book's $\widetilde{K}^*$ is not: theorem 2.3.6 reaches only compact Hausdorff and finite CW pairs, and box 2.2.1 records that K^0 is undefined on $\mathbb{C}\mathrm{P}^\infty$ here. So $\widetilde{h}^*$ is the spectrum-level theory represented by KU, [14, The K-theory Spectrum KU], and everything below is a report on that extension rather than a consequence of anything proved in this book. Take $X_m = \mathbb{C}\mathrm{P}^m$. The $\varprojlim^1$ term the sequence asks about is the one on $\widetilde{h}^{-1}(X_m) = \widetilde{K}^{-1}(\mathbb{C}\mathrm{P}^m)$, and every one of those groups vanishes by theorem 2.4.5, so that tower is trivially Mittag-Leffler and $\varprojlim^1$ vanishes by [14, Mittag-Leffler Collapse of the Milnor Sequence]. (The degree-0 transition maps $\mathbb{Z}[t]/(t^{n+1}) \to \mathbb{Z}[t]/(t^n)$, $t \mapsto t$, are surjective too, but it is the degree -1 tower that the theorem needs.) Hence

$$K^0(\mathbb{C}\mathrm{P}^\infty) \cong \varprojlim_n \mathbb{Z}[t]/(t^{n+1}) \cong \mathbb{Z}[[t]]$$

the formal power series ring in $t = [H]-1$. This is the home of the universal Chern class: a complex line bundle on any space is classified by a map to $\mathbb{C}\mathrm{P}^\infty = BU(1)$, and its K-theory invariant is a power series in t. The same computation in cohomology gives only $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \cong \mathbb{Z}[x]$, a polynomial ring: ordinary cohomology is graded, so the limit stabilizes in each fixed degree, whereas in K-theory every power of t lands in the one group K^0 and genuine power series appear. The formal-group-law structure on $\mathbb{Z}[[t]]$ coming from the tensor product of line bundles, $t(L_1 \otimes L_2) = t(L_1) + t(L_2) + t(L_1)t(L_2)$, is the multiplicative formal group, in contrast to the additive formal group $x(L_1 \otimes L_2) = x(L_1) + x(L_2)$ of ordinary cohomology.

The third computation concerns products. Computing K^* of a product reduces, when it works, to the tensor product of the factors' K-theory. The mechanism is the external product, a pairing $K^0(X) \otimes K^0(Y) \to K^0(X \times Y)$ built from the external tensor product of bundles. After recording it and a worked instance, the section closes with the Künneth statement, proved by comparing the two sides as cohomology theories in one variable.

Example 2.4.7: The External Product on $S^2 \times S^2$

For vector bundles $E \to X$ and $F \to Y$, the *external tensor product* is $E \boxtimes F = \pi_X^*(E) \otimes \pi_Y^*(F) \to X \times Y$, where π_X, π_Y are the two projections. On classes this defines the external product

$$\times \colon K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y), \qquad a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$$

which is a ring homomorphism out of the tensor-product ring. Take $X = Y = S^2$ with $\widetilde{K}^0(S^2) = \mathbb{Z}\beta$, $\beta = [H] - 1$ and $\beta^2 = 0$ (example 2.4.1). Write $\beta_1 = \pi_1^*(\beta)$ and $\beta_2 = \pi_2^*(\beta)$ in $K^0(S^2 \times S^2)$. Since pullback is a ring map and $\beta^2 = 0$ on each factor, $\beta_1^2 = \beta_2^2 = 0$, while the cross term $\beta_1\beta_2 = \beta \times \beta$ is the external square. The external product gives a ring isomorphism

$$K^0(S^2 \times S^2) \cong K^0(S^2) \otimes_{\mathbb{Z}} K^0(S^2) \cong \frac{\mathbb{Z}[\beta_1, \beta_2]}{(\beta_1^2,\ \beta_2^2)}$$

a free abelian group of rank 4 on the basis $1, \beta_1, \beta_2, \beta_1\beta_2$. The reduced group $\widetilde{K}^0(S^2 \times S^2)$ is the rank-3 ideal (β_1, β_2), and one checks directly that $\beta_1\beta_2$ generates the top class supported on the

4-cell $S^2 \wedge S^2 = S^4$: indeed under the collapse $S^2 \times S^2 \to S^2 \wedge S^2 = S^4$ the class $\beta_1\beta_2$ pulls back from a generator of $\widetilde{K}^0(S^4) \cong \mathbb{Z}$, consistent with $\widetilde{K}^0(S^4) \cong \widetilde{K}^0(S^2)$ under Bott periodicity. The element $\beta_1\beta_2$ is q^* of the external Bott class $\beta^{\times 2} \in \widetilde{K}^0(S^4)$, the same η that generated $\widetilde{K}^0(S^4)$ in example 2.4.1; the apparent clash with $\beta^2 = 0$ is resolved by noting that the *internal* square on a single S^2 vanishes while the *external* square on $S^2 \times S^2$ does not, because the diagonal $S^2 \to S^2 \times S^2$ kills it but the product space retains it.

The example exhibits the general pattern: the K-theory of a product is the tensor product of the K-theories, provided no torsion or extension obstructions intervene. The precise statement is the Künneth theorem in K-theory, which holds cleanly when one factor has torsion-free, finitely generated K-theory.

Theorem 2.4.8: Künneth Theorem in K-Theory

Let X and Y be finite CW complexes, and suppose at least one of $K^*(X)$, $K^*(Y)$ is a finitely generated free abelian group (for instance whenever that factor is S^n or $\mathbb{C}\mathrm{P}^n$). Then the external product is a $\mathbb{Z}/2$-graded ring isomorphism

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

In particular, if both $K^*(X)$ and $K^*(Y)$ are finitely generated and free, then so is $K^*(X \times Y)$, with rank equal to the product of the ranks.

In words: when the K-theory of one factor is free, computing K^* of a product is pure algebra, the tensor product of graded rings. The freeness hypothesis is what avoids a Tor correction term; in full generality there is a short exact Künneth sequence with a $\mathrm{Tor}_1^{K^*(\{*\})}$ summand, exactly as in ordinary cohomology, but the spaces of this chapter (spheres, projective spaces, and their products) all have free K-theory, so the clean statement applies to every example here.

***Proof*:**
Step 1: The external product is well-defined and natural. For $E \to X$ and $F \to Y$ the bundle $E \boxtimes F = \pi_X^* E \otimes \pi_Y^* F$ depends only on the isomorphism classes of E and F by functoriality of pullback (proposition 1.2.3) and bilinearity of the tensor product over $\oplus$ (definition 1.1.16). Hence $a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$ extends to a homomorphism $K^0(X) \otimes K^0(Y) \to K^0(X \times Y)$, and using the negative groups (definition 2.3.1) to a graded homomorphism $K^*(X) \otimes K^*(Y) \to K^*(X \times Y)$. It is a ring map because pullback is a ring map and the cup product on $K^*(X \times Y)$ is graded-commutative.

Step 2: Both sides are cohomology theories in the inducting variable. By symmetry of the external product (the two factors enter on equal footing, since $X \times Y \cong Y \times X$) it is harmless to label the factors so that Y is the one whose K-theory is finitely generated and free; this is exactly the hypothesis. Fix this free factor Y and regard both sides as cohomology theories in the remaining variable X, over whose cells the induction will run. The functor $X \mapsto K^*(X \times Y)$ is a generalized cohomology theory, though not because $- \times Y$ preserves cofiber sequences, which it does not: define its relative groups on a pair (X, A) to be $K^*(X \times Y, A \times Y)$, note that $(X \times Y, A \times Y)$ is again a finite CW pair, and the axioms follow from theorems 2.3.6 and 2.3.7 applied to those pairs. Meanwhile $X \mapsto K^*(X) \otimes_{\mathbb{Z}} K^*(Y)$ is a cohomology theory because the fixed factor $K^*(Y)$ is a *free* $\mathbb{Z}$-module, so tensoring with it is exact and preserves the long exact sequences (theorem 2.3.7). The external product is a natural transformation between these two theories.

Step 3: The transformation is an isomorphism on a point, hence everywhere. Regard K^* as $\mathbb{Z}/2$-graded via Bott periodicity, so that the coefficient ring is $K^*(\{*\}) = \mathbb{Z}$ concentrated in even degree. On $X = \{*\}$ the external product is

$$K^*(\{*\}) \otimes_{\mathbb{Z}} K^*(Y) = \mathbb{Z} \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(Y)$$

the unit isomorphism of the tensor product, since $K^*(\{*\}) = \mathbb{Z}$. A natural transformation of cohomology theories on finite CW complexes that is an isomorphism on the point (together with the additivity axiom, which handles disjoint unions and hence S^0) is an isomorphism on all finite CW complexes: this is the standard comparison via the five lemma and induction over cells, run on the skeletal filtration. Concretely, suppose the transformation is an isomorphism for all complexes with fewer cells than X. Attaching a cell to a subcomplex X' gives a cofiber sequence $X' \hookrightarrow X \to X/X' = S^m$, and naturality produces a map of long exact sequences (theorem 2.3.7) in which the maps for X' and for S^m are isomorphisms (the latter because $K^*(S^m) \otimes K^*(Y) \to K^*(S^m \times Y)$ is an iterated suspension of the point case, using Bott periodicity theorem 2.2.15). The five lemma then forces the map for X to be an isomorphism.

Step 4: Conclusion. The induction starts from $X = \{*\}$ (Step 3) and covers every finite CW complex X, so for the fixed free factor Y the external product

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

is an isomorphism for *every* finite CW complex X, with no freeness assumed on $K^*(X)$. The freeness that was used is that of the fixed factor $K^*(Y)$, which is what made $X \mapsto K^*(X) \otimes K^*(Y)$ a cohomology theory in Step 2; by the relabelling at the start of that step this covers the stated hypothesis that at least one of $K^*(X)$, $K^*(Y)$ is finitely generated and free. When *both* factors are finitely generated and free, the rank and freeness statements follow since a tensor product of finitely generated free abelian groups is finitely generated and free of the product rank, completing the proof.

With the Künneth theorem the K-theory of every product of spheres and projective spaces is now mechanical. For instance $K^0((\mathbb{C}\mathrm{P}^n)^{\times m}) \cong \mathbb{Z}[t_1, \ldots, t_m]/(t_1^{n+1}, \ldots, t_m^{n+1})$, the tensor power of $\mathbb{Z}[t]/(t^{n+1})$, with each $t_i = \pi_i^*([H] - 1)$ the pulled-back generator from the i-th factor. This is the K-theoretic analogue of the cohomology of a product of projective spaces, and it is the computation that underlies the splitting principle of theorem 2.5.16: a sum of line bundles over a base B is classified by a map to $(\mathbb{C}\mathrm{P}^\infty)^{\times m} = BU(1)^m$, and its K-theory class is a polynomial in the pulled-back generators.

Exercise 2.4.1

1. Using only the suspension identity $\widetilde{K}^{-1}(X) = \widetilde{K}^0(\Sigma X)$ and the values $\widetilde{K}^0(S^0) \cong \mathbb{Z}$, $\widetilde{K}^0(S^1) = 0$, together with Bott periodicity, compute $\widetilde{K}^0(S^n)$ and $\widetilde{K}^{-1}(S^n)$ for $n = 3, 4, 5, 6$. State the unreduced groups $K^0(S^n)$ and $K^{-1}(S^n)$ in each case.
2. Let $t = [H] - 1 \in K^0(\mathbb{C}\mathrm{P}^2)$. Compute, in the basis $1, t, t^2$ of $\mathbb{Z}[t]/(t^3)$, the products $t \cdot t$, $t \cdot t^2$, and $(1 + t)^2$. Deduce the K-theory class of $H^{\otimes 2} = H^2$ as an element of $\mathbb{Z}[t]/(t^3)$.
3. Use the Atiyah-Hirzebruch spectral sequence to compute $K^*(S^2 \times S^4)$. State the E_2 page explicitly, explain why all differentials vanish, and give the ranks of K^0 and K^{-1}. Confirm

the answer agrees with the Künneth theorem applied to $K^*(S^2) \otimes K^*(S^4)$.

4. Let $H \to \mathbb{C}\mathrm{P}^1 = S^2$ be the tautological line bundle and $\beta = [H] - 1$. Show directly, without invoking the smash-product argument, that $\beta^2 = 0$ in $K^0(\mathbb{C}\mathrm{P}^1)$ by using the relation $t^{n+1} = 0$ of theorem 2.4.5 for $n = 1$. Then explain why the external square $\beta \times \beta$ is nonzero in $K^0(S^2 \times S^2)$.

5. Compute the rank of $\widetilde{K}^0(\mathbb{C}\mathrm{P}^n \times \mathbb{C}\mathrm{P}^m)$ and write down a $\mathbb{Z}$-basis in terms of the pulled-back generators $t_1 = \pi_1^*([H] - 1)$ and $t_2 = \pi_2^*([H] - 1)$. Identify the single class supported on the top cell.

6. The reduced suspension formula gives $\widetilde{K}^0(\Sigma X) \cong \widetilde{K}^{-1}(X)$. Using $\mathbb{C}\mathrm{P}^2/\mathbb{C}\mathrm{P}^1 = S^4$ and the cofiber long exact sequence, recompute $\widetilde{K}^0(\mathbb{C}\mathrm{P}^2)$ and verify it is free of rank 2, matching theorem 2.4.5.

2.5 The Projective-Bundle Theorem and the Splitting Principle

The computations of section 2.4 determine the K-theory of projective space over a point. This section proves the relative statement: for a vector bundle V over a finite complex, K^* of the projectivization $P(V)$ is a free module over K^* of the base, on the powers of the tautological line bundle. The corollary is the splitting principle in K-theory, theorem 2.5.16, the tool chapter 4 uses to reduce every statement about Adams operations to the case of a sum of line bundles. The geometric construction of the flag bundle is cited from [14, Splitting Principle]. What is proved here is the K^*-injectivity, which does not follow from the cohomological statement, since the Chern character kills torsion.

Two pieces of foundation have to be laid first, and both are dictated by the same obstacle. The induction behind the projective-bundle theorem runs over a cell structure on $P(V)$, and $P(V)$ is not a CW complex: its cells are products of cells of the base with cells of $\mathbb{C}\mathrm{P}^{n-1}$, attached in an order that need not be monotone in dimension. The first piece is therefore the notion of a *finite cell complex*, a filtration by single-cell attachments with no dimension condition, together with the fact that a projectivization over a finite cell complex is again one. The second piece is an exact-sequence engine that works on such spaces. The sequences of section 2.3 are stated for finite CW pairs because their leftward extension is cited to the Puppe sequence of [14, The Puppe Sequence and its Coexactness], which is proved there for pointed CW complexes. Rather than widen that citation, this section rebuilds the extension for arbitrary *compact pairs* out of two results this book already proved at exactly that generality, lemma 2.2.3 and lemma 2.2.4, following Atiyah's construction. The finite-CW sequences of section 2.3 are unchanged and remain in force. The compact versions below extend them and are used only where CW structures are unavailable.

Point-set preliminaries come first: the spaces built below are cones, quotients and cell attachments glued from compact pieces, and each gluing must be known to produce a compact Hausdorff space before lemma 2.2.3 may be applied to it.

Lemma 2.5.1: Attaching Preserves Compact Hausdorff

Let X and Y be compact Hausdorff spaces, $A \subseteq X$ a closed subspace, and $f\colon A \to Y$ a continuous map. The adjunction space

$$Y \cup_f X = (Y \sqcup X)\big/\,(a \sim f(a) \text{ for } a \in A)$$

is compact Hausdorff, the canonical map $Y \to Y \cup_f X$ is a closed embedding, and the canonical map $X \setminus A \to Y \cup_f X$ is an open embedding. In particular, for A nonempty and closed the quotient X/A, which is the case $Y = \{*\}$, is compact Hausdorff.

***Proof*:**
Write $Z = Y \cup_f X$ and $\pi\colon Y \sqcup X \to Z$ for the quotient map. Z is compact as the continuous image of the compact space $Y \sqcup X$. Each fiber of π is either a single point of $X \setminus A$ or a set $\{y\} \cup f^{-1}(y)$ with $y \in Y$, and both are closed in $Y \sqcup X$, so points of Z are closed. For the Hausdorff property, separate two distinct points of Z by saturated open sets in three cases. Throughout, X and Y are normal, being compact Hausdorff ([8, Compact Hausdorff Spaces Are Normal]).

Case 1: both points come from $X \setminus A$. Since X is Hausdorff and $X \setminus A$ is open, the two preimage points have disjoint open neighbourhoods contained in $X \setminus A$. These are saturated, so their images are disjoint open neighbourhoods in Z.

Case 2: one point is $\pi(x)$ with $x \in X \setminus A$, the other is $\pi(y)$ with $y \in Y$. By normality of X choose disjoint open sets $U \ni x$ and $W \supseteq A$ in X. Then U is saturated, and $Y \sqcup W$ is a saturated open set (its trace on X is $W \supseteq A$ and its trace on Y is everything) whose image contains $\pi(y)$ and misses $\pi(U)$.

Case 3: both points are $\pi(y_1), \pi(y_2)$ with $y_1 \neq y_2 \in Y$. Choose disjoint open sets $V_1' \ni y_1$, $V_2' \ni y_2$ in Y, and by normality of Y shrink them to open sets $V_i \ni y_i$ with $\overline{V_i} \subseteq V_i'$, so that $\overline{V_1} \cap \overline{V_2} = \emptyset$. The sets $C_i = f^{-1}(\overline{V_i})$ are disjoint and closed in A, hence in X. By normality of X choose disjoint open sets $W_i \supseteq C_i$ in X. Since $f^{-1}(V_i)$ is open in the subspace A, write $f^{-1}(V_i) = G_i \cap A$ with G_i open in X, and set $U_i = G_i \cap W_i$. Then $U_i \cap A = f^{-1}(V_i) \cap W_i = f^{-1}(V_i)$, because $f^{-1}(V_i) \subseteq C_i \subseteq W_i$. Consequently $V_i \sqcup U_i$ is saturated: its trace on X is $U_i = (U_i \setminus A) \cup f^{-1}(V_i)$, open, and its trace on Y is V_i, open. The images $O_i = \pi(V_i \sqcup U_i)$ are open, contain $\pi(y_i)$, and are disjoint: their Y-parts lie in the disjoint V_1, V_2, and their X-parts lie in the disjoint W_1, W_2.

Finally, $\pi|_Y$ is a continuous injection from a compact space to a Hausdorff space, hence a closed embedding, and $\pi|_{X \setminus A}$ is an injective open map onto the image of the saturated open set $X \setminus A$, hence an open embedding, completing the proof.

Definition 2.5.2: Finite Cell Complex

A *finite cell structure* on a space X is a finite filtration by closed subspaces

$$X_0 \subseteq X_1 \subseteq \cdots \subseteq X_k = X$$

in which X_0 is a finite discrete space and each X_{i+1} is obtained from X_i by attaching a single cell: there are an integer $n_i \geq 0$ and a map $\varphi_i \colon S^{n_i-1} \to X_i$ with $X_{i+1} = X_i \cup_{\varphi_i} D^{n_i}$ in the sense of lemma 2.5.1, where $S^{-1} = \emptyset$, so that attaching a 0-cell adjoins a disjoint point. The induced map $\Phi_i \colon D^{n_i} \to X_{i+1}$ is the cell's *characteristic map.* A *finite cell complex* is a space admitting a finite cell structure, and the X_i are its *stages.*
No condition relates n_i to the dimensions of earlier cells: a cell may attach to cells of lower and higher dimension alike. This is the one respect in which the notion is wider than that of a finite CW complex, and it is not decorative: the cell structures produced by proposition 2.5.5 genuinely attach cells out of dimension order, so a finite CW structure is not what that argument delivers. This is a statement about the proof, not about the conclusion: for particular X and V the total space may well admit a CW structure with the same cells (it does for $X = S^2$, $V = \underline{\mathbb{C}}^2$, where $P(V) = S^2 \times S^2$), and no claim is made here either way.

Lemma 2.5.3: Basic Properties of Finite Cell Complexes

1. Every finite cell complex is compact Hausdorff, each stage is a closed subspace, and each characteristic map Φ_i restricts to a homeomorphism of the open cell $\mathrm{int}D^{n_i}$ onto $X_{i+1} \setminus X_i$.

2. Every finite CW complex admits a finite cell structure, obtained by attaching its cells one at a time in any order of non-decreasing dimension.

3. $\mathbb{C}\mathrm{P}^n$ is a finite CW complex with exactly one cell in each of the dimensions $0, 2, \ldots, 2n$, the $2m$-cell attached to $\mathbb{C}\mathrm{P}^{m-1}$ by the quotient map $S^{2m-1} \to \mathbb{C}\mathrm{P}^{m-1}$. In particular $\mathbb{C}\mathrm{P}^n$ is a finite cell complex and $\mathbb{C}\mathrm{P}^m/\mathbb{C}\mathrm{P}^{m-1} \cong S^{2m}$.

***Proof*:**
(1) By induction on the number of cells. A finite discrete space is compact Hausdorff, and each attachment preserves compactness and the Hausdorff property by lemma 2.5.1, which also makes $X_i \to X_{i+1}$ a closed embedding and $\mathrm{int}D^{n_i} \to X_{i+1}$ an open embedding, and a composite of closed embeddings is one, so every stage is closed in X.

(2) List the cells of a finite CW complex in non-decreasing order of dimension. Each k-cell's attaching map has image in the $(k-1)$-skeleton, which is contained in the union of the earlier-listed cells, so the list is a valid attaching order. It remains to see the resulting space is the CW complex itself. For a finite complex each skeleton $X^{(k)}$ carries the quotient topology from $X^{(k-1)} \sqcup \coprod_j D^k_j$, and performing the identifications of the finitely many k-cells one at a time yields the same set with the same quotient topology, because both are quotients of the same space by the same relation and a quotient of a quotient is the total quotient. Concatenating over k gives the finite cell structure.

(3) Define $\Phi\colon D^{2m} \to \mathbb{C}\mathrm{P}^m$ by

$$\Phi(z_1, \ldots, z_m) = \left[z_1 : \cdots : z_m : \sqrt{1 - |z|^2}\,\right]$$

This is continuous, and surjective: a point of $\mathbb{C}\mathrm{P}^m$ has a representative w on the unit sphere whose last coordinate is real and non-negative, obtained by scaling any unit representative by the inverse phase of its last coordinate when that coordinate is nonzero, and lying in $\Phi(\partial D^{2m})$ when it is zero. On $\mathrm{int} D^{2m}$ the last coordinate is positive and such a representative is unique, so Φ is injective there with image the complement of $\mathbb{C}\mathrm{P}^{m-1}$, the locus where the last coordinate vanishes. On $\partial D^{2m} = S^{2m-1}$ the map Φ is the quotient map $S^{2m-1} \to \mathbb{C}\mathrm{P}^{m-1}$. Being a continuous surjection from a compact space to a Hausdorff space, $\Phi \sqcup (\mathbb{C}\mathrm{P}^{m-1} \hookrightarrow \mathbb{C}\mathrm{P}^m)$ is a closed map, hence a quotient map, and its fibers are exactly the identifications of the adjunction $\mathbb{C}\mathrm{P}^{m-1} \cup_{\Phi|_{S^{2m-1}}} D^{2m}$, so $\mathbb{C}\mathrm{P}^m$ is that adjunction. Induction on m builds the CW structure, the attaching maps landing in the full lower complex, which is the $(2m-2)$-skeleton. Collapsing $\mathbb{C}\mathrm{P}^{m-1}$ collapses the boundary sphere of the one top cell, so $\mathbb{C}\mathrm{P}^m / \mathbb{C}\mathrm{P}^{m-1} \cong D^{2m}/S^{2m-1} \cong S^{2m}$, and (2) makes $\mathbb{C}\mathrm{P}^n$ a finite cell complex, completing the proof.

Lemma 2.5.4: Compact Convex Bodies Are Disks

Let $K \subseteq \mathbb{R}^N$ be compact and convex with 0 in its interior. Then the map $\varphi\colon K \to D^N$ given by $\varphi(0) = 0$ and $\varphi(x) = \frac{g(x)}{|x|}\, x$ for $x \neq 0$, where $g(x) = \inf\{t > 0 \big| x \in tK\}$ is the Minkowski functional of K, is a homeomorphism. In particular $D^a \times D^b \cong D^{a+b}$ for all $a, b \geq 0$, by a homeomorphism carrying $(S^{a-1} \times D^b) \cup (D^a \times S^{b-1})$ onto S^{a+b-1}.

***Proof*:**
Choose $r, R > 0$ with $rD^N \subseteq K \subseteq RD^N$. Then $|x|/R \leq g(x) \leq |x|/r$ for all x. The functional g is positively homogeneous and, by convexity of K, subadditive, so $|g(x) - g(y)| \leq \max\{g(x-y), g(y-x)\} \leq |x-y|/r$ and g is continuous. Since K is closed, $K = \{x \big| g(x) \leq 1\}$. The map φ is continuous away from 0 and at 0 because $|\varphi(x)| = g(x) \leq |x|/r$. It preserves each ray from the origin and on the ray through a unit vector u sends the point su to $s\,g(u)\,u$, which is strictly increasing in s, so φ is injective. It is surjective because for $0 \neq v \in D^N$ the point $x = \frac{|v|}{g(v)} v$ satisfies $g(x) = |v| \leq 1$ and $\varphi(x) = v$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Since φ carries $\partial K = \{g = 1\}$ onto S^{N-1}, the final claim follows by taking $K = D^a \times D^b \subseteq \mathbb{R}^{a+b}$, whose boundary is $(S^{a-1} \times D^b) \cup (D^a \times S^{b-1})$, completing the proof.

The next proposition is the specialization to projectivizations of a general principle of Hatcher (*Vector Bundles and K-Theory* §2.3, Appendix, Proposition 2.28): a fiber bundle whose base and fiber are finite cell complexes is one too, with cells the products of cells. The general statement needs the triviality of an arbitrary fiber bundle over a disk, which this book has not proved. For the projectivization of a vector bundle that input is theorem 1.2.5, and projectivizations are the only case the splitting principle uses, so the proposition is stated for them.

Proposition 2.5.5: Cell Structure on a Projective Bundle

Let X be a finite cell complex and $V \to X$ a rank-n complex vector bundle, $n \geq 1$, with projectivization $\pi\colon P(V) \to X$, the fiber bundle with fiber $\mathbb{C}\mathrm{P}^{n-1}$ of [14, Projective Bundles over a Paracompact Base]. Then $P(V)$ is a finite cell complex, with one cell of dimension $m + 2j$ for each m-cell of X and each $j \in \{0, 1, \ldots, n-1\}$.

***Proof*:**
The cited construction applies since a finite cell complex is compact Hausdorff (lemma 2.5.3) and a compact Hausdorff space is paracompact ([8, Compact Spaces, Discrete Spaces, and Euclidean Space](1)), and it exhibits $P(V)$ as a fiber bundle over X with fiber $\mathbb{C}\mathrm{P}^{n-1}$. Its total space is Hausdorff by [8, Locally Trivial Maps with Compact Fiber Preserve Paracompactness], whose last clause gives exactly that from X and $\mathbb{C}\mathrm{P}^{n-1}$ Hausdorff with $\mathbb{C}\mathrm{P}^{n-1}$ compact. The projective-bundle proposition itself says nothing about the total space. The proof is an induction on the number of cells of X.

Step 1: Base of the induction. If $X = X_0$ is finite discrete, a bundle over it is a finite disjoint union of vector spaces and $P(V)$ is a finite disjoint union of copies of $\mathbb{C}\mathrm{P}^{n-1}$, which is a finite cell complex: each copy is one by lemma 2.5.3(3), and a disjoint union of finite cell complexes is one, by taking the union of the discrete bottom stages and then attaching all cells of both.

Step 2: Pulling back over the new cell. Let $X = X' \cup_\varphi e^m$ with characteristic map $\Phi\colon D^m \to X$, and set $E = P(V)$, $E' = \pi^{-1}(X')$, which is the projectivization of $V|_{X'}$ and by induction a finite cell complex. For $m = 0$ the cell is a disjoint point, E is the disjoint union of E' with one copy of $\mathbb{C}\mathrm{P}^{n-1}$, and the argument of Step 1 applies, so assume $m \geq 1$. The pullback $\Phi^*V \to D^m$ is trivial: D^m is paracompact and contractible, so theorem 1.2.5 applied to the identity and a constant map makes Φ^*V isomorphic to the pullback of one fiber. A trivialization $\Phi^*V \cong D^m \times \mathbb{C}^n$ is fiberwise linear, so it carries lines to lines and descends to a homeomorphism $P(\Phi^*V) \cong D^m \times \mathbb{C}\mathrm{P}^{n-1}$ over D^m. Moreover $P(\Phi^*V)$ is canonically the pullback of $P(V)$: both have as points the pairs (d, ℓ) with $d \in D^m$ and ℓ a line in $V_{\Phi(d)}$. Write $g\colon D^m \times \mathbb{C}\mathrm{P}^{n-1} \to E$ for the composite of the trivialization with the bundle map $P(\Phi^*V) \to P(V)$ over Φ.

Step 3: E is an adjunction. The map

$$E' \sqcup \left(D^m \times \mathbb{C}\mathrm{P}^{n-1}\right) \longrightarrow E$$

given by the inclusion and by g is a continuous surjection from a compact space to a Hausdorff space, hence closed, hence a quotient map. Over $X \setminus X' = \Phi(\mathrm{int}D^m)$ it is injective, since Φ is injective on $\mathrm{int}D^m$ and g is a homeomorphism on each fiber. Over X' its identifications are exactly $(d, \ell) \sim g(d, \ell) \in E'$ for $d \in S^{m-1}$. Therefore $E \cong E' \cup_\alpha \left(D^m \times \mathbb{C}\mathrm{P}^{n-1}\right)$ with $\alpha = g|_{S^{m-1} \times \mathbb{C}\mathrm{P}^{n-1}}\colon S^{m-1} \times \mathbb{C}\mathrm{P}^{n-1} \to E'$.

Step 4: The product pair is built by cell attachments. Let F be a finite cell complex with stages $F_0 \subseteq \cdots \subseteq F_k = F$, where $F = \mathbb{C}\mathrm{P}^{n-1}$. Set $Z_{-1} = S^{m-1} \times F$ and $Z_i = (S^{m-1} \times F) \cup (D^m \times F_i)$ for $0 \leq i \leq k$, closed subspaces of $D^m \times F$. First, Z_0 is obtained from Z_{-1} by attaching one m-cell $D^m \times \{v\}$ for each point v of the finite discrete F_0, along the inclusion $S^{m-1} \times \{v\} \hookrightarrow S^{m-1} \times F$. Next, suppose $F_{i+1} = F_i \cup_{\varphi_F} D^d$ with characteristic map $\Phi_F\colon D^d \to F_{i+1}$, and consider

$$\Psi\colon D^m \times D^d \longrightarrow D^m \times F_{i+1} \subseteq D^m \times F, \qquad (u, w) \longmapsto (u, \Phi_F(w))$$

By lemma 2.5.4, $D^m \times D^d$ is an $(m+d)$-disk whose boundary sphere is $(S^{m-1} \times D^d) \cup (D^m \times S^{d-1})$, and Ψ carries that boundary into Z_i: the first piece into $S^{m-1} \times F$ and the second into $D^m \times F_i$.

The set difference is

$$Z_{i+1} \setminus Z_i = (\mathrm{int} D^m) \times (F_{i+1} \setminus F_i)$$

and Ψ restricts to a homeomorphism of $\mathrm{int} D^m \times \mathrm{int} D^d$ onto it, by lemma 2.5.3(1) applied to F. The induced map $Z_i \sqcup (D^m \times D^d) \to Z_{i+1}$ is a continuous surjection from a compact space to a Hausdorff space, hence a quotient map, and its identifications are exactly those of attaching the $(m+d)$-cell along $\Psi|_\partial$. So each Z_{i+1} is obtained from Z_i by attaching one cell, and $D^m \times F$ is obtained from $S^{m-1} \times F$ by finitely many cell attachments, one $(m + \dim)$-cell per cell of F, counting the points of F_0 as 0-cells.

Step 5: Gluing the attachments along α. Attaching cells commutes with gluing: with $W_i = E' \cup_\alpha Z_i$, each W_{i+1} is obtained from W_i by attaching the same cell along the composite of its attaching map with $Z_i \to W_i$, since $W_i \sqcup D^{(\cdot)} \to W_{i+1}$ is again a closed surjection with exactly those identifications, all spaces being compact Hausdorff by lemma 2.5.1. Starting from $W_{-1} = E'$ and running through Step 4's attachments produces E with the finite cell structure claimed. The count of cells is one cell of dimension $m + 2j$ per m-cell of X and per $2j$-cell of $\mathbb{C}\mathrm{P}^{n-1}$, and lemma 2.5.3(3) lists the latter, completing the proof.

Definition 2.5.6: Tautological Line Bundle on a Projective Bundle

Let $V \to X$ be a rank-n complex vector bundle over a compact Hausdorff space, $\pi\colon P(V) \to X$ its projectivization. The *tautological line bundle* is

$$L_V = \{(\ell, v) \in P(V) \times V \big| v \in \ell\}$$

with projection $(\ell, v) \mapsto \ell$, a rank-1 subbundle of π^*V. Over a trivializing open set U for V, the identification $\pi^{-1}(U) \cong U \times \mathbb{C}\mathrm{P}^{n-1}$ of [14, Projective Bundles over a Paracompact Base] carries L_V to the pullback of the tautological line bundle $H \to \mathbb{C}\mathrm{P}^{n-1}$ of section 1.2, which is locally trivial, hence L_V is locally trivial. On each fiber $P(V_x)$, the bundle L_V restricts to the tautological line bundle of the projective space $P(V_x)$, and any linear isomorphism $V_x \cong \mathbb{C}^n$ carries it to $H \to \mathbb{C}\mathrm{P}^{n-1}$.

The exact-sequence engine comes next. The input is a *pointed compact pair*: a compact Hausdorff space P with basepoint, and a closed subspace $Q \subseteq P$ containing the basepoint. Reduced cones and suspensions are those of section 2.2: $CQ = (Q \times [0,1])/(Q \times \{1\} \cup \{+\} \times [0,1])$, glued to P along $Q \times \{0\} = Q$, and $\Sigma Q = CQ/Q$, with all cone points and basepoints identified to the single basepoint $+$. By lemma 2.5.1 every space formed below is compact Hausdorff.

Lemma 2.5.7: The Five-Term Exact Sequence of a Compact Pair

Let (P, Q) be a pointed compact pair, with inclusion $\iota\colon Q \hookrightarrow P$ and collapse $q\colon P \to P/Q$. There is a homomorphism

$$\delta = \delta_{(P,Q)}\colon \widetilde{K}^0(\Sigma Q) \longrightarrow \widetilde{K}^0(P/Q)$$

natural in the pair, such that the five-term sequence

$$\widetilde{K}^0(\Sigma P) \xrightarrow{(\Sigma\iota)^*} \widetilde{K}^0(\Sigma Q) \xrightarrow{\delta} \widetilde{K}^0(P/Q) \xrightarrow{q^*} \widetilde{K}^0(P) \xrightarrow{\iota^*} \widetilde{K}^0(Q)$$

is exact at $\widetilde{K}^0(\Sigma Q)$, at $\widetilde{K}^0(P/Q)$, and at $\widetilde{K}^0(P)$. No CW hypothesis is imposed, and the proof does not use the Puppe sequence.

***Proof*:**
Step 1: The two cone spaces. Set $W_1 = P \cup_Q CQ$ and $W_2 = W_1 \cup_P CP$, compact Hausdorff by lemma 2.5.1, with all cone points equal to the basepoint $+$, since the reduced cone collapses the segment over the basepoint. Both cones are closed, contractible subspaces, the contraction being the straight-line slide to the cone point. Collapsing gives, by continuous bijections from compact spaces to Hausdorff spaces, the identifications

$$W_1/CQ = P/Q, \qquad W_1/P = CQ/Q = \Sigma Q, \qquad W_2/CP = W_1/P = \Sigma Q, \qquad W_2/W_1 = CP/P = \Sigma P$$

where the suspension coordinate of ΣQ is the cone coordinate t of CQ and that of ΣP is the cone coordinate s of CP. Write $k\colon W_1 \to P/Q$, $c_1\colon W_1 \to \Sigma Q$, $k_2\colon W_2 \to \Sigma Q$, $c_2\colon W_2 \to \Sigma P$ for the four collapse maps. By lemma 2.2.4, $k^*\colon \widetilde{K}^0(P/Q) \to \widetilde{K}^0(W_1)$ and $k_2^*\colon \widetilde{K}^0(\Sigma Q) \to \widetilde{K}^0(W_2)$ are isomorphisms.

Step 2: Exactness at $\widetilde{K}^0(P)$. This is lemma 2.2.3 applied to the pair (P, Q).

Step 3: The connecting map, and exactness at $\widetilde{K}^0(P/Q)$. Apply lemma 2.2.3 to the pair (W_1, P): with $W_1/P = \Sigma Q$ this gives exactness of

$$\widetilde{K}^0(\Sigma Q) \xrightarrow{c_1^*} \widetilde{K}^0(W_1) \xrightarrow{\mathrm{res}^*} \widetilde{K}^0(P)$$

Define $\delta = (k^*)^{-1} \circ c_1^*$. The composite of the inclusion $P \hookrightarrow W_1$ with k is q, since k collapses CQ and its trace on P collapses Q, hence $\mathrm{res}^* \circ k^* = q^*$, and transporting the exact sequence along the isomorphism k^* yields $\mathrm{im}\delta = \ker q^*$.

Step 4: Exactness at $\widetilde{K}^0(\Sigma Q)$, with a transported map. Apply lemma 2.2.3 to the pair (W_2, W_1): with $W_2/W_1 = \Sigma P$ this gives exactness of

$$\widetilde{K}^0(\Sigma P) \xrightarrow{c_2^*} \widetilde{K}^0(W_2) \xrightarrow{\mathrm{res}_2^*} \widetilde{K}^0(W_1)$$

The composite of the inclusion $W_1 \hookrightarrow W_2$ with k_2 is c_1, since k_2 collapses CP and its trace on W_1 collapses P. hence $\mathrm{res}_2^* \circ k_2^* = c_1^*$. Transporting along k_2^* and then along k^* yields the exact sequence

$$\widetilde{K}^0(\Sigma P) \xrightarrow{T} \widetilde{K}^0(\Sigma Q) \xrightarrow{\delta} \widetilde{K}^0(P/Q), \qquad T = (k_2^*)^{-1} \circ c_2^*$$

so $\mathrm{im}T = \ker \delta$, and it remains to identify $\mathrm{im}T$ with $\mathrm{im}(\Sigma\iota)^*$.

Step 5: The pinch homotopy. Let $\mathrm{fl}\colon \Sigma Q \to \Sigma Q$ be the flip of the suspension coordinate, $[y,t] \mapsto [y, 1-t]$, a pointed homeomorphism with $\mathrm{fl} \circ \mathrm{fl} = \mathrm{id}$. The claim is

$$c_2 \simeq (\Sigma\iota) \circ \mathrm{fl} \circ k_2 \colon W_2 \longrightarrow \Sigma P$$

as pointed maps. Both sides are described on the two cones, which cover W_2 and meet in Q: writing $[x,s]$ for the class in $\Sigma P = CP/P$ of $(x,s) \in P \times [0,1]$, the map c_2 sends $(x,s) \in CP$ to $[x,s]$ and all of $W_1 = P \cup CQ$ to the basepoint, while $(\Sigma\iota) \circ \mathrm{fl} \circ k_2$ sends $(y,t) \in CQ$ to $[\iota(y), 1-t]$ and all of $CP \supseteq P$ to the basepoint. Define $H\colon W_2 \times [0,1] \to \Sigma P$ by

$$H\big((x,s),\tau\big) = \big[x,\ s + \tau(1-s)\big] \quad \text{on } CP, \qquad H\big((y,t),\tau\big) = \big[\iota(y),\ \tau(1-t)\big] \quad \text{on } CQ$$

At $\tau = 0$ this is c_2, the second formula landing at $[\,\cdot\,,0]$, which is the basepoint. At $\tau = 1$ it is $(\Sigma\iota) \circ \mathrm{fl} \circ k_2$, the first formula landing at $[\,\cdot\,,1]$, the basepoint. The two formulas agree where the cones meet: a point of $Q = CQ \cap P$ is $(y,0)$ in the cone coordinates of CQ and $(\iota(y),0)$ in those of CP, and both formulas send it to $[\iota(y),\tau]$. The formulas respect every identification made in forming W_2: cone points and basepoint segments land in $\{[x,1]\} \cup \{[+,\cdot]\} = \{+\}$ at every τ. Hence H descends from $(P \sqcup Q \times I \sqcup P \times I) \times [0,1]$. The descent is continuous because the product of the defining quotient map with $\mathrm{id}_{[0,1]}$ is a quotient map, $[0,1]$ being locally compact ([8, Products of Quotient Maps]). So H is a pointed homotopy and the claim holds.

Step 6: Conclusion. By homotopy invariance of $\widetilde{K}^0$ (Axiom 1 of theorem 2.3.6, restricted to the kernels of basepoint restriction along pointed maps), $c_2^* = k_2^* \circ \mathrm{fl}^* \circ (\Sigma\iota)^*$, so $T = \mathrm{fl}^* \circ (\Sigma\iota)^*$. The flip commutes with suspended maps, $\mathrm{fl}_{\Sigma P} \circ \Sigma\iota = \Sigma\iota \circ \mathrm{fl}_{\Sigma Q}$ as maps $\Sigma Q \to \Sigma P$, both sending $[y,t]$ to $[\iota(y), 1-t]$, hence

$$T = (\Sigma\iota)^* \circ \mathrm{fl}_{\Sigma P}^*$$

and since $\mathrm{fl}_{\Sigma P}^*$ is an automorphism of $\widetilde{K}^0(\Sigma P)$, being induced by a homeomorphism, $\mathrm{im} T = \mathrm{im}(\Sigma\iota)^*$. With Step 4, $\ker\delta = \mathrm{im}(\Sigma\iota)^*$, which is the remaining exactness.

Step 7: Naturality. A map of pointed compact pairs $f\colon (P,Q) \to (P',Q')$ induces maps of all four cone spaces commuting strictly with the collapses k, c_1, k_2, c_2 and with the inclusions, hence a commuting ladder after applying $\widetilde{K}^0$. The isomorphisms k^*, k_2^* commute with the ladder, so δ and the whole five-term sequence are natural in the pair, completing the proof.

The first dividend is the smash identification of proposition 2.2.5 without its CW hypothesis: the one step of that proof which cited the Puppe sequence is replaced by lemma 2.5.7.

Proposition 2.5.8: The Smash Identification for Compact Pairs

1. Let X and Y be compact Hausdorff pointed spaces, $r\colon X \vee Y \hookrightarrow X \times Y$ the inclusion and $q\colon X \times Y \to X \wedge Y$ the collapse. Then $q^*\colon \widetilde{K}^0(X \wedge Y) \to \widetilde{K}^0(X \times Y)$ is injective, with image exactly $\ker r^*$.

2. For compact Hausdorff pointed X, Y, Z, the pullback along the collapse $X \wedge (Y \times Z) \to X \wedge Y \wedge Z$ is injective; consequently the pullback along the total collapse $X \times Y \times Z \to X \wedge Y \wedge Z$ is injective.

***Proof*:**
(1) The image is $\ker r^*$ by lemma 2.2.3 applied to the pair $(X \times Y, X \vee Y)$, which is a pointed compact pair by lemma 2.5.1. For injectivity, apply lemma 2.5.7 to that pair: $\ker q^* = \mathrm{im}\delta$ and $\ker \delta = \mathrm{im}(\Sigma\iota)^*$, so it suffices to show $(\Sigma\iota)^* \colon \widetilde{K}^0(\Sigma(X \times Y)) \to \widetilde{K}^0(\Sigma(X \vee Y))$ is surjective, for then $\delta = 0$ and $\ker q^* = 0$. Suspension carries the wedge to a wedge, $\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$, and by lemma 2.2.2 a class there is a pair (a, b) of reduced classes on the summands. The class $(\Sigma p_X)^* a + (\Sigma p_Y)^* b$ on $\Sigma(X \times Y)$ restricts to (a, b): on the ΣX summand the projection p_X composed with the inclusion is the identity while p_Y composed with it is constant, whose suspension pulls reduced classes to zero, and symmetrically on the other summand. Hence $(\Sigma\iota)^*$ is surjective.

(2) Apply lemma 2.5.7 to the pointed compact pair $\big(X \wedge (Y \times Z),\ X \wedge (Y \vee Z)\big)$, whose quotient is $X \wedge Y \wedge Z$: collapsing $X \wedge (Y \vee Z)$ in $X \wedge (Y \times Z)$ collapses, fiberwise in X, the wedge inside the product, and the resulting continuous bijection of compact Hausdorff spaces is a homeomorphism. As in (1) it suffices that restriction

$$\widetilde{K}^0\big(\Sigma(X \wedge (Y \times Z))\big) \longrightarrow \widetilde{K}^0\big(\Sigma(X \wedge (Y \vee Z))\big)$$

be surjective. Smashing distributes over the wedge, $X \wedge (Y \vee Z) = (X \wedge Y) \vee (X \wedge Z)$, again by a continuous bijection of compact Hausdorff spaces, and the two maps $\mathrm{id}_X \wedge p_Y$ and $\mathrm{id}_X \wedge p_Z$ out of $X \wedge (Y \times Z)$ restrict on the two wedge summands to the identity and a constant exactly as in (1), so the same splitting argument applies. For the consequence, factor the total collapse as

$$X \times Y \times Z \xrightarrow{q_{X, Y \times Z}} X \wedge (Y \times Z) \longrightarrow X \wedge Y \wedge Z$$

and compose the injectivity of (1), applied to the pair $(X, Y \times Z)$, with the injectivity just proved, completing the proof.

Corollary 2.5.9: The Reduced Product and Bott Map on Compact Spaces

Let X and Y be compact Hausdorff pointed spaces. The external product of two reduced classes restricts to zero on $X \vee Y$, hence is q^* of a unique class on $X \wedge Y$ by proposition 2.5.8. The assignment

$$\widetilde{\mu} \colon \widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \longrightarrow \widetilde{K}^0(X \wedge Y)$$

so defined extends the reduced external product of definition 2.2.6 to all compact Hausdorff pointed spaces, is natural in both variables, and satisfies $q^*\widetilde{\mu}(a \otimes b) = p_X^* a \cdot p_Y^* b$. In particular the Bott map $\beta_X \colon \widetilde{K}^0(X) \to \widetilde{K}^0(\Sigma^2 X)$, $\alpha \mapsto \widetilde{\mu}(\beta \otimes \alpha)$, of definition 2.2.7 is defined for every compact Hausdorff pointed X and is natural in pointed maps.

Moreover $\widetilde{\mu}$ is *associative*, in the sense that the two composites

$$\widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \otimes \widetilde{K}^0(Z) \longrightarrow \widetilde{K}^0(X \wedge Y \wedge Z)$$

built by bracketing to the left and to the right agree under the canonical associativity homeomorphism, and *unital*: $\widetilde{\mu}(1_{S^0} \otimes -)$ is the identity under the canonical identification $S^0 \wedge W \cong W$, where 1_{S^0} generates $\widetilde{K}^0(S^0) \cong \mathbb{Z}$.

***Proof*:**
Existence and uniqueness are lemma 2.2.3 and proposition 2.5.8(1), exactly as in definition 2.2.6, whose construction used its CW hypothesis only through the injectivity clause of proposition 2.2.5. Naturality: for pointed maps f, g the defining property gives $q^*(f \wedge g)^* \widetilde{\mu}(a \otimes b) = (f \times g)^* q^* \widetilde{\mu}(a \otimes b) = (f \times g)^*(p^*a \cdot p^*b) = p^*f^*a \cdot p^*g^*b$, and injectivity of q^* identifies $(f \wedge g)^* \widetilde{\mu}(a \otimes b)$ with $\widetilde{\mu}(f^*a \otimes g^*b)$.

For associativity, let $R\colon X \times Y \times Z \to X \wedge Y \wedge Z$ be the triple collapse, and note first that the two bracketings land in the same space: $(X \wedge Y) \wedge Z$ and $X \wedge (Y \wedge Z)$ are both the quotient of $X \times Y \times Z$ by the fat wedge, so the evident continuous bijection between them is a homeomorphism, being a continuous bijection of compact Hausdorff spaces, and it carries one bracketed collapse to the other. Now R factors as a composite of two double collapses in either bracketing, so R^* is injective by proposition 2.5.8(2), and both bracketed composites satisfy

$$R^* \widetilde{\mu}\big(\widetilde{\mu}(a \otimes b) \otimes c\big) = p_X^* a \cdot p_Y^* b \cdot p_Z^* c = R^* \widetilde{\mu}\big(a \otimes \widetilde{\mu}(b \otimes c)\big)$$

by two applications of the defining property, the middle expression being unambiguous because $K^0(X \times Y \times Z)$ is a commutative ring (proposition 2.1.10, which holds over any compact Hausdorff space) and products in a ring are associative. Injectivity of R^* gives the two classes equal.

For the unit, write $S^0 = \{*, 1\}$ with $*$ the basepoint and let $1_{S^0} = [\underline{\mathbb{C}}_-] \in \widetilde{K}^0(S^0)$ be the class of the line bundle supported on the non-basepoint, which is the generator fixed by the convention of example 2.4.1. *Which* generator is used matters: the clause is false for -1_{S^0}, and proposition 2.5.13 below needs $c^*1_{S^0} = 1$ rather than -1. The collapse $q\colon S^0 \times W \to S^0 \wedge W$ restricts on $\{1\} \times W$ to the canonical identification $S^0 \wedge W \cong W$. Writing $\iota\colon W \cong \{1\} \times W \hookrightarrow S^0 \times W$ for that inclusion and evaluating the defining property directly,

$$\iota^* q^* \widetilde{\mu}(1_{S^0} \otimes w) = \iota^*\left(p_{S^0}^* 1_{S^0} \cdot p_W^* w\right) = 1 \cdot w = w$$

since $p_{S^0} \circ \iota$ is constant at the non-basepoint. This is a direct evaluation, not a cancellation, so no injectivity is used, and $\widetilde{\mu}(1_{S^0} \otimes -)$ is the identity, completing the proof.

Corollary 2.5.10: Bott Periodicity for Compact Pointed Spaces

For every compact Hausdorff pointed space X the Bott map

$$\beta_X\colon \widetilde{K}^0(X) \longrightarrow \widetilde{K}^0(\Sigma^2 X)$$

is an isomorphism, naturally in X. Consequently $\widetilde{K}^{-n}(X)$ of definition 2.3.1 depends only on the parity of n, and the periodic grading of box 2.3.5 extends verbatim to compact Hausdorff pointed spaces and compact pairs.

***Proof*:**
Step 4 of the proof of theorem 2.2.15 derives the reduced clause from the unreduced ring clause, which holds for every compact Hausdorff X, in three moves: the basepoint splittings $K^0 = \widetilde{K}^0 \oplus \mathbb{Z}$ on X and on S^2, which use only the basepoint inclusion and collapse. The identification of $\ker\big(\widetilde{K}^0(X \times S^2) \to \widetilde{K}^0(X \vee S^2)\big)$ with $\widetilde{K}^0(X) \cdot \beta$, a computation inside $K^0(X \times S^2)$. The

identification of that kernel with $q^*\widetilde{K}^0(X \wedge S^2) = q^*\widetilde{K}^0(\Sigma^2 X)$ with q^* injective. Only the last move used the CW hypothesis, through proposition 2.2.5. proposition 2.5.8(1) gives it for compact Hausdorff pointed X, and the argument is otherwise unchanged. Naturality is corollary 2.5.9. The parity statement and the extension of the grading follow by iterating β, as in box 2.3.5, completing the proof.

Theorem 2.5.11: The Long Exact Sequence of a Compact Pair

Let (X, A) be a compact pair (definition 2.3.3), A nonempty, with inclusions $i\colon A \hookrightarrow X$ and $j\colon (X, \emptyset) \hookrightarrow (X, A)$. There are natural homomorphisms $\delta_n\colon K^{-n-1}(A) \to K^{-n}(X, A)$, $n \geq 0$, such that the sequence, infinite to the left,

$$\cdots \to K^{-1}(X, A) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(A) \xrightarrow{\delta_0} K^0(X, A) \xrightarrow{j^*} K^0(X) \xrightarrow{i^*} K^0(A)$$

is exact. Setting $K^1 = K^{-1}$ and $\delta^1 = \delta_1 \circ \beta_A\colon K^0(A) \to K^1(X, A)$, where $\beta_A\colon K^0(A) \to K^{-2}(A)$ is the periodicity isomorphism of corollary 2.5.10, the sequence folds into the exact six-term sequence

$$\begin{tikzcd} K^0(X,A) \arrow[r, "j^*"] & K^0(X) \arrow[r, "i^*"] & K^0(A) \arrow[d, "\delta^1"] \\ K^1(A) \arrow[u, "\delta_0"] & K^1(X) \arrow[l, "i^*"] & K^1(X,A) \arrow[l, "j^*"] \end{tikzcd}$$

Both sequences are natural in the pair. No CW hypothesis is imposed, on finite CW pairs, theorem 2.3.7 remains available and is the sequence used elsewhere in this book.

***Proof*:**
Step 1: The tower of pointed pairs. For $m \geq 0$ let $(P_m, Q_m) = (\Sigma^m(X_+), \Sigma^m(A_+))$, a pointed compact pair: $\Sigma^m(A_+)$ is the image of the compact $S^m \times A_+$-cylinder inside $\Sigma^m(X_+)$, hence closed, and contains the basepoint. The quotient satisfies

$$P_m/Q_m = \Sigma^m(X_+)/\Sigma^m(A_+) \cong \Sigma^m(X_+/A_+) = \Sigma^m(X/A)$$

naturally, the middle map being a continuous bijection of compact Hausdorff spaces induced by collapsing in either order, and $X_+/A_+ = X/A$ because adjoined basepoint and collapsed subspace merge. So the groups of definitions 2.3.1 and 2.3.3 appear as

$$\widetilde{K}^0(P_m) = K^{-m}(X), \qquad \widetilde{K}^0(Q_m) = K^{-m}(A), \qquad \widetilde{K}^0(P_m/Q_m) = K^{-m}(X, A)$$

Step 2: Splicing the five-term windows. Apply lemma 2.5.7 to (P_m, Q_m) for each m, and define δ_m to be its connecting map, transported along the identifications of Step 1. The window for (P_m, Q_m) reads

$$K^{-m-1}(X) \xrightarrow{(\Sigma^{m+1}i)^*} K^{-m-1}(A) \xrightarrow{\delta_m} K^{-m}(X, A) \xrightarrow{j^*} K^{-m}(X) \xrightarrow{i^*} K^{-m}(A)$$

using $\Sigma P_m = P_{m+1}$ and $\Sigma\iota_m = \iota_{m+1}$ under the natural identification $\Sigma\Sigma^m = \Sigma^{m+1}$. Consecutive windows overlap in the two terms $K^{-m-1}(X)$ and $K^{-m-1}(A)$, and the maps in the overlap agree: the first map of window m is $(\Sigma^{m+1}i)^*$, which is the fourth map of window $m + 1$. Exactness of the spliced sequence at each spot is exactness of one of the windows there: at $K^{-m}(X)$ and $K^{-m}(X,A)$ by window m, and at $K^{-m-1}(A)$ by window m again, whose exactness at its second term was proved in lemma 2.5.7. Naturality is Step 7 there, applied levelwise.

Step 3: Folding. All terms in degrees $-m$ and $-m-2$ are identified by the natural isomorphisms of corollary 2.5.10. The hexagon's maps are those of the sequence in degrees 0 and -1, except for $\delta^1 = \delta_1 \circ \beta_A$. Exactness at the six corners: at $K^0(X)$, $K^0(X,A)$, $K^1(A)$, $K^1(X)$ it is the exactness of Step 2 read off directly. At $K^1(X,A) = K^{-1}(X,A)$, $\mathrm{im}\delta^1 = \mathrm{im}\delta_1$ since β_A is an isomorphism, and $\mathrm{im}\delta_1 = \ker j^*$ by Step 2. At $K^0(A)$, $\ker \delta^1 = \beta_A^{-1}(\ker \delta_1) = \beta_A^{-1}(\mathrm{im}\, i^*_{-2})$, and naturality of β in the map i gives $\beta_A^{-1} \circ i^*_{-2} = i^*_0 \circ \beta_X^{-1}$, so $\ker \delta^1 = \mathrm{im} i^*_0$, completing the proof.

2.5.12 Note: Two Connecting Maps, Not Compared A finite CW pair is in particular a compact pair, so theorem 2.3.7 and theorem 2.5.11 both apply to it, and each gives a connecting map. This book does not compare them, and no result in it depends on their agreeing: section 2.5 uses the compact sequence throughout, and every other chapter uses the finite CW one. A future result that runs one of them into a statement proved with the other owes the comparison first.

For the induction to come, the sequence has to be a sequence of modules. Only multiplication by classes of degree zero is ever used, which keeps the bookkeeping free of signs. The module structure is defined through a single device: every space in the tower of theorem 2.5.11 for the pair (X, A) carries a canonical *marking*, a pointed map to $X_+\wedge$ itself remembering the X-coordinate, and multiplication is smash product with a fixed class followed by pullback along the marking.

Proposition 2.5.13: Module Structure on the Compact-Pair Sequence

Let (X, A) be a compact pair, A nonempty, and $\gamma \in K^0(X) = \widetilde{K}^0(X_+)$. For a pointed compact space W equipped with a pointed map $\lambda\colon W \to X_+ \wedge W$ (a *marking*), set

$$\gamma \cdot_\lambda w = \lambda^* \,\widetilde{\mu}(\gamma \otimes w) \qquad (w \in \widetilde{K}^0(W))$$

with $\widetilde{\mu}$ as in corollary 2.5.9. Equip each term of the sequence of theorem 2.5.11 with the marking induced by its X-coordinate: on $\Sigma^m(X_+)$ and on $\Sigma^m(X/A)$ the map sending the class of (s, x) to $x \wedge [(s, x)]$, and on $\Sigma^m(A_+)$ the map sending the class of (s, a) to $i(a) \wedge [(s, a)]$. Then:

1. each $\gamma \cdot_\lambda (-)$ is a homomorphism, and on the absolute groups $K^{-m}(X)$ and $K^{-m}(A)$ it agrees with the ring-theoretic action: under the injection $K^{-m}(X) \hookrightarrow K^0(S^m \times X)$ of Step 1 below, it is multiplication by the pullback of γ along the projection to X, and the analogous statement holds over A with $i^*\gamma$;
2. every map of the long exact sequence and of the six-term sequence of theorem 2.5.11, including δ_m, δ^1, and the periodicity isomorphisms used in the folding, commutes with the action of γ;
3. the actions make $\bigoplus_m K^{-m}(X, A)$, $\bigoplus_m K^{-m}(X)$ and $\bigoplus_m K^{-m}(A)$ modules over the ring $K^0(X)$, unital and associative;
4. the action is natural: for a map of compact pairs $f\colon (Y, B) \to (X, A)$ and any term w of the sequence of (X, A), the induced map sends $\gamma \cdot w$ to $f^*\gamma \cdot f^*w$.

Clauses (1), (3) and (4), restricted to the *absolute* terms, hold for every compact Hausdorff X with no pair structure and no nonemptiness hypothesis: their proofs below use only the marking on $\Sigma^m(X_+)$ and a map $f\colon Y \to X$. Only the clauses naming A or δ need the pair.

***Proof*:**
Step 1: The absolute groups sit in rings. Fix m and realize $K^{-m}(X)$ inside the K^0 of an actual compact space, via the collapse

$$\rho\colon (S^m \times X) \big/ (\{s_0\} \times X) \;\cong\; \Sigma^m(X_+)$$

is a homeomorphism, both sides having the points (s, x) with $s \neq s_0$ together with a basepoint, by a continuous bijection of compact Hausdorff spaces. The pair $\big(S^m \times X, \{s_0\} \times X\big)$ admits the retraction $(s, x) \mapsto (s_0, x)$, so in its five-term sequence (lemma 2.5.7) the map $(\Sigma\iota)^*$ is split surjective and hence $\delta = 0$ and the collapse pullback

$$\widetilde{K}^0\big(\Sigma^m(X_+)\big) \hookrightarrow K^0(S^m \times X)$$

is injective, with image the ideal of classes restricting to zero on $\{s_0\} \times X$ by lemma 2.2.3. Under this injection, the marking action of γ is multiplication by $\pi_X^*\gamma$: writing $Q\colon S^m \times X \to \Sigma^m(X_+)$ for the collapse and λ for the marking, the composite $\lambda \circ Q$ factors as

$$S^m \times X \xrightarrow{(\pi_X, Q)} X_+ \times \Sigma^m(X_+) \xrightarrow{q} X_+ \wedge \Sigma^m(X_+)$$

so $Q^*(\gamma\cdot_\lambda w) = (\pi_X, Q)^* q^*\widetilde{\mu}(\gamma\otimes w) = (\pi_X, Q)^*\big(p_1^*\gamma\cdot p_2^* w\big) = \pi_X^*\gamma\cdot Q^* w$, by the defining property of $\widetilde{\mu}$ in corollary 2.5.9. This proves (1), the A-case being identical.

Clause (3) is proved directly from the marking, uniformly on absolute and relative terms, with no injectivity used anywhere. Let $\lambda\colon W\to X_+\wedge W$ be any of the markings above, so that $\gamma\cdot_\lambda w = \lambda^*\widetilde{\mu}(\gamma\otimes w)$.

Unitality. Let $c\colon X_+\to S^0$ collapse X to the non-basepoint. Then $1\in K^0(X)=\widetilde{K}^0(X_+)$ is $c^*(1_{S^0})$, and $(c\wedge \mathrm{id}_W)\circ\lambda = \mathrm{id}_W$ under $S^0\wedge W\cong W$, because λ sends w to $x\wedge w$ for a point $x\in X$ and c sends every such x to the non-basepoint. Naturality of $\widetilde{\mu}$ in the first variable and its unit clause (corollary 2.5.9) give

$$1\cdot_\lambda w = \lambda^*\widetilde{\mu}\big(c^*1_{S^0}\otimes w\big) = \lambda^*(c\wedge\mathrm{id}_W)^* w = \big((c\wedge\mathrm{id}_W)\circ\lambda\big)^* w = w$$

Associativity. Write $\Delta_+\colon X_+\to X_+\wedge X_+$ for the based diagonal. Then $(\mathrm{id}_{X_+}\wedge\lambda)\circ\lambda = (\Delta_+\wedge\mathrm{id}_W)\circ\lambda$, both sending w to $x\wedge x\wedge w$. Using naturality of $\widetilde{\mu}$ in the second variable, then its associativity, then naturality in the first variable, and finally $\Delta_+^*\widetilde{\mu}(\gamma\otimes\gamma')=\gamma\gamma'$, which is not quoted but derived: $\Delta_+ = q\circ d$ with $d\colon X_+\to X_+\times X_+$ the diagonal $z\mapsto(z,z)$ and q the collapse, so by the defining property of $\widetilde{\mu}$ and the fact that a pullback is a ring homomorphism (proposition 2.1.10),

$$\Delta_+^*\widetilde{\mu}(\gamma\otimes\gamma') = d^*q^*\widetilde{\mu}(\gamma\otimes\gamma') = d^*\big(p_1^*\gamma\cdot p_2^*\gamma'\big) = (p_1 d)^*\gamma\cdot(p_2 d)^*\gamma' = \gamma\gamma'$$

since $p_i\circ d = \mathrm{id}_{X_+}$. The product on the right is the one on $\widetilde{K}^0(X_+)$, which agrees with that on $K^0(X)$ because $\widetilde{K}^0(X_+)$ is the kernel of restriction to the added basepoint and restriction to X is a ring map bijective on it. With that in hand,

$$\gamma\cdot_\lambda(\gamma'\cdot_\lambda w) = \big((\mathrm{id}\wedge\lambda)\circ\lambda\big)^*\widetilde{\mu}\big(\gamma\otimes\widetilde{\mu}(\gamma'\otimes w)\big) = \lambda^*(\Delta_+\wedge\mathrm{id}_W)^*\widetilde{\mu}\big(\widetilde{\mu}(\gamma\otimes\gamma')\otimes w\big) = (\gamma\gamma')\cdot_\lambda w$$

Neither computation refers to q^* or j^*, so both hold verbatim on the relative terms. Additivity of each $\gamma\cdot_\lambda(-)$ is additivity of $\widetilde{\mu}$ in its second variable, and additivity in γ is additivity in its first: $\widetilde{\mu}$ is defined on a tensor product, hence bilinear. That completes clause (3).

Step 2: The maps induced by marked maps are equivariant. Call a pointed map $g\colon W\to W'$ of marked spaces *marked* if $\lambda'\circ g = (\mathrm{id}_{X_+}\wedge g)\circ\lambda$. For marked g,

$$g^*(\gamma\cdot_{\lambda'} w') = g^*\lambda'^*\widetilde{\mu}(\gamma\otimes w') = \lambda^*(\mathrm{id}\wedge g)^*\widetilde{\mu}(\gamma\otimes w') = \lambda^*\widetilde{\mu}(\gamma\otimes g^*w') = \gamma\cdot_\lambda g^*(w')$$

by naturality of $\widetilde{\mu}$ in its second variable. Now every map in the construction of theorem 2.5.11 is induced by marked maps: the inclusions $Q_m\hookrightarrow P_m$ and collapses $P_m\to P_m/Q_m$ are marked by inspection of the three markings, and in the cone tower of lemma 2.5.7 for (P_m, Q_m), the spaces W_1, W_2 carry the markings that remember the X-coordinate along the cone, $(w,t)\mapsto x(w)\wedge(w,t)$, for which the collapses k, c_1, k_2, c_2 and the inclusions are all marked, again by inspection: each either forgets cone coordinates or collapses a marked subspace, and the marking formulas match on the nose. Hence every map in every five-term window is γ-equivariant, including the transported ones: if g^* is equivariant and an isomorphism, so is $(g^*)^{-1}$, so $\delta_m = (k^*)^{-1}\circ c_1^*$ is equivariant. One map in Step 5 of lemma 2.5.7 was replaced up to homotopy. Equivariance of the maps that appear in the sequence, namely $(\Sigma^m i)^*$, j^*, δ_m, does not depend on that replacement.

Step 3: The periodicity transport is equivariant. The remaining maps of the folded sequence are the isomorphisms $\beta_W\colon\widetilde{K}^0(W)\to\widetilde{K}^0(\Sigma^2 W)$ on absolute terms $W=\Sigma^m(X_+)$ or $\Sigma^m(A_+)$, with

$\Sigma^2 W$ marked through its X-coordinate as before. Treat the X-case. The A-case is identical with $i^*\gamma$. Write $W = S^m \wedge X_+$, so that $\Sigma^2 W = S^2 \wedge S^m \wedge X_+$, and let

$$R\colon S^2 \times S^m \times X_+ \longrightarrow S^2 \wedge S^m \wedge X_+$$

be the total collapse, whose pullback R^* is injective by proposition 2.5.8(2). The claim $\beta_W(\gamma\cdot w) = \gamma\cdot\beta_W(w)$ is checked after R^*. For the left side, R factors through the collapse $q_{S^2,W}\colon S^2\times W \to S^2\wedge W$, so by the defining property of $\widetilde{\mu}$ and then the factorization argument of Step 1, applied with X_+ in place of X and pulled back along the remaining collapse,

$$R^*\beta_W(\gamma\cdot w) = \pi_{S^2}^*\beta\cdot\pi_X^*\gamma\cdot\pi_{S^m\times X_+}^*\big(Q_+^* w\big)$$

in $K^0(S^2\times S^m\times X_+)$, where $Q_+\colon S^m\times X_+\to W$ is the collapse. For the right side, the marking $\lambda_{\Sigma^2 W}$ composed with R factors as (π_X, R) followed by the collapse of $X_+\times\Sigma^2 W$ onto $X_+\wedge\Sigma^2 W$, so the same two moves give

$$R^*\big(\gamma\cdot\beta_W(w)\big) = \pi_X^*\gamma\cdot R^*\beta_W(w) = \pi_X^*\gamma\cdot\pi_{S^2}^*\beta\cdot\pi_{S^m\times X_+}^*\big(Q_+^* w\big)$$

and the two right-hand sides agree by commutativity of the ring $K^0(S^2\times S^m\times X_+)$ (proposition 2.1.10). Injectivity of R^* gives the claim. Hence $\delta^1 = \delta_1\circ\beta_A$ is equivariant, proving (2).

Step 4: Naturality. Let $f\colon (Y,B)\to(X,A)$ be a map of compact pairs and let g be the induced map on the corresponding term of the sequence. What has to be checked is the square

$$\lambda^X\circ g = (f_+\wedge g)\circ\lambda^Y$$

relating the markings over the two bases. This is *not* the marked-map condition of Step 2, which compares two maps over one fixed X. Here the base of the marking changes, which is why the computation needs naturality of $\widetilde{\mu}$ in both variables at once. On the absolute X-terms, with $g = \Sigma^m(f_+)$, both routes send $s\wedge y_+$ to $f(y)\wedge(s\wedge f(y)_+)$, and the A-terms are the same computation with $i_X\circ f|_B = f\circ i_Y$. On the relative terms, with $g = \Sigma^m\bar{f}$ and y such that $f(y)\notin A$, both routes give $f(y)\wedge(s\wedge[f(y)])$. The one case that is not immediate is $y\notin B$ with $f(y)\in A$, which $f(B)\subseteq A$ does not exclude: there $s\wedge[f(y)]$ is the basepoint downstairs, so $\lambda^X\circ g$ lands on the basepoint, while $(f_+\wedge g)\big(y\wedge(s\wedge[y])\big) = f(y)\wedge(\text{basepoint})$ is the basepoint too. With the square established,

$$g^*(\gamma\cdot_{\lambda^X} w) = g^*\lambda^{X*}\widetilde{\mu}(\gamma\otimes w) = \lambda^{Y*}(f_+\wedge g)^*\widetilde{\mu}(\gamma\otimes w) = \lambda^{Y*}\widetilde{\mu}(f_+^*\gamma\otimes g^* w) = (f^*\gamma)\cdot_{\lambda^Y} g^* w$$

termwise, proving (4) and completing the proof.

Theorem 2.5.14: The Leray-Hirsch Theorem in K-Theory

Let B be a finite cell complex with a chosen finite cell structure, and let $p\colon E \to B$ be a fiber bundle whose fiber F is compact Hausdorff, and suppose:

1. for every characteristic map $\Phi\colon D^m \to B$ of that cell structure, the pullback Φ^*E is isomorphic to $D^m \times F$ as a bundle over D^m;

2. there are classes $c_1, \ldots, c_k \in K^0(E)$ whose restrictions to each fiber E_b form a $\mathbb{Z}$-basis of $K^0(E_b)$, and $K^{-1}(E_b) = 0$ for every $b \in B$.

Then for every $n \geq 0$ the homomorphism

$$\Phi_n\colon K^{-n}(B)^{\oplus k} \longrightarrow K^{-n}(E), \qquad (b_1, \ldots, b_k) \longmapsto \sum_{i=1}^{k} p^*(b_i) \cdot c_i$$

with the module action of proposition 2.5.13, is an isomorphism. Equivalently, $K^*(E) = K^0(E) \oplus K^1(E)$ is a free $\mathbb{Z}/2$-graded $K^*(B)$-module with basis $c_1, \ldots, c_k$, the grading on E being that of corollary 2.5.10.

***Proof*:**
E is compact Hausdorff: Hausdorff because points in distinct fibers are separated by preimages of disjoint open sets of B and points in one fiber inside a trivializing chart, and compact because finitely many closed trivializing sets cover B, each with preimage homeomorphic to a closed set times F. Throughout, for a compact subspace $B'' \subseteq B$ write $E'' = p^{-1}(B'')$ and $\Phi_n^{B''}$ for the map of the statement over B'' built from the restricted classes. Hypothesis (2) restricts to every subspace. The relative version, for a compact pair (B'', B''') in B, is

$$\Phi_n^{(B'',B''')}\colon K^{-n}(B'', B''')^{\oplus k} \to K^{-n}(E'', E'''), \qquad (b_i) \mapsto \sum_i \bar{p}^*(b_i) \cdot c_i$$

with $\bar{p}$ the induced map of pairs and the relative action of proposition 2.5.13. By parts (2) and (4) of that proposition, the maps Φ_n and $\Phi_n^{(\cdot,\cdot)}$ commute with every map of the six-term ladders of the pairs (B'', B''') and (E'', E'''): with i^* and j^* by equivariance and naturality, and with the connecting maps because, writing $p'\colon E''' \to B'''$ for the restriction of p and δ_E, δ_B for the connecting maps of the two ladders,

$$\delta_E\big(p'^*(b) \cdot c_i\big) = \delta_E\big(p'^* b\big) \cdot c_i = \bar{p}^*\big(\delta_B\, b\big) \cdot c_i$$

the first equality by proposition 2.5.13(2) applied on the total-space pair with $\gamma = c_i$, whose action on the subspace term is through $c_i|_{E'''}$ as the marking there prescribes, and the second by naturality of δ in the map of pairs induced by p (theorem 2.5.11).

Step 1: Fibers, points, and disks. For $B'' = \{b\}$ a point, Φ_0 is an isomorphism by hypothesis (2), Φ_n for n even follows by proposition 2.5.13(2)-equivariant transport along the periodicity isomorphisms, and for n odd both sides vanish: $K^{-n}(\mathrm{pt}) = 0$ for odd n (example 2.3.2) and $K^{-n}(E_b) \cong K^{-1}(E_b) = 0$ by hypothesis and corollary 2.5.10. For B'' finite discrete, both sides are finite direct sums (Axiom 5 of theorem 2.3.6) and the point case applies summandwise. For $B'' = D^m$ carrying a trivialized bundle $D^m \times F$: the fiber inclusion over the center b_0 and the projections induce, by homotopy invariance (Axiom 1), isomorphisms on every K^{-n} of base and total space, the squares comparing $\Phi_n^{D^m}$ with $\Phi_n^{\{b_0\}}$ commute by proposition 2.5.13(4), and two-out-of-three in each square gives that $\Phi_n^{D^m}$ is an isomorphism.

Step 2: The double induction. Order the statement $S(d,c)$: for every finite cell complex B'' of dimension at most d (the maximum of its cell dimensions) with at most c cells, every bundle over B'' satisfying (1) and (2) has all Φ_n isomorphisms. Induct lexicographically on (d,c). The base, $d=0$, is the finite discrete case of Step 1. For the inductive step let $B = B' \cup_\varphi e^m$, $m \geq 1$, the top stage of the structure (an attached 0-cell is the disjoint-union case, handled by additivity as in Step 1). By induction S holds for B'.

Step 3: The relative term over the cell. Let $\Phi\colon (D^m, S^{m-1}) \to (B, B')$ be the characteristic map and $\widetilde{\Phi}\colon D^m \times F \cong \Phi^*E \to E$ the bundle map over it from hypothesis (1). Both quotient comparisons

$$D^m/S^{m-1} \longrightarrow B/B', \qquad (D^m \times F)/(S^{m-1} \times F) \longrightarrow E/E'$$

are continuous bijections of compact Hausdorff spaces, hence homeomorphisms, since Φ and $\widetilde{\Phi}$ are injective over the open cell and collapse everything else. Hence restriction along $(\Phi, \widetilde{\Phi})$ identifies the six-term ladder of (E, E') over (B, B') at the relative terms with that of $(D^m \times F, S^{m-1} \times F)$ over (D^m, S^{m-1}), compatibly with the Φ-maps by proposition 2.5.13(4), the comparison classes being $\widetilde{\Phi}^* c_i$, which restrict to fiberwise bases. The pair (D^m, S^{m-1}) satisfies the hypotheses with the sphere carrying its standard CW structure, of dimension $m-1 \leq d-1$, and the restricted bundle over S^{m-1} inherits (1) since every pullback of a trivial bundle is trivial. By induction ($S(m-1,\cdot)$, first coordinate smaller) all $\Phi_n^{S^{m-1}}$ are isomorphisms, and by Step 1 all $\Phi_n^{D^m}$ are. In the ladder of (D^m, S^{m-1}), unrolled two-periodically by theorem 2.5.11, each window of five consecutive terms centered on a relative term has its four outer vertical maps isomorphisms, so the five lemma ([14, The Five Lemma]) makes every $\Phi_n^{(D^m, S^{m-1})}$, hence every $\Phi_n^{(B,B')}$, an isomorphism. The top row of each window is exact because it is a direct sum of k copies of the exact sequence of theorem 2.5.11 for the pair of bases.

Step 4: Closing the induction. In the unrolled ladder of (B, B') and (E, E'), each window of five consecutive terms centered on an absolute B-term has outer verticals $\Phi_\cdot^{(B,B')}$ (twice) and $\Phi_\cdot^{B'}$ (twice), all isomorphisms by Step 3 and the inductive hypothesis ($S(d, c-1)$). The five lemma gives that Φ_n^B is an isomorphism for every n, closing the induction.

Step 5: The module reformulation. By proposition 2.5.13(1) and (3) the graded group $K^*(E)$ is a $K^*(B)$-module via p^* and the ring structure, the action agreeing with the one in Φ_n. Bijectivity of Φ_0 and Φ_1 says exactly that $c_1, \ldots, c_k$ is a basis, completing the proof.

Theorem 2.5.15: The Projective-Bundle Theorem in K-Theory

Let X be a finite cell complex, in particular any finite CW complex (lemma 2.5.3), and $V \to X$ a rank-n complex vector bundle with projectivization $\pi\colon P(V) \to X$ and tautological line bundle L_V (definition 2.5.6). Then $K^*(P(V))$ is a free $\mathbb{Z}/2$-graded $K^*(X)$-module with basis

$$1,\ [L_V],\ [L_V]^2,\ \ldots,\ [L_V]^{n-1}$$

In particular $\pi^*\colon K^*(X) \to K^*(P(V))$ is a split injection of $K^*(X)$-modules.

***Proof*:**
Apply theorem 2.5.14 to $p = \pi$, whose fiber $\mathbb{C}\mathrm{P}^{n-1}$ is compact Hausdorff, with $c_i = [L_V]^{i-1}$ for $1 \leq i \leq n$. Hypothesis (1): over a characteristic map Φ of a cell of X the bundle Φ^*V is trivial, as in Step 2 of proposition 2.5.5, and a trivialization descends to $\Phi^*P(V) \cong D^m \times \mathbb{C}\mathrm{P}^{n-1}$. Hypothesis (2): the restriction of L_V to the fiber $P(V_x)$ is the tautological line bundle of that projective space (definition 2.5.6), so under a linear identification $V_x \cong \mathbb{C}^n$ the restrictions of the c_i become $1, [H], \ldots, [H]^{n-1} \in K^0(\mathbb{C}\mathrm{P}^{n-1})$. By theorem 2.4.5, $K^0(\mathbb{C}\mathrm{P}^{n-1})$ is free abelian on $1, t, \ldots, t^{n-1}$ with $t = [H] - 1$, and $K^{-1}(\mathbb{C}\mathrm{P}^{n-1}) = 0$. The change of basis from $(1, t, \ldots, t^{n-1})$ to $(1, [H], \ldots, [H]^{n-1})$ is given by $[H]^j = (1+t)^j$, a unipotent upper-triangular integer matrix, hence invertible over $\mathbb{Z}$, so the restrictions of the c_i form a $\mathbb{Z}$-basis of every fiber's K^0. The conclusion of theorem 2.5.14 is the statement. The splitting of π^* is the $K^*(X)$-linear projection onto the coefficient of the basis element 1, completing the proof.

The ring structure of $K^*(P(V))$, the relation $\sum_i (-1)^i \lambda^i(V)\,[L_V]^{n-i} = 0$ presenting it as a quotient of a polynomial algebra, requires the multiplicativity of the total λ-operation and is developed with the λ-ring structure in chapter 4. Only the free-module statement is used below.

Theorem 2.5.16: The Splitting Principle in K-Theory

Let X be a finite cell complex, in particular any finite CW complex (lemma 2.5.3(2)), and $V \to X$ a rank-n complex vector bundle. There is a space $F(V)$, the iterated projectivization built in the proof below, and a map $p\colon F(V) \to X$, such that

1. $p^*\colon K^*(X) \to K^*(F(V))$ is a split injection of $K^*(X)$-modules, and
2. the pulled-back bundle splits as a sum of line bundles, $p^*V \cong L_1 \oplus \cdots \oplus L_n$.

Moreover $F(V)$ and every intermediate stage of its construction is a finite cell complex.

***Proof*:**
The space $F(V)$ is built as a tower of iterated projectivizations. This is the same construction [14, Splitting Principle] runs for cohomology, and that theorem is the attribution for it. The statement there produces a space with two properties and does not say how it is filtered, so nothing below is imported from it. Set $Y_0 = X$ and $Y_{j+1} = P(V_j)$, where $V_0 = V$ and V_{j+1} is the orthogonal complement of the tautological line subbundle $L_{j+1} \subseteq \pi_{j+1}^* V_j$ over Y_{j+1}, given by proposition 1.1.14 using an inner product from [14, Inner Products on a Vector Bundle]. Both require a paracompact base, and each Y_j is one: the base Y_0 is a finite cell complex by hypothesis, by proposition 2.5.5 each Y_{j+1} is again one, and a finite cell complex is compact Hausdorff (lemma 2.5.3(1)) hence paracompact ([8, Compact Spaces, Discrete Spaces, and Euclidean Space](1)). This also proves the last claim of the statement.

Clause (2) now follows from the tower itself, by induction on j, without appeal to the upstream theorem's conclusion. Writing $p_j\colon Y_j \to X$ for the composite of the projections, the claim is that

$$p_j^* V \;\cong\; \Lambda_1 \oplus \cdots \oplus \Lambda_j \oplus W_j, \qquad W_j = V_j$$

with each Λ_i a line bundle on Y_j and V_j of rank $n - j$. Carrying the identification $W_j = V_j$ in the inductive claim is what makes the step go through, since the bundle the next stage projectivizes is V_j. For $j = 0$ the claim is V itself with the empty sum and $W_0 = V_0 = V$. Given it at stage j, pull back along $\pi_{j+1}\colon Y_{j+1} \to Y_j$: the summand $W_j = V_j$ becomes $\pi_{j+1}^* V_j$, which by construction splits as $L_{j+1} \oplus V_{j+1}$. Pullback preserves direct sums and carries line bundles to line bundles, so the claim holds at stage $j+1$ with Λ_i replaced by $\pi_{j+1}^*\Lambda_i$ for $i \leq j$, with $\Lambda_{j+1} = L_{j+1}$, and with $W_{j+1} = V_{j+1}$ of rank $n-j-1$. At $j = n-1$ the remainder W_{n-1} has rank 1, so $p^*V \cong \Lambda_1 \oplus \cdots \oplus \Lambda_{n-1} \oplus W_{n-1}$ is a sum of n line bundles, where $p = p_{n-1}$.

For clause (1), theorem 2.5.15 applies at every stage, since its base hypothesis is a finite cell complex: each $\pi_{j+1}^*\colon K^*(Y_j) \to K^*(Y_{j+1})$ is a split injection of $K^*(Y_j)$-modules, the splitting being projection onto the coefficient of 1. The composite $p^* = \pi_{n-1}^* \circ \cdots \circ \pi_1^*$ is therefore injective, and composing the splittings, each restricted along the earlier pullbacks, gives a $K^*(X)$-linear left inverse, completing the proof.

3

The Chern Character

The previous chapter turned the monoid of vector bundles into the ring $K^0(X)$ and showed it is periodic. Ordinary cohomology is the other invariant attached to a bundle, and this chapter is about the bridge from one to the other. A bundle determines cohomology classes, the *characteristic classes*, which measure how far it is from trivial: the Stiefel-Whitney classes $w_i(E) \in H^i(B;\mathbb{Z}/2\mathbb{Z})$ of a real bundle, the Chern classes $c_i(E) \in H^{2i}(B;\mathbb{Z})$ of a complex one, the Pontryagin classes $p_i(E) \in H^{4i}(B;\mathbb{Z})$ of a real bundle, and the Euler class $e(E) \in H^n(B;\mathbb{Z})$ of an oriented bundle of rank n. Their construction, and every property of them used here, belongs to algebraic topology and is owned by [14]. This chapter states what it imports, computes with it, and then builds the object that is K-theoretic.

In the other direction the whole ring $K^0(X)$ maps to cohomology. That map is the Chern character, and theorem 3.3.7 shows it is an isomorphism after tensoring with $\mathbb{Q}$: rationally, K-theory carries exactly the information of even-dimensional cohomology. Integrally it does not, and the gap is where the arithmetic of the subject lives. Section 3.1 sets up the four classes and reads them off the skeleta of the base; section 3.2 records what is imported from [14] and computes with it; section 3.3 builds ch and proves the rational isomorphism; section 3.4 measures how far that isomorphism is from integral; and section 3.5 evaluates characteristic classes against a fundamental class, producing the numerical invariants that decide when a manifold bounds.

3.1 Definition and Intuitions

A characteristic class assigns cohomology to a bundle, naturally in the base. Two pieces of data go into the definition and must be kept apart: the *structure group* G, which says what kind of bundle is being classified, and the *coefficient group* A, which says where the class lives. Conflating them is easy and produces nonsense, since the four classes below use three different structure groups and two different coefficient groups.

Definition 3.1.1: Characteristic Classes

Let G be a topological group and let A be an abelian group. Write $b_G(-)$ for the contravariant functor sending a space B to the set of isomorphism classes of principal G-bundles over B, and a map f to the pullback f^*. A *characteristic class for G-bundles with coefficients in A* is a natural transformation

$$c\colon b_G \Rightarrow H^*(-;A)$$

Concretely, c assigns to each principal G-bundle $P \to X$ a class $c(P) \in H^*(X;A)$ subject to

$$c(f^*P) = f^*c(P)$$

the f^* on the left being pullback of bundles and the one on the right the induced map in cohomology. The component landing in $H^i(-;A)$ is written c_i.

A rank-n real vector bundle has structure group $O(n)$ and is the bundle associated to its frame bundle, so $b_{O(n)}(B)$ is identified with $\mathrm{Vect}^n_{\mathbb{R}}(B)$; the complex case replaces $O(n)$ by $U(n)$ and the oriented real case by $SO(n)$. Throughout this section $\mathrm{Vect}^n(B)$ means $\mathrm{Vect}^n_{\mathbb{R}}(B)$, and $\mathrm{Vect}^n(B) \cong [B, \mathrm{Grass}_n]$ by [14, Classification of Vector Bundles over a Paracompact Base]. In particular, if $f\colon B \to \mathrm{Grass}_n$ is a classifying map and $E = f^*(E_n)$ for the universal bundle E_n, then

$$c_i(E) = c_i(f^*(E_n)) = f^*(c_i(E_n))$$

so if f induces the trivial map on cohomology with the relevant coefficients, every $c_i(E)$ vanishes.

The definitions below name which (G, A) each class belongs to. Naming the pair does *not* single out one class: there are many natural transformations $\mathrm{Vect}^n(-) \Rightarrow H^*(-;\mathbb{Z}/2\mathbb{Z})$, so each definition names a type and then says which member of that type it means. In each case the member is named by its construction upstream, as the pullback along a classifying map of a fixed universal class: w_i and c_i from the polynomial generators of $H^*(BO(n);\mathbb{Z}/2\mathbb{Z})$ and $H^*(BU(n);\mathbb{Z})$, the Euler class from the transgression class on $BSO(n)$. The axiom lists of [14, Stiefel-Whitney Classes and the Total Class] and [14, Chern Classes and the Total Chern Class] record what those families satisfy, and are what the computations below use; they are not offered here as characterizations, since neither theorem is stated with a uniqueness clause. The Pontryagin classes are of type $(O(n), \mathbb{Z})$ like the rest, but definition 3.1.3 names its member by an explicit formula in the Chern classes of the complexification rather than by a universal class.

The four classes are the following instances.

Definition 3.1.2: Stiefel-Whitney Classes

Take $G = O(n)$, so that $b_G(B) = \mathrm{Vect}^n(B)$, and coefficients $A = \mathbb{Z}/2\mathbb{Z}$. A characteristic class of this type is a *Stiefel-Whitney class*, written $w_i(E) \in H^i(B;\mathbb{Z}/2\mathbb{Z})$, with total class $w(E) = 1 + w_1(E) + w_2(E) + \cdots$. The particular family so named is the one constructed in [14, Stiefel-Whitney Classes and the Total Class] as $w_i(E) = f_E^* w_i$, the pullback along a classifying map $f_E\colon B \to BO(n)$ of the degree-i polynomial generator of $H^*(BO(n);\mathbb{Z}/2\mathbb{Z})$.

These classes matter here because they generate the mod-2 cohomology of the infinite Grassmannian freely:

$$H^*(\mathrm{Grass}_n;\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[w_1(E_n), \ldots, w_n(E_n)]$$

This is proved in [14, The Cohomology of $BO(n)$]. It fails with $\mathbb{Z}$ coefficients, and modulo torsion the Pontryagin classes are what characterize the integral case:

Definition 3.1.3: Pontryagin Classes

For a real vector bundle $E \to B$, the *Pontryagin classes* are the integral characteristic classes

$$p_i(E) = (-1)^i \, c_{2i}(E \otimes_{\mathbb{R}} \mathbb{C}) \;\in\; H^{4i}(B;\mathbb{Z})$$

obtained from the Chern classes of the complexification, with $p_0(E) = 1$. The total Pontryagin class is $p(E) = 1 + p_1(E) + p_2(E) + \cdots$. Only the even Chern classes of $E \otimes_{\mathbb{R}} \mathbb{C}$ are used: the odd ones are 2-torsion, because $E \otimes_{\mathbb{R}} \mathbb{C}$ is isomorphic to its own conjugate.

The classes of interest sit in H^{4i} because the odd Chern classes of $E \otimes_{\mathbb{R}} \mathbb{C}$ are 2-torsion, so they are the only ones of infinite order; they need not vanish, and over a base with 2-torsion in its integral cohomology they generally do not. For an oriented bundle the classifying map may be taken as $f\colon B \to \widetilde{\mathrm{Grass}}_n$ into the oriented Grassmannian, which yields the following.

Definition 3.1.4: Euler Class

Let B be paracompact. Take $G = SO(n)$, so that $b_G(B) = \widetilde{\mathrm{Vect}}^n(B)$ is the set of isomorphism classes of *oriented* real rank-n bundles, classified by maps into the oriented Grassmannian $\widetilde{\mathrm{Grass}}_n = BSO(n)$, and coefficients $A = \mathbb{Z}$. The *Euler class* is the characteristic class $e(E) \in H^n(B;\mathbb{Z})$ of this type constructed in [14, Euler Class of an Oriented Bundle], namely the pullback

$$e(E) = f_E^*\, e, \qquad e \in H^n(BSO(n);\mathbb{Z})$$

of the universal class e along a classifying map $f_E\colon B \to BSO(n)$ for E, where e is the transgression of the oriented fiber generator in the Gysin sequence of the universal sphere bundle $S^{n-1} \to BSO(n-1) \to BSO(n)$. The orientation is part of the data: reversing it replaces the classifying map by its composite with the deck involution of $BSO(n) \to BO(n)$, which negates the universal class and hence $e(E)$, so e is not a function on orientable bundles.

For an oriented real vector bundle, f induces the trivial map exactly when the Stiefel-Whitney, Pontryagin, and Euler classes all vanish.

The complex case is separated out because removing 0 from $\mathbb{C}$ leaves a path-connected space, so complex bundles carry no orientability obstruction:

Definition 3.1.5: Chern Class

Take $G = U(n)$, so that $b_G(B) = \mathrm{Vect}^n_{\mathbb{C}}(B)$, and coefficients $A = \mathbb{Z}$. A characteristic class of this type is a *Chern class*, written $c_i(E) \in H^{2i}(B;\mathbb{Z})$, with total class $c(E) = 1 + c_1(E) + c_2(E) + \cdots$. The family so named is the one constructed in [14, Chern Classes and the Total Chern Class] as $c_i(E) = f_E^* c_i$, the pullback along a classifying map $f_E\colon B \to BU(n)$ of the degree-$2i$ polynomial generator of $H^*(BU(n);\mathbb{Z})$.

The complex Grassmannian satisfies

$$H^*(\mathrm{Grass}_n(\mathbb{C});\mathbb{Z}) \cong \mathbb{Z}[c_1, \ldots, c_n]$$

where $c_i = c_i(E_n)$ for the universal complex bundle $E_n \to \mathrm{Grass}_n(\mathbb{C})$

Several classes were needed above to detect triviality, because vanishing characteristic classes are necessary but not sufficient for it. The simplest nontrivial example is the tangent bundle of S^5, which is nontrivial (S^5 is not parallelizable) yet has every characteristic class zero. Two separate reasons combine. The Stiefel-Whitney, Chern and Pontryagin classes are *stably invariant*, meaning unchanged by direct sum with a trivial bundle, and TS^5 is stably trivial, so all of them vanish. The Euler class is *not* stably invariant (indeed $e(E \oplus \underline{\mathbb{R}}) = 0$ for every oriented E over a finite-type CW base, by [14, Naturality of the Euler Class] together with [14, Whitney Sum Formula for the Euler Class]: the trivial line bundle is pulled back from a bundle over a point, so $e(\underline{\mathbb{R}}) = 0$, and the Whitney formula multiplies by it), and it vanishes here for the different reason that $\langle e(TS^5), [S^5]\rangle = \chi(S^5) = 0$. Every tangent bundle of a sphere is stably trivial, since TS^n plus the trivial normal line is $\underline{\mathbb{R}}^{n+1}$, so the stably invariant classes cannot distinguish odd-dimensional spheres.

Let us now think about for a moment what characteristic classes are detecting. The first "obstruction" it detects is orientability. For a vector bundle $E \to B$ with B path-connected there is a map

$$\pi_1(B) \to \mathbb{Z}/2\mathbb{Z}$$

where it assigns 0 or 1 to each loop according to whether orientation of fibers is preserved or reversed (as one goes around the loop). As $\mathbb{Z}/2\mathbb{Z}$ is abelian, the homomorphism factors through the abelianization $H_1(B)$ of $\pi_1(B)$, and the homomorphisms $H_1(B) \to \mathbb{Z}/2\mathbb{Z}$ are elements of $H^1(B; \mathbb{Z}/2\mathbb{Z})$. From this one obtains an element of $H^1(B; \mathbb{Z}/2\mathbb{Z})$ associated to E, and it vanishes exactly when E is orientable. This shall be the interpretation of the first Stiefel-Whitney class $w_1(E)$.

To generalize to higher Stiefel-Whitney classes, let us re-interpret this result when B is a CW complex. Then E is orientable if and only if the restriction of the 1-skeleton B^1 of B is orientable, as all loops in B can be deformed to lie in B^1. Then a vector bundle over a 1-dimensional CW complex is trivial if and only if it is orientable, so $w_1(E)$ measures whether E is trivial over the 1-skeleton B^1. The fibers may be higher dimensional: over each loop the object in question is a vector bundle on S^1. The next question is whether triviality over the 1-skeleton extends to the 2-skeleton. Assuming E is trivial over B^1 and its fibers are n-dimensional, choose n orthonormal sections over B^1 and ask whether they extend to orthonormal sections over each 2-cell. Each 2-cell is given by a characteristic map $D^2 \to B$, so E pulls back over the disk D^2, where the sections over B^1 pullback to sections over ∂D^2. As D^2 is contractible, this bundle is trivial. Choosing a trivialization, the sections over ∂D^2 determine a map:

$$\partial D^2 \to O(n)$$

Choosing a trivialization with the same orientation as the trivialization of E over B^1 puts the image in $SO(n)$. Such a map is an element of $\pi_1(SO(n))$, so each 2-cell is assigned an element of $\pi_1(SO(n))$. Then B^1 extends to orthonormal sections over B^2 if and only if this element of $\pi_1(SO(n))$ is zero for each 2-cell.

This groups is also relatively easy to compute : when $n = 1$ it is 0, when $n = 2$ it is $\mathbb{Z}$ (as $SO(2)$ is homeomorphic to S^1). For $n \geq 3$, it is $\mathbb{Z}/2\mathbb{Z}$ (exercise). Thus, the inclusion $SO(2) \hookrightarrow SO(n)$ induces a surjection $\pi_1(SO(2)) \twoheadrightarrow \pi_1(SO(n))$, so the generator of $\pi_1(SO(n))$ can be represented by the loop given by rotating the plane through a full turn, fixing the orthogonal complement of the plane. Trivializing E over the 1-skeleton and comparing the two trivializations across each 2-cell assigns to that cell the class in $\pi_1(SO(n))$ of the resulting clutching loop. That assignment is a 2-dimensional cellular cochain with coefficients in $\pi_1(SO(n))$, it is a cocycle, and so it defines an element of $H^2(B; \pi_1(SO(n)))$. This element shall be called $\omega_2(E)$, and will not depend on the choice of sections over B^1. So E is trivial over B^2 if and only if $\omega_2(E)$ is zero in $H^2(B; \pi_1(SO(n)))$. The name is deliberately not w_2: the obstruction takes its coefficients in $\pi_1(SO(n))$, which varies with n, while the Stiefel-Whitney class of definition 3.1.2 always lives in $H^2(B;\mathbb{Z}/2\mathbb{Z})$. If $n=1$ then $\omega_2(E)=0$. If $n=2$ then $\omega_2(E)\in H^2(B;\mathbb{Z})$ and is the Euler class $e(E)$ covered below, whose mod-2 reduction is $w_2(E)$ by [14, Top Class is the Euler Class](1). Only for $n\geq 3$ is $\omega_2(E)$ itself the class $w_2(E)\in H^2(B;\mathbb{Z}/2\mathbb{Z})$.

The natural next question to ask if if E is trivial over B^2, can it be extended to be trivial over B^3 which again is given by looking at elements of $\pi_2(SO(n))$. Fortunately for us, $\pi_2(SO(n))=0$ for all $n\in\mathbb{N}_{>0}$ and so the sections automatically extend over B^3. For extension over 4-cells, $\pi_3(SO(n))$ is relatively simple to compute (see [10, p.75]). For higher dimensional extension B^k for $k\geq 5$, things become a good deal more complicated as the groups $\pi_{k-1}(SO(n))$ become increasingly difficult to compute. They are finitely generated abelian groups whose free part is fairly easy to compute, but whose torsion part is complicated[1]. In the range of $n>k$, then Bott periodicity gives the value of $\pi_{k-1}(SO(n))$. For $k=4i$,

$$\pi_{4i-1}(SO(n))\cong\mathbb{Z}$$

which is why the Pontryagin class lies in $H^{4i}(B;\mathbb{Z})$, and it detects the non-triviality of bundles through the obstruction

$$\omega_{4i}(E)\in H^{4i}(B;\pi_{4i-1}(SO(n)))$$

which is detected by the Pontryagin class $p_i(E)$ of definition 3.1.3 rather than by a Stiefel-Whitney class, the coefficient group being $\mathbb{Z}$. The detection is not an identification: $p_i(E)$ is a nonzero integer multiple of $\omega_{4i}(E)$, the multiple being 2 for $i=1$ and 6 for $i=2$, so $p_i(E)\neq 0$ forces $\omega_{4i}(E)\neq 0$, but not conversely: the multiple can annihilate torsion in $H^{4i}(B;\mathbb{Z})$. The two vanishing loci agree when $H^{4i}(B;\mathbb{Z})$ is torsion-free.

In the stable range $n>k$ the other nontrivial groups are

$$\pi_{k-1}(SO(n))\cong\mathbb{Z}/2\mathbb{Z}\qquad k=8i+1\ (i\geq 1),\qquad k=8i+2\ (i\geq 0)$$

The restriction on the first family matters, and it is the only one. At $i=0$, $k=1$ the group is $\pi_0(SO(n))=0$, since $SO(n)$ is connected, whereas Bott's $\mathbb{Z}/2$ in degree 0 is $\pi_0(O)=\mathbb{Z}/2\mathbb{Z}$, which belongs to the full orthogonal group; that discrepancy is confined to degree 0, and in the positive degrees $8i$ with $i\geq 1$ the group $\pi_{8i}(SO(n))\cong\mathbb{Z}/2\mathbb{Z}$ is present for $SO(n)$ as well. The second family runs from $i=0$: there $k=2$ and the group is $\pi_1(SO(n))\cong\mathbb{Z}/2\mathbb{Z}$ for $n\geq 3$, the case computed above.

In degree $k=2$ the obstruction is the class named $\omega_2(E)$ above, which for $n\geq 3$ is the second Stiefel-Whitney class. In every other degree of the two families, that is for $k=8i+1$ and $k=8i+2$ with $i\geq 1$, the obstruction $\omega_k(E)$ is not detected by the Stiefel-Whitney or Pontryagin classes; **??** takes up that claim and identifies these obstructions through KO-theory instead.

A second way of measuring triviality leads to the Euler class. Consider an n-dimensional vector bundle $E\to B$ and the obstruction to the existence of n orthonormal sections. Take first the obstruction to a single orthonormal section, that is, to a section of unit vectors. The induction again runs over the skeleta of B. Given a section over B^{k-1}, extending over a k-cell pulls E back to the trivial bundle over D^k via the characteristic map, leaving a unit-vector section on ∂D^k. This time the relevant homotopy group is that of the fiber sphere rather than of the structure group. Hence, if $k<n$, such an extension always is possible, so the first (possibly) nontrivial obstruction occurs when $k=n$ where

$$\pi_{k-1}(S^{n-1})\cong\mathbb{Z}$$

[1] Interestingly, this contrasts with elliptic curves over rationals $E(\mathbb{Q})$ for which it is relatively easy to compute the torsion but a million dollar problem (literally) to compute the free part

Identifying $\pi_{n-1}(S^{n-1})$ with $\mathbb{Z}$ consistently across the cells *requires* E to be orientable. This class is the Euler class, denoted $e(E)$. Dropping the orientability condition, the coefficients must be $\mathbb{Z}/2\mathbb{Z}$, giving the Stiefel-Whitney class $w_n(E) \in H^n(B; \mathbb{Z}/2\mathbb{Z})$.

3.2 The Classes in Use

The skeleton-by-skeleton picture of section 3.1 says what the classes mean but computes almost nothing: it needs the homotopy groups $\pi_{k-1}(SO(n))$, which are not available in general. The route that computes is the classifying-space route, and it is carried out in full in *Algebraic Topology* [14]. This section does not repeat it. What follows is a statement of what is imported and at what generality, and then the computations this book needs.

Four facts do most of the work below. They are not everything the computations use: the Euler characteristic formula, the identification of the top class with the Euler class, the Pontryagin classes of a complex bundle and the three obstruction theorems are all drawn on as well, and are cited where they appear. Their hypotheses differ and are worth reading before use: naturality, the Whitney sum formula and the splitting principle hold over a paracompact Hausdorff base, which by box 1.0.1 is the standing hypothesis here, while the Euler-class versions are stated over a CW complex of finite type.

- *Naturality.* $w_i(f^*E) = f^*w_i(E)$ and $c_i(f^*E) = f^*c_i(E)$, by [14, Naturality of Characteristic Classes]; the corresponding statement for the Euler class is [14, Naturality of the Euler Class].
- *The Whitney sum formula.* $w(E \oplus F) = w(E) \smile w(F)$ and $c(E \oplus F) = c(E) \smile c(F)$, by [14, Whitney Sum Formula], with the Euler-class version [14, Whitney Sum Formula for the Euler Class].
- *Normalization.* The value on the tautological line bundle, [14, Normalization on the Universal Line Bundle], which is what pins the classes down rather than merely constraining them.
- *The splitting principle.* [14, Splitting Principle], together with [14, The Flag Bundle is Paracompact], which is what makes the classes defined on the flag bundle in the first place.

One boundary matters here and is easy to trip over, because this book's bases are paracompact while three upstream statements are not. The obstruction-theoretic characterizations are proved over a CW complex and not over a general paracompact space: that $w_1(E) = 0$ *implies* orientability is [14, First Stiefel-Whitney Class is the Obstruction to Orientability] clause (2), CW; the converse, that an orientable bundle has $w_1(E) = 0$, is clause (1) and holds over any paracompact Hausdorff base. The other two, [14, Second Stiefel-Whitney Class is the Obstruction to a Spin Structure] and [14, Top Class is the Obstruction to a Nowhere-Zero Section], are CW-only in both directions, with no paracompact half. The reason is not incidental: those three identify a class with a homotopy classification of maps, which needs cohomology to be representable, and representability is a CW theorem. Wherever one of the three is used below the base is a closed manifold, a projective space or an explicitly CW base, and the hypothesis is discharged at that point rather than globally here; the distinction is recorded because it does not survive to arbitrary paracompact bases and the convention of this book might suggest otherwise.

The computations follow. Each applies the imported machinery to a bundle this book cares about, and together they are what section 3.3 and the K-theoretic chapters draw on.

Example 3.2.1: Stiefel-Whitney and Chern Classes

1. By naturality ([14, Naturality of Characteristic Classes]), the trivial bundle $B \times \mathbb{R}^n$ has
$$w_i(E) = 0 \qquad \forall i > 0$$
as the product is the pullback of the bundle over a point.

2. By the Whitney sum formula ([14, Whitney Sum Formula]), the direct sum of a bundle with a product bundle does not change its Stiefel-Whitney classes; this property is called *stable*. The tangent bundle TS^n is stably trivial, and hence
$$w_i(TS^n) = 0 \qquad \forall i > 0$$
More generally, as :
$$(1 + w_1 + \cdots)(1 + w_1' + \cdots) = 1 + (w_1 + w_1') + (w_2 + w_1 w_1' + w_2') + \cdots$$
given $w(E_1)$ and $w(E_1 \oplus E_2)$, the class $w(E_2)$ is determined by
$$\begin{aligned} w_1 + w_1' &= a_1 \\ w_2 + w_1 w_1' + w_2' &= a_2 \\ &\vdots \\ \sum_i w_{n-i} w_i' &= a_n \end{aligned}$$
If $E_1 \oplus E_2$ is the trivial bundle, $a_i = 0$ for all $i > 0$, so $w(E_1)$ determines $w(E_2)$ uniquely by the above formulas. The same applies to Chern classes.

3. There is no bundle $E \to \mathbb{R}\mathrm{P}^\infty$ whose sum with the line bundle $E_1(\mathbb{R}^\infty)$ is trivial. For
$$w(E_1(R^\infty)) = 1 + \omega$$
where ω is a generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2\mathbb{Z})$. Over $\mathbb{Z}/2\mathbb{Z}$ coefficients this gives
$$w(E) = (1 + \omega)^{-1} = 1 + \omega + \omega^2 + \cdots$$
So $w_i(E) = \omega^i$ is nonzero for all i, contradicting $w_i(E) = 0$ for all $i > \dim E$. This is why compactness is necessary in proposition 1.1.15.

4. [10, p.82] constructs an n-dimensional vector bundle $E \to B$ with $w_i(E)$ nonzero for each $i \leq \dim E$.

Example 3.2.2: Euler Class of the Tangent Bundle of S^2

The tangent bundle TS^2 is an oriented real rank-2 bundle. Its Euler class lies in $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$, so it is determined by the Euler number $\langle e(TS^2), [S^2] \rangle$. Two routes compute it.

First, $S^2 = \mathbb{C}\mathrm{P}^1$ and $T\mathbb{C}\mathrm{P}^1$ is a complex line bundle. As recorded in section 1.2, TS^2 is the oriented plane bundle $E_2 \to S^2$ with clutching function $z \mapsto z^2$ of degree 2. For an oriented plane bundle over S^2 the Euler number is exactly that degree. This is the obstruction picture of section 3.1 in its smallest case: give S^2 its two-cell structure $D^2_- \cup D^2_+$, take the section constant over D^2_-, and the obstruction to extending it without a zero across the top cell is the class in $\pi_1(SO(2)) \cong \mathbb{Z}$ of the clutching function, which is its degree m. That obstruction is the Euler class by definition 3.1.4, so $\langle e(E_m), [S^2]\rangle = m$ and in particular

$$\langle e(TS^2), [S^2]\rangle = 2$$

By [14, Top Class is the Euler Class](2) this also equals $\langle c_1(T\mathbb{C}\mathrm{P}^1), [\mathbb{C}\mathrm{P}^1]\rangle$.

Second, the Euler number of the tangent bundle of any closed connected oriented surface equals its Euler characteristic ([14, Euler Class of the Tangent Bundle Computes the Euler Characteristic]; the differential-geometric form is the Poincaré-Hopf theorem of [16, Poincaré-Hopf]). Since $\chi(S^2) = 2$, this agrees: $\langle e(TS^2), [S^2]\rangle = 2$. In particular $e(TS^2) \neq 0$, which is the cohomological statement of the Hairy Ball theorem: TS^2 has no nowhere-zero section.

Example 3.2.3: Pontryagin Class of the Tangent Bundle of $\mathbb{C}\mathrm{P}^2$

Let $a \in H^2(\mathbb{C}\mathrm{P}^2; \mathbb{Z})$ be the generator with $\langle a^2, [\mathbb{C}\mathrm{P}^2]\rangle = 1$, normalized so that $a = c_1$ of the hyperplane bundle. The cohomology ring is $H^*(\mathbb{C}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}[a]/(a^3)$. The total Chern class of the tangent bundle is

$$c(T\mathbb{C}\mathrm{P}^2) = (1+a)^3 = 1 + 3a + 3a^2$$

using the Euler sequence $T\mathbb{C}\mathrm{P}^2 \oplus \underline{\mathbb{C}} \cong (H^\vee)^{\oplus 3}$ and [14, Whitney Sum Formula], where $H^\vee$ is the dual of the tautological bundle with $c_1(H^\vee) = a$. Thus $c_1 = 3a$ and $c_2 = 3a^2$.

To compute p_1, complexify the real tangent bundle. By [14, Pontryagin Classes of a Complex Bundle], $(T\mathbb{C}\mathrm{P}^2)_\mathbb{R} \otimes \mathbb{C} \cong T\mathbb{C}\mathrm{P}^2 \oplus \overline{T\mathbb{C}\mathrm{P}^2}$, with

$$c(\overline{T\mathbb{C}\mathrm{P}^2}) = 1 - 3a + 3a^2$$

since $c_k(\overline{F}) = (-1)^k c_k(F)$. Multiplying,

$$c\big((T\mathbb{C}\mathrm{P}^2)_\mathbb{R} \otimes \mathbb{C}\big) = (1 + 3a + 3a^2)(1 - 3a + 3a^2) = 1 + (3 + 3 - 9)a^2 = 1 - 3a^2$$

where the degree-6 and higher terms vanish since $a^3 = 0$. Hence $c_2((T\mathbb{C}\mathrm{P}^2)_\mathbb{R} \otimes \mathbb{C}) = -3a^2$, and by [14, Pontryagin Classes],

$$p_1(T\mathbb{C}\mathrm{P}^2) = (-1)^1 c_2\big((T\mathbb{C}\mathrm{P}^2)_\mathbb{R} \otimes \mathbb{C}\big) = 3a^2$$

So the total Pontryagin class is $p(T\mathbb{C}\mathrm{P}^2) = 1 + 3a^2$.

Example 3.2.4: The Tangent Bundle of S^{2n} Has No Nowhere-Zero Section

Let $M = S^{2n}$ with its standard orientation, $n \geq 1$. The Euler characteristic of an even-dimensional sphere is

$$\chi(S^{2n}) = 1 + (-1)^{2n} = 2$$

(its homology is $\mathbb{Z}$ in degrees 0 and $2n$ and 0 otherwise). By [14, Euler Class of the Tangent Bundle Computes the Euler Characteristic],

$$\langle e(TS^{2n}), [S^{2n}]\rangle = \chi(S^{2n}) = 2 \neq 0$$

so $e(TS^{2n})$ is twice a generator of $H^{2n}(S^{2n};\mathbb{Z}) \cong \mathbb{Z}$ and in particular is nonzero. By [14, Top Class is the Obstruction to a Nowhere-Zero Section] the tangent bundle TS^{2n} admits no nowhere-zero section, that is, S^{2n} carries no nowhere-vanishing vector field. For $n = 1$ this is the hairy-ball theorem recalled after definition 1.1.9, now obtained from the Euler class.

The contrast with odd spheres is sharp: $\chi(S^{2n+1}) = 1+(-1)^{2n+1} = 0$, consistent with the standard nowhere-zero field $x \mapsto (-x_2, x_1, -x_4, x_3, \dots)$ on $S^{2n+1} \subset \mathbb{R}^{2n+2}$. Here the vanishing of χ, hence of the integral pairing, is necessary for the field to exist, and on spheres it is also sufficient because the only obstruction is the primary one.

Example 3.2.5: The Möbius Bundle Is Non-Orientable

Let $E \to S^1$ be the Möbius bundle (example 1.1.4), the rank-1 bundle $I \times \mathbb{R}/\!\sim$ with (s,t) for $s \in I = [0,1]$ the base coordinate and $t \in \mathbb{R}$ the fiber coordinate, glued by $(0,t) \sim (1,-t)$. Transporting a chosen fiber orientation once around the base circle reverses it: an orientation is a choice of sign on $\mathbb{R}$, and traversing the base from $s = 0$ to $s = 1$ identifies the fiber at $s = 1$ with the fiber at $s = 0$ through $t \mapsto -t$, which carries the orientation $+$ to $-$. Hence the orientation-transport homomorphism $\pi_1(S^1) = \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ of section 3.1 sends the generator to 1, so it is the nonzero homomorphism and

$$w_1(E) \neq 0 \quad \text{in} \quad H^1(S^1;\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

Therefore E is non-orientable. Equivalently, the orientation double cover $\mathrm{Or}(E) \to S^1$ is the connected double cover $S^1 \xrightarrow{z \mapsto z^2} S^1$, which has no section. This recovers, through the orientability obstruction, the fact recorded after definition 1.1.9 that the Möbius bundle is not isomorphic to the trivial bundle $S^1 \times \mathbb{R}$: a trivial bundle is orientable, while $w_1(E) \neq 0$.

By contrast, the trivial line bundle has $w_1 = 0$ and the cylinder $S^1 \times \mathbb{R}$ admits the constant orientation; the two bundles are distinguished by $w_1 \in H^1(S^1;\mathbb{Z}/2\mathbb{Z})$, which takes both of its values.

Example 3.2.6: Spin Structures on Surfaces and on $\mathbb{C}\mathrm{P}^2$

1. Let Σ be a closed connected orientable surface of genus g and $E \to \Sigma$ an oriented rank-n bundle with $n \geq 3$. Since $H^2(\Sigma;\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, the bundle admits a spin structure if and only if the single class $w_2(E) \in \mathbb{Z}/2\mathbb{Z}$ vanishes. For the (stabilized) tangent bundle, $w_2(T\Sigma)$ is the mod-2 reduction of the Euler class, and $\langle e(T\Sigma), [\Sigma]\rangle = \chi(\Sigma) = 2 - 2g$ is even, so $w_2(T\Sigma) = 0$ and every such surface is spin. The spin structures then form a torsor under $H^1(\Sigma;\mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^{2g}$, giving 2^{2g} of them, matching the classical count.

2. Consider $\mathbb{C}\mathrm{P}^2$, whose tangent bundle (as a rank-4 real bundle) has total Stiefel-Whitney class computed from the complex structure. The mod-2 reduction of the total Chern class of $T\mathbb{C}\mathrm{P}^2$ is $(1+a)^3 = 1 + 3a + 3a^2 = 1 + a + a^2$ in $H^*(\mathbb{C}\mathrm{P}^2;\mathbb{Z}/2\mathbb{Z})$, where a is the generator of $H^2(\mathbb{C}\mathrm{P}^2;\mathbb{Z}/2\mathbb{Z})$. By [14, Relating Chern and Stiefel-Whitney Classes], $w_2(T\mathbb{C}\mathrm{P}^2)$ is the mod-2 reduction of $c_1(T\mathbb{C}\mathrm{P}^2) = 3a$, namely $a \neq 0$. Hence $w_2(T\mathbb{C}\mathrm{P}^2) \neq 0$ and $\mathbb{C}\mathrm{P}^2$ admits no spin structure on its tangent bundle, a closed orientable 4-manifold that is not spin.

Example 3.2.7: The Top Chern Class of a Bundle With a Section

Let B be a CW complex of finite type and let $E \to B$ be a complex rank-n bundle that splits off a trivial line summand, $E \cong E' \oplus \underline{\mathbb{C}}$ for some complex rank-$(n-1)$ bundle E'. The trivial summand gives a nowhere-zero section (the constant section $b \mapsto (0,1)$), so by [14, Top Class is the Obstruction to a Nowhere-Zero Section] applied to $E_{\mathbb{R}}$, whose Euler class is $c_n(E)$ by [14, Top Class is the Euler Class](2),

$$c_n(E) = 0$$

This also follows directly from the Whitney formula: $c(\underline{\mathbb{C}}) = 1$, so $c(E) = c(E') \smile 1 = c(E')$, and $c(E')$ has no component in degree $2n$ because E' has complex rank $n-1$, forcing $c_n(E) = 0$. The two computations agree, illustrating that the obstruction-theoretic vanishing and the multiplicative vanishing are the same phenomenon.

The tangent bundle of $\mathbb{C}\mathrm{P}^n$ sits on both sides of this and is worth following, because it is a *contrast* to the example rather than an instance of it. The Euler sequence $0 \to \underline{\mathbb{C}} \to (H^\vee)^{\oplus(n+1)} \to T\mathbb{C}\mathrm{P}^n \to 0$, with $H^\vee$ the dual of the tautological bundle, splits as a sequence of bundles over the paracompact $\mathbb{C}\mathrm{P}^n$ by proposition 1.1.14, so

$$(H^\vee)^{\oplus(n+1)} \;\cong\; T\mathbb{C}\mathrm{P}^n \oplus \underline{\mathbb{C}}$$

The bundle the example applies to is therefore $(H^\vee)^{\oplus(n+1)}$, of rank $n+1$, and it duly has $c_{n+1} = 0$: writing $a = c_1(H^\vee)$, the Whitney formula gives $c\big((H^\vee)^{\oplus(n+1)}\big) = (1+a)^{n+1}$, whose degree-$2(n+1)$ term is $a^{n+1} = 0$ in $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$.

It is $T\mathbb{C}\mathrm{P}^n$ that fails the hypothesis, and the same computation shows why. From the displayed isomorphism, $c(T\mathbb{C}\mathrm{P}^n) = (1+a)^{n+1}$ as well, so

$$c_n(T\mathbb{C}\mathrm{P}^n) = \binom{n+1}{n} a^n = (n+1)\, a^n \neq 0$$

since a^n generates $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$. Reading the example contrapositively, $T\mathbb{C}\mathrm{P}^n$ splits off no trivial line summand, and by [14, Top Class is the Obstruction to a Nowhere-Zero Section] its underlying real bundle has no nowhere-zero section. Note the direction: $c_n(T\mathbb{C}\mathrm{P}^n)$ is *computed* from the Euler sequence, not read off from $\chi(\mathbb{C}\mathrm{P}^n)$. That the two agree, $\langle c_n(T\mathbb{C}\mathrm{P}^n), [\mathbb{C}\mathrm{P}^n]\rangle = n+1 = \chi(\mathbb{C}\mathrm{P}^n)$, is the content of [14, Euler Class of the Tangent Bundle Computes the Euler Characteristic] and is a check on the computation rather than an input to it.

Exercise 3.2.1

1. Let $E \to B$ be an oriented real vector bundle of odd rank n over a CW complex B of finite type with $H^n(B;\mathbb{Z})$ free of torsion (for instance $B = S^n$ with n odd). Show that $e(E) = 0$.

2. Using the Euler sequence $T\mathbb{C}\mathrm{P}^n \oplus \underline{\mathbb{C}} \cong (H^\vee)^{\oplus(n+1)}$, compute the Euler number $\langle e(T\mathbb{C}\mathrm{P}^n), [\mathbb{C}\mathrm{P}^n]\rangle$ of the tangent bundle of $\mathbb{C}\mathrm{P}^n$, and check that it equals $\chi(\mathbb{C}\mathrm{P}^n)$.

3. Let $E \to B$ be a real vector bundle that is the underlying real bundle of a complex line bundle L. Show that $p_1(E) = c_1(L)^2$.

4. Show that the Pontryagin classes are stable: $p_i(E \oplus \underline{\mathbb{R}}^k) = p_i(E)$ for the trivial bundle $\underline{\mathbb{R}}^k$. Deduce that $p_i(TS^n) = 0$ for all $i > 0$, using that $TS^n \oplus NS^n \cong \underline{\mathbb{R}}^{n+1}$ with NS^n *trivial.* Then explain why this argument does *not* extend to an arbitrary summand of a trivial bundle (a bundle E with $E \oplus F \cong \underline{\mathbb{R}}^N$ for *some* F, not necessarily trivial), and exhibit a counterexample

using $T\mathbb{C}\mathrm{P}^2$ from example 3.2.3.

5. Let $E \to B$ be an oriented real rank-n bundle over a connected CW complex with $\dim B < n$. Using [14, Top Class is the Obstruction to a Nowhere-Zero Section], show that E admits a nowhere-zero section and that $e(E) = 0$. Explain why this is consistent with $e(E) \in H^n(B;\mathbb{Z})$.

6. Compute w_1 of the Möbius bundle pulled back along the degree-2 map $f\colon S^1 \to S^1$, $z \mapsto z^2$, and decide whether the pullback is orientable. (Use [14, First Stiefel-Whitney Class is the Obstruction to Orientability] and naturality.)

7. Let M be a closed connected oriented 3-manifold. Show that $\chi(M) = 0$ and conclude that M admits a nowhere-zero vector field. Where does the argument use [14, Euler Class of the Tangent Bundle Computes the Euler Characteristic], and where does it use that the only obstruction is primary?

8. [14, Top Class is the Euler Class](1) says the mod-2 reduction of $e(E)$ is the top Stiefel-Whitney class. For an oriented real rank-2 bundle $E \to B$, deduce that $\rho(e(E)) = w_2(E)$. Then let Σ be a closed connected orientable surface and $E \to \Sigma$ an oriented rank-2 bundle, and use [14, Top Class is the Obstruction to a Nowhere-Zero Section] to show that E admits a nowhere-zero section if and only if $e(E) = 0$. Deduce that $w_2(E) = 0$ is necessary but not sufficient, and exhibit a bundle witnessing the gap.

9. Let $E \to B$ be a complex rank-n bundle over a CW complex B of finite type and suppose $c_n(E) \neq 0$. Prove that E is not isomorphic to a bundle of the form $E' \oplus \underline{\mathbb{C}}$, and that the underlying real bundle $E_{\mathbb{R}}$ has no nowhere-zero section. Which theorem of this section gives each conclusion?

10. Explain why $\mathbb{C}\mathrm{P}^2$ admits no spin structure on its tangent bundle while S^4 does, using [14, Second Stiefel-Whitney Class is the Obstruction to a Spin Structure]. Compute the number of spin structures on S^4.

3.3 The Chern Character

The computations of section 2.4 leave a coincidence unexplained. For complex projective space, theorem 2.4.5 gives $K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[t]/(t^{n+1})$ with $t = [H] - 1$, while ordinary cohomology gives $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$. Two truncated polynomial rings on one generator, of the same rank. The same coincidence of ranks appeared for spheres, where $\widetilde{K}^0(S^{2k}) \cong \mathbb{Z} \cong \widetilde{H}^{2k}(S^{2k};\mathbb{Z})$.

The coincidence is structural, and this section identifies the map that explains it. There is a natural ring homomorphism from $K^0(X)$ to rational cohomology, the Chern character, which becomes an isomorphism after tensoring the source with $\mathbb{Q}$. It is not an isomorphism over $\mathbb{Z}$, and the failure is not a defect of the construction: it is the integrality phenomenon that section 2.4 already stumbled onto in $\mathbb{C}\mathrm{P}^2$, and the subject of the next section.

Two conventions before beginning. Throughout, $H^{\mathrm{even}}(X;\mathbb{Q}) = \bigoplus_{k\geq 0} H^{2k}(X;\mathbb{Q})$ and $H^{\mathrm{odd}}(X;\mathbb{Q}) = \bigoplus_{k\geq 0} H^{2k+1}(X;\mathbb{Q})$, both finite sums for a finite complex. And H continues to denote the *tautological* line bundle, with $H^\vee$ its dual, so that $c_1(H) = -x$ where $x = c_1(H^\vee)$ is the positive generator. *Algebraic Topology* [14, The First Chern Class of a Line Bundle, The Tautological Line Bundle on $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$] makes the opposite choice, writing γ^1 for the tautological bundle and reserving H for the hyperplane bundle with $c_1(H) = +x$. Every sign below is stated in this book's convention, and a reader carrying a formula across should dualize.

One input is needed that the series does not give in the form required, and it is proved here: a statement about symmetric polynomials, and an integrality statement at that: the power sums are *integral* polynomial expressions in the elementary symmetric functions, not merely rational ones. The other input, that c_1 converts tensor products into sums, is [14, The First Chern Class is Additive on Tensor Products].

Lemma 3.3.1: Power Sums as Integral Polynomials

For each $k \geq 1$ there is a polynomial $s_k \in \mathbb{Z}[y_1, \ldots, y_k]$, independent of n, such that for all n and all $x_1, \ldots, x_n$,

$$\sum_{j=1}^{n} x_j^k \;=\; s_k\big(\sigma_1, \ldots, \sigma_k\big)$$

where σ_i is the i-th elementary symmetric polynomial in $x_1, \ldots, x_n$, with the convention $\sigma_i = 0$ for $i > n$. The first three are

$$s_1 = y_1, \qquad s_2 = y_1^2 - 2y_2, \qquad s_3 = y_1^3 - 3y_1y_2 + 3y_3$$

***Proof*:**
Write $p_k = \sum_j x_j^k$ and let $E(T) = \prod_{j=1}^n (1 + x_jT) = \sum_{i\geq 0} \sigma_i T^i$, a polynomial identity in $\mathbb{Z}[x_1, \ldots, x_n][T]$. Differentiating and dividing,

$$\frac{E'(T)}{E(T)} \;=\; \sum_{j=1}^{n} \frac{x_j}{1 + x_jT} \;=\; \sum_{j=1}^{n} \sum_{m\geq 0} (-1)^m x_j^{m+1} T^m \;=\; \sum_{m\geq 0} (-1)^m p_{m+1} T^m$$

as formal power series, the middle step being the geometric expansion of $(1+x_jT)^{-1}$. Multiplying by $E(T)$ and comparing coefficients of T^{k-1} gives Newton's identity

$$k\,\sigma_k \;=\; \sum_{j=1}^{k} (-1)^{j-1} \sigma_{k-j}\, p_j \tag{3.1}$$

valid for every $k \geq 1$, with $\sigma_0 = 1$. Solving (3.1) for p_k, whose coefficient is $\sigma_0 = 1$, gives

$$p_k \;=\; \sigma_1 p_{k-1} - \sigma_2 p_{k-2} + \cdots + (-1)^k \sigma_{k-1} p_1 + (-1)^{k-1} k\,\sigma_k$$

and induction on k expresses p_k as a polynomial with integer coefficients in $\sigma_1, \ldots, \sigma_k$. Since the coefficient of p_k in (3.1) is 1 rather than k, no division occurs at any stage, which is what makes the expression integral. The recursion does not mention n, so the resulting polynomial s_k is independent of n. The convention $\sigma_i = 0$ for $i > n$ is what makes the same s_k serve every n. Evaluating for $k = 1, 2, 3$ gives $p_1 = \sigma_1$, $p_2 = \sigma_1^2 - 2\sigma_2$ and $p_3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3$, as claimed.

Core Algebra [5] proves a fundamental theorem on symmetric functions, but in the form that a symmetric *rational function* over a field is a rational function of the elementary symmetric functions. That statement is about a different ring and does not give the integrality above, which is the property the Chern character needs: the denominators appearing in ch must come from the explicit $1/k!$ and from nowhere else.

The second input concerns line bundles. The Chern character is built so as to convert tensor products into products, and that rests on the first Chern class converting tensor products into sums.

With both inputs in hand the definition can be made. The guiding formula is that for a sum of line bundles $L_1 \oplus \cdots \oplus L_n$ the Chern character should be $\sum_j e^{c_1(L_j)}$, since that is the unique expression converting $\oplus$ to $+$ and $\otimes$ to $\times$ simultaneously. The splitting principle reduces the general case to that one, and lemma 3.3.1 is what turns the resulting symmetric expression back into a formula in the Chern classes of E itself.

Definition 3.3.2: [Topological] Chern Character

Let X be a finite CW complex and $E \to X$ a complex vector bundle. The *Chern character* of E is

$$\mathrm{ch}(E) \;=\; \rho_{\mathbb{Q}}(\mathrm{rk}E) \;+\; \sum_{k\geq 1} \frac{1}{k!}\, s_k\big(c_1(E), \ldots, c_k(E)\big) \;\in\; H^{\mathrm{even}}(X;\mathbb{Q})$$

where $\mathrm{rk}E \in H^0(X;\mathbb{Z})$ is the rank read as a locally constant function and $\rho_{\mathbb{Q}}$ denotes the coefficient map $H^*(X;\mathbb{Z}) \to H^*(X;\mathbb{Q})$. with s_k the integral polynomials of lemma 3.3.1. The k-th summand lies in $H^{2k}(X;\mathbb{Q})$ and is written $\mathrm{ch}_k(E)$, so $\mathrm{ch}(E) = \sum_{k\geq 0} \mathrm{ch}_k(E)$ with $\mathrm{ch}_0(E) = \mathrm{rk}E$. The sum is finite because X is finite dimensional. Written out,

$$\mathrm{ch}(E) = \mathrm{rk}E + c_1 + \tfrac{1}{2}\big(c_1^2 - 2c_2\big) + \tfrac{1}{6}\big(c_1^3 - 3c_1c_2 + 3c_3\big) + \cdots$$

Equivalently, in terms of Chern roots: if $p\colon Y \to X$ is a map from a paracompact Y for which $p^*E \cong L_1 \oplus \cdots \oplus L_r$ splits into line bundles with Chern roots $x_j = c_1(L_j)$, then $p^*\big(k!\,\mathrm{ch}_k(E)\big) = \sum_{j=1}^r x_j^k$ in rational cohomology for every $k \geq 1$, which is the degreewise reading of the familiar $p^*\mathrm{ch}(E) = \sum_j e^{x_j}$, an expression that is purely formal on Y. The degreewise form is the one that applies: the splitting base need not be a finite complex, so the exponential series has not been given a meaning there.

The rank is carried as a class in $H^0(X;\mathbb{Z})$ rather than as an integer because it is only locally constant. Over a base with several components the bundles of non-constant rank are exactly the ones generating K^0, and the last clause of theorem 3.3.4 turns on that case: it is the reason ch_0 need not vanish on a reduced class there.

The bracket in the title marks the word this book drops. Three books in the series define a Chern character and all three are correct: *Intersection Theory* [6] defines an operator on Chow groups $A_*(X)_{\mathbb{Q}}$, and *Riemannian Geometry* [7] defines a de Rham class $\mathrm{tr}\exp(\frac{i}{2\pi}\Omega)$ built from the curvature of a connection. Neither construction is available here, since a compact space carries neither an algebraic structure nor a connection, and neither book can state the theorem this section is aimed at, which is a statement about K^0.

The two descriptions in the definition agree by construction: the elementary symmetric polynomials in the Chern roots are the Chern classes, so $\sum_j x_j^k = s_k(c_1, \ldots, c_k)$ by lemma 3.3.1, and dividing by $k!$ gives the degreewise identities just displayed. What needs checking is that the formula on the left, which involves no choices, is the one that governs, and the splitting principle gives the injectivity that makes an identity verified on $F(E)$ an identity on X.

Proposition 3.3.3: The Chern Character of a Line Bundle

For a complex line bundle L over a finite CW complex,

$$\mathrm{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \tfrac{1}{2}c_1(L)^2 + \tfrac{1}{6}c_1(L)^3 + \cdots$$

and for line bundles L, L' one has $\mathrm{ch}(L \otimes L') = \mathrm{ch}(L)\,\mathrm{ch}(L')$.

***Proof*:**
A line bundle has $c_i(L) = 0$ for $i \geq 2$ by [14, Chern Classes and the Total Chern Class], so $s_k(c_1, 0, \ldots, 0) = c_1^k$: the recursion of lemma 3.3.1 reduces to $p_k = \sigma_1 p_{k-1}$ when every σ_i with $i \geq 2$ vanishes, and $p_1 = \sigma_1$. Hence $\mathrm{ch}(L) = 1 + \sum_{k\geq 1} c_1(L)^k/k! = e^{c_1(L)}$. For the product, $c_1(L \otimes L') = c_1(L) + c_1(L')$ by [14, The First Chern Class is Additive on Tensor Products], so

$$\mathrm{ch}(L \otimes L') = e^{c_1(L)+c_1(L')} = e^{c_1(L)}e^{c_1(L')} = \mathrm{ch}(L)\mathrm{ch}(L')$$

the middle equality being the identity $e^{a+b} = e^a e^b$ for commuting elements of the rational cohomology ring, which holds because the exponential series is finite here, completing the proof.

Theorem 3.3.4: The Chern Character is a Natural Ring Homomorphism

Let X be a finite CW complex and let E, F be complex vector bundles over X. Then

1. $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$;
2. $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$;
3. $\mathrm{ch}(f^*E) = f^*\mathrm{ch}(E)$ for every map $f\colon Y \to X$ of finite CW complexes.

Consequently ch extends uniquely to a natural homomorphism of unital rings

$$\mathrm{ch}\colon K^0(X) \longrightarrow H^{\mathrm{even}}(X;\mathbb{Q}), \qquad \mathrm{ch}([E]-[F]) = \mathrm{ch}(E) - \mathrm{ch}(F)$$

which, for X connected, restricts to $\widetilde{K}^0(X) \to \bigoplus_{k\geq 1} H^{2k}(X;\mathbb{Q})$ on reduced classes. (Connectedness is needed: $\widetilde{K}^0$ is the kernel of restriction to the basepoint, which for disconnected X constrains the rank on one component only, so ch_0 need not vanish. See example 2.4.1 at S^0.)

***Proof*:**
Step 1: Naturality. The Chern classes are natural ([14, Naturality of Characteristic Classes]) and s_k has constant coefficients, so $\mathrm{ch}(f^*E) = \sum_k \frac{1}{k!}s_k(c_\bullet(f^*E)) = f^* \sum_k \frac{1}{k!}s_k(c_\bullet(E)) = f^*\mathrm{ch}(E)$. This is (3).

Step 2: Reduction to split bundles. Apply [14, Splitting Principle](2) to E: there is $q\colon F(E) \to X$ with q^*E a sum of line bundles and $q^*\colon H^*(X;\mathbb{Z}) \hookrightarrow H^*(F(E);\mathbb{Z})$ injective, and then $r\colon F(q^*F) \to F(E)$ splitting q^*F with r^* injective as well. Write $p = q \circ r$, so that p^*E and p^*F *both* split as sums of line bundles and p^* is injective on $H^*(-;\mathbb{Z})$. The two bundles have to be split one after the other in this way: splitting $E \oplus F$ in one step and then refining does not separate the summands, because vector bundles admit no cancellation.

The base X is paracompact, so [14, Splitting Principle] applies and [14, The Flag Bundle is Paracompact] makes each stage paracompact in turn; the Chern classes are therefore defined upstairs by [14, Classification of Vector Bundles over a Paracompact Base], and are natural under p^* by the same proposition. The Chern character is not: definition 3.3.2 requires a finite CW complex, the flag tower need not be one, and no vanishing of its cohomology in high degrees has been established that would make the exponential sums finite there. The proof therefore never forms ch upstairs. Instead, clauses (1) and (2) are proved one degree at a time, as identities among the classes $s_k(c_1,\dots,c_k)$ of lemma 3.3.1: each is a single class in $H^{2k}(-;\mathbb{Z})$, defined on every stage of the tower by [14, Classification of Vector Bundles over a Paracompact Base] and involving no infinite sum. The identities are verified after pulling back along p, where both bundles split; they descend to $H^{2k}(X;\mathbb{Z})$ along the injective p^*; and only then, on the finite complex X, do the passage to rational coefficients and the division by $k!$ produce the statements about ch.

Step 3: The split case, degree by degree. For a bundle V over a paracompact base write $s_k(V) = s_k\big(c_1(V),\dots,c_k(V)\big) \in H^{2k}(-;\mathbb{Z})$ for $k \geq 1$, the integral polynomial of lemma 3.3.1 evaluated on the Chern classes of [14, Classification of Vector Bundles over a Paracompact Base], and set $s_0(V) = \mathrm{rk}V$ in $H^0(-;\mathbb{Z})$, so that on the finite complex X the definition reads $k!\,\mathrm{ch}_k(V) = \rho_{\mathbb{Q}}\big(s_k(V)\big)$, with $\rho_{\mathbb{Q}}\colon H^{2k}(X;\mathbb{Z}) \to H^{2k}(X;\mathbb{Q})$ the coefficient homomorphism, a ring homomorphism. Since the polynomials s_k have constant coefficients and the Chern classes are natural, $p^*s_k(V) = s_k(p^*V)$ for every bundle V over X.

Upstairs write $p^*E \cong L_1 \oplus \cdots \oplus L_r$ and $p^*F \cong M_1 \oplus \cdots \oplus M_s$ with Chern roots $x_j = c_1(L_j)$ and $y_l = c_1(M_l)$. The Whitney formula ([14, Whitney Sum Formula]) makes the total Chern class of a direct sum the product of the totals, so $c_i(p^*E) = \sigma_i(x_1,\dots,x_r)$, and lemma 3.3.1 evaluates

$$s_k(p^*E) = \sum_{j=1}^{r} x_j^k, \qquad s_k(p^*F) = \sum_{l=1}^{s} y_l^k$$

while $p^*(E\oplus F) = p^*E\oplus p^*F$ has Chern roots the x_j together with the y_l, whence $s_k\big(p^*(E\oplus F)\big) = \sum_j x_j^k + \sum_l y_l^k = s_k(p^*E)+s_k(p^*F)$. By naturality this reads $p^*\big(s_k(E\oplus F)\big) = p^*\big(s_k(E)+s_k(F)\big)$, and the injectivity of p^* on $H^*(-;\mathbb{Z})$ gives

$$s_k(E\oplus F) = s_k(E) + s_k(F) \quad \text{in } H^{2k}(X;\mathbb{Z}) \text{ for every } k \geq 1$$

while the case $k = 0$ is the additivity of the rank. Applying $\rho_{\mathbb{Q}}$ and dividing by $k!$ in $H^{2k}(X;\mathbb{Q})$ yields $\mathrm{ch}_k(E\oplus F) = \mathrm{ch}_k(E) + \mathrm{ch}_k(F)$ for every $k \geq 0$, and the sum over the finitely many k with $H^{2k}(X;\mathbb{Q}) \neq 0$ is (1).

For (2) the identity to be established is integral: for every $k \geq 0$,

$$s_k(E\otimes F) \;=\; \sum_{i+i'=k} \binom{k}{i} s_i(E)\, s_{i'}(F) \quad \text{in } H^{2k}(X;\mathbb{Z}) \tag{3.2}$$

whose coefficients are integers, so both sides are defined without denominators. Upstairs $p^*(E\otimes F) = p^*E\otimes p^*F \cong \bigoplus_{j,l} L_j\otimes M_l$, and $c_1(L_j\otimes M_l) = x_j + y_l$ by [14, The First Chern Class is Additive on Tensor Products](1), whose paracompactness hypothesis is met by [14, The Flag Bundle is Paracompact], so the Chern roots of $p^*(E\otimes F)$ are the rs classes $x_j + y_l$ and

$$s_k\big(p^*(E\otimes F)\big) = \sum_{j,l}(x_j+y_l)^k = \sum_{i=0}^{k}\binom{k}{i}\Big(\sum_j x_j^i\Big)\Big(\sum_l y_l^{k-i}\Big) = \sum_{i+i'=k}\binom{k}{i} s_i(p^*E)\, s_{i'}(p^*F)$$

by the binomial theorem, which applies because the classes x_j, y_l have even degree and therefore commute; the extreme terms use $\sum_j x_j^0 = r = s_0(p^*E)$ and $\sum_l y_l^0 = s = s_0(p^*F)$. By naturality the two sides of (3.2) pull back along p^* to the two ends of this chain of equalities, so they agree after applying the injective p^*, hence on X. Applying $\rho_{\mathbb{Q}}$ and dividing by $k!$ turns $\binom{k}{i}/k!$ into $1/(i!\,i'!)$, giving

$$\mathrm{ch}_k(E \otimes F) = \sum_{i+i'=k} \mathrm{ch}_i(E)\,\mathrm{ch}_{i'}(F)$$

for every $k \geq 0$; these are exactly the homogeneous components of $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, which is (2). Both clauses were proved upstairs among finite integral expressions and descended to X by the injectivity of Step 2; at no point was ch formed on the tower.

Step 4: Extension to K^0. By (1), ch is a homomorphism of monoids from $(\mathrm{Vect}(X), \oplus)$ to the additive group of $H^{\mathrm{even}}(X;\mathbb{Q})$. The Grothendieck group is universal among such (definition 2.1.1), so ch extends uniquely to a group homomorphism on $K^0(X)$ with $\mathrm{ch}([E]-[F]) = \mathrm{ch}(E) - \mathrm{ch}(F)$; (2) makes the extension multiplicative for the ring structure of proposition 2.1.10, and $\mathrm{ch}(\mathbb{C}) = 1$ makes it unital. For the last clause, X is connected, so a reduced class has rank 0 by proposition 2.1.3, and $\mathrm{ch}_0 = \mathrm{rk}$, so its Chern character has vanishing degree-zero component, completing the proof.

So far ch is a ring map out of K^0 of a space. To compare it with cohomology as a *theory*, it has to be defined on the relative and negative groups and shown to respect the structure maps that tie those groups together. That is the content of the next proposition, and it is what the comparison theorem uses.

Proposition 3.3.5: Compatibility with Suspension and the Bott Map

Let (X, A) be a finite CW pair.

1. ch is defined on relative groups, $\mathrm{ch}\colon K^0(X, A) \to H^{\mathrm{even}}(X, A; \mathbb{Q})$, naturally in the pair, and commutes with the maps i^* and j^* of theorem 2.3.7.

(1′) ch is defined on the odd group. By definition 2.3.1, $K^{-1}(X) = \widetilde{K}^0(\Sigma X_+)$, and the suspension isomorphism of reduced rational cohomology gives $\widetilde{H}^{\mathrm{even}}(\Sigma X_+; \mathbb{Q}) \cong H^{\mathrm{odd}}(X; \mathbb{Q})$. Define

$$\mathrm{ch}\colon K^{-1}(X) \longrightarrow H^{\mathrm{odd}}(X; \mathbb{Q})$$

as the composite of ch on $\widetilde{K}^0(\Sigma X_+)$ with that isomorphism. This is the map meant whenever ch is applied to an odd class below, in corollary 3.3.6 and in theorem 3.3.7; with it, ch is a map of $\mathbb{Z}/2$-graded theories.

2. For the Bott generator $\beta = [H] - 1 \in \widetilde{K}^0(S^2)$ of definition 2.2.7,

$$\mathrm{ch}(\beta) = c_1(H) = -x$$

a generator of $H^2(S^2; \mathbb{Z})$, where $x = c_1(H^\vee)$.

3. ch intertwines the Bott map with the double suspension: for every finite pointed CW complex X the square formed by $\beta_X : \widetilde{K}^0(X) \to \widetilde{K}^0(\Sigma^2 X)$, the suspension isomorphism σ^2 of reduced rational cohomology, and ch on both rows commutes. Here σ^2 is normalized by the generator $\mathrm{ch}(\beta) = -x$ of $\widetilde{H}^2(S^2; \mathbb{Z})$, not by x. With the other normalization the square commutes up to the sign -1, which is why the choice is recorded rather than left implicit. Consequently ch commutes with the connecting map δ of theorem 2.3.7.

***Proof*:**

(1). By definition 2.3.3, $K^0(X, A) = \widetilde{K}^0(X/A)$, and X/A is again a finite CW complex, so definition 3.3.2 applies to it and lands in $\widetilde{H}^{\mathrm{even}}(X/A; \mathbb{Q}) \cong H^{\mathrm{even}}(X, A; \mathbb{Q})$, the identification being the one for a CW pair. Naturality and commutation with i^*, j^* are theorem 3.3.4(3) applied to the maps of spaces inducing them, since both i^* and j^* are induced by maps.

(2). By proposition 3.3.3, $\mathrm{ch}(\beta) = \mathrm{ch}(H) - \mathrm{ch}(\underline{\mathbb{C}}) = e^{c_1(H)} - 1 = c_1(H) + \frac{1}{2}c_1(H)^2 + \cdots$. On S^2 every term beyond the first vanishes, since $c_1(H)^2 \in H^4(S^2; \mathbb{Q}) = 0$. So $\mathrm{ch}(\beta) = c_1(H)$, which is $-x$ in this book's convention (example 2.1.12) and generates $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$.

(3). The external product in cohomology is $a \times b = \pi_X^* a \smile \pi_Y^* b$, and the same formula defines the external product in K-theory (definition 2.2.6). By theorem 3.3.4(2),(3),

$$\mathrm{ch}\big(\mu(a \otimes b)\big) = \mathrm{ch}(\pi_X^* a)\,\mathrm{ch}(\pi_Y^* b) = \pi_X^* \mathrm{ch}(a) \smile \pi_Y^* \mathrm{ch}(b) = \mathrm{ch}(a) \times \mathrm{ch}(b)$$

so ch carries the external product to the external product. Taking $a = \beta$ and restricting to reduced classes on the smash, which is where both products land by proposition 2.2.5, gives $\mathrm{ch}(\beta_X(\alpha)) = \mathrm{ch}(\beta) \times \mathrm{ch}(\alpha)$. Since $\mathrm{ch}(\beta)$ generates $\widetilde{H}^2(S^2; \mathbb{Z})$ by (2), external multiplication by it is a double suspension isomorphism on reduced rational cohomology, and it is the one this proposition calls σ^2: the normalization is fixed by $\mathrm{ch}(\beta)$ itself, so the square commutes on the nose. Normalizing instead by x would replace σ^2 with $-\sigma^2$ throughout, since $\mathrm{ch}(\beta) = -x$.

For the last clause, δ in K-theory is induced by the map $\partial\colon X/A \to \Sigma A_+$ of the Puppe sequence appearing in Step 1 of theorem 2.3.7, and the connecting map in ordinary cohomology is induced by the same map of spaces. Naturality, which is (1), therefore makes ch commute with both, completing the proof.

Corollary 3.3.6: The Chern Character on Even Spheres

For $k \geq 1$ the Chern character carries $\widetilde{K}^0(S^{2k}) \cong \mathbb{Z}$ isomorphically onto the integral lattice $\widetilde{H}^{2k}(S^{2k};\mathbb{Z})$ inside $\widetilde{H}^{2k}(S^{2k};\mathbb{Q})$, and $\widetilde{K}^{-1}(S^{2k}) = 0 = \widetilde{H}^{\mathrm{odd}}(S^{2k};\mathbb{Q})$. On odd spheres the two are interchanged.

***Proof*:**
By example 2.4.1, $\widetilde{K}^0(S^{2k})$ is infinite cyclic generated by the k-fold external power $\beta^{\times k}$, under $S^{2k} \cong S^2 \wedge \cdots \wedge S^2$, and that example obtains it as $\beta^{\times k} = \beta_{S^{2k-2}}(\beta^{\times(k-1)})$, an iterated Bott map. Applying proposition 3.3.5(3) $k-1$ times therefore gives $\mathrm{ch}(\beta^{\times k}) = \sigma^{2k-2}\mathrm{ch}(\beta)$, where σ^2 is the double suspension isomorphism; being induced by the same maps of spaces as its integral counterpart, it carries integral generators to integral generators. By (2), $\mathrm{ch}(\beta) = -x$ generates $H^2(S^2;\mathbb{Z})$, so $\mathrm{ch}(\beta^{\times k})$ generates $H^{2k}(S^{2k};\mathbb{Z})$. So ch sends a generator to a generator, which is the claim. The vanishing statements are example 2.4.1 and the cohomology of a sphere. The odd case follows by replacing S^{2k} with its suspension, completing the proof.

The comparison theorem now follows by the standard method for two cohomology theories agreeing on a point: build up the space one cell at a time and force the agreement along the exact sequences.

Theorem 3.3.7: The Chern Character is a Rational Isomorphism

For every finite CW complex X the Chern character induces isomorphisms

$$\mathrm{ch}\otimes\mathbb{Q}\colon K^0(X)\otimes_{\mathbb{Z}}\mathbb{Q} \xrightarrow{\sim} H^{\mathrm{even}}(X;\mathbb{Q}), \qquad K^{-1}(X)\otimes_{\mathbb{Z}}\mathbb{Q} \xrightarrow{\sim} H^{\mathrm{odd}}(X;\mathbb{Q})$$

the first an isomorphism of rings. Taken together they give an isomorphism of $\mathbb{Z}/2$-graded groups $K^*(X)\otimes\mathbb{Q} \cong H^*(X;\mathbb{Q})$.

***Proof*:**
Step 1: Both sides are cohomology theories, and the map is a transformation. On finite CW pairs the family $\{K^{-n}\}$ with its long exact sequence is a cohomology theory by theorem 2.3.6 and theorem 2.3.7, folded into two degrees by box 2.3.5. Tensoring with $\mathbb{Q}$ preserves exactness, since $\mathbb{Q}$ is flat over $\mathbb{Z}$, so $\{K^*(-)\otimes\mathbb{Q}\}$ is again one. Ordinary rational cohomology, folded by parity into H^{even} and H^{odd}, is a cohomology theory on the same pairs ([14, Singular Cohomology as an Ordinary Theory]). By proposition 3.3.5, $\mathrm{ch}\otimes\mathbb{Q}$ is a natural transformation between them commuting with the connecting maps.

Step 2: Agreement on a point. $K^0(\{*\}) = \mathbb{Z}$ by example 2.1.2 and $\mathrm{ch}_0 = \mathrm{rk}$ (proposition 2.1.3), so $\mathrm{ch}\otimes\mathbb{Q}\colon \mathbb{Q} \to H^0(\{*\};\mathbb{Q}) = \mathbb{Q}$ is the identity. In odd degree $K^{-1}(\{*\}) = 0 = H^{\mathrm{odd}}(\{*\};\mathbb{Q})$ by example 2.3.2. So the transformation is an isomorphism on a point, and by corollary 3.3.6 it is an isomorphism on every sphere.

Step 3: Induction over cells. Argue by induction on the number of cells of X. If X is a point, Step 2 applies. Otherwise write $X = X' \cup e^m$ for a subcomplex X' with one fewer cell, so that $X/X' \cong S^m$. The pair (X, X') has a long exact sequence in each theory, and by Step 1 the Chern character maps one to the other, giving a commutative ladder

$$\begin{tikzcd}
K^*(X,X')\otimes\mathbb{Q} \arrow[r] \arrow[d, "\mathrm{ch}"] & K^*(X)\otimes\mathbb{Q} \arrow[r] \arrow[d, "\mathrm{ch}"] & K^*(X')\otimes\mathbb{Q} \arrow[r] \arrow[d, "\mathrm{ch}"] & {} \\
H^*(X,X';\mathbb{Q}) \arrow[r] & H^*(X;\mathbb{Q}) \arrow[r] & H^*(X';\mathbb{Q}) \arrow[r] & {}
\end{tikzcd}$$

extending in both directions. The relative groups are those of $X/X' \cong S^m$, where the vertical map is an isomorphism by Step 2, and the groups of X' are handled by the inductive hypothesis. Four of every five consecutive vertical maps are therefore isomorphisms, and the five lemma ([14, The Five Lemma]) forces the fifth, which is the map for X. A disjoint union is handled by additivity, Axiom 5 of theorem 2.3.6, on both sides.

Step 4: The ring structure. The map is multiplicative by theorem 3.3.4(2), and tensoring with $\mathbb{Q}$ preserves that, so the isomorphism in even degree is one of rings. The $\mathbb{Z}/2$-graded statement is the two displayed isomorphisms taken together as groups; multiplicativity is claimed in even degree only, no product of two classes of $K^{-1}(X)$ having been computed above, completing the proof.

Corollary 3.3.8: Kernel, Ranks, and the Euler Characteristic

Let X be a finite CW complex. The kernel of $\mathrm{ch}\colon K^0(X) \to H^{\mathrm{even}}(X;\mathbb{Q})$ is exactly the torsion subgroup of $K^0(X)$, and

$$\mathrm{rk}_{\mathbb{Z}} K^0(X) = \dim_{\mathbb{Q}} H^{\mathrm{even}}(X;\mathbb{Q}), \qquad \mathrm{rk}_{\mathbb{Z}} K^{-1}(X) = \dim_{\mathbb{Q}} H^{\mathrm{odd}}(X;\mathbb{Q})$$

Consequently $\mathrm{rk}_{\mathbb{Z}} K^0(X) - \mathrm{rk}_{\mathbb{Z}} K^{-1}(X) = \chi(X)$, the Euler characteristic.

***Proof*:**
For a finitely generated abelian group A the kernel of $A \to A \otimes \mathbb{Q}$ is the torsion subgroup, and $K^0(X)$ is finitely generated because X is a finite complex. Since $\mathrm{ch} \otimes \mathbb{Q}$ is injective by theorem 3.3.7, a class dies under ch exactly when it dies in $K^0(X) \otimes \mathbb{Q}$, which is exactly when it is torsion. The rank statements are the dimensions of the two sides of the same theorem. For the last, $\chi(X) = \sum_i (-1)^i \dim_{\mathbb{Q}} H^i(X;\mathbb{Q})$ splits by parity into $\dim H^{\mathrm{even}} - \dim H^{\mathrm{odd}}$, completing the proof.

Example 3.3.9: The Chern Character on $\mathbb{C}\mathrm{P}^2$
Take $n = 2$, so $K^0(\mathbb{C}\mathrm{P}^2) \cong \mathbb{Z}[t]/(t^3)$ with $t = [H] - 1$, free on $1, t, t^2$ (theorem 2.4.5), and $H^*(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^3)$ with $|x| = 2$. By proposition 3.3.3 the Chern character of a line bundle is $\mathrm{ch}(L) = e^{c_1(L)}$, and $c_1(H) = -x$ because H is the tautological bundle and $x = c_1(H^\vee)$, so

$$\mathrm{ch}(H) = e^{-x} = 1 - x + \tfrac{1}{2}x^2$$

using $x^3 = 0$. Therefore

$$\mathrm{ch}(t) = \mathrm{ch}([H] - 1) = e^{-x} - 1 = -x + \tfrac{1}{2}x^2$$

and squaring (with $x^3 = 0$),

$$\mathrm{ch}(t^2) = \mathrm{ch}(t)^2 = \left(-x + \tfrac{1}{2}x^2\right)^2 = x^2$$

Thus on the basis $1, t, t^2$ the Chern character is

$$\mathrm{ch}(1) = 1, \qquad \mathrm{ch}(t) = -x + \tfrac{1}{2}x^2, \qquad \mathrm{ch}(t^2) = x^2$$

The matrix of ch whose columns express $\mathrm{ch}(1), \mathrm{ch}(t), \mathrm{ch}(t^2)$ in the basis $\{1, x, x^2\}$ is triangular,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & \frac{1}{2} & 1 \end{pmatrix}$$

with determinant -1, so ch is a rational isomorphism. Because the determinant is a unit, the image $\mathrm{ch}(K^0(\mathbb{C}\mathrm{P}^2))$ is a full-rank lattice in $H^{\mathrm{even}}(\mathbb{C}\mathrm{P}^2; \mathbb{Q})$ of the same covolume as $H^{\mathrm{even}}(\mathbb{C}\mathrm{P}^2; \mathbb{Z})$. The half-integer entry $\frac{1}{2}$ shows that the two lattices are nonetheless different, neither containing the other. Solving $a \cdot 1 + b(-x + \frac{1}{2}x^2) + c\,x^2 = x$ forces $b = -1$ and $c = \frac{1}{2} \notin \mathbb{Z}$, so the cohomology generator x is not in the image. This is the integrality phenomenon: K^0 and H^{even} are abstractly the same free group, but the Chern character identifies them only after inverting 2, and the half-integral discrepancy is the first appearance of the Todd genus that corrects the Riemann-Roch theorem.

The contrast with corollary 3.3.6 is the point. On an even sphere the Chern character hits the integral lattice exactly. On $\mathbb{C}\mathrm{P}^2$ it does not, and the obstruction is a single factor of 2.

3.3.10 Remark: What the Rational Isomorphism Does Not Say Three things are worth separating.
First, theorem 3.3.7 is an isomorphism only after tensoring with $\mathbb{Q}$. Example 3.3.9 exhibits a space where the integral map is not surjective, so the rational hypothesis is not an artifact of the proof.
Second, the theorem says nothing about torsion. By corollary 3.3.8 the whole torsion subgroup of $K^0(X)$ lies in the kernel of ch, so any information carried by torsion classes is invisible to cohomology through this map. For $X = \mathbb{R}\mathrm{P}^2$, for instance, $\widetilde{K}^0(\mathbb{R}\mathrm{P}^2)$ is a nonzero finite group while $\widetilde{H}^*(\mathbb{R}\mathrm{P}^2; \mathbb{Q}) = 0$, and ch is identically zero on reduced classes.
Third, the failure of integrality is measured rather than repaired. Section 3.4 bounds the denominators that can occur, chapter 4 measures the failure intrinsically through the γ-filtration, whose associated graded is compared to H^{2k} by the Adams operations, and **??** corrects for it with the Todd genus in the Riemann-Roch formula. The half-integer $\frac{1}{2}$ in example 3.3.9 is the first term of that correction.

Exercise 3.3.1

1. The Euler sequence gives $T\mathbb{C}\mathrm{P}^n \oplus \underline{\mathbb{C}} \cong (H^\vee)^{\oplus(n+1)}$. Use theorem 3.3.4 and proposition 3.3.3 to show $\mathrm{ch}(T\mathbb{C}\mathrm{P}^n) = (n+1)e^x - 1$ with $x = c_1(H^\vee)$, and write out the terms through degree 6. Check that $\mathrm{ch}_0(T\mathbb{C}\mathrm{P}^n) = n$ and that $\mathrm{ch}_1(T\mathbb{C}\mathrm{P}^n)$ agrees with the first Chern class computed in exercise 3.2.1 (2).
2. From lemma 3.3.1, derive $\mathrm{ch}_1 = c_1$, $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ and $\mathrm{ch}_3 = \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3)$. Deduce

that $k!\,\mathrm{ch}_k(E)$ is an integral class for every bundle E, and exhibit a space and a line bundle for which the denominator of ch_k is exactly $k!$.

3. Using theorem 3.3.7, compute $\mathrm{rk}_{\mathbb{Z}} K^0$ and $\mathrm{rk}_{\mathbb{Z}} K^{-1}$ for $S^2 \times S^4$, for $\mathbb{C}\mathrm{P}^n \times \mathbb{C}\mathrm{P}^m$, and for a closed orientable surface Σ_g of genus g. Reconcile the first with what theorem 2.4.8 gives directly.
4. Show that ch carries $\widetilde{K}^0(S^{2k})$ *onto* the integral lattice $\widetilde{H}^{2k}(S^{2k};\mathbb{Z})$, but that $\mathrm{ch}(K^0(\mathbb{C}\mathrm{P}^2))$ does not contain $H^{\mathrm{even}}(\mathbb{C}\mathrm{P}^2;\mathbb{Z})$. Determine the smallest n for which the cohomology generator $x \in H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ fails to lie in $\mathrm{ch}(K^0(\mathbb{C}\mathrm{P}^n))$, and say what goes right for smaller n.
5. Prove that $\ker\big(\mathrm{ch}\colon K^0(X) \to H^{\mathrm{even}}(X;\mathbb{Q})\big)$ is the torsion subgroup, for X a finite CW complex. Deduce that if $H^{\mathrm{odd}}(X;\mathbb{Q}) = 0$ then $K^{-1}(X)$ is a finite group. Then give a space for which ch is *not* injective on K^0, and identify the classes it kills.
6. Using [14, Splitting Principle](2) and $c_1(L^\vee) = -c_1(L)$, prove that $\mathrm{ch}_k(E^\vee) = (-1)^k \mathrm{ch}_k(E)$ for every complex bundle E over a finite CW complex. Deduce that the degree-2 component of $\mathrm{ch}(E \otimes E^\vee)$ vanishes, and say what that means about $c_1(E \otimes E^\vee)$.
7. From proposition 3.3.5, show that $\mathrm{ch}(\beta)$ generates $H^2(S^2;\mathbb{Z})$ and that ch carries external multiplication by β to the double suspension isomorphism. Use this to recompute ch on $\widetilde{K}^0(S^{2k})$ without appealing to corollary 3.3.6, and state which sign your answer carries under this book's convention for H.

3.4 Integrality and the Image of the Chern Character

Theorem 3.3.7 identifies $K^0(X) \otimes \mathbb{Q}$ with $H^{\mathrm{even}}(X;\mathbb{Q})$, and example 3.3.9 shows the identification does not descend to the integral lattices. That failure is the subject here. It is not a defect of the construction: ch is built out of the polynomials s_k of lemma 3.3.1 divided by $k!$, so denominators are present from the definition, and the question is which ones are realized. The answer bounds them by $n!$ on a complex of dimension $2n$, exhibits a space where that bound is attained, and isolates the first degree in which integrality can fail.

The starting point is that although ch_k is a rational combination of Chern classes, its leading term is controlled: modulo products of lower classes, only one denominator survives.

Proposition 3.4.1: The Leading Term of the Chern Character

Let $E \to X$ be a complex vector bundle over a finite CW complex and let $k \geq 1$. Then

$$\mathrm{ch}_k(E) \;\equiv\; \frac{(-1)^{k-1}}{(k-1)!}\, c_k(E)$$

modulo the $\mathbb{Q}$-subspace of $H^{2k}(X;\mathbb{Q})$ spanned by products $c_{i_1}(E)\cdots c_{i_r}(E)$ with $r \geq 2$, each $i_j \geq 1$, and $i_1 + \cdots + i_r = k$. (The span must be taken over $\mathbb{Q}$, not over $\mathbb{Z}$: the error term carries the denominator $k!$, as ch_2 on a bundle with $c_2 = 0$ already shows.) In particular $\mathrm{ch}_1(E) = c_1(E)$ and $\mathrm{ch}_2(E) = \frac{1}{2}c_1(E)^2 - c_2(E)$.

***Proof*:**
By definition 3.3.2, $\mathrm{ch}_k(E) = \frac{1}{k!}s_k(c_1(E), \ldots, c_k(E))$ with s_k the integral polynomial of lemma 3.3.1 expressing the k-th power sum p_k in the elementary symmetric functions. Solving Newton's identity (3.1) for p_k, whose coefficient there is $(-1)^{k-1}\sigma_0 = \pm 1$ and in particular a unit, gives

$$p_k \;=\; \sigma_1 p_{k-1} - \sigma_2 p_{k-2} + \cdots + (-1)^k \sigma_{k-1} p_1 + (-1)^{k-1} k\, \sigma_k$$

Every term but the last carries a factor σ_i with $1 \le i \le k-1$, and p_{k-i} is itself an integral polynomial in $\sigma_1, \ldots, \sigma_{k-i}$ by lemma 3.3.1, so every such term is a $\mathbb{Z}$-linear combination of monomials $\sigma_{i_1} \cdots \sigma_{i_r}$ with $r \ge 2$. Hence $s_k = (-1)^{k-1} k\, y_k + (\text{decomposable monomials})$. Substituting $c_i(E)$ for y_i and dividing by $k!$,

$$\mathrm{ch}_k(E) = \frac{(-1)^{k-1} k\, c_k(E)}{k!} + (\text{decomposables}) = \frac{(-1)^{k-1}}{(k-1)!} c_k(E) + (\text{decomposables})$$

For $k = 1$ the decomposable part is empty and $s_1 = y_1$, giving $\mathrm{ch}_1(E) = c_1(E)$; for $k = 2$, $s_2 = y_1^2 - 2y_2$ gives $\mathrm{ch}_2(E) = \frac{1}{2}(c_1(E)^2 - 2c_2(E)) = \frac{1}{2}c_1(E)^2 - c_2(E)$, as we sought to show.

The proposition says the Chern character and the Chern classes carry the same information one degree at a time, but with a denominator that grows: in degree $2k$ the comparison costs a factor $(k-1)!$. Since $k!\,\mathrm{ch}_k(E) = s_k(c_1(E), \ldots, c_k(E))$ is an integral class outright, the denominators are bounded by $k!$, and on a finite complex only finitely many degrees occur. That is the content of theorem 3.4.2, which also extends the statement from bundles to arbitrary K-theory classes.

Theorem 3.4.2: The Denominators of the Chern Character are Bounded

Let X be a finite CW complex with $H^{2k}(X;\mathbb{Q}) = 0$ for all $k > n$. Then for every $\xi \in K^0(X)$ the class $n! \cdot \mathrm{ch}(\xi)$ lies in the image of the coefficient map $H^{\mathrm{even}}(X;\mathbb{Z}) \to H^{\mathrm{even}}(X;\mathbb{Q})$. The same holds with n replaced by any $m \ge n$.

***Proof*:**
Step 1: Bundles. Let $E \to X$ be a complex vector bundle and $1 \le k \le n$. By definition 3.3.2, $k!\,\mathrm{ch}_k(E) = s_k(c_1(E), \ldots, c_k(E))$, and s_k has integer coefficients by lemma 3.3.1, so $k!\,\mathrm{ch}_k(E)$ is the image of an integral class, namely the same polynomial evaluated on the integral Chern classes of [14, Chern Classes and the Total Chern Class]. Since $k \le n$, the integer $n!/k!$ is a positive integer, and

$$n!\,\mathrm{ch}_k(E) \;=\; \frac{n!}{k!} \cdot k!\,\mathrm{ch}_k(E)$$

is again the image of an integral class. For $k > n$ the group $H^{2k}(X;\mathbb{Q})$ vanishes, so $\mathrm{ch}_k(E) = 0$ there and contributes nothing. Summing over $0 \le k \le n$, and noting $\mathrm{ch}_0(E) = \mathrm{rk}E$ is an integer, gives that $n!\,\mathrm{ch}(E)$ is integral.

Step 2: Arbitrary classes. Every $\xi \in K^0(X)$ is a difference $[E] - [F]$ of bundle classes by the Grothendieck construction of definition 2.1.1, and ch is additive by theorem 3.3.4(1), so $n!\,\mathrm{ch}(\xi) = n!\,\mathrm{ch}(E) - n!\,\mathrm{ch}(F)$ is a difference of two integral classes, hence integral. The final clause holds because $k!$ divides $m!$ for every $k \le n \le m$, completing the proof.

Two remarks on the shape of this statement. It is a bound on denominators, not a vanishing statement: it does not say ch is integral after any smaller multiple, and example 3.4.6 shows $n!$ cannot be improved in general. And it is stated over the image of integral cohomology rather than over $H^{\mathrm{even}}(X;\mathbb{Z})$ itself, because ch takes values in rational cohomology, so "integral" can only mean lying in the image of the coefficient map.

With the denominators bounded, the image of ch becomes a lattice that can be compared with the integral one, and the comparison is by a single finite number.

Definition 3.4.3: Integrality Defect

Let X be a finite CW complex. Write $V = H^{\mathrm{even}}(X;\mathbb{Q})$, let

$$L(X) = \mathrm{ch}\big(K^0(X)\big) \subseteq V, \qquad L_0(X) = \mathrm{im}\left(H^{\mathrm{even}}(X;\mathbb{Z}) \to V\right) \subseteq V$$

The *integrality defect* of X is the index

$$\delta(X) = \left[\, L(X) + L_0(X) \ :\ L_0(X) \,\right]$$

which proposition 3.4.4 shows is finite. It equals 1 exactly when $L(X) \subseteq L_0(X)$, that is, exactly when the Chern character of every class in $K^0(X)$ is integral.

Proposition 3.4.4: The Image of the Chern Character is a Full-Rank Lattice

Let X be a finite CW complex. Then $L(X)$ and $L_0(X)$ are finitely generated free subgroups of $V = H^{\mathrm{even}}(X;\mathbb{Q})$, both of rank $\dim_{\mathbb{Q}} V$, and $\delta(X)$ is finite.

***Proof*:**
X is a finite complex, so $H^{\mathrm{even}}(X;\mathbb{Z})$ is finitely generated and $L_0(X)$, being its image in a $\mathbb{Q}$-vector space, is finitely generated and torsion-free, hence free. Its rank is $\dim_{\mathbb{Q}} V$ because $H^{\mathrm{even}}(X;\mathbb{Z}) \otimes \mathbb{Q} \cong V$. For $L(X)$, choose n with $H^{2k}(X;\mathbb{Q}) = 0$ for $k > n$, which is possible since X is finite dimensional. By theorem 3.4.2, $L(X) \subseteq \frac{1}{n!}L_0(X)$, a free abelian group of rank $\dim_{\mathbb{Q}} V$; a subgroup of a free abelian group of finite rank is free of no larger rank, so $L(X)$ is free of rank at most $\dim_{\mathbb{Q}} V$. It has rank exactly $\dim_{\mathbb{Q}} V$ because theorem 3.3.7 makes $\mathrm{ch} \otimes \mathbb{Q}$ surjective, so $L(X)$ spans V over $\mathbb{Q}$. (This route avoids any appeal to finite generation of $K^0(X)$.)

Both lattices have rank equal to $\dim_{\mathbb{Q}} V$, so $L(X) + L_0(X)$ is a finitely generated subgroup of V containing $L_0(X)$ with the same rank. A containment of free abelian groups of equal finite rank has finite index, so $\delta(X) < \infty$, as we sought to show.

The defect is a single integer measuring how far the Chern character is from carrying $K^0(X)$ into integral cohomology. It is finite by proposition 3.4.4 and bounded by the denominators of theorem 3.4.2, and it is a homotopy invariant, since ch is natural (theorem 3.3.4(3)) and both K^0 and H^* are homotopy functors. The two extremes are worth seeing side by side.

Example 3.4.5: The Defect of an Even Sphere
Let $X = S^{2k}$ with $k \geq 1$. Then $H^{\mathrm{even}}(S^{2k};\mathbb{Q})$ has basis the classes 1 and a generator u of H^{2k}, and $K^0(S^{2k}) \cong \mathbb{Z} \oplus \widetilde{K}^0(S^{2k}) \cong \mathbb{Z} \oplus \mathbb{Z}$. By corollary 3.3.6 the Chern character carries $\widetilde{K}^0(S^{2k})$

isomorphically onto the integral lattice $\widetilde{H}^{2k}(S^{2k};\mathbb{Z})$, and on the rank summand it is the identity $\mathrm{ch}_0 = \mathrm{rk}$. Hence $L(S^{2k}) = L_0(S^{2k})$ and

$$\delta(S^{2k}) = 1$$

Every K-theory class on an even sphere has integral Chern character. What proposition 3.4.1 contributes is a reduction, not the answer: on S^{2k} every decomposable product $c_{i_1}\cdots c_{i_r}$ with $r \geq 2$ and each $i_j \geq 1$ vanishes, since each factor lies in a group $H^{2i_j}(S^{2k};\mathbb{Z})$ with $0 < i_j < k$, which is zero. So the leading term is exact, $\mathrm{ch}_k(E) = (-1)^{k-1}c_k(E)/(k-1)!$, and integrality is equivalent to the divisibility $(k-1)!\big|c_k(E)$. That divisibility is not formal, and for $k \geq 3$ it is a genuine constraint on which bundles exist over S^{2k}; what gives it here is corollary 3.3.6, which rests on Bott periodicity.

The projective spaces behave in the opposite way, and they show that the bound of theorem 3.4.2 is attained.

Example 3.4.6: The Defect of $\mathbb{C}\mathrm{P}^n$

Let $x = c_1(H^\vee)$ be the generator of $H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ used in theorem 2.4.5, so $H^{\mathrm{even}}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1})$ with basis $1, x, \ldots, x^n$, and $K^0(\mathbb{C}\mathrm{P}^n) = \mathbb{Z}[t]/(t^{n+1})$ with basis $1, t, \ldots, t^n$ for $t = [H] - 1$. Since $c_1(H) = -x$, proposition 3.3.3 gives

$$\mathrm{ch}(t) = e^{-x} - 1 = -x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots + \frac{(-1)^n x^n}{n!}$$

and $\mathrm{ch}(t^m) = \left(e^{-x} - 1\right)^m$ by theorem 3.3.4(2).

The two lattices have the same covolume. The lowest-degree term of $(e^{-x}-1)^m$ is $(-1)^m x^m$, so the matrix of ch from the basis $1, t, \ldots, t^n$ to the basis $1, x, \ldots, x^n$ is triangular with diagonal entries $(-1)^m$ for $m = 0, \ldots, n$. Its determinant is $(-1)^{0+1+\cdots+n} = (-1)^{n(n+1)/2}$, a unit. So $L(\mathbb{C}\mathrm{P}^n)$ and $L_0(\mathbb{C}\mathrm{P}^n)$ are full-rank lattices of equal covolume, and neither need contain the other. For $n = 2$ this is the matrix computed in example 3.3.9.

The defect is divisible by $n!$. Work in the quotient $V/L_0(\mathbb{C}\mathrm{P}^n)$, which is $\bigoplus_{k=0}^n (\mathbb{Q}/\mathbb{Z})x^k$. This group is infinite, but it is a torsion group, and that is all the argument uses: every element has finite order. The class of $\mathrm{ch}(t)$ there has x-component $-1 \equiv 0$ and x^k-component $(-1)^k/k!$ for $2 \leq k \leq n$, so its order is $\mathrm{lcm}(2!, 3!, \ldots, n!) = n!$, since $k!$ divides $n!$ for every $k \leq n$. That order divides $\delta(\mathbb{C}\mathrm{P}^n)$, so

$$n! \mid \delta(\mathbb{C}\mathrm{P}^n)$$

In particular $\mathrm{ch}(t)$ has x^n-coefficient $(-1)^n/n!$, so no integer smaller than $n!$ clears the denominators of every class: the bound in theorem 3.4.2 is sharp.

The divisibility is strict in general. Equality $\delta(\mathbb{C}\mathrm{P}^n) = n!$ holds for $n \leq 3$ but fails after that: a direct computation gives $\delta(\mathbb{C}\mathrm{P}^4) = 72$ against $4! = 24$, the extra factor of 3 coming from the x^4-coefficient $\frac{7}{12}$ of $\mathrm{ch}(t^2)$. So $n!$ is a lower bound for the defect and an upper bound for the denominators, and the two need not agree.

Small cases. For $n = 1$, $\mathrm{ch}(t) = -x$ and $\delta(\mathbb{C}\mathrm{P}^1) = 1$, consistent with $\mathbb{C}\mathrm{P}^1 = S^2$ and example 3.4.5. For $n = 2$, $\mathrm{ch}(t) = -x + x^2/2$ and $\mathrm{ch}(t^2) = x^2$, so $L = \mathbb{Z}\langle 1, -x + \frac{1}{2}x^2, x^2\rangle$. Modulo L_0 the generator $\mathrm{ch}(t)$ has order 2 and the others are integral, giving $\delta(\mathbb{C}\mathrm{P}^2) = 2$. This is the "single factor of 2" recorded in example 3.3.9.

The contrast between the two examples is not about the size of the space but about the shape of its cohomology ring. An even sphere has nothing to multiply; $\mathbb{C}\mathrm{P}^n$ has a generator in degree 2 whose powers fill every even degree, and it is exactly the powers of a degree-2 class that proposition 3.3.3 divides by $k!$. The next proposition makes the mildest form of this precise: when there are no degrees to fill, the defect vanishes.

Proposition 3.4.7: Integrality in Low Cohomological Dimension

Let X be a finite CW complex with $H^{2k}(X;\mathbb{Q}) = 0$ for all $k \geq 2$. Then $\delta(X) = 1$: the Chern character of every class in $K^0(X)$ is integral. Explicitly, writing $\xi = [E] - [F]$,

$$\mathrm{ch}(\xi) = \big(\mathrm{rk}E - \mathrm{rk}F\big) + \big(c_1(E) - c_1(F)\big)$$

both terms being images of integral classes.

***Proof*:**
Let $\xi \in K^0(X)$. Its Chern character decomposes as $\mathrm{ch}(\xi) = \sum_{k\geq 0} \mathrm{ch}_k(\xi)$ with $\mathrm{ch}_k(\xi) \in H^{2k}(X;\mathbb{Q})$ by definition 3.3.2. By hypothesis $H^{2k}(X;\mathbb{Q}) = 0$ for $k \geq 2$, so every term with $k \geq 2$ vanishes and $\mathrm{ch}(\xi) = \mathrm{ch}_0(\xi) + \mathrm{ch}_1(\xi)$. Writing $\xi = [E] - [F]$, the degree-zero term is $\mathrm{rk}E - \mathrm{rk}F \in \mathbb{Z}$, and by proposition 3.4.1 the degree-two term is $c_1(E) - c_1(F)$, the image of an integral class under $H^2(X;\mathbb{Z}) \to H^2(X;\mathbb{Q})$. Hence $\mathrm{ch}(\xi) \in L_0(X)$ for every ξ, so $L(X) \subseteq L_0(X)$ and $\delta(X) = 1$, completing the proof.

Beyond that range the first obstruction appears in degree four, where proposition 3.4.1 gives $\mathrm{ch}_2 = \frac{1}{2}c_1^2 - c_2$ and the only possible denominator is a single factor of 2. It is worth recording exactly when that factor is real, since this is the case that occurs in practice.

Corollary 3.4.8: The Degree-Four Criterion

Let X be a finite CW complex with $H^*(X;\mathbb{Z})$ torsion-free, and let $E \to X$ be a complex vector bundle. Then $\mathrm{ch}_2(E)$ is integral if and only if

$$c_1(E)^2 \in 2\,H^4(X;\mathbb{Z})$$

***Proof*:**
Since $H^*(X;\mathbb{Z})$ is torsion-free, the coefficient map $H^4(X;\mathbb{Z}) \to H^4(X;\mathbb{Q})$ is injective and its image is L_0 in degree four, so "integral" means lying in that image and no torsion ambiguity arises. By proposition 3.4.1, $2\,\mathrm{ch}_2(E) = c_1(E)^2 - 2c_2(E)$. If $\mathrm{ch}_2(E)$ is integral then $c_1(E)^2 = 2\mathrm{ch}_2(E) + 2c_2(E) \in 2H^4(X;\mathbb{Z})$. Conversely if $c_1(E)^2 = 2\alpha$ for some $\alpha \in H^4(X;\mathbb{Z})$ then $\mathrm{ch}_2(E) = \alpha - c_2(E)$ is integral, as we sought to show.

The criterion separates the two examples above at a glance. On S^4 the group H^2 vanishes, so $c_1(E) = 0$ and the condition holds trivially, recovering example 3.4.5 in the first degree where it could fail. On $\mathbb{C}\mathrm{P}^2$ the tautological bundle H has $c_1(H) = -x$ and $c_1(H)^2 = x^2$, which generates $H^4(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}$ and so is not twice anything. This is the failure computed in example 3.3.9, now identified as the only way integrality can break in degree four.

Exercise 3.4.1

1. Show that the congruence in proposition 3.4.1 needs the $\mathbb{Q}$-span and would be false over the $\mathbb{Z}$-span: exhibit a space X, a bundle E and a degree k for which $\mathrm{ch}_k(E) - \frac{(-1)^{k-1}}{(k-1)!}c_k(E)$ is not a $\mathbb{Z}$-linear combination of products of Chern classes. (Try $k = 2$ on $\mathbb{C}\mathrm{P}^2$ with a bundle whose c_2 vanishes.)

2. Compute the integrality defect $\delta(\mathbb{C}\mathrm{P}^3)$. (Expand $\mathrm{ch}(t^m)$ for $m = 1, 2, 3$ modulo x^4, identify the lattice $L(\mathbb{C}\mathrm{P}^3)$, and compute the order of its image in V/L_0.) Compare your answer with the divisibility $n!\big|\delta(\mathbb{C}\mathrm{P}^n)$ established in example 3.4.6.

3. Let X be a finite CW complex with $H^*(X;\mathbb{Z})$ torsion-free and let $L \to X$ be a complex line bundle. Prove that $\mathrm{ch}(L)$ is integral if and only if $c_1(L)^k \in k!\, H^{2k}(X;\mathbb{Z})$ for every $k \geq 1$. Deduce that for $n \geq 2$ the tautological bundle $H \to \mathbb{C}\mathrm{P}^n$ fails this at $k = 2$, and say why $n = 1$ is different.

4. Show that $\delta(S^2 \times S^2) = 1$. (Use theorem 3.3.4(2) and corollary 3.3.6 to compute ch on a basis of $K^0(S^2 \times S^2)$, then compare with the basis $1, a, b, ab$ of $H^{\mathrm{even}}(S^2 \times S^2; \mathbb{Z})$.) Contrast this with $\mathbb{C}\mathrm{P}^2$: both are closed simply connected 4-manifolds with cohomology in even degrees only, yet the defect is 1 here and 2 there. Explain the difference in terms of how each degree-4 generator is built from degree-2 classes.

5. Prove that the integrality defect is a homotopy invariant: if $f\colon X \to Y$ is a homotopy equivalence of finite CW complexes then $\delta(X) = \delta(Y)$. Then show that $\delta(X) = 1$ for every finite wedge of even-dimensional spheres, using lemma 2.2.2.

3.5 Characteristic Numbers and Cobordism

The classes of section 3.2 live in a cohomology group that depends on the base, so comparing two bundles means comparing classes in two different groups. On a closed manifold there is a way out: pair a product of classes of the tangent bundle against the fundamental class and read off a single number. Those numbers are [14, Stiefel-Whitney Number] and [14, Pontryagin Number], written $w_I[M] \in \mathbb{F}_2$ for a partition I of $\dim M$ and $p_I[M] \in \mathbb{Z}$ for a partition of $\dim M/4$; the fundamental class they are paired against is [14, [Manifold] Fundamental Class].

Both are cobordism invariants, and the reason is short: over a manifold W with $\partial W = M$ the relevant classes extend, while $[M]$ is a boundary, so the pairing vanishes by exactness. What is deep is the converse. For unoriented cobordism the Stiefel-Whitney numbers are a *complete* invariant, which is [14, Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant]; in the oriented case the corresponding statement is only rational, and is read off the structure of Ω_*^{SO} in [14, Oriented Bordism and the Spectrum MSO]. Both rest on Thom's theorem, [14, Thom's Theorem].

This is where characteristic classes stop being local measurements of twisting and become global invariants strong enough to classify manifolds, and it is the last the book has to say about the classes themselves.

Example 3.5.1: Pontryagin Number of $\mathbb{C}\mathrm{P}^2$

The manifold $\mathbb{C}\mathrm{P}^2$ is closed, oriented, of real dimension $4 = 4 \cdot 1$, so it has a single Pontryagin number, indexed by the partition $I = (1)$ of $k = 1$. From example 3.2.3, $p_1(T\mathbb{C}\mathrm{P}^2) = 3a^2$ with $\langle a^2, [\mathbb{C}\mathrm{P}^2] \rangle = 1$. Therefore

$$p_1[\mathbb{C}\mathrm{P}^2] = \langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2] \rangle = \langle 3a^2, [\mathbb{C}\mathrm{P}^2] \rangle = 3$$

This is consistent with the Hirzebruch signature theorem ([13]), which in dimension 4 reads $\sigma(M) = \frac{1}{3} p_1[M]$: the signature of $\mathbb{C}\mathrm{P}^2$ is $\sigma(\mathbb{C}\mathrm{P}^2) = 1$ (its intersection form on H^2 is the rank-one form $\langle a, a \rangle = 1$), and indeed $\frac{1}{3} \cdot 3 = 1$. Since $p_1[\mathbb{C}\mathrm{P}^2] = 3 \neq 0$, the manifold $\mathbb{C}\mathrm{P}^2$ is not the boundary of any compact oriented 5-manifold: it represents a nonzero class in the oriented cobordism group $\Omega_4^{SO} \cong \mathbb{Z}$.

Exercise 3.5.1

1. Let M be a closed oriented 4-manifold all of whose Stiefel-Whitney numbers and Pontryagin numbers vanish. What does [14, Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant] allow you to conclude about M, and what can it not conclude in general? Then show that in dimension 4 the gap closes, and identify what would have to be true in higher dimensions for it to reopen.
2. Compute every Stiefel-Whitney number of $\mathbb{R}\mathrm{P}^2$, and deduce that $\mathbb{R}\mathrm{P}^2$ does not bound a compact 3-manifold. (Use $T\mathbb{R}\mathrm{P}^2 \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus 3}$ with $w(\gamma^1) = 1 + a$ and $H^*(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}[a]/(a^3)$.)
3. Show that every closed orientable 3-manifold has all Stiefel-Whitney numbers zero, and conclude from [14, Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant] that it bounds. Explain why the Pontryagin numbers give no information here.
4. Compute the total Pontryagin class $p(T\mathbb{C}\mathrm{P}^3)$ and all Pontryagin numbers of $\mathbb{C}\mathrm{P}^3$. Reconcile your answer with the fact that $\mathbb{C}\mathrm{P}^3$ has odd complex dimension.

4

Adams Operations and the Hopf Invariant

Chapter 3 compared K^0 with ordinary cohomology through the Chern character, and the comparison became an isomorphism only after tensoring with $\mathbb{Q}$. Everything of finite order was lost in the passage. What survives integrally, and what makes K-theory sharper than cohomology rather than a repackaging of it, is a family of operations on $K^0(X)$ that the ring structure alone does not see. They all come from one source, the exterior powers of a bundle, and this chapter is the unfolding of that source into a tool strong enough to settle a question about $\mathbb{R}^n$ that no cohomological argument reaches.

Section 4.1 carries the exterior powers from bundles to virtual classes. Section 4.2 converts them into the Adams operations ψ^k, which are ring homomorphisms and compose by $\psi^k\psi^l = \psi^{kl}$, and which act on a line bundle by $L \mapsto L^{\otimes k}$. Section 4.3 builds a filtration from a reindexing of the same operations and measures, intrinsically and integrally, the failure of the Chern character to be onto. **??** computes ψ^k on spheres and projective spaces, where it turns out to be multiplication by an explicit power of k. **??** spends that computation: the Adams-Atiyah theorem, the resulting list of spheres admitting an H-space structure, and the dimension theorem for real division algebras.

4.1 Exterior Powers and the λ-Ring Structure on K^0

Definition 1.1.17 already builds $\Lambda^k E$ for a vector bundle E, and the assignment $E \mapsto \Lambda^k E$ respects isomorphism, so it descends to a function on isomorphism classes. The difficulty is that it does *not* descend to $K^0(X)$ in the obvious way, because it is not additive: $\Lambda^k(E \oplus F)$ is not $\Lambda^k E \oplus \Lambda^k F$, and there is no sensible value to assign to Λ^k of a formal difference by working one degree at a time.

What resolves this is a change of target. The individual Λ^k are not additive, but the *total* exterior power, assembled into a power series

$$\lambda_t(E) \;=\; \sum_{k\geq 0}[\Lambda^k E]\, t^k$$

is *exponential*: it converts $\oplus$ into multiplication. Power series with constant term 1 form a group under multiplication, and a monoid homomorphism into a group is exactly what the Grothendieck construction extends. So λ_t extends to $K^0(X)$ for the same formal reason that the rank does, and the operations λ^k on virtual classes are read off as its coefficients. The construction is forced rather than chosen.

Two facts about exterior powers proved in chapter 1 are used throughout and are worth recalling by name: Λ^k does not distinguish isomorphic bundles (lemma 1.1.18), so it descends to isomorphism classes, and the Whitney formula $\Lambda^k(E\oplus F)\cong\bigoplus_{i+j=k}\Lambda^i E\otimes\Lambda^j F$ is an isomorphism of bundles (proposition 1.1.19). The second is what this section computes with. The first is what lets it be applied to a class, not a bundle.

Definition 4.1.1: The λ-Operations on Bundles

Let X be a compact Hausdorff space and $E \to X$ a complex vector bundle. For $k \geq 0$ write

$$\lambda^k(E) \;=\; [\Lambda^k E] \in K^0(X)$$

for the class of the k-th exterior power of definition 1.1.17, and define the *total λ-operation*

$$\lambda_t(E) \;=\; \sum_{k\geq 0}\lambda^k(E)\, t^k \;\in\; K^0(X)[[t]]$$

Three values are immediate. The bundle $\Lambda^0 E$ is *trivial*. The reason is not that its fibres are one-dimensional, since the Möbius bundle has one-dimensional fibres and is not trivial. It is that the gluing functions $\Lambda^0 g_{\beta\alpha}$ of definition 1.1.17 are all the identity of the ground field, which is exactly a trivialization. Hence $\lambda^0(E)=1$. Next, $\Lambda^1 E = E$, so $\lambda^1(E)=[E]$. Last, $\Lambda^k V = 0$ once k exceeds $\dim V$ ([5, Bases of the Exterior Powers]). This last fact bounds the degree, but the bookkeeping deserves a word, because the rank of a bundle over a compact Hausdorff space need not be constant.

Lemma 4.1.2: The Total λ-Operation is a Polynomial

Let $E\to X$ be a complex vector bundle over a compact Hausdorff space. The rank function $x\mapsto \dim_{\mathbb{C}} E_x$ is locally constant and takes finitely many values. Let N be its maximum. Then $\Lambda^k E = 0$ for every $k > N$, so $\lambda_t(E)$ is a polynomial in t of degree at most N. If X is connected, the rank is a constant n and the degree is exactly n, with leading coefficient the class of the determinant line bundle $\Lambda^n E$.

***Proof*:**
Local triviality makes the rank locally constant, so its level sets are open and cover X. Compactness extracts a finite subcover, so the rank takes finitely many values and attains a maximum N. Over a point x with $\dim_{\mathbb{C}} E_x = m$ one has $\Lambda^k(E_x)=0$ whenever $k>m$, and in particular whenever $k>N$ ([5, Bases of the Exterior Powers]). A bundle all of whose fibres vanish is the zero bundle. For connected X the rank is constant at n, and the same citation makes each $\Lambda^n(E_x)$ one-dimensional, so $\Lambda^n E$ is the line bundle $\det E$. That the degree is exactly n is a

statement about the *class*, not the bundle: $\Lambda^n E$ is a line bundle, so the rank homomorphism $K^0(X) \to \mathbb{Z}$ sends $[\Lambda^n E]$ to 1, and in particular $[\Lambda^n E] \neq 0$.

Corollary 4.1.3: The Total λ-Operation is Exponential

For complex vector bundles E, F over a compact Hausdorff space X,

$$\lambda_t(E \oplus F) \;=\; \lambda_t(E)\,\lambda_t(F) \;\in\; K^0(X)[[t]]$$

Moreover $\lambda_t(E)$ is a unit in $K^0(X)[[t]]$.

***Proof*:**
Taking classes in $K^0(X)$ in proposition 1.1.19 and using that $[\,\cdot\,]$ carries $\oplus$ to $+$ and $\otimes$ to the product of proposition 2.1.10,

$$\lambda^k(E \oplus F) \;=\; \sum_{i+j=k} \lambda^i(E)\,\lambda^j(F)$$

which is the coefficient of t^k in $\lambda_t(E)\lambda_t(F)$. For the second claim, $\lambda_t(E)$ has constant term $\lambda^0(E) = 1$, and a power series over a commutative ring with unit constant term is invertible: the coefficients of the inverse are determined recursively by $b_0 = 1$ and $b_k = -\sum_{i=1}^{k} \lambda^i(E)\,b_{k-i}$.

This is the whole content of the extension. Write

$$\Lambda^+(X) \;=\; 1 + t\,K^0(X)[[t]]$$

for the set of power series with constant term 1, regarded as a group under multiplication of power series. It is a group by the recursion just used, and it is *abelian* because $K^0(X)$ is a commutative ring. Commutativity is the property that the universal property below requires. Corollary 4.1.3 says that λ_t is a homomorphism from the monoid $(\mathrm{Vect}_{\mathbb{C}}(X), \oplus)$ to the *group* $\Lambda^+(X)$, and the Grothendieck construction was built to extend exactly such a map.

Theorem 4.1.4: The λ-Operations on K^0

Let X be a compact Hausdorff space. There is a unique group homomorphism

$$\lambda_t \colon \big(K^0(X), +\big) \longrightarrow \Lambda^+(X)$$

whose restriction along $\mathrm{Vect}_{\mathbb{C}}(X) \to K^0(X)$ is the total λ-operation of definition 4.1.1. Writing $\lambda^k(x)$ for the coefficient of t^k in $\lambda_t(x)$, the resulting operations $\lambda^k \colon K^0(X) \to K^0(X)$ satisfy

1. $\lambda^0(x) = 1$ and $\lambda^1(x) = x$ for every $x \in K^0(X)$
2. $\lambda^k(x+y) = \sum_{i+j=k} \lambda^i(x)\, \lambda^j(y)$ for all $x, y \in K^0(X)$ and $k \geq 0$
3. $\lambda^k([E]) = [\Lambda^k E]$ for the class of a bundle E, and for a difference, $\lambda_t([E]-[F]) = \lambda_t(E)\, \lambda_t(F)^{-1}$.

The operations are natural: for a map $f \colon Y \to X$ of compact Hausdorff spaces, $f^*\lambda^k(x) = \lambda^k(f^*x)$.

***Proof*:**
By corollary 4.1.3 the map $\lambda_t \colon \mathrm{Vect}_{\mathbb{C}}(X) \to \Lambda^+(X)$ is a homomorphism of commutative monoids, from the monoid of isomorphism classes under $\oplus$ to a group. The Grothendieck group $K^0(X)$ of definition 2.1.1 is the universal such target: any monoid homomorphism from $\mathrm{Vect}_{\mathbb{C}}(X)$ to an abelian group factors uniquely through it. This gives existence and uniqueness of the extension, and clause (3) is the statement that the factorization restricts correctly together with the homomorphism property applied to $[E]-[F]$.

For (1), $\lambda^0(x) = 1$ holds for every x because the target $\Lambda^+(X)$ consists by definition of series with constant term 1. For λ^1, comparing coefficients of t^1 in $\lambda_t(x+y) = \lambda_t(x)\lambda_t(y)$ gives $\lambda^1(x+y) = \lambda^1(x) + \lambda^1(y)$, so λ^1 is additive and agrees with the identity on the generating classes $[E]$, hence everywhere. Clause (2) is the coefficient of t^k in the homomorphism property $\lambda_t(x+y) = \lambda_t(x)\lambda_t(y)$.

Naturality holds on bundles, since $f^*\Lambda^k E \cong \Lambda^k f^* E$ (both are glued from the functions $\Lambda^k g_{\beta\alpha} \circ f$), so $f^* \circ \lambda_t$ and $\lambda_t \circ f^*$ are two monoid homomorphisms $\mathrm{Vect}_{\mathbb{C}}(X) \to \Lambda^+(Y)$ that agree. By the uniqueness half of the universal property their extensions to $K^0(X)$ agree.

Two cautions are worth stating explicitly, because both are places where the formal extension behaves unlike the geometric operation it came from.

4.1.5 Note: λ^k of a Virtual Class Need Not Vanish For the class of a bundle, lemma 4.1.2 makes $\lambda_t(E)$ a polynomial, so $\lambda^k(E) = 0$ for large k. This *fails* for virtual classes. If $x = [E] - [F]$ then $\lambda_t(x) = \lambda_t(E)\lambda_t(F)^{-1}$, and the inverse of a polynomial is in general an infinite series, so $\lambda^k(x)$ can be nonzero for arbitrarily large k. The simplest witness is $x = -[L]$ for a line bundle L on a nonempty X, where $\lambda_t(x) = (1 + [L]t)^{-1} = \sum_{k\geq 0}(-1)^k[L]^k t^k$, so $\lambda^k(-[L]) = (-1)^k[L^{\otimes k}]$ for every k. These really are nonzero: $L^{\otimes k}$ is a line bundle, so $[L^{\otimes k}]$ has rank 1 at every point of X and therefore cannot be 0 in $K^0(X)$. Only the operations γ^i of section 4.3, which are built from the λ^i by a substitution, restore a finiteness statement on virtual classes.

The second caution concerns what the word λ-ring is being made to mean, and it is a genuine fork in the literature rather than a matter of taste.

Definition 4.1.6: λ-Ring

A *λ-ring* is a commutative ring R equipped with functions $\lambda^k\colon R \to R$ for $k \geq 0$ such that $\lambda^0(x) = 1$, $\lambda^1(x) = x$, and

$$\lambda^k(x+y) \;=\; \sum_{i+j=k} \lambda^i(x)\,\lambda^j(y)$$

for all $x, y \in R$ and all $k \geq 0$. A morphism of λ-rings is a ring homomorphism commuting with every λ^k.

Corollary 4.1.7: $K^0(X)$ is a λ-Ring

For X compact Hausdorff, $K^0(X)$ with the operations of theorem 4.1.4 is a λ-ring, and $f \mapsto f^*$ makes K^0 a contravariant functor into λ-rings.

***Proof*:**
The axioms are clauses (1) and (2) of theorem 4.1.4, and the functoriality statement is its naturality clause together with the fact that f^* is a ring homomorphism (proposition 2.1.10).

4.1.8 Convention: λ-Ring Versus Special λ-Ring Part of the literature reserves *λ-ring* for a ring satisfying two further axioms, which prescribe λ^k on a product and on an iterate,

$$\lambda^k(xy) = P_k\big(\lambda^1 x, \ldots, \lambda^k x; \lambda^1 y, \ldots, \lambda^k y\big), \qquad \lambda^k\big(\lambda^l(x)\big) = P_{k,l}\big(\lambda^1 x, \ldots, \lambda^{kl} x\big)$$

for certain universal polynomials P_k and $P_{k,l}$ with integer coefficients, defined through symmetric functions. Such a ring is a *special* λ-ring, and what definition 4.1.6 calls a λ-ring is then a *pre-*λ-ring. This book uses the weaker convention throughout, following Husemoller.
$K^0(X)$ *is* a special λ-ring, but nothing this chapter builds needs that fact, and the weaker structure is what gets used. The Adams operations of section 4.2 obtain their multiplicativity and composition laws $\psi^k(xy) = \psi^k(x)\psi^k(y)$ and $\psi^k\psi^l = \psi^{kl}$ from the splitting principle theorem 2.5.16, which verifies them where every class is a sum of line bundles and transports them back along an injection. The γ-operations of section 4.3 are defined from the λ^i by the substitution $\gamma_t(x) = \lambda_{t/(1-t)}(x)$ and inherit exactly the axioms of definition 4.1.6.
Neither does the third, though the reason is worth seeing, because the obvious argument fails. That ψ^k acts on gr^i_γ by k^i (section 4.3) cannot be proved by checking a *congruence* upstairs and pushing it down. Injectivity of p^* gives no way to conclude $a \in F^{i+1}(X)$ from $p^*a \in F^{i+1}(F(V))$, because a filtration is not detected by an injection of groups. What is transported instead is an *identity.* Over $F(V)$ one has

$$\psi^k\big(\gamma^i(p^*x)\big) - k^i\,\gamma^i(p^*x) \;=\; Q\big(\gamma^1(p^*x), \ldots, \gamma^N(p^*x)\big)$$

where Q is a universal polynomial with integer coefficients, every monomial of which has weight at least $i+1$, which the fundamental theorem of symmetric polynomials over $\mathbb{Z}$ produces. Both sides are p^* of classes on X, so the identity itself descends, and its right-hand side is by construction a $\mathbb{Z}$-combination of generators of $F^{i+1}_\gamma(X)$. Nothing has to be detected inside a filtration. A reader who wants the two special axioms themselves can obtain them from the same splitting principle.

Finally, the computation that makes the splitting principle worth invoking at all. On a line bundle the total λ-operation is as short as it could be.

Lemma 4.1.9: λ-Operations on a Line Bundle

Let $L \to X$ be a complex line bundle over a compact Hausdorff space. Then

$$\lambda_t(L) \;=\; 1 + [L]\,t$$

that is, $\lambda^1(L) = [L]$ and $\lambda^k(L) = 0$ for all $k \geq 2$. Consequently, if $E \cong L_1 \oplus \cdots \oplus L_n$ is a sum of line bundles, then

$$\lambda_t(E) \;=\; \prod_{i=1}^{n}\big(1 + [L_i]\,t\big)$$

so that $\lambda^k(E)$ is the k-th elementary symmetric polynomial in the classes $[L_1], \ldots, [L_n]$.

***Proof*:**
A line bundle has rank 1, so lemma 4.1.2 gives $\Lambda^k L = 0$ for $k \geq 2$, while $\lambda^0(L) = 1$ and $\lambda^1(L) = [L]$. The product formula is corollary 4.1.3 applied $n-1$ times, and expanding $\prod_i(1 + [L_i]t)$ exhibits the coefficient of t^k as $e_k([L_1], \ldots, [L_n])$.

This last identity is the reason the splitting principle is the organizing tool of section 4.2. Any expression built from the λ^k becomes, on a sum of line bundles, a symmetric polynomial in the line-bundle classes. A statement about symmetric polynomials is a statement that can be checked.

Section 2.5 proved that $K^*(P(V))$ is a free module over $K^*(X)$ on the powers $1, [L_V], \ldots, [L_V]^{n-1}$ and deferred the ring structure to this section, on the grounds that the missing relation needs the multiplicativity of the total λ-operation. That debt is paid here. Two preliminary facts are needed, and the first is the tensor analogue of the Whitney formula.

Lemma 4.1.10: Exterior Powers of a Tensor With a Line Bundle

Let $W \to B$ be a vector bundle and $L \to B$ a line bundle. Then for every $k \geq 0$

$$\Lambda^k(W \otimes L) \;\cong\; \Lambda^k W \otimes L^{\otimes k}$$

***Proof*:**
Over a point, send $(w_1 \otimes l_1) \wedge \cdots \wedge (w_k \otimes l_k)$ to $(w_1 \wedge \cdots \wedge w_k) \otimes (l_1 \cdots l_k)$, the product taken in $L^{\otimes k}$. This is multilinear in the pairs (w_i, l_i), and it is alternating: interchanging two entries reverses the sign of $w_1 \wedge \cdots \wedge w_k$ and leaves $l_1 \cdots l_k$ fixed, since $L^{\otimes k}$ is commutative in its factors. So it descends to a linear map on $\Lambda^k(W_x \otimes L_x)$, natural in W and L. When $\dim L_x = 1$ it is surjective, because a basis vector s of L_x writes every element of $W_x \otimes L_x$ as $w \otimes s$, and both sides have dimension $\binom{\dim W_x}{k}$, so it is an isomorphism. Naturality makes the coordinate description of the assembled map independent of the point, exactly as in proposition 1.1.19, so the map is continuous and fibrewise a linear isomorphism, hence an isomorphism of bundles (lemma 1.1.6).

The second fact is where the multiplicativity does its work. Recall from lemma 4.1.2 that $\lambda_t(W)$ is a polynomial, so it may be evaluated at $t = -1$. Write $\lambda_{-1}(W) = \sum_k (-1)^k [\Lambda^k W]$.

Proposition 4.1.11: A Nowhere-Zero Section Kills λ_{-1}

Let $W \to B$ be a complex vector bundle over a compact Hausdorff space. If W admits a nowhere-zero section, then $\lambda_{-1}(W) = 0$ in $K^0(B)$.

***Proof*:**
A nowhere-zero section s spans a line subbundle $\langle s \rangle \subseteq W$. It is a subbundle because s extends to a local frame: in a trivialization near x, complete $s(x)$ to a basis of the fiber, and the completed family stays a basis at nearby points, so $\langle s \rangle$ is locally a direct summand of rank one. It is trivial because s trivializes it. A compact Hausdorff space is paracompact ([8, Compact Spaces, Discrete Spaces, and Euclidean Space](1)), so proposition 1.1.14 splits it off: $W \cong \mathbb{C} \oplus W'$ for some W'. By corollary 4.1.3, $\lambda_t(W) = \lambda_t(\mathbb{C})\, \lambda_t(W')$, and $\lambda_t(\mathbb{C}) = 1 + t$ by lemma 4.1.9. Evaluating at $t = -1$ makes the first factor 0, so $\lambda_{-1}(W) = 0$.

Theorem 4.1.12: The Tautological Relation on a Projective Bundle

Let $V \to X$ be a rank-n complex vector bundle over a compact Hausdorff space, $\pi\colon P(V) \to X$ its projectivization and L_V the tautological line bundle of definition 2.5.6. Then in $K^0(P(V))$

$$\sum_{k=0}^{n}(-1)^k\,\pi^*\lambda^k(V)\,[L_V]^{\,n-k} \;=\; 0$$

***Proof*:**
By definition 2.5.6 the bundle L_V is a subbundle of π^*V, so the inclusion is a nowhere-vanishing bundle map $L_V \to \pi^*V$, that is, a nowhere-zero section of $\mathrm{Hom}(L_V, \pi^*V) \cong \pi^*V \otimes L_V^\vee$. The base $P(V)$ is again compact Hausdorff. It is Hausdorff because π is a locally trivial map with compact fiber $\mathbb{C}\mathrm{P}^{n-1}$ ([14, Projective Bundles over a Paracompact Base]) over a Hausdorff base, so [8, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] applies. It is compact because X is compact Hausdorff, hence normal: a finite trivializing cover $\{U_i\}$ admits a shrinking $\{V_i\}$ with $\overline{V_i} \subseteq U_i$, each $\pi^{-1}(\overline{V_i}) \cong \overline{V_i} \times \mathbb{C}\mathrm{P}^{n-1}$ is compact, and finitely many of them cover $P(V)$. So proposition 4.1.11 gives

$$\lambda_{-1}\big(\pi^*V \otimes L_V^\vee\big) \;=\; 0$$

Now expand the left side. By lemma 4.1.10, $\Lambda^k(\pi^*V \otimes L_V^\vee) \cong \Lambda^k(\pi^*V) \otimes (L_V^\vee)^{\otimes k}$, while at the level of classes $[\Lambda^k(\pi^*V)] = \pi^*\lambda^k(V)$ by the naturality clause of theorem 4.1.4, which is all that is used. So

$$0 \;=\; \sum_{k=0}^{n}(-1)^k\,\pi^*\lambda^k(V)\,[L_V^\vee]^{\,k}$$

the sum stopping at n because π^*V has rank n. Finally $[L_V][L_V^\vee] = [L_V \otimes L_V^\vee] = 1$, since $L_V \otimes L_V^\vee \cong \mathrm{Hom}(L_V, L_V)$ is trivialized by the identity, so $[L_V]$ is a unit with inverse $[L_V^\vee]$. Multiplying the displayed identity by $[L_V]^n$ replaces $[L_V^\vee]^k$ by $[L_V]^{n-k}$ and gives the relation.

The relation holds over any compact Hausdorff base. Turning it into a *presentation* needs the free-module statement, which is proved only for finite cell complexes, so the corollary inherits that hypothesis and no more.

Corollary 4.1.13: The Ring $K^*(P(V))$

Let X be a finite cell complex and $V \to X$ a rank-n complex vector bundle. Then sending t to $[L_V]$ induces an isomorphism of $K^*(X)$-algebras

$$K^*(X)[t]\big/\Big(\textstyle\sum_{k=0}^{n}(-1)^k\lambda^k(V)\,t^{\,n-k}\Big) \;\xrightarrow{\;\cong\;}\; K^*(P(V))$$

***Proof*:**
Write $f(t) = \sum_{k=0}^{n}(-1)^k\lambda^k(V)t^{n-k}$. Its leading coefficient is $\lambda^0(V) = 1$, so f is monic of degree n. Its coefficients $\lambda^k(V)$ lie in $K^0(X)$, which is central in the graded-commutative ring $K^*(X)$, so f is a monic polynomial over a commutative subring and division by it behaves as usual: $K^*(X)[t]/(f)$ is a free $K^*(X)$-module with basis $1, t, \ldots, t^{n-1}$. By theorem 4.1.12 the class

$f([L_V])$ vanishes, so $t \mapsto [L_V]$ descends to a $K^*(X)$-algebra map on the quotient. That map carries the basis $1, t, \dots, t^{n-1}$ to $1, [L_V], \dots, [L_V]^{n-1}$, which is a basis of $K^*(P(V))$ as a free $K^*(X)$-module by theorem 2.5.15. A module map carrying a basis bijectively onto a basis is an isomorphism.

Taking X a point and $V = \mathbb{C}^{n+1}$ recovers a result already proved in chapter 2. There $\lambda^k(V) = \binom{n+1}{k}$, so the relation reads $\sum_k (-1)^k \binom{n+1}{k} [L]^{n+1-k} = ([L]-1)^{n+1} = 0$ in $K^0(\mathbb{C}\mathrm{P}^n)$, which is lemma 2.4.3. That lemma is proved where it stands because chapter 2 needs it, by a variant of this argument rather than this argument itself: it splits a trivial summand off $\mathrm{Hom}(V, H)$, the dual of the $\mathrm{Hom}(H, V)$ used here, which reaches the relation with no final multiplication.

4.2 The Adams Operations

The λ^k are not homomorphisms, and theorem 4.1.4(2) says exactly how badly they fail: $\lambda^k(x+y)$ is a convolution rather than a sum. That makes them awkward to compute with, and it is the reason a second family of operations is extracted from them. The Adams operations ψ^k are additive by construction, multiplicative as a theorem, and on a line bundle they are the k-th tensor power. The last property is what makes them computable, because theorem 2.5.16 reduces any identity among them to line bundles.

Two preliminaries are needed. The splitting principle as stated splits a single bundle of constant rank, while the multiplicativity of ψ^k involves two classes at once, over a base that need not be connected.

Lemma 4.2.1: Componentwise Reduction

Let X be a finite cell complex with connected components $X_1, \dots, X_r$. Each X_j is again a finite cell complex, r is finite, and restriction

$$K^0(X) \xrightarrow{\cong} \bigoplus_{j=1}^{r} K^0(X_j)$$

is a ring isomorphism commuting with every λ^k and every ψ^k. Consequently an identity built from the ring operations and these operations holds on X if and only if it holds on each X_j.

***Proof*:**
A finite cell complex is compact, so it has finitely many components, and each is clopen. Every cell is connected, hence lies in a single component, so the cell filtration of X restricts to one of each X_j, making X_j a finite cell complex. Restriction to a finite disjoint union of clopen pieces is an isomorphism by the additivity axiom of theorem 2.3.6, and it is a ring map because restriction along the inclusion $X_j \hookrightarrow X$ is (proposition 2.1.10). It commutes with the λ^k and ψ^k by their naturality (theorem 4.1.4, proposition 4.2.5(2)).

Lemma 4.2.2: Simultaneous Splitting

Let X be a *connected* finite cell complex and $x_1, \ldots, x_r \in K^0(X)$. There are a finite cell complex Y and a map $p\colon Y \to X$ such that $p^*\colon K^*(X) \to K^*(Y)$ is a split injection of $K^*(X)$-modules and each p^*x_i is a $\mathbb{Z}$-linear combination of classes of line bundles on Y. Moreover, if $\mathrm{rk} x_i = 0$ then p^*x_i has the sharper form $p^*x_i = \sum_r \big([L_{i,r}] - 1\big)$ for line bundles $L_{i,r}$ on Y.

***Proof*:**
Connectedness makes the rank of a bundle a single integer, so each x_i is a difference of a bundle and a trivial bundle: writing $x_i = [E_i'] - [F_i]$ and choosing G_i with $F_i \oplus G_i \cong \underline{\mathbb{C}}^{a_i}$ (proposition 1.1.15, available since a finite cell complex is compact Hausdorff by lemma 2.5.3(1)), one has $x_i = [E_i' \oplus G_i] - [\underline{\mathbb{C}}^{a_i}]$. Set $E_i = E_i' \oplus G_i$, and assume each E_i has rank at least 1, which costs nothing: replacing E_i by $E_i \oplus \underline{\mathbb{C}}$ and a_i by $a_i + 1$ leaves x_i unchanged.

Now split the E_i one at a time. Apply theorem 2.5.16 to E_1 over X, giving $p_1\colon Y_1 \to X$, then to $p_1^*E_2$ over Y_1, giving $p_2\colon Y_2 \to Y_1$, and so on for r stages. This is legal at every stage because theorem 2.5.16 takes a finite cell complex and returns one, so each Y_j is again an admissible base, and because pullback preserves the rank of a bundle, so each stage is applied to a bundle of constant rank ≥ 1. Splitting a bundle is preserved by further pullback, so after the last stage every p^*E_i is a sum of line bundles, where $p = p_1 \circ \cdots \circ p_r$ and $Y = Y_r$.

Each p_j^* is a split injection of modules, and a composite of split injections is a split injection, so p^* is one. Finally $[\underline{\mathbb{C}}^{a_i}] = a_i[\underline{\mathbb{C}}]$ with $\underline{\mathbb{C}}$ a line bundle, so each p^*x_i is a $\mathbb{Z}$-linear combination of line-bundle classes.

For the last clause, suppose $\mathrm{rk} x_i = 0$. Then $\mathrm{rk} E_i = a_i$, and pullback preserves rank, so the splitting $p^*E_i \cong L_{i,1} \oplus \cdots \oplus L_{i,m}$ has exactly $m = a_i$ summands. Hence $p^*x_i = \sum_{r=1}^{a_i} [L_{i,r}] - a_i = \sum_{r=1}^{a_i} \big([L_{i,r}] - 1\big)$. This is sharper than the previous sentence: a $\mathbb{Z}$-combination of line-bundle classes of total rank 0 need not be a sum of terms $[L] - 1$, so the form is read off from the construction rather than deduced from the rank.

The definition itself uses no splitting. It is a formal manipulation of the series λ_t, and the one thing to be careful about is integrality.

Definition 4.2.3: The Adams Operations

Let X be a compact Hausdorff space and $x \in K^0(X)$. Write $\lambda_{-t}(x) = \sum_{k\geq 0}(-1)^k \lambda^k(x)\, t^k$, which has constant term 1 and is therefore a unit in $K^0(X)[[t]]$, since theorem 4.1.4 lands λ_t in the group $\Lambda^+(X)$ of series with constant term 1. The *total Adams operation* is the logarithmic derivative

$$\psi_t(x) \;=\; -t\,\frac{d}{dt}\big(\lambda_{-t}(x)\big)\cdot \lambda_{-t}(x)^{-1} \;\in\; K^0(X)[[t]]$$

and the *Adams operations* $\psi^k \colon K^0(X) \to K^0(X)$ for $k \geq 1$ are its coefficients, $\psi_t(x) = \sum_{k\geq 1} \psi^k(x)\, t^k$.

4.2.4 Note: Why the Logarithmic Derivative and Not the Logarithm The additivity of ψ_t comes from turning a product into a sum, which is what a logarithm does, and it is tempting to define $\psi_t(x)$ as $-t\,\frac{d}{dt}\log \lambda_{-t}(x)$. That formula is correct and useless here: $\log(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$ divides by every integer, so it lands in $K^0(X)\otimes \mathbb{Q}[[t]]$ and destroys exactly the integral information these operations exist to carry.
The logarithmic *derivative* $f'f^{-1}$ needs no denominators. It is defined whenever f is a unit, and it is additive on products for the same formal reason the logarithm is, since $(fg)' = f'g + fg'$ gives $(fg)'(fg)^{-1} = f'f^{-1} + g'g^{-1}$ in any commutative ring. The whole construction stays over $\mathbb{Z}$.

Proposition 4.2.5: Basic Properties of the Adams Operations

Let X be a compact Hausdorff space. For every $k \geq 1$ the operation $\psi^k \colon K^0(X) \to K^0(X)$ is

1. *additive*: $\psi^k(x+y) = \psi^k(x) + \psi^k(y)$
2. *natural*: $f^*\psi^k(x) = \psi^k(f^*x)$ for a map $f \colon Y \to X$ of compact Hausdorff spaces
3. given by *Newton's recursion*

$$\psi^k(x) = \lambda^1(x)\psi^{k-1}(x) - \lambda^2(x)\psi^{k-2}(x) + \cdots + (-1)^k\lambda^{k-1}(x)\psi^1(x) + (-1)^{k+1}k\,\lambda^k(x)$$

so that in particular $\psi^1 = \mathrm{id}$ and $\psi^2(x) = x^2 - 2\lambda^2(x)$

***Proof*:**
For (1), theorem 4.1.4(2) says $\lambda_t(x+y) = \lambda_t(x)\lambda_t(y)$, and substituting $-t$ for t gives $\lambda_{-t}(x+y) = \lambda_{-t}(x)\lambda_{-t}(y)$. All three series are units, and the logarithmic derivative carries products to sums, as recorded in box 4.2.4. Multiplying by $-t$ preserves that, so $\psi_t(x+y) = \psi_t(x) + \psi_t(y)$, and comparing coefficients of t^k gives the claim. Part (2) holds because f^* is a ring homomorphism commuting with every λ^k (proposition 2.1.10 and theorem 4.1.4 respectively), hence commutes with the formal operations of differentiating, inverting and multiplying by $-t$ that produce ψ_t from λ_{-t}.

For (3), clear the inverse in definition 4.2.3: the definition says

$$-t\,\frac{d}{dt}\big(\lambda_{-t}(x)\big) \;=\; \psi_t(x)\,\lambda_{-t}(x)$$

an identity in $K^0(X)[[t]]$ with no denominators. The left side is $-t\sum_{m\geq 1}(-1)^m m\lambda^m(x)t^{m-1} = \sum_{m\geq 1}(-1)^{m+1}m\lambda^m(x)t^m$, so its t^k coefficient is $(-1)^{k+1}k\lambda^k(x)$. The right side has t^k coefficient $\sum_{m=1}^{k}(-1)^{k-m}\lambda^{k-m}(x)\psi^m(x)$. Equating and multiplying by $(-1)^k$,

$$-k\,\lambda^k(x) \;=\; \sum_{m=1}^{k}(-1)^m\lambda^{k-m}(x)\,\psi^m(x)$$

The term $m=k$ is $(-1)^k\psi^k(x)$, since $\lambda^0=1$. Solving for it and reindexing the remaining sum by $j=k-m$ gives the displayed recursion. Taking $k=1$ leaves $\psi^1(x)=\lambda^1(x)=x$, and $k=2$ gives $\psi^2(x)=\lambda^1(x)\psi^1(x)-2\lambda^2(x)=x^2-2\lambda^2(x)$.

Lemma 4.2.6: Adams Operations on Line Bundles

Let $L\to X$ be a complex line bundle over a compact Hausdorff space. Then

$$\psi^k([L]) \;=\; [L^{\otimes k}] \;=\; [L]^k$$

for every $k\geq 1$. Consequently, if $x=\sum_i c_i\,[L_i]$ is any $\mathbb{Z}$-linear combination of classes of line bundles, then $\psi^k(x)=\sum_i c_i\,[L_i]^k$. In the special case $x=\sum_{i=1}^n[L_i]-a$ this says ψ^k is the k-th power sum.

***Proof*:**
By lemma 4.1.9, $\lambda_t(L)=1+[L]t$, so $\lambda_{-t}(L)=1-[L]t$ and $\frac{d}{dt}\lambda_{-t}(L)=-[L]$. The inverse is $(1-[L]t)^{-1}=\sum_{m\geq 0}[L]^m t^m$, so

$$\psi_t([L]) \;=\; -t\cdot(-[L])\cdot\sum_{m\geq 0}[L]^m t^m \;=\; \sum_{m\geq 0}[L]^{m+1}t^{m+1} \;=\; \sum_{k\geq 1}[L]^k t^k$$

and $[L]^k=[L^{\otimes k}]$ by proposition 2.1.10. For the second statement, ψ^k is additive by proposition 4.2.5(1), so it respects $\mathbb{Z}$-linear combinations, and on the class of the trivial line bundle it gives $[\underline{\mathbb{C}}]^k=1$, so $\psi^k(a\cdot 1)=a$.

Everything so far holds over a compact Hausdorff base. The three results that follow do not. Their proofs run through theorem 2.5.16 and inherit its hypothesis of a finite cell complex.

Theorem 4.2.7: The Adams Operations are Ring Homomorphisms

Let X be a finite cell complex. For every $k\geq 1$ and all $x,y\in K^0(X)$,

$$\psi^k(xy) \;=\; \psi^k(x)\,\psi^k(y)$$

and $\psi^k(1)=1$, so ψ^k is a ring homomorphism.

***Proof*:**
By lemma 4.2.1 it is enough to prove the identity on each connected component, so assume X connected. By lemma 4.2.2 there is $p\colon Y \to X$ with p^* a split injection and $p^*x = \sum_i \ell_i - a$, $p^*y = \sum_j m_j - b$ with all ℓ_i, m_j classes of line bundles and $a, b \in \mathbb{Z}$. Expanding the product,

$$p^*(xy) \ = \ \sum_{i,j} \ell_i m_j \ - \ b \sum_i \ell_i \ - \ a \sum_j m_j \ + \ ab$$

and each $\ell_i m_j$ is again the class of a line bundle, the tensor product of two line bundles being a line bundle. So $p^*(xy)$ is of the shape lemma 4.2.6 handles, and applying it,

$$\psi^k\big(p^*(xy)\big) \ = \ \sum_{i,j} \ell_i^k m_j^k - b \sum_i \ell_i^k - a \sum_j m_j^k + ab \ = \ \Big(\sum_i \ell_i^k - a\Big)\Big(\sum_j m_j^k - b\Big)$$

using $(\ell_i m_j)^k = \ell_i^k m_j^k$. The right side is $\psi^k(p^*x)\,\psi^k(p^*y)$, again by lemma 4.2.6. Now naturality (proposition 4.2.5(2)) turns this into

$$p^*\big(\psi^k(xy)\big) \ = \ p^*\big(\psi^k(x)\psi^k(y)\big)$$

since p^* is a ring homomorphism, and p^* is injective, so the two classes agree on X. For the unit, $1 = [\mathbb{C}]$ is the class of a line bundle, so $\psi^k(1) = 1^k = 1$ by lemma 4.2.6.

Theorem 4.2.8: Composition of Adams Operations

Let X be a finite cell complex. Then $\psi^k \circ \psi^l = \psi^{kl}$ on $K^0(X)$ for all $k, l \geq 1$.

***Proof*:**
By lemma 4.2.1 assume X connected. Fix x and take $p\colon Y \to X$ as in lemma 4.2.2, with $p^*x = \sum_i \ell_i - a$. All three operations are additive and natural, so it suffices to compare them on p^*x, and by additivity on each summand. On the class of a line bundle, lemma 4.2.6 gives $\psi^l(\ell) = \ell^l$, which is again the class of a line bundle, so applying the same lemma a second time,

$$\psi^k\big(\psi^l(\ell)\big) \ = \ \psi^k(\ell^l) \ = \ (\ell^l)^k \ = \ \ell^{kl} \ = \ \psi^{kl}(\ell)$$

On the integer summand $-a$ both sides give $-a$. Hence $\psi^k\psi^l(p^*x) = \psi^{kl}(p^*x)$, and naturality plus injectivity of p^* give the identity on X.

The next property is the one that makes the ψ^k visible to torsion, and it is the reason the splitting principle's conclusion was recorded as a *split* injection rather than merely an injection.

Proposition 4.2.9: The Adams Operations Modulo a Prime

Let X be a finite cell complex and p a prime. Then for every $x \in K^0(X)$

$$\psi^p(x) \ \equiv \ x^p \pmod{p\,K^0(X)}$$

***Proof*:**
By lemma 4.2.1 assume X connected, the decomposition being compatible with reduction modulo p since it is a ring isomorphism. Take $q\colon Y \to X$ as in lemma 4.2.2, so $q^*x = \sum_{i=1}^n \ell_i - a$. In the commutative ring $K^0(Y)/p$ the p-th power map is additive, since the binomial coefficients $\binom{p}{j}$ for $0 < j < p$ are divisible by p. So modulo p,

$$(q^*x)^p \;\equiv\; \sum_{i=1}^n \ell_i^p - a^p \;\equiv\; \sum_{i=1}^n \ell_i^p - a \;=\; \psi^p(q^*x)$$

the middle step by Fermat's little theorem, $a^p \equiv a \pmod p$, and the last by lemma 4.2.6. So $\psi^p(q^*x) - (q^*x)^p$ lies in $p\,K^0(Y)$.

It remains to descend. The class $z = \psi^p(x) - x^p$ satisfies $q^*z \in p\,K^0(Y)$, and injectivity of q^* alone does not give $z \in p\,K^0(X)$, since an injection of abelian groups need not stay injective after reduction modulo p. What does give it is that q^* is a *split* injection of $K^*(X)$-modules (theorem 2.5.16(1), preserved by the composite in lemma 4.2.2): a split injection stays injective after applying $- \otimes_{\mathbb{Z}} \mathbb{Z}/p$, because a retraction r with $r \circ q^* = \mathrm{id}$ reduces to a retraction modulo p. So $\bar{z} = 0$ in $K^0(X)/p$, which is the claim.

Exercise 4.2.1

1. Let $h = [H] - 1$ be the reduced generator of $\widetilde{K}^0(S^2)$ of example 2.1.12, so that $h^2 = 0$. Show that $\psi^k(h) = k\,h$ for every $k \geq 1$.

2. With $K^0(S^2) \cong \mathbb{Z}\{1, h\}$ as in example 2.1.12, write down the matrix of ψ^k in the basis $\{1, h\}$, and check directly from it that $\psi^k\psi^l = \psi^{kl}$ on $K^0(S^2)$.

3. Let $L \to X$ be a line bundle over a compact Hausdorff space and put $u = [L] - 1$. Prove that
$$\lambda^k(u) \;=\; (-1)^{k-1}u \qquad \text{for every } k \geq 1$$
(Compute $\lambda_t(u) = (1 + [L]t)(1+t)^{-1}$ and read off the coefficients.) Note in particular that $\lambda^k(u)$ need *not* vanish for $k \geq 2$, even though u is the reduced class of a line bundle. It vanishes exactly when $[L] = 1$.

4. Using $K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[h]/(h^{n+1})$ with $h = [H] - 1$ (section 2.4), show that
$$\psi^k(h) \;=\; (1+h)^k - 1 \;=\; \sum_{j=1}^n \binom{k}{j} h^j$$
and deduce that ψ^k acts on the subgroup generated by h^n by multiplication by k^n.

5. Show that ψ^k is a polynomial with integer coefficients in $\lambda^1, \ldots, \lambda^k$ by unwinding Newton's recursion, and compute it explicitly for $k = 3$: verify $\psi^3 = (\lambda^1)^3 - 3\lambda^1\lambda^2 + 3\lambda^3$.

4.3 The γ-Filtration and Integrality

The Adams operations are additive and multiplicative, which makes them easy to apply but says nothing yet about *where* in $K^0(X)$ a class sits. Ordinary cohomology has a grading for this, and K^0

does not: it is $\mathbb{Z}/2$-graded and otherwise shapeless. What replaces the missing grading is a filtration, built by reindexing the operations of section 4.1, and the point of the construction is that ψ^k can see it. On the i-th graded piece ψ^k acts by k^i, which turns a question about position in the filtration into an eigenvalue computation.

Definition 4.3.1: The γ-Operations

Let X be a compact Hausdorff space. The γ-*operations* are defined by the substitution

$$\gamma_t(x) \;=\; \lambda_{t/(1-t)}(x) \;=\; \sum_{i\geq 0} \gamma^i(x)\, t^i$$

The substitution is legitimate over $\mathbb{Z}$: the series $t/(1-t) = t + t^2 + \cdots$ has zero constant term, so composing it into $\lambda_s(x) = \sum_k \lambda^k(x) s^k$ produces a well-defined element of $K^0(X)[[t]]$, and expanding $(1-t)^{-k} = \sum_{m\geq 0} \binom{k+m-1}{m} t^m$ gives the closed formula

$$\gamma^i(x) \;=\; \sum_{k=1}^{i} \binom{i-1}{i-k} \lambda^k(x) \qquad (i \geq 1), \qquad \gamma^0(x) = 1$$

Proposition 4.3.2: Properties of the γ-Operations

Let X be a compact Hausdorff space. Then

1. $\gamma^0(x) = 1$, $\gamma^1(x) = x$, and $\gamma^i(x+y) = \sum_{a+b=i} \gamma^a(x)\gamma^b(y)$, so $K^0(X)$ is a λ-ring in the sense of definition 4.1.6 with the γ^i in place of the λ^i
2. $\gamma^i(x) = \lambda^i(x+i-1)$ for $i \geq 1$
3. if L is a line bundle and $u = [L]-1$, then $\gamma_t(u) = 1 + ut$, so that $\gamma^1(u) = u$ and $\gamma^i(u) = 0$ for all $i \geq 2$
4. the γ^i are natural, and $\gamma^i(x)$ has rank 0 whenever x does and $i \geq 1$
5. if $x = [E] - n$ with $\mathrm{rk}E = n$ constant, then $\gamma^i(x) = 0$ for every $i > n$, so $\gamma_t(x)$ is a polynomial even though $\lambda_t(x)$ is not

***Proof*:**
For (1), γ^0 is the constant term of $\lambda_s(x)$ at $s=0$, which is 1, and the t-coefficient is $\lambda^1(x) = x$ from the closed formula with $i=1$. Substituting a fixed series with zero constant term is a ring homomorphism $K^0(X)[[s]] \to K^0(X)[[t]]$, so applying it to $\lambda_s(x+y) = \lambda_s(x)\lambda_s(y)$ (theorem 4.1.4(2)) gives $\gamma_t(x+y) = \gamma_t(x)\gamma_t(y)$, which is the addition law.

For (2), $\lambda_t(1) = 1+t$ by lemma 4.1.9, so $\lambda_t(m) = (1+t)^m$ for a positive integer m and $\lambda^b(m) = \binom{m}{b}$. Hence

$$\lambda^i(x+i-1) \;=\; \sum_{k+b=i} \lambda^k(x)\binom{i-1}{b} \;=\; \sum_{k=0}^{i} \binom{i-1}{i-k}\lambda^k(x)$$

and the $k=0$ term is $\binom{i-1}{i}=0$ for $i\geq 1$, leaving the formula of definition 4.3.1.

For (3), $\lambda_s(u)=\lambda_s(L)\lambda_s(1)^{-1}=(1+[L]s)(1+s)^{-1}$ by theorem 4.1.4(3) and lemma 4.1.9. Substituting $s=t/(1-t)$ gives $1+s=(1-t)^{-1}$, hence $(1+s)^{-1}=1-t$, and $1+[L]s=\big(1-t+[L]t\big)(1-t)^{-1}$. The two factors of $(1-t)^{-1}$ and $(1-t)$ cancel, leaving $\gamma_t(u)=1-t+[L]t=1+ut$.

For (4), naturality is inherited from theorem 4.1.4, since γ^i is an integral polynomial in the λ^k and f^* is a ring homomorphism. The rank claim is naturality again rather than multiplicativity of rk: choosing a point x_j in each component X_j and writing $i_j\colon\{x_j\}\hookrightarrow X$, the j-th component of the rank homomorphism of proposition 2.1.3 is i_j^* read through $K^0(\{x_j\})\cong\mathbb{Z}$ (example 2.1.2). So $\mathrm{rk}_j\gamma^i(x)=\gamma^i(\mathrm{rk}_jx)$, and when $\mathrm{rk}_jx=0$ the right-hand side is $\gamma^i(0)$, which vanishes for $i\geq 1$ because $\gamma_t(0)=\lambda_{t/(1-t)}(0)=1$.

For (5), use (2): $\gamma^i(x)=\lambda^i(x+i-1)$. If $i>n$ then $i-1\geq n$, so $x+i-1=[E]-n+(i-1)=[E\oplus\underline{\mathbb{C}}^{\,i-1-n}]$ is the class of an genuine bundle, of rank $n+(i-1-n)=i-1$. A bundle of rank $i-1$ has $\Lambda^i=0$ (lemma 4.1.2), so λ^i of its class vanishes. This is the finiteness statement promised in box 4.1.5: the λ^i of a virtual class run forever, and the γ^i do not.

Definition 4.3.3: The γ-Filtration

Let X be a compact Hausdorff space and write $K^0(X)_{\mathrm{rk}=0}=\ker\big(\mathrm{rk}\colon K^0(X)\to\mathbb{Z}^r\big)$ for the classes of rank zero on every component (proposition 2.1.3). Set $F^0_\gamma=K^0(X)$ and, for $i\geq 1$, let $F^i_\gamma(X)$ be the subgroup of $K^0(X)$ generated by all products

$$\gamma^{i_1}(x_1)\,\gamma^{i_2}(x_2)\cdots\gamma^{i_m}(x_m),\qquad m\geq 1,\quad x_j\in K^0(X)_{\mathrm{rk}=0},\quad i_1+\cdots+i_m\geq i$$

The number $i_1+\cdots+i_m$ is the *weight* of such a product.

Proposition 4.3.4: The γ-Filtration is a Ring Filtration

For X compact Hausdorff the subgroups $F^i_\gamma(X)$ satisfy

$$K^0(X)=F^0_\gamma\supseteq F^1_\gamma\supseteq F^2_\gamma\supseteq\cdots,\qquad F^i_\gamma\cdot F^j_\gamma\subseteq F^{i+j}_\gamma$$

and $F^1_\gamma(X)=K^0(X)_{\mathrm{rk}=0}$.

***Proof*:**
A product of weight at least $i+1$ has weight at least i, so the filtration is decreasing. Multiplying a generator of weight $\geq i$ by one of weight $\geq j$ produces a product of the same shape of weight $\geq i+j$, which gives multiplicativity. For the last claim, F^1_γ contains $\gamma^1(x)=x$ for every rank-zero x, so it contains $K^0(X)_{\mathrm{rk}=0}$. Conversely every generator of F^1_γ is a product of classes $\gamma^{i_j}(x_j)$ with $i_j\geq 1$ and $\mathrm{rk}x_j=0$, each of which has rank 0 by proposition 4.3.2(4), and rk is a ring homomorphism, so the product has rank 0.

The theorem below needs one fact about symmetric polynomials that the series does not currently carry in the form required. What is available is lemma 3.3.1, which handles power sums only, and [5, The Fundamental Theorem on Symmetric Functions], which is stated over a field, for rational functions, and without the grading. The graded integral statement is short enough to prove where it is used.

Lemma 4.3.5: Symmetric Polynomials, Integrally and by Weight

Let $P \in \mathbb{Z}[u_1, \ldots, u_n]$ be symmetric and homogeneous of degree d. Then P is a $\mathbb{Z}$-linear combination of monomials $e_{j_1} e_{j_2} \cdots e_{j_m}$ in the elementary symmetric polynomials $e_j = e_j(u_1, \ldots, u_n)$, in which every occurring monomial satisfies $j_1 + j_2 + \cdots + j_m = d$.

***Proof*:**
Order monomials $u^a = u_1^{a_1} \cdots u_n^{a_n}$ of degree d lexicographically by their exponent vectors. There are finitely many, so any strictly decreasing chain terminates, and induction on this order is legitimate.

Let u^a be the lexicographically largest monomial occurring in P, with coefficient $c \in \mathbb{Z}$. Because P is symmetric, every permutation of a also occurs, so maximality forces $a_1 \geq a_2 \geq \cdots \geq a_n$. Consider

$$E \; = \; e_1^{a_1 - a_2} \, e_2^{a_2 - a_3} \cdots e_{n-1}^{a_{n-1} - a_n} \, e_n^{a_n}$$

all exponents being non-negative by the monotonicity just established. The lexicographically largest monomial of e_j is $u_1 u_2 \cdots u_j$, and the largest monomial of a product is the product of the largest ones, so the largest monomial of E is $u_1^{a_1 - a_2}(u_1 u_2)^{a_2 - a_3} \cdots (u_1 \cdots u_n)^{a_n} = u^a$, with coefficient 1. Hence $P - cE$ is symmetric, homogeneous of degree d, and its largest monomial is strictly smaller than u^a, so the induction applies to it.

It remains to check the weight. Assigning e_j the weight j, the monomial E has weight

$$\sum_{j=1}^{n} j \, (a_j - a_{j+1}) \; = \; \sum_{j=1}^{n} a_j \; = \; d$$

with the convention $a_{n+1} = 0$, the middle equality being Abel summation. So every monomial produced at every stage has weight exactly d.

Theorem 4.3.6: The Adams Operations on the Associated Graded

Let X be a finite cell complex, $i \geq 1$ and $k \geq 1$. Then for every $x \in F_\gamma^i(X)$

$$\psi^k(x) \; \equiv \; k^{\,i} \, x \pmod{F_\gamma^{i+1}(X)}$$

That is, ψ^k preserves the γ-filtration and acts on $\mathrm{gr}_\gamma^i(X) = F_\gamma^i(X)/F_\gamma^{i+1}(X)$ as multiplication by k^i.

***Proof*:**
Step 1: reduction to a single γ^i. Both sides are additive in x, so it is enough to treat a generator $y = \gamma^{i_1}(x_1)\cdots\gamma^{i_m}(x_m)$ of weight $w = \sum_l i_l \geq i$. Suppose the case $m = 1$ is known, so that $\psi^k(\gamma^{i_l}(x_l)) = k^{i_l}\gamma^{i_l}(x_l) + z_l$ with $z_l \in F_\gamma^{i_l+1}$. Since ψ^k is multiplicative (theorem 4.2.7),

$$\psi^k(y) \;=\; \prod_{l=1}^{m}\Big(k^{i_l}\gamma^{i_l}(x_l) + z_l\Big) \;=\; k^w\, y \;+\; (\text{terms with at least one factor } z_l)$$

and every such term lies in F_γ^{w+1}, because proposition 4.3.4 adds weights and replacing one factor of weight i_l by one of weight $\geq i_l + 1$ raises the total. As $w \geq i$, both $k^w y$ and the error lie in the right places: $y \in F_\gamma^w \subseteq F_\gamma^i$, and if $w > i$ then $y \in F_\gamma^{i+1}$ and the congruence is trivial, so the only case to prove is $w = i$, which is $k^i y$ plus something in F_γ^{i+1}.

Step 2: splitting. So let x have rank 0 and consider $\gamma^i(x)$. By lemma 4.2.2 there is $p\colon Y \to X$ with p^* a split injection, and since $\mathrm{rk} x = 0$ that lemma's last clause gives the sharper form $p^*x = \sum_{r=1}^n\big([L_r] - 1\big) = \sum_{r=1}^n u_r$ with $u_r = [L_r] - 1$. The sharper form is needed and does not follow from the weaker one: a rank-zero $\mathbb{Z}$-combination of line-bundle classes need not be a sum of terms $[L] - 1$, as $-h$ on $\mathbb{C}\mathrm{P}^2$ shows. By proposition 4.3.2(3) each $\gamma_t(u_r) = 1 + u_r t$, so by the addition law

$$\gamma_t(p^*x) \;=\; \prod_{r=1}^{n}\big(1 + u_r t\big), \qquad \text{hence} \qquad \gamma^j(p^*x) \;=\; e_j(u_1, \ldots, u_n)$$

the j-th elementary symmetric polynomial in the u_r.

Step 3: an identity upstairs. By lemma 4.2.6, $\psi^k(u_r) = [L_r]^k - 1 = (1 + u_r)^k - 1 =: v_r$, and $v_r = k\,u_r + \binom{k}{2}u_r^2 + \cdots$ has no constant term and lowest term $k\,u_r$. Since ψ^k is a ring homomorphism it commutes with the polynomial e_i, so

$$\psi^k\big(\gamma^i(p^*x)\big) \;=\; \psi^k\big(e_i(u)\big) \;=\; e_i(v_1, \ldots, v_n)$$

Expanding, $e_i(v) = \sum_{|S|=i}\prod_{r\in S}\big(ku_r + O(u_r^2)\big) = k^i e_i(u) + R(u)$, where $R \in \mathbb{Z}[u_1, \ldots, u_n]$ is symmetric and every monomial of R has degree at least $i + 1$. Decomposing R into homogeneous pieces and applying lemma 4.3.5 to each,

$$e_i(v) - k^i e_i(u) \;=\; Q\big(e_1(u), \ldots, e_n(u)\big)$$

for a polynomial Q with integer coefficients, every monomial of which has weight at least $i + 1$.

Step 4: descending an identity, not a congruence. Rewriting Step 3 with $e_j(u) = \gamma^j(p^*x) = p^*\gamma^j(x)$ (proposition 4.3.2(4)) and using naturality of ψ^k,

$$p^*\Big(\psi^k\big(\gamma^i(x)\big) - k^i\gamma^i(x)\Big) \;=\; p^*\Big(Q\big(\gamma^1(x), \ldots, \gamma^n(x)\big)\Big)$$

Both sides are p^* of a class on X, and p^* is injective, so the identity holds in $K^0(X)$:

$$\psi^k\big(\gamma^i(x)\big) - k^i\gamma^i(x) \;=\; Q\big(\gamma^1(x), \ldots, \gamma^n(x)\big)$$

This is the step the naive argument gets wrong. Had Step 3 produced only a *congruence* modulo $F_\gamma^{i+1}(Y)$, injectivity of p^* would not transport it, because a filtration is not detected by an injection of groups: $p^*z \in F_\gamma^{i+1}(Y)$ does not imply $z \in F_\gamma^{i+1}(X)$. What is transported here is an equality of classes, which injectivity does handle.

Step 5: reading off the conclusion. Every monomial of Q has weight at least $i+1$, so the right-hand side is a $\mathbb{Z}$-combination of products $\gamma^{j_1}(x)\cdots\gamma^{j_m}(x)$ with $\sum j_l \geq i+1$ and $\mathrm{rk} x = 0$. By definition 4.3.3 each such product lies in $F_\gamma^{i+1}(X)$. Hence $\psi^k(\gamma^i(x)) \equiv k^i\gamma^i(x)$ modulo $F_\gamma^{i+1}(X)$, which is the case $m=1$ assumed in Step 1.

Corollary 4.3.7: The Graded Pieces as Eigenspaces

Let X be a finite cell complex whose γ-filtration is finite, so that $F_\gamma^N(X)=0$ for some N. Then for each $k \geq 2$ the operator $\psi^k \otimes \mathbb{Q}$ on $K^0(X)\otimes\mathbb{Q}$ is diagonalizable with eigenvalues among $1, k, k^2, \dots, k^{N-1}$, and its k^i-eigenspace maps isomorphically onto $\mathrm{gr}_\gamma^i(X)\otimes\mathbb{Q}$.

***Proof*:**
First the case $i=0$, which theorem 4.3.6 does not cover. On $K^0(\mathrm{pt}) \cong \mathbb{Z}$ the operation ψ^k is the identity: for an integer m, $\lambda_{-t}(m) = (1-t)^m$, so $-t\,\frac{d}{dt}(1-t)^m \cdot (1-t)^{-m} = mt(1-t)^{-1} = m(t+t^2+\cdots)$ and every $\psi^k(m)=m$. By the naturality argument of proposition 4.3.2(4) this makes $\mathrm{rk}\psi^k(x) = \mathrm{rk}x$, so ψ^k acts as the identity on $\mathrm{gr}_\gamma^0 = K^0(X)/F_\gamma^1$, which is the statement $(\psi^k - k^0)F_\gamma^0 \subseteq F_\gamma^1$.

Now set $P(T) = \prod_{i=0}^{N-1}(T-k^i)$. For each i in that range, $(\psi^k - k^i)$ carries F_γ^i into F_γ^{i+1}, by theorem 4.3.6 for $i \geq 1$ and by the previous paragraph for $i=0$. The factors are polynomials in ψ^k and so commute, and composing them in the order $i = 0, 1, \dots, N-1$ carries $F_\gamma^0 = K^0(X)$ into $F_\gamma^N = 0$. So P annihilates ψ^k on $K^0(X)$, hence on $K^0(X)\otimes\mathbb{Q}$.

For $k \geq 2$ the scalars $k^0, k^1, \dots, k^{N-1}$ are pairwise distinct, so P is a squarefree polynomial split over $\mathbb{Q}$. An operator annihilated by such a polynomial is diagonalizable, with no finiteness hypothesis needed: the partial-fraction decomposition of $1/P$ writes 1 as a combination of the $P/(T-k^i)$, which exhibits the identity as a sum of commuting projections onto the eigenspaces. Hence $K^0(X)\otimes\mathbb{Q} = \bigoplus_i V_{k^i}$.

Finally, the eigenspace decomposition refines the filtration. Each V_{k^i} lies in $F_\gamma^i \otimes \mathbb{Q}$, since applying $\prod_{j<i}(\psi^k - k^j)$ to it is multiplication by the nonzero scalar $\prod_{j<i}(k^i - k^j)$ while carrying F_γ^0 into F_γ^i. Comparing $\bigoplus_{j\geq i} V_{k^j} \subseteq F_\gamma^i \otimes \mathbb{Q}$ with the fact that the two sides have the same total dimension count across i forces equality at every stage, so $F_\gamma^i\otimes\mathbb{Q} = \bigoplus_{j\geq i} V_{k^j}$ and V_{k^i} maps isomorphically onto $\mathrm{gr}_\gamma^i(X)\otimes\mathbb{Q}$.

4.3.8 Note: The γ-Filtration Against the Skeletal One The γ-filtration is the K-theoretic stand-in for the grading of ordinary cohomology, and it is not the only candidate. K^* also carries the *skeletal* filtration, in which $F^i_{\text{skel}}(X)$ is the kernel of restriction to the $(i-1)$-skeleton. That is the filtration the Atiyah-Hirzebruch spectral sequence of [14, The Atiyah-Hirzebruch Spectral Sequence] converges to, built from the skeletal exact couple of [14, The Skeletal Exact Couple of a Generalized Theory], and its E_2 page is $H^p(X; K^q(\text{pt}))$. Both filtrations are available here, the second by citation upstream.

The two are compared as follows, and the boundary of what is proved sits closer than it looks. What is proved here is an eigenspace decomposition: corollary 4.3.7 splits $K^0(X) \otimes \mathbb{Q}$ into the k^i-eigenspaces of ψ^k, with the k^i-eigenspace isomorphic to $\mathrm{gr}^i_\gamma(X) \otimes \mathbb{Q}$. Separately, theorem 3.3.7 identifies $K^0(X) \otimes \mathbb{Q}$ with $\bigoplus_j H^{2j}(X; \mathbb{Q})$. Both are decompositions of the same rational vector space, and it is tempting to conclude that they agree piece by piece.

They do agree, but not for any reason proved in this book. Matching them needs the compatibility $\mathrm{ch}_j \circ \psi^k = k^j \,\mathrm{ch}_j$ between the Adams operations and the graded pieces of the Chern character, and that identity is not among the results of chapter 3, which develops ch before the ψ^k are available. So the sentence to avoid is "the γ-filtration is the Chern character's grading". What is available is two decompositions of the same space, and a dictionary between them that this book does not establish.

Integrally the two filtrations are *not* equal, and the theorem that their associated gradeds agree up to torsion is Atiyah-Tall's, which this book does not prove. Nothing below depends on it: **??** uses only the eigenvalue statement of theorem 4.3.6, on spaces where the filtration is computed by hand. The failure of integrality itself is the same phenomenon section 3.4 bounds and **??** corrects with the Todd genus.

Exercise 4.3.1

1. Compute γ^i on $\widetilde{K}^0(S^2)$. With $h = [H] - 1$ and $h^2 = 0$, show $\gamma^1(h) = h$ and $\gamma^i(h) = 0$ for $i \geq 2$, and deduce that $F^1_\gamma(S^2) = \mathbb{Z}h$ and $F^2_\gamma(S^2) = 0$.

2. Verify theorem 4.3.6 directly on S^2: check that ψ^k acts on $\mathrm{gr}^1_\gamma(S^2)$ by k, and reconcile this with the matrix computed in exercise 4.2.1 (2).

3. Let $X = \mathbb{C}\mathrm{P}^n$ and $h = [H] - 1$. Show that $\gamma^i(h) = 0$ for $i \geq 2$ but that $F^i_\gamma(\mathbb{C}\mathrm{P}^n) \supseteq \mathbb{Z}h^i$, so the filtration is generated by *products* rather than by single γ-operations. Deduce $F^i_\gamma(\mathbb{C}\mathrm{P}^n) = \mathbb{Z}h^i \oplus \cdots \oplus \mathbb{Z}h^n$ and identify $\mathrm{gr}^i_\gamma(\mathbb{C}\mathrm{P}^n)$. (For the inclusion $\subseteq$, use that the classes $[H^{\otimes m}] - 1$ generate $\ker \mathrm{rk}$.)

4. Use proposition 4.3.2(5) to show that a single class $x = [E] - n$ with $\mathrm{rk} E = n$ has only finitely many nonzero $\gamma^i(x)$. Then explain why this does *not* by itself give $F^N_\gamma(X) = 0$ for some N, which is the hypothesis corollary 4.3.7 requires, and show that $F^N_\gamma = 0$ is strictly stronger than nilpotence of $\ker \mathrm{rk}$ by considering the λ-ring $\mathbb{Z}[x]/(x^2)$ with $\lambda_t(x) = 1 + xt$.

5. Fix $k \geq 2$ and show that the conclusion of theorem 4.3.6 fails if k^i is replaced by k^{i+1}, by exhibiting a class in $F^1_\gamma(S^2)$ on which ψ^k acts by k and not by k^2. Where does $k \geq 2$ enter?